Atiyah-Macdonald Exercises

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1 Rings and Ideals (Done)

1.1 Exercise

Proof. By assumption, $x^m = 0$ for some m. Then,

$$(1+x)\sum_{i=0}^{m-1}(-1)^i x^i = 1 + (-1)^{m-1}x^m = 1,$$

so $1+x$ is a unit.

Let x be nilpotent and u be a unit. Then, $u + x = u(1 + u^{-1}x)$. But $u^{-1}x$ is nilpotent since x is. By the previous result, $1 + u^{-1}x$ is a unit and hence $u(1 + u^{-1}x) = u + x$ is also a unit. □

1.2 Exercise

(i) Proof. We will prove this by induction. Suppose f has degree 0 with inverse $g = b_0 + \cdots + b_m x^m$. Then, $1 = fb_0 + \cdots + fb_m x^m$. Therefore, since $f \in A$ and $fb_0 = 1$, f is a unit. The converse is trivial since $f = a_0$, proving the base case.

Now, suppose that for all polynomials of degree $n-1$, the proposition holds. Suppose f has degree n and has an inverse $g = b_0 + \cdots + b_m x^m$. Then,

$$\begin{aligned} 1 &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} b_j x^j\right) \\ &= \sum_{k=0}^{n+m}\left(\sum_{i=0}^{k} a_i b_{k-i}\right) x^i. \end{aligned}$$

This implies $a_n b_m = 0$. We'll show by induction on r that $a_n^{r+1} b_{m-r} = 0$, the base case already proven. We assume this is true for all $r < l$. Multiplying by a_n^l,

$$\begin{aligned} a_n^l &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m} a_n^l b_j x^j\right) \\ &= \left(\sum_{i=0}^{n} a_i x^i\right) \cdot \left(\sum_{j=0}^{m-l} a_n^l b_j x^j\right), \end{aligned}$$

where we used the inductive hypothesis to remove higher terms: if $j > m-l$, then $l-(m-j+1) \geq 0$, so $a_n^l b_j = a_n^{l-(m-j+1)}(a_n^{m-j+1} b_{m-(m-j)}) = 0$. The coefficient of highest degree is $a_n \cdot a_n^l b_{m-l}$, so $a_n^{l+1} b_{m-l} = 0$. By induction, $a_n^{r+1} b_{m-r} = 0$ for all r.

From the expansion of $f \cdot g = 1$, we know that $a_0 b_0 = 1$. Then, using our result,

$$a_n^{m+1} = a_0(a_n^{m+1} b_{m-m}) = a_0 \cdot 0 = 0,$$

so a_n is nilpotent. Then, $-a_n x^n$ is nilpotent. Since f is a unit, by Exercise 1.1, $f + (-a_n x^n)$ is also a unit. However, it has degree $n-1$. By the inductive hypothesis, a_0 is a unit and a_i is nilpotent for $1 \leq i < n$. We've shown a_n is nilpotent, so this direction is complete.

Conversely, suppose a_0 is a unit and that a_i is nilpotent for $1 \leq i \leq n$. Then, $f - a_n x^n$ has degree $n-1$, so by the inductive hypothesis, $f - a_n x^n$ is a unit since its coefficients have the right properties. However, $a_n x^n$ is nilpotent since a_n is, so by Exercise 1.1, $(f - a_n x^n) + a_n x^n = f$ is a unit, so both directions are complete. By induction, the proposition follows for f of any degree. □

(ii) Proof. We'll prove this by induction on the degree. The base case is trivial since $f = a_0$ in degree 0. Now, suppose for all degrees less than n, the proposition holds.

If f has degree n and is nilpotent, there is an m such that $f^m = 0$. Expanding, the top term is $a_n^m x^{nm}$, which is zero. Therefore, a_n is nilpotent. However, $g = f + (-a_n x^n)$ is also nilpotent since it is the sum of two nilpotent elements (if a and b are the required nilpotent powers, take $a + b$; this power will annihilate all terms in the binomial expansion of g). g has degree at most $n - 1$, so by the inductive hypothesis $a_0, \ldots, a_{n-1}$ are all nilpotent. Since a_n is nilpotent, this direction is complete.

Conversely, if f has degree n and all coefficients are nilpotent, take $f = g + a_n x^n$, where g has degree at most $n - 1$. By the inductive hypothesis, g is nilpotent, so f is the sum of two nilpotent elements. Thus, f is nilpotent. By induction, the proposition follows for f of any degree. □

(iii) Proof. The reverse implication is by definition. For the forward implication, suppose f is a zero-divisor. The set of all possible g such that $fg = 0$ has minimal degree m since it is non-empty. Choose g of degree m such that $fg = 0$. If the highest terms are $a_n x^n$ and $b_m x^m$, then $a_n b_m x^{n+m} = 0$ by $fg = 0$, so $a_n b_m = 0$. Note that $a_n g$ annihilates the term of degree m and annihilates f, so $a_n g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $a_{n-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $a_{n-r} g = 0$ for all $r < k$. Then,

$$0 = g \sum_{i=0}^{n} a_i x^i = \sum_{i=0}^{n} (g a_i) x^i = \sum_{i=0}^{n-k} (g a_i) x^i.$$

The top term is $b_m a_{n-k} x^{m+n-k}$, so $b_m a_{n-k} = 0$. Thus, $a_{n-k} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $a_{n-k} g = 0$. By induction, $a_{n-r} g = 0$ for all r.

For any a_i, we have $a_i g = 0$. Thus,

$$0 = a_i \left(b_0 + \cdots + b_m x^m\right) = a_i b_0 + \cdots + a_i b_m x^m.$$

By definition, $b_m \neq 0$, yet $a_i b_m x^m = 0$. So, $a_i b_m = 0$ for all i. Thus,

$$b_m f = b_m \sum_{i=0}^{n} (b_m a_i) x^i = 0,$$

proving the claim. □

Remark. *For use in Exercise 1.3, we just proved: if $fg = 0$, g annihilates every coefficient of f.*

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x] \to (A/\mathfrak{m})[x]$, $0 = \overline{fg} = \overline{f}\,\overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then,

$$fg = \sum_{k=0}^{n+m} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k.$$

Obviously, $a_i b_{k-i} \in \mathfrak{m}$ since $a_i \in \mathfrak{m}$, so any finite sum of these terms is also. Therefore, the coefficients of fg all lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.3 Exercise

Here, we will frequently use the fact that

$$A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n],$$

which is easy to see by simply grouping terms on the left-hand side by x_n-degree.

(ii) Proof. We'll prove this by induction, using Exercise 1.2(ii), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Let $f \in A[x_1, \ldots, x_n]$. We can write $f = f_0 + \cdots + f_m x_n^m \in A[x_1, \ldots, x_{n-1}][x_n]$. Assume f is nilpotent. By Exercise 1.2(ii), each f_i is nilpotent. Then, by the inductive hypothesis for (ii), all coefficients are nilpotent for each f_i and thus for f. Conversely, if all coefficients of f are nilpotent, then all coefficients of each f_i are nilpotent. By the inductive hypothesis for (ii), each f_i is nilpotent. By Exercise 1.2(ii), f is nilpotent. By induction, the result follows. □

(i) Proof. We'll prove this by induction, using Exercise 1.2(i), which establishes the base case. Now, assume the inductive hypothesis up to $n-1$.

Suppose f is a unit in $A[x_1, \ldots, x_n] = A[x_1, \ldots, x_{n-1}][x_n]$. Then, $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, where $f_i \in A[x_1, \ldots, x_{n-1}]$. By the inductive hypothesis for (i), f_0 is a unit and f_i is nilpotent for $i > 0$. By the inductive hypothesis for (ii), all the coefficients of f_i are nilpotent for $i > 0$. By the inductive hypothesis for f_0, the constant coefficient is a unit, and all other coefficients are nilpotent. Thus, viewing f in $A[x_1, \ldots, x_n]$, the constant term is a unit and all other coefficients are nilpotent.

Conversely, suppose f in $A[x_1, \ldots, x_n]$ with the constant term a unit and all other coefficients nilpotent. Then, we can group terms so that $f = f_0 + f_1 x_n + \cdots + f_m x_n^m$, viewing $f \in A[x_1, \ldots, x_{n-1}][x_n]$. Then, f_0 has a unit for a constant term, with all other coefficients being nilpotent. Thus, f_0 is a unit by the inductive hypothesis for (i). Likewise, f_i for $i > 0$ has all nilpotent coefficients, so Exercise 1.3(ii) shows that f_i is nilpotent. By Exercise 1.2(i), f is a unit. By induction, (i) is proven in the general case. □

(iii) Proof. First, we start with a lemma: we want to show by induction that for $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$, we have that g annihilates every coefficient of f. The base case is shown in Exercise 1.2(iii).

Suppose f is a zero-divisor and $f \in A[x_1, \ldots, x_n] \cong A[x_1, \ldots, x_{n-1}][x_n]$. The set of all possible g such that $fg = 0$ has minimal degree m in x_n since it is non-empty. Choose g of degree m in x_n such that $fg = 0$. If the highest terms are $f_k x_n^k$ and $g_l x_n^l$, then $f_k g_l x^{k+l} = 0$ by $fg = 0$, so $f_k g_l = 0$. Note that $f_k g$ annihilates the term of degree m and annihilates f, so $f_k g = 0$; otherwise, we would be violating the minimality of m.

We will show by induction that $f_{k-r} g = 0$, the base case just proven. For the inductive hypothesis, suppose $f_{k-r} g = 0$ for all $r < q$. Then,

$$0 = g \sum_{i=0}^{k} f_i x_n^i = \sum_{i=0}^{k} (g f_i) x_n^i = \sum_{i=0}^{k-q} (g f_i) x_n^i.$$

The top term is $g_l f_{k-q} x_n^{l+k-q}$, so $g_l f_{k-q} = 0$. Thus, $f_{k-q} g$ has degree less than m and annihilates f. Again, to preserve the minimality of m, we have $f_{k-q} g = 0$. By induction, $f_{k-r} g = 0$ for all r.

For any f_i, we have $f_i g = 0$. Thus,

$$0 = f_i \left(g_0 + \cdots + g_m x_n^m\right) = f_i g_0 + \cdots + f_i g_m x_n^m.$$

For any g_j, we have $f_i g_j x_n^j = 0$. So, $f_i g_j = 0$ for all i and j. By induction, since $f_i, g_j \in A[x_1, \ldots, x_{n_1}]$, g_j annihilates every coefficient of f_i. Since this is true for all j, g annihilates every coefficient of f_i for all i. By induction, for all n, if $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then g annihilates every coefficient of f individually. This proves the lemma.

With this lemma, we can prove (iii). The reverse implication is by definition. For the forward implication, we prove (iii) by induction, using the base case established by Exercise 1.2(iii). Assume the inductive hypothesis up to $n-1$ and that $f, g \in A[x_1, \ldots, x_n]$, $fg = 0$. By our lemma, and since f and g are symmetrically defined, f annihilates every coefficient of g individually; since g is nonzero, it has some nonzero coefficient $g_\beta \in A$. Thus, $g_\beta f = 0$. By induction, if for any n, $f, g \in A[x_1, \ldots, x_n]$ and $fg = 0$, then $af = 0$ for some nonzero $a \in A$. □

(iv) Proof. Suppose fg is not primitive. Then, the ideal generated by its coefficients lie in a maximal ideal $\mathfrak{m}$. Thus, under the ring homomorphism $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$, $0 = \overline{fg} = \overline{f} \cdot \overline{g}$. If $\overline{f}$ is a zero divisor, then by (iii), there is a nonzero $a \in A/\mathfrak{m}$ such that $a\overline{f} = 0$. But, a is a unit since $A/\mathfrak{m}$ is a field, so $\overline{f} = 0$. The same argument applies to $\overline{g}$. Thus, either $\overline{f}$ or $\overline{g}$ is 0; pulling back whichever is zero, its coefficients must all lie in $\mathfrak{m}$, violating primitivity.

Conversely, suppose f or g is not primitive. Without loss of generality, we can say f is not primitive, so its coefficients lie in a maximal ideal $\mathfrak{m}$. Then, consider the same map as before, $A[x_1, \ldots, x_n] \to (A/\mathfrak{m})[x_1, \ldots, x_n]$. Then, $\overline{fg} = \overline{f} \cdot \overline{g} = 0 \cdot \overline{g} = 0$. In particular, all the coefficients of fg lie in $\mathfrak{m}$, so fg is not primitive, concluding the proof. □

1.4 Exercise

Proof. Since all maximal ideals are prime ideals, the nilradical (intersection of all prime ideals) is contained in the Jacobson radical (intersection of all maximal ideals) for any ring. Thus, we need to show that any $f(x)$ in the Jacobson radical is nilpotent.

By Proposition 1.9, if $f(x)$ is in the Jacobson radical, then $1 + xf(x)$ is a unit in $A[x]$. We can write this as

$$1 + xf(x) = 1 + a_0 x + \cdots + a_n x^{n+1}.$$

By Exercise 1.2(i), $a_0, \ldots, a_n$ are nilpotent. Thus, by Exercise 1.2(ii), f is nilpotent. □

1.5 Exercise

(i) Proof. Suppose $f = \sum_i a_i x^i \in A[[x]]$ is a unit so that there is some $g = \sum_i b_i x^i \in A[[x]]$ yielding $fg = 1$. Expanding, the constant term is $a_0 b_0 = 1$. Thus, a_0 is a unit in A.

We will prove the converse by induction; we will show that for all n, we can construct some $g_n \in A[x]$ (realized as a subring of $A[[x]]$) of degree $n-1$ such that $fg_n - 1 \in (x^n)$ and $g_n \equiv g_r \mod (x^r)$ for $r < n$. Suppose $f = \sum_i a_i x^i$ and a_0 is a unit so that there is a b_0 yielding $a_0 b_0 = 1$. Let $g_0 = b_0$. Then,

$$fg_0 - 1 = \sum_{i=0}^{\infty} a_i b_0 x^i - 1 = \sum_{i=1}^{\infty} a_i b_0 x^i + (a_0 b_0 - 1) = \sum_{i=1}^{\infty} a_i b_0 x^i \in (x^1),$$

the ideal generated by x. This proves the base case. Our inductive hypothesis is that there is some $g_{n-1} = \sum_{i=0}^{n-1} b_i x^i$ such that $fg_{n-1} - 1 \equiv 0 \mod (x^n)$. Define $b_n = -a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i}$, recalling that

a_0 is a unit. We claim that $g_n = \sum_{i=0}^{n} b_i x^i$ satisfies the proper congruence. We have

$$\begin{aligned}
fg_n &= \sum_{k=0}^{\infty} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k \\
&= 1 + \left(\sum_{i=0}^{n} a_i b_{n-i} \right) x^n + \sum_{k=n+1}^{\infty} \left(\sum_{i=0}^{k} a_i b_{k-i} \right) x^k \\
&\equiv 1 + \left(a_0 b_n + \sum_{i=1}^{n} a_i b_{n-i} \right) x^n \mod (x^{n+1}) \\
&\equiv 1 + \left(a_0 \left(-a_0^{-1} \sum_{i=1}^{n} a_i b_{n-i} \right) + \sum_{i=1}^{n} a_i b_{n-i} \right) x^n \mod (x^{n+1}) \\
&\equiv 1 + 0 \cdot x^n \mod (x^{n+1}) \\
&\equiv 1 \mod (x^{n+1}).
\end{aligned}$$

Note that since only the x^n coefficient was changed, the compatibility requirement holds: i.e., $g_n \equiv g_r \mod (x^r)$ for $r < n$. Thus, if we let $g = \sum_{i=0}^{\infty} b_i x^i$, then $fg - 1 \equiv 0 \mod (x^n)$ for all n. Once we show that $\bigcap_n (x^n) = (0)$, then we'll have $fg = 1$, proving the claim. But this is clear since any nonzero element $h \in A[[x]]$ has some nonzero coefficient h_j, so $h \notin (x^{j+1})$. □

(ii) Proof. Suppose $f = \sum_i a_i x^i$ is nilpotent. We will show that each a_i is nilpotent by induction. For the base case, note that if $f^m = 0$, then the constant term is $a_0^m = 0$, so $f_0 = a_0$ is nilpotent. For the inductive step, suppose that a_i is nilpotent for $i < n$. Then, since the nilradical is an ideal, $f - \sum_{i=0}^{n-1} a_i x^i = \sum_{i=n}^{\infty} a_i x^i$ is nilpotent. Thus, there is some m such that

$$0 = \left(\sum_{i=n}^{\infty} a_i x^i \right)^m \equiv a_n^m x^{nm} \mod (x^{nm+1}),$$

so a_n is nilpotent. The result follows by induction. □

Remark. *Consider the direct product ring $A = \Pi_{i=2}^{\infty} (\mathbb{Z}/2^i\mathbb{Z})$. Then, construct*

$$f = \sum_i a_i x^i,$$

where $a_i = (2, \ldots, 2, 0, \ldots)$; 2 appears until slot i, whereafter it is 0 indefinitely. Let $a_0 = a_1 = 0$ since there is no map into the ring. So, the minimal exponent that annihilates a_i is i whenever $i > 1$. Each a_i is therefore nilpotent.

Note that we have principal ideals $\mathfrak{q}_n = (1, \ldots, 1, 0, \ldots)$ that project the ring onto $\Pi_{i=2}^{n} (\mathbb{Z}/2^i\mathbb{Z})$; $a_i \in \mathfrak{q}_i$.

On the other hand, for any m, consider f^m. Then, consider $a_{m(m+1)}$. Note that the coefficient summands are the ordered partitions of $m(m+1)$ that have size m. There is precisely one summand that is not in the ideal $\mathfrak{q}_m$ and that is a_{m+1}^m, so the projection of the $x^{m(m+1)}$ coefficient onto $A/\mathfrak{q}_m$ is a_{m+1}^m. But, this is nonzero since a_{m+1}^m projects onto $\mathbb{Z}/2^{m+1}\mathbb{Z}$ as 2^m. Thus, f^m has a nonzero $x^{m(m+1)}$ coefficient.

In particular, for general commutative rings with unit (not necessarily Noetherian), if a_i is nilpotent for all i, this does not imply that $f = \sum_{i=0}^{\infty} a_i x^i$ is nilpotent.

(iii) Proof. Consider the ring homomorphism $\varphi : A[[x]] \to A$ that sends the power series to its constant term.

For every maximal ideal $\mathfrak{m}$ of A, $\varphi^{-1}(\mathfrak{m}) = \mathfrak{m} + xA[[x]]$ is an ideal, the contraction of $\mathfrak{m}$. Each of these are contained in maximal ideals of $A[[x]]$, so

$$\bigcap_{\mathfrak{m}} (\mathfrak{m} + xA[[x]]) \subset \mathfrak{R}(A[[x]])$$
$$\bigcap_{\mathfrak{m}} \mathfrak{m} + xA[[x]] \subset \mathfrak{R}(A[[x]])$$
$$\mathfrak{R}(A) + xA[[x]] \subset \mathfrak{R}(A[[x]])$$

In particular, if $a_0 \in \mathfrak{R}(A)$ and f has constant term a_0, then $f \in \mathfrak{R}(A[[x]])$.

Conversely, assume a_0 is not in the Jacobson radical of A. Then, there is some maximal ideal $\mathfrak{m}$ in A that does not contain a_0. But, $\mathfrak{m} + xA[[x]]$ is a non-trivial ideal of $A[[x]]$ under the contraction $\varphi^{-1}(\mathfrak{m})$. If $\mathfrak{m} + xA[[x]]$ is maximal, we'd have that f is not in the maximal ideal $\mathfrak{m} + xA[[x]]$ (since its constant term $a_0 \notin \mathfrak{m}$), so $f \notin \mathfrak{R}(A[[x]])$.

So, the proof is complete once we show that $\mathfrak{m} + xA[[x]]$ is maximal. Let $\mathfrak{r}$ be the maximal ideal containing $\mathfrak{m} + xA[[x]]$. In order for these ideals to be unequal, there is some $g \in \mathfrak{r}$ such that $g_0 \notin \mathfrak{m}$. However, projecting to A (inverse map of homomorphism $A \hookrightarrow A[[x]]$), we have an ideal $\mathfrak{r}'$ containing $\mathfrak{m}$ properly, so by the maximality of $\mathfrak{m}$, $\mathfrak{r} = A$. Thus, $\mathfrak{r} = A + xA[[x]] = A[[x]]$, violating maximality, so $\mathfrak{m} + A[[x]]$ is maximal. □

(iv) Proof. First, let's show that for any maximal ideal $\mathfrak{m}$ of A, $\mathfrak{m} + xA[[x]]$ is maximal. Let $\mathfrak{r}$ be the maximal ideal containing $\mathfrak{m} + xA[[x]]$. In order for these ideals to be unequal, there must be some $g \in \mathfrak{r}$ such that $g_0 \notin \mathfrak{m}$. However, projecting to A (inverse map of homomorphism $A \hookrightarrow A[[x]]$), we have an ideal $\mathfrak{r}'$ containing $\mathfrak{m}$ properly, so by the maximality of $\mathfrak{m}$, $\mathfrak{r} = A$. Thus, $\mathfrak{r} = A + xA[[x]] = A[[x]]$, violating maximality, so $\mathfrak{m} + A[[x]]$ is maximal.

Suppose $\mathfrak{m}$ is a maximal ideal of $A[[x]]$. Then, under the ring homomorphism $\varphi : A \hookrightarrow A[[x]]$, $\mathfrak{m}^c = \varphi^{-1}(\mathfrak{m})$ is an ideal of A. First of all, if $\mathfrak{m}^c = A$, then $\mathfrak{m}$ contains some series with constant term 1; by (i), it contains a unit, so $\mathfrak{m} = A[[x]]$, violating maximality.

If $\mathfrak{m}^c = 0$, then $\mathfrak{m}$ is contained in the ideal $xA[[x]]$, so $\mathfrak{m} = xA[[x]]$. Note that $xA[[x]]$ is maximal when A only has the ideals 0 and (1) (i.e., A is a field) since for any ideal I of A (not 0 or A), $I + xA[[x]]$ is a proper ideal containing $\mathfrak{m}$, violating the maximality of $\mathfrak{m}$. Otherwise, the only ideal containing $xA[[x]]$ is $A + xA[[x]] = A[[x]]$, the whole ring. For our purposes, just note that $\mathfrak{m}$ is indeed generated by x and $\mathfrak{m}^c$.

Thus, we are left with the case where $\mathfrak{m}^c$ is neither 0 nor A. Since it is a nontrivial ideal, it is contained in some maximal ideal $\mathfrak{r}$ of A. Thus, $\mathfrak{m} \subset \mathfrak{m}^c + xA[[x]] \subset \mathfrak{r} + xA[[x]]$. By the maximality of $\mathfrak{m}$, $\mathfrak{m}^c = \mathfrak{r}$, and $\mathfrak{m} = \mathfrak{m}^c + xA[[x]]$, completing the proof. □

(v) Proof. Let $\mathfrak{p}$ be a prime ideal of A. Consider the ring homomorphism $\varphi : A[[x]] \to A$ that maps each series to its constant term. Then, $\varphi^{-1}(\mathfrak{p}) = \mathfrak{p} + xA[[x]]$ is a prime ideal of $A[[x]]$ since contraction preserves primality. Under the ring homomorphism $\psi : A \hookrightarrow A[[x]]$, $\mathfrak{p}$ is the contraction of the prime ideal $\mathfrak{p} + xA[[x]]$, proving the claim. □

1.6 Exercise

Proof. Since all maximal ideals are prime, we know that $\mathfrak{N} \subset \mathfrak{R}$. Suppose to the contrary that the ideal $\mathfrak{R}$ is not contained in $\mathfrak{N}$. Then, by assumption, there is some nonzero idempotent $e \in \mathfrak{R}$.

Since $e^2 = e$, a simple inductive argument reveals that e cannot be nilpotent.

Let $\mathfrak{p}$ be any prime ideal. Consider the map $A \to A/\mathfrak{p}$. Because $A/\mathfrak{p}$ is an integral domain and $0 = \overline{e^2 - e} = \overline{e}(\overline{e} - 1)$, we have $e \in \mathfrak{p}$ or $e \in 1 + \mathfrak{p}$. If $e \in \mathfrak{p}$ for all prime ideals, then e must also be in nilradical, as it is the intersection of prime ideals. Thus, there is some prime ideal $\mathfrak{p}'$ such that $e \in 1 + \mathfrak{p}'$.

There is some maximal ideal $\mathfrak{m}'$ containing $\mathfrak{p}'$, so $e \in 1 + \mathfrak{m}'$. However, since $e \in \mathfrak{R}$, which is the intersection of all maximal ideals, $e \in \mathfrak{m}'$ also. Together, these two facts imply $1 \in \mathfrak{m}'$, yielding the desired contradiction. □

1.7 Exercise

Proof. Let $\mathfrak{p}$ be a prime ideal so that $A/\mathfrak{p}$ is an integral domain. For any $y \in A/\mathfrak{p}$ such that $y \neq 0$, there is some $x \in A$, such that $\overline{x} = y$. By assumption, there is an $n > 1$ such that $x^n = x$, so $y^n = y$. Note that $y^{n-1} - 1 \in A/\mathfrak{p}$ and

$$y(y^{n-1} - 1) = y^n - y = 0.$$

Since $A/\mathfrak{p}$ is an integral domain and $y \neq 0$, $y^{n-1} = 1$. In particular, y is a unit. Thus, all nonzero elements are nonzero, implying that $A/\mathfrak{p}$ is a field and $\mathfrak{p}$ is maximal. □

1.8 Exercise

Proof. Let $\mathcal{S}$ be the set of prime ideals of $\mathcal{S}$. If $\mathfrak{p} \subset \mathfrak{q}$, we'll define a partial ordering by $\mathfrak{q} \leq \mathfrak{p}$. Now, suppose we have a chain $\mathcal{T} \subset \mathcal{S}$. Let

$$Q = \bigcap_{\mathfrak{q} \in \mathcal{T}} \mathfrak{q},$$

which, as an intersection of ideals, is itself an ideal.

Suppose $a, b \in A$ and $ab \in Q$. If $a \notin Q$, then there is some $\mathfrak{r} \in \mathcal{T}$ such that $a \notin \mathfrak{r}$. Since $a \notin \mathfrak{r}$, for any $\mathfrak{q}$ such that $\mathfrak{q} \subset \mathfrak{r}$, $a \notin \mathfrak{q}$. Since $\mathcal{T}$ is a chain,

$$Q = \bigcap_{\mathfrak{q} \leq \mathfrak{r}} \mathfrak{q} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \mathfrak{r} \cap \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q} = \bigcap_{\mathfrak{q} \geq \mathfrak{r}} \mathfrak{q}.$$

Let $\mathfrak{q} \geq \mathfrak{r}$. Then, since $ab \in Q \subset \mathfrak{q}$ and $\mathfrak{q}$ is prime, we must have $b \in \mathfrak{q}$, as we've already shown a cannot be in $\mathfrak{r}$. Thus, $b \in \mathfrak{q}$ for all $\mathfrak{q} \geq \mathfrak{r}$, so $b \in Q$. Thus, Q is a prime ideal.

For all $\mathfrak{q} \in \mathcal{T}$, $Q \subset \mathfrak{q}$ ($Q \geq \mathfrak{q}$) and is thus a maximal element. Using Zorn's lemma, $\mathcal{S}$ has at least one maximal element. By our notation, these are minimal elements with respect to inclusion since we used $\geq$ for $\subset$. □

1.9 Exercise

Proof. If $\mathfrak{a} = \bigcap_{\mathfrak{p}} \mathfrak{p}$ is an intersection of prime ideals, then

$$r(\mathfrak{a}) = r\left(\bigcap_{\mathfrak{p}} \mathfrak{p}\right) = \bigcap_{\mathfrak{p}} r\left(\mathfrak{p}\right) = \bigcap_{\mathfrak{p}} \mathfrak{p} = \mathfrak{a}.$$

Conversely, suppose $r(\mathfrak{a}) = \mathfrak{a}$. By Proposition 1.14, $r(\mathfrak{a}) = \bigcap_{\mathfrak{p}} \mathfrak{p}$, where $\mathfrak{p}$ ranges over the prime ideals containing $\mathfrak{a}$. Therefore, $\mathfrak{a}$ is an intersection of prime ideals. □

1.10 Exercise

Proof. (i) $\implies$ (ii). Since $\mathfrak{N}$ is the intersection of all prime ideals, $\mathfrak{N}$ is the unique prime ideal. By Theorem 1.3, A must have a maximal ideal, but since every maximal ideal is prime and $\mathfrak{N}$ is the only prime ideal, $\mathfrak{N}$ must be maximal. Hence, it is also the only maximal ideal. By Corollary 1.5, every non-unit is contained in a maximal ideal, so every non-unit must be in $\mathfrak{N}$, hence nilpotent.

(ii) $\implies$ (iii). Consider the ring homomorphism $A \to A/\mathfrak{N}$. Every element $x \in A$ is either a unit or nilpotent; if x is a unit, $\overline{x}$ is a unit, and if x is nilpotent, $x \in \mathfrak{N}$, so $\overline{x} = 0$. Since the projection is surjective, every nonzero element of the ring $A/\mathfrak{N}$ is a unit, i.e. $A/\mathfrak{N}$ is a field.

(iii) $\implies$ (i). Since $A/\mathfrak{N}$ is a field, $\mathfrak{N}$ a maximal ideal. All maximal ideals are prime. However, $\mathfrak{N}$ is also the intersection of all prime ideals. Suppose there is another prime ideal $\mathfrak{p}$, not equal to $\mathfrak{N}$. Thus, there is an element $x \in \mathfrak{p} - \mathfrak{N}$ since $\mathfrak{N} \subset \mathfrak{p}$. Under the map $A \to A/\mathfrak{N}$, $\mathfrak{x}$ is a unit since it is non-zero. There is a y in A such that $xy = 1 + \mathfrak{N}$. But, $xy \in \mathfrak{p}$ since $x \in \mathfrak{p}$, so $\mathfrak{p} + \mathfrak{N} = (1)$ and are thus coprime. But $\mathfrak{N} \subset \mathfrak{p}$, so $\mathfrak{p} = (1)$, a contradiction. □

1.11 Exercise

(i) Proof. For every $x \in A$,

$$\begin{aligned}(1+x)^2 &= 1+x\\ 1+2x+x^2 &= 1+x\\ 1+2x+x &= 1+x\\ 2x &= 0.\end{aligned}$$

□

(ii) Proof. Suppose $x \in A$ represents the class $\bar{x} \in A/\mathfrak{p}$. Then,

$$\bar{x}^2 = (\mathfrak{p}+x)(\mathfrak{p}+x) = \mathfrak{p}+x^2 = \mathfrak{p}+x = \bar{x}.$$

Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain. Hence, $\bar{x}^2 = \bar{x} \implies \bar{x}(\bar{x}-1) = 0 \implies \bar{x} \in \{0,1\}$. Thus, $A/\mathfrak{p}$ is the field of size 2, making $\mathfrak{p}$ a maximal ideal. □

(iii) Proof. For I finitely-generated, $I = (x_1, \ldots, x_n)$ for $x_i \in A$. We'll show that I is principal.

We do this by induction, proving that an ideal with n-many generators can be generated by a single element. The base case is trivial. For the inductive step, suppose that for all ideals generated by $n-1$ elements (or fewer), $(a_1, \ldots, a_{n-1}) = (a)$. Consider $(x_1, \ldots, x_n)$. By the inductive step, $(x_1, \ldots, x_{n-1}) = (a)$, so

$$(x_1, \ldots, x_n) = (a, x_n).$$

Consider the element $b = a + x_n + ax_n$. Note that

$$ab = a^2 + ax_n + a^2x_n = a + ax_n + ax_n = a + 2ax_n = a$$

and

$$x_nb = ax_n + x_n^2 + ax_n^2 = ax_n + x_n + ax_n = x_n + 2ax_n = x_n.$$

Thus, $a, x_n \in (b)$, so $(x_1, \ldots, x_n) = (a, x_n) = (b)$ since, by the construction of b, $b \in (a, x_n)$. By induction, any finitely generated ideal of a Boolean ring A is principal. □

1.12 Exercise

Proof. If x is a unit, $x^2 = x \implies x^2x^{-1} = xx^{-1} \implies x = 1$. If x is not a unit, then $x \in \mathfrak{m}$. That means that $1 - x$ is not in $\mathfrak{m}$ (otherwise $x + (1 - x) = 1 \in \mathfrak{m}$) and consequently that $1 - x$ is a unit. $x^2 = x \implies x(1 - x) = 0$, but since $1 - x$ is a unit, we have $x = 0$. □

1.13 Exercise

Here, we construct an algebraic closure of a field K.

Let K be a field, and let Σ be the set of all irreducible monic polynomials f in one indeterminate with coefficients in K. Let A be the polynomial ring over K generated by indeterminates x_f, one for each $f \in \Sigma$. Let $\mathfrak{a}$ be the ideal of A generated by the polynomials $f(x_f)$ for all $f \in \Sigma$. We now show that $\mathfrak{a} \neq (1)$.

Proof. Let R be any nonzero commutative ring with unit. Consider $R[x]/(f(x))$, where $f(x)$ is a monic polynomial over R. Suppose $R[x] = (f(x))$. Then, $1 = f(x)r(x)$ for some $r(x) \in R[x]$. However, by Exercise 1.2(i), $f(x)$ cannot be a unit since its highest coefficient is 1, a unit. In particular, $(f(x))$ is a proper ideal of $R[x]$. Thus, for a nonzero ring R, $R[x]/(f(x))$ is also nonzero.

We'll show that $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ is nonzero if $\{f_i\}$ are monic polynomials over K. We do this by induction. We proved the base case above, setting $K = R$.

Now, suppose $K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1}))$ is nonzero. Then,

$$\begin{aligned} K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n)) &\cong K[x_1, \ldots, x_{n-1}][x_n]/(f_1(x_1), \ldots, f_n(x_n)) \\ &\cong (K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1})))\,[x_n]/(f_n(x_n)) \\ &\cong R[x_n]/(f_n(x_n)), \end{aligned}$$

where $R = K[x_1, \ldots, x_{n-1}]/(f_1(x_1), \ldots, f_{n-1}(x_{n-1}))$. Since R is nonzero and $f_n(x_n)$ is a monic polynomial over R, we have that $R[x_n]/(f_n(x_n))$ is nonzero by our initial proof. By induction, $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ is nonzero.

We turn to the main point. Suppose to the contrary that $\mathfrak{a} = (1)$. Then, there is some finite collection of pairs $(h_i, f_i) \in A \times \Sigma$ such that

$$1 = \sum_{i=1}^{n} h_i f_i(x_{f_i}).$$

Write $x_i = x_{f_i}$ for convenience. For the indeterminates x_g not equal to some f_i, we can map $x_g \mapsto 0$. This is a ring homomorphism of A. This maps the $h_i \mapsto k_i \in K[x_1, \ldots, x_n]$ so that

$$1 = \sum_{i=1}^{n} k_i f_i(x_i).$$

But then, we have $(f_1(x_1), \ldots, f_n(x_n)) = (1)$, violating $K[x_1, \ldots, x_n]/(f_1(x_1), \ldots, f_n(x_n))$ being nonzero. Thus, $\mathfrak{a} \neq (1)$. □

By Corollary 1.4, A has a maximal ideal $\mathfrak{m}$ containing $\mathfrak{a}$ since $\mathfrak{a} \neq (1)$. Now, let $K_1 = A/\mathfrak{m}$, which is a field since $\mathfrak{m}$ is maximal. Since $\mathfrak{m}$ cannot contain any units and $K \hookrightarrow A$, K is a subfield of K_1. Further, for every $f \in \Sigma$, $x_f \in K_1$ and satisifies $f(x_f) = 0$.

Since K_1 is a field, we can repeat the construction with K_1 in place of K, obtaining a field K_2, and so on. Now, let $L = \bigcup_{n=1}^{\infty} K_n$, which is a field since all operations take place in some subfield K_n.

Suppose $f \in \Sigma$ has degree n. In K_1, if x_f maps to the residue class α_f in $A/\mathfrak{m}$, then $f(x) = (x - \alpha_f)g(x)$, where $g(x) \in K_1[x]$ has degree $n-1$. In the worse case, g is irreducible here. By a simple inductive argument, $f(x)$ factors completely over $K_n \subset K$. Thus, every $f \in \Sigma$ splits completely in K.

By the same logic, let $f \in L[x]$ be a monic polynomial; suppose f has degree n. Since f has finitely-many coefficients, all coefficients lie in some common subfield K_k. Then, by construction, f splits completely in K_{k+n}, so L is algebraically closed.

Now, let $\overline{K}$ be the set of all elements of L that are algebraic over K. Then, $\overline{K}$ is an algebraic closure of K.

1.14 Exercise

Proof. Since 0 is such an ideal, Σ is non-empty. For $I, J \in \Sigma$, we say that $I \leq J$ if $I \subset J$, endowing Σ with a partial ordering. Let $\mathcal{T}$ be a chain of Σ. Then, for all $I \in \mathcal{T}$, let $T = \bigcup_{I \in \mathcal{T}} I$. T is an ideal (because the chain is ordered by inclusion) and contains only zero-divisors, so $T \in \Sigma$. Further, $I \subset T$ for all $I \in \mathcal{T}$. Thus, every chain has an upper bound in Σ. By Zorn's lemma, Σ has at least one maximal element.

Now that we've proven existence, let M be a maximal element of Σ. We now show that M is prime. Note that the elements of A that are not zero divisors form a multiplicatively-closed set S. Suppose $x, y \in A - M$. By the maximality of M, $M + (x)$ and $M + (y)$ must both contain elements of S. Thus, there are $m_1, m_2 \in M$ and $a_1, a_2 \in A$ so that $m_1 + a_1 x, m_2 + a_2 y \in S$. Since S is multiplicatively-closed,

$$(m_1 + a_1 x)(m_2 + a_2 y) = m_1 m_2 + m_1 a_2 y + m_2 a_1 x + a_1 a_2 xy \in S.$$

If $xy \in M$, then all terms would be in M so that the sum would be in M. However, M only contains zero divisors and S contains no zero divisors, so this would be a contradiction. Thus, $xy \notin M$, proving the primality of M.

If z is a zero divisor, then $(z) \in \Sigma$. Thus, (z) is contained in some maximal element M of Σ, which we've just shown is prime. Thus, the set of zero divisors in A is a union of prime ideals. □

1.15 Exercise

(i) Proof. Clearly, $V(\mathfrak{a}) \subset V(E)$ since $E \subset \mathfrak{a}$. However, if $\mathfrak{p}$ contains E, then the ideal $\mathfrak{a}$ that E generates must also be contained in $\mathfrak{p}$, so $V(E) = V(\mathfrak{a})$.

Since $\mathfrak{a} \subset r(\mathfrak{a})$, we have $V(r(\mathfrak{a})) \subset V(\mathfrak{a})$. Let $\mathfrak{p}$ be any prime containing $\mathfrak{a}$; consider the ring homomorphism $\varphi : A \to A/\mathfrak{p}$. Then, $\varphi(\mathfrak{a}) = 0$. If $x \in r(\mathfrak{a})$, then $x^n \in \mathfrak{a}$ for some $n > 0$. Under the map, we have $\overline{x}^n = 0$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain, so $\overline{x} = 0$. Thus, $x \in \mathfrak{p}$ and $r(\mathfrak{a}) \subset \mathfrak{p}$. Consequently, $V(\mathfrak{a}) = V(r(\mathfrak{a}))$. □

(ii) Proof. $V(0) = X$ since for any prime ideal $\mathfrak{p} \in X$, $0 \subset \mathfrak{p}$. Likewise, for any prime ideal $\mathfrak{p} \in X$, $\mathfrak{p}$ is a proper ideal of A, so $\mathfrak{p}$ does not contain (1). Thus, $V(1) = \emptyset$. □

(iii) Proof. Let $(E_i)_{i\in I}$ be a family of subsets of A. Suppose $\mathfrak{p}$ is a prime ideal containing $\bigcup_{i\in I} E_i$. Then, $\mathfrak{p}$ contains each E_i; i.e.,

$$V\left(\bigcup_{i\in I} E_i\right) \subset \bigcap_{i\in I} V(E_i).$$

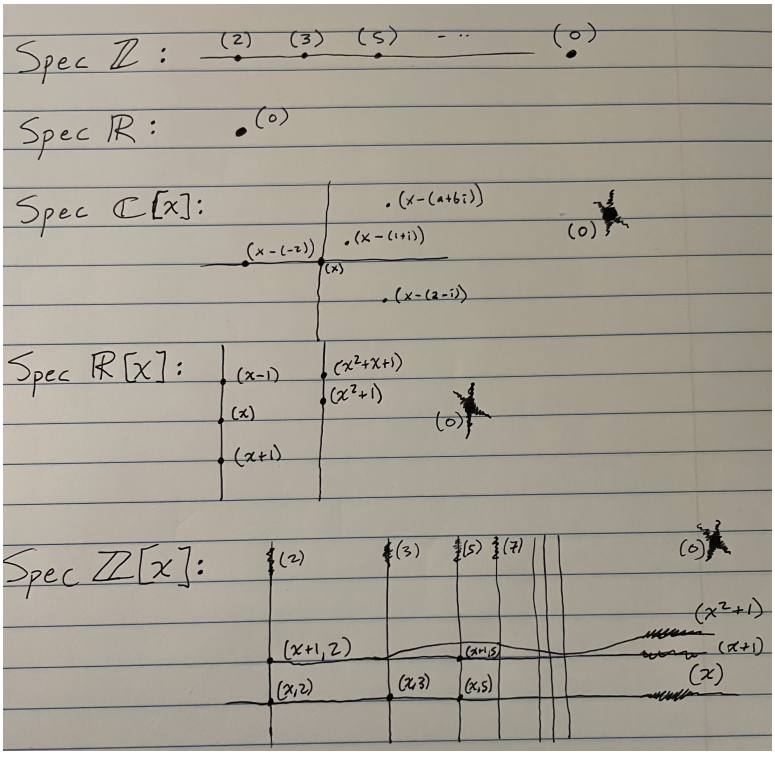
Conversely, suppose $\mathfrak{p}$ contains each E_i; then clearly, it contains the union. So, we have equality. □

(iv) Proof. If a prime ideal $\mathfrak{p} \in V(\mathfrak{a})\cup V(\mathfrak{b})$, then it is in one of the two sets. Assume without loss of generality that $\mathfrak{p} \in V(\mathfrak{a})$. Then, $\mathfrak{p}$ contains $\mathfrak{a}$ and thus also contains $\mathfrak{a}\cap\mathfrak{b}$, so $V(\mathfrak{a})\cup V(\mathfrak{b}) \subset V(\mathfrak{a}\cap\mathfrak{b})$. Since $\mathfrak{ab} \subset \mathfrak{a}\cap\mathfrak{b}$, we have $V(\mathfrak{a}\cap\mathfrak{b}) \subset V(\mathfrak{ab})$. Thus, if we can show that $V(\mathfrak{ab}) \subset V(\mathfrak{a})\cup V(\mathfrak{b})$, we will have shown full equality.

Let $\mathfrak{p}$ be a prime ideal containing $\mathfrak{ab}$. Consider the homomorphism $\varphi : A \to A/\mathfrak{p}$. Since φ is surjective, the ideals $\mathfrak{a}$ and $\mathfrak{b}$ map to ideals $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{b}}$ of $A/\mathfrak{p}$. Suppose to the contrary that both $\overline{\mathfrak{a}}$ and $\overline{\mathfrak{b}}$ contain nonzero elements, say a and b. Then, ab is nonzero since $A/\mathfrak{p}$ is an integral domain ($\mathfrak{p}$ is prime). However, $\overline{\mathfrak{a}}\cdot\overline{\mathfrak{b}} = \overline{\mathfrak{ab}} = 0$ since $\mathfrak{ab} \subset \mathfrak{p}$. Thus, either $\overline{\mathfrak{a}} = 0$ or $\overline{\mathfrak{b}} = 0$. In other words, $\mathfrak{a} \subset \mathfrak{p}$ or $\mathfrak{b} \subset \mathfrak{p}$. In either case, $\mathfrak{p} \in V(\mathfrak{a}) \cup V(\mathfrak{b})$, completing the proof. □

Remark. *We've shown that $V(E)$ satisfy the axioms for closed sets in a topological space. The resulting topology is called the Zariski topology. We denote this topological space* $\mathrm{Spec}(A) := X$*, the prime spectrum of A.*

1.16 Exercise



1.17 Exercise

Before proving properties (i)-(vii), we must show that the sets X_f form a basis of open sets for the Zariski topology.

Proof. Any open set U, by definition, is the complement of a closed set V of $\operatorname{Spec} A$. In Exercise 1.15, we showed that each closed set of $\operatorname{Spec} A$ can be represented by $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$ (since arbitrary intersections and finite unions have the same form), so $U = V(\mathfrak{a})^c$ for some ideal $\mathfrak{a}$. By Exercise 1.15(iii),

$$V(\mathfrak{a}) = V\left(\bigcup_{f\in\mathfrak{a}}\right) = \bigcap_{f\in\mathfrak{a}} V(f).$$

Thus,

$$U = V(\mathfrak{a})^c = \left(\bigcap_{f\in\mathfrak{a}} V(f)\right)^c = \bigcup_{f\in\mathfrak{a}} V(f)^c = \bigcup_{f\in\mathfrak{a}} X_f,$$

so $\{X_f\}_{f\in A}$ forms a basis of open sets in $\operatorname{Spec} A$. □

(i) Proof. Note that $V(f) = V(\{f\})$, the prime ideals containing the element $f \in A$. Then, $X_f = V(\{f\})^c$, where $-^c$ denotes the complement. By Exercise 1.15(i), $V(f) = V((f))$ for any $f \in A$. If $f, g \in A$, then using Exercise 1.15(iv), we have

$$X_f \cap X_g = V((f))^c \cap V((g))^c = (V((f)) \cup V((g)))^c = V((f)(g))^c = V((fg))^c = V(fg)^c = X_{fg},$$

completing the proof. □

(ii) Proof. If $X_f = \emptyset$, then $V(f) = X$. Thus, f is in the intersection of all prime ideals, meaning $f \in \mathfrak{N}_A$ (i.e., f is nilpotent). Conversely, if f is nilpotent, it is in the nilradical and thus in every prime ideals, so $X_f = V(f)^c = X^c = \emptyset$. □

(iii) Proof. If $X_f = X$, then $V(f) = \emptyset$. Thus, f is not in any prime ideal and thus not in any maximal ideal. In particular, $(f) = A$, so f is a unit. Conversely, if f is a unit, then there is some $g \in A$ such that $fg = 1$. Thus, for any prime ideal $\mathfrak{p}$, under $A \to A/\mathfrak{p}$, we have $\overline{f} \cdot \overline{g} = 1$. So, $\overline{f} \neq 0$ and $f \notin \mathfrak{p}$. Hence, $V(f) = \emptyset$ and $X_f = X$. □

(iv) Proof. If $X_f = X_g$, then $V(f) = V(g)$. By Proposition 1.14,

$$r(f) = \bigcap_{\mathfrak{p}:(f)\subset\mathfrak{p}} \mathfrak{p} = \bigcap_{\mathfrak{p}\in V(f)} \mathfrak{p} = \bigcap_{\mathfrak{p}\in V(g)} \mathfrak{p} = r(g).$$

Conversely, if $r(f) = r(g)$, then $V(f) = V(r(f)) = V(r(g)) = V(g)$, so $X_f = X_g$. □

(v) Proof. Suppose $\{U_j\}_{j\in J}$ is an open cover of X. Since we showed that $\{X_f\}_{f\in A}$ form a basis of open sets, we can write $U_j = \bigcup_{f\in I_j} X_f$ for each $j \in J$. Thus, if $I = \bigcup_{j\in J} I_j$, we have an open cover

$$X = \bigcup_{f\in I} X_f = \bigcup_{f\in I} V(f)^c = \left(\bigcap_{f\in I} V(f)\right)^c$$

$$\implies \emptyset = \bigcap_{f\in I} V(f) = V\left(\bigcup_{f\in I}(f)\right),$$

using Exercise 1.15(iii). Let $\mathfrak{a}$ be the ideal generated by $f \in I$. Then, $V(\mathfrak{a}) = \emptyset$, so $V(r(\mathfrak{a})) = \emptyset$ by Exercise 1.15(i). No prime ideal contains $r(\mathfrak{a})$, so $r(\mathfrak{a}) = (1)$. Thus, $\mathfrak{a} = (1)$.

In particular, there is some finite collection $\{f_s\}_{s\in S}$ of $\{f_i\}_{i\in I}$ such that

$$\sum_{s\in S} a_s f_s = 1$$

for some $a_s \in A$. Thus, there is no prime ideal that divides every (f_i), so $\bigcap_{s\in S} V(f_s) = \emptyset$ and $X = \bigcup_{s\in S} X_{f_s}$. Since each f_s is contained in some U_s, we have

$$X = \bigcup_{s\in S} U_s,$$

where S is finite and $U_s \in \{U_j\}_{j\in J}$. Hence, X is quasi-compact. □

(vi) Proof. For $f \in A$, let $\{U_j\}_{j\in J}$ be an open covering of X_f. Then, for each j, $U_j = \bigcup_{g\in I_j} X_g$ since $\{X_g\}_{g\in A}$ form a basis of open sets. If $I = \bigcup_{j\in J} I_j$, then

$$X_f \subset \bigcup_{g\in I} X_g.$$

Taking the intersection with X_f and using (i), we have

$$X_f \subset X_f \cap \bigcup_{g\in I} X_g = \bigcup_{g\in I} (X_f \cap X_g) = \bigcup_{g\in I} X_{fg} \subset X_f,$$

so $X_f = \bigcup_{g\in I} X_{fg}$. Taking complements and using results of Exercise 1.15, we have

$$V(f) = \bigcap_{g\in I} V(fg) = V\left(\bigcup_{g\in I} (fg)\right) = V\left(\sum_{g\in I} (fg)\right).$$

By (iv), $r(f) = r\left(\sum_{g\in I} fg\right)$. Therefore, for some power $n > 0$ of f, we have

$$f^n = \sum_{s\in S} a_s f g_s$$

for a finite subset $S \subset I$ and $a_s \in A$. From the sum, if there is some prime ideal $\mathfrak{p}$ that does not contain f (and thus not f^n), it does not contain at least one element $a_s f g_s$ on the right-hand side. So, it does not contain g_s for some $s \in S$. In particular,

$$V(f)^c \subset \bigcup_{s\in S} V(g_s)^c$$
$$X_f \subset \bigcup_{s\in S} X_{g_s}.$$

Since each X_{g_s} is contained in some $U \in \{U_j\}_{j\in J}$ by the definition of I, X_f has a finite subcover and is thus quasi-compact. □

(vii) Proof. Suppose U is an open set of X and that U is a finite union of sets X_f. Now, let S be an open covering of U. Then, it is an open covering of each X_f. By (vi), each X_f has a finite subcover. The union of these finite subcovers is finite and covers U, so U is quasi-compact.

Conversely, suppose U is quasi-compact. By Exercise 1.15, $U = V(\mathfrak{a})^c$ for some ideal $\mathfrak{a}$. Thus, by Exercise 15(iii),

$$U = V(\mathfrak{a})^c = V\left(\bigcup_{f\in\mathfrak{a}} (f)\right)^c = \left(\bigcap_{f\in\mathfrak{a}} V(f)\right)^c = \bigcup_{f\in\mathfrak{a}} V(f)^c = \bigcup_{f\in\mathfrak{a}} X_f.$$

Since U is quasi-compact, there is a finite subcover, so U is the union of finitely-many X_f. □

1.18 Exercise

(i) Proof. Let $x \in \operatorname{Spec} A$. If $\mathfrak{p}_x$ is maximal, then $V(\mathfrak{p}_x)$ contains only the prime $\mathfrak{p}_x$. Otherwise, there is a prime ideal containing $\mathfrak{p}_x$ properly, violating maximality. So in this case, x is a closed point ($\{x\}$ is closed). Conversely, if $\mathfrak{p}_x$ is not maximal, there is some maximal ideal $\mathfrak{p}_y$ properly containing $\mathfrak{p}_x$. In order for $\{x\}$ to be closed, there must be some subset $E \subset A$ such that $V(E) = \{x\}$. But in this case, $E \subset \mathfrak{p}_x \subset \mathfrak{p}_y$, so $\{x, y\} \subset V(E)$. Thus, x is not a closed point when $\mathfrak{p}_x$ is not maximal. □

(ii) Proof. Any closed set containing x is, by definition, of the form $V(E)$ for some subset $E \subset A$, and by assumption, $x \in V(E)$. Thus, $E \subset \mathfrak{p}_x$. By Exercise 1.15(iii),

$$\overline{\{x\}} = \bigcap_{E \subset \mathfrak{p}_x} V(E) = V\left(\bigcup_{E \subset \mathfrak{p}_x} E\right) = V(\mathfrak{p}_x).$$

□

(iii) Proof. By (ii), $\overline{\{x\}} = V(\mathfrak{p}_x)$, so we need to show that $y \in V(\mathfrak{p}_x) \iff \mathfrak{p}_x \subset \mathfrak{p}_y$. But, this holds by definition. □

(iv) Proof. We aim to show that $\operatorname{Spec} A$ is T_0, so for any two elements $x, y \in \operatorname{Spec} A$, we must show that there is an open set that contains only one of x and y. Equivalently, we can produce a closed set that contains exactly one.

Let $x, y \in \operatorname{Spec} A$ be distinct points. If $y \notin V(\mathfrak{p}_x)$, then we are done since $x \in V(\mathfrak{p}_x)$. So, suppose $y \in V(\mathfrak{p}_x)$. By (iii), $\mathfrak{p}_x \subset \mathfrak{p}_y$. Now, consider $V(\mathfrak{p}_y)$. If $x \notin V(\mathfrak{p}_y)$, then we are done since $y \in V(\mathfrak{p}_y)$. So, suppose $x \in V(\mathfrak{p}_y)$. By (iii), $\mathfrak{p}_y \subset \mathfrak{p}_x$, so $\mathfrak{p}_x = \mathfrak{p}_y$ (or $x = y$), providing the necessary contradiction. □

1.19 Exercise

Proof. A topological space is irreducible if $X \neq \emptyset$ and every pair of non-empty open sets in X intersect. Since $\{X_f\}_{f \in A}$ form an open basis of X, X is irreducible if and only if $X_f \cap X_g \neq \emptyset$ for all $f, g \in A$ satisyfing $X_f \neq \emptyset$ and $X_g \neq \emptyset$.

Suppose that the nilradical $\mathfrak{N}$ is prime, and suppose $f, g \in A$ satisfy $X_f \neq \emptyset$ and $X_g \neq \emptyset$. Then, $V(f) \neq X$ and $V(g) \neq X$. Since $\mathfrak{N}$ is the intersect of all primes in X, $\mathfrak{N} \notin V(f) \cup V(g)$ since, by assumption, neither $V(f)$ nor $V(g)$ contain all primes. Thus, $\mathfrak{N} \in V(f)^c \cap V(g)^c = X_f \cap X_g$, so $X_f \cap X_g$ is non-empty.

If the nilradical is not prime, then there are $f, g \in A$ such that $fg \in \mathfrak{N}$ yet neither f nor g are nilpotent. By Exercise 1.17(ii), X_f and X_g are thus non-empty. However, by Exercise 1.17(i), $X_f \cap X_g = X_{fg}$. Since $fg \in \mathfrak{N}$ and $\mathfrak{N}$ is the intersection of all prime ideals, each prime ideal contains fg, so $V(fg) = X$ and $X_{fg} = \emptyset$. $X_f \cap X_g = \emptyset$ even though both X_f and X_g are non-empty, so X is not irreducible. □

1.20 Exercise

(i) Proof. Let Y be an irreducible subspace of X. Because $\overline{Y}$ is closed, any closed set V of X yields a closed set $V \cap \overline{Y}$ in X. Since all closed sets of the subspace $\overline{Y}$ have this form (definition of subspace topology), any closed set of $\overline{Y}$ is inherited from a closed set of X.

Since Y is irreducible, $Y \neq \emptyset$ (thus $\overline{Y} \neq \emptyset$) and every pair of non-empty open sets in Y intersect. Equivalently, every pair of closed sets V_1, V_2 not equal to all of Y satisfy $V_1 \cup V_2 \neq Y$.

Let V_1, V_2 be any two closed sets of $\overline{Y}$, neither alone containing all of $\overline{Y}$. Then, for some closed sets A_1, A_2 of X, we have $V_1 = A_1 \cap \overline{Y}$ and $V_2 = A_2 \cap \overline{Y}$. Thus, $B_1 = A_1 \cap Y$ and $B_2 = A_2 \cap Y$ are closed in Y. If, for instance, $A_1 \cap Y$ contains all of Y, then $A_1 \cap \overline{Y}$ is a closed set containing Y, so $\overline{Y} \subset A_1 \cap \overline{Y}$. But this is a contradiction since we assumed $A_1 \cap \overline{Y}$ does not contain all of $\overline{Y}$. Thus, neither B_1 nor B_2 alone contain all of Y. By the irreducibility of Y, $B_1 \cup B_2 \neq Y$.

Now, assume for the sake of contradiction that $V_1 \cup V_2 = \overline{Y}$. Then,

$$\overline{Y} = V_1 \cup V_2 = (A_1 \cap \overline{Y}) \cup (A_2 \cap \overline{Y}) = (A_1 \cup A_2) \cap \overline{Y}.$$

Since $Y \subset \overline{Y}$, we can intersect with Y:

$$Y = (A_1 \cup A_2) \cap \overline{Y} \cap Y = (A_1 \cup A_2) \cap Y = (A_1 \cap Y) \cup (A_2 \cap Y) = B_1 \cup B_2,$$

yielding our contradiction. Thus, $\overline{Y}$ is irreducible when Y is irreducible. □

(ii) Proof. Let Σ be the set of irreducible subspaces of X; $\subset$ gives Σ a partial ordering. Let $\mathcal{T}$ be a chain, and define $U = \bigcup_{Y \in \mathcal{T}} Y$. Let A_1, A_2 be two non-empty open sets of U. Choose some $x \in A_1$ and $y \in A_2$. Then, there is some Y_1 that contains x and some Y_2 that contains y. Since $\mathcal{T}$ is a chain, either $Y_1 \subset Y_2$ or $Y_2 \subset Y_1$. Without loss of generality, assume $Y_1 \subset Y_2$. Then, $x \in Y_1 \subset Y_2$, so $x, y \in Y_2$. For convenience, write $Y = Y_2$. Note that since A_1, A_2 are open in U, $A_1 = V_1 \cap U$ and $A_2 = V_2 \cap U$ for open sets V_1, V_2 in X. But also, $A_1^Y = V_1 \cap Y \subset A_1$ and $A_2^Y = V_2 \cap Y \subset A_2$ since $Y \subset U$. Since x is in A_1 and Y, $x \in A_1^Y$, and since y is in A_2 and Y, $y \in A_2^Y$, so both sets are open and non-empty. By the irreducibility of Y, $A_1^Y \cap A_2^Y \neq \emptyset$. Since $A_1^Y \subset A_1$ and $A_2^Y \subset A_2$, $A_1^Y \cap A_2^Y \subset A_1 \cap A_2$, so $A_1 \cap A_2$ is also non-empty. Thus, U is irreducible and $U \in \Sigma$. In particular, all chains in Σ have maximal elements in Σ. By Zorn's lemma, every irreducible subspace of X is contained in a maximal irreducible subspace. □

(iii) Proof. Let Y be a maximal irreducible subspace of X. By (i), $\overline{Y}$ is irreducible, but since Y is maximal and $Y \subset \overline{Y}$, we have $Y = \overline{Y}$, so Y is closed.

Let $x \in X$; consider the subspace $\{x\}$. If U_1 and U_2 are any non-empty open sets of $\{x\}$, then $U_1 = U_2 = \{x\}$. Thus, $U_1 \cap U_2 = \{x\} \neq \emptyset$. In particular, $\{x\}$ is an irreducible subspace. By (ii), it is contained in some maximal irreducible subspace. Thus, the maximal irreducible subspaces cover X.

Now, suppose X is Hausdorff. Let Y be an irreducible subspace of X, and let $x, y \in Y$ (Y is non-empty by irreducibility). If x and y are distinct, then since X is Hausdorff, there are open sets U_x and U_y such that $x \in U_x$, $y \notin U_x$, $y \in U_y$, $x \notin U_y$, and $U_x \cap U_y = \emptyset$. Thus, $Y \cap U_x$ and $Y \cap U_y$ are non-empty open sets of Y since they contain x and y, respectively. However,

$$(Y \cap U_x) \cap (Y \cap U_y) = Y \cap (U_x \cap U_y) = Y \cap \emptyset = \emptyset.$$

Thus, Y is not irreducible, yielding a contradiction. Thus, Y cannot have more than one point. Conversely, we've already shown that one-point sets are irreducible subspaces. Thus, the irreducible components of Hausdorff spaces are one-point sets. □

(iv) Proof. First, we show that $V(\mathfrak{p})$ is irreducible for any $\mathfrak{p} \in X$. In (iii), we showed that one-point sets are irreducible, i.e. $\{\mathfrak{p}\}$ is irreducible for every $\mathfrak{p} \in \operatorname{Spec} A$. Since $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ by Exercise 1.18(ii), (i) tells us that $\overline{\{\mathfrak{p}\}} = V(\mathfrak{p})$ inherits the irreducibility of $\{\mathfrak{p}\}$.

By (iii), every irreducible component is closed, so each is of the form $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$. If $\mathfrak{a} = (1)$, then $V(\mathfrak{a}) = \emptyset$, violating irreducibility. Thus, there is some prime ideal $\mathfrak{p}$ such that $\mathfrak{p} \subset \mathfrak{a}$, so $V(\mathfrak{a}) \subset V(\mathfrak{p})$. Since $V(\mathfrak{p})$ is irreducible, $V(\mathfrak{a}) = V(\mathfrak{p})$ by the maximality of $V(\mathfrak{a})$. Thus, every irreducible component is of the form $V(\mathfrak{p})$ for some $\mathfrak{p} \in X$.

Now, suppose we have an irreducible component; then, we can write it as $V(\mathfrak{p})$ for some $\mathfrak{p} \in X$. Then, if $\mathfrak{q} \in X$ and $\mathfrak{q} \subset \mathfrak{p}$ properly, then $V(\mathfrak{p}) \subset V(\mathfrak{q})$ properly since $V(\mathfrak{q})$ contains $\mathfrak{q}$ but $V(\mathfrak{p})$ does not contain $\mathfrak{q}$. Since $\mathfrak{q}$ is prime, $V(\mathfrak{q})$ is irreducible, violating the maximality of $V(\mathfrak{p})$. Thus, there is no such $\mathfrak{q}$, and $\mathfrak{p}$ is a minimal prime ideal of A.

Conversely, let $\mathfrak{p}$ be a minimal prime ideal of A. Then, $V(\mathfrak{p})$ is contained in some irreducible component, which we've shown has the form $V(\mathfrak{q})$ for some $\mathfrak{q} \in X$. Thus, $V(\mathfrak{p}) \subset V(\mathfrak{q})$, so $\mathfrak{q} \subset \mathfrak{p}$. Since $\mathfrak{p}$ is a minimal prime ideal, $\mathfrak{p} = \mathfrak{q}$, so $V(\mathfrak{p}) = V(\mathfrak{q})$, and $V(\mathfrak{p})$ is an irreducible component. □

1.21 Exercise

(i) Proof. Let $f \in A$. If $\mathfrak{q} \in Y_{\phi(f)}$, then $\mathfrak{q} \notin V_Y(\phi(f))$, so $(\phi(f)) \not\subset \mathfrak{q}$. Thus, $\phi(f) \notin \mathfrak{q}$, so $f \notin \phi^{-1}(\mathfrak{q})$. Hence, $\phi^{-1}(\mathfrak{q}) \notin V_X(f)$, so $\phi^{-1}(\mathfrak{q}) \in X_f$ and $\mathfrak{q} \in \phi^{*-1}(X_f)$.

Conversely, if $\mathfrak{q} \notin Y_{\phi(f)}$, then $\mathfrak{q} \in V_Y(\phi(f))$, so $\phi(f) \in \mathfrak{q}$. Thus, $f \in \phi^{-1}(\mathfrak{q})$, so $\phi^{-1}(\mathfrak{q}) \in V_X(f)$. Hence, $\phi^{-1}(\mathfrak{q}) \notin X_f$. Then, by definition, $\mathfrak{q} \notin \phi^{*-1}(X_f)$. Thus, $\phi^{*-1}(X_f) = Y_{\phi(f)}$ for any $f \in A$.

Let U be an open set of X, so $U = \bigcup_{f \in I} X_f$. Then,

$$\phi^{*-1}(U) = \phi^{*-1}\left(\bigcup_{f \in I} X_f\right) = \bigcup_{f \in I} \phi^{*-1}(X_f) = \bigcup_{f \in I} Y_{\phi(f)},$$

which is an open set of Y. Thus, $\phi^* : Y \to X$ is continuous. □

(ii) Proof. If $\mathfrak{q} \in \phi^{*-1}(V_X(\mathfrak{a}))$, then $\phi^*(\mathfrak{q}) = \phi^{-1}(\mathfrak{q}) \in V_X(\mathfrak{a})$. Hence, $\mathfrak{a} \subset \phi^{-1}(\mathfrak{q})$, so $\phi(\mathfrak{a}) \subset \mathfrak{q}$, so $\mathfrak{q} \in V_Y(\phi(\mathfrak{a})) = V_Y(\mathfrak{a}^e)$.

Conversely, if $\mathfrak{q} \in V_Y(\mathfrak{a}^e)$, then $\phi(\mathfrak{a}) \subset \mathfrak{a}^e \subset \mathfrak{q}$, so $\mathfrak{a} \subset \phi^{-1}(\mathfrak{q})$, hence $\phi^{-1}(\mathfrak{q}) \in V_X(\mathfrak{a})$. Thus, $\mathfrak{q} \in \phi^{*-1}(V_X(\mathfrak{a}))$. Combining both directions, we have $\phi^{*-1}(V_X(\mathfrak{a})) = V_Y(\mathfrak{a}^e)$. □

(iii) Proof. Note that $\phi^*(V(\mathfrak{b})) = \{\mathfrak{q}^c : \mathfrak{b} \subset \mathfrak{q}\}$. So, $\mathfrak{a}$ must contain all of these $\mathfrak{q}^c$. Since all closed sets containing $\phi^*(V(\mathfrak{b}))$ must contain each $\mathfrak{q}^c$, all must contain $\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c$. Since this is itself an ideal, it gives the closure:

$$\overline{\phi^*(V(\mathfrak{b}))} = V\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c\right).$$

Note that

$$\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c = \bigcap_{\mathfrak{b} \subset \mathfrak{q}} \phi^{-1}(\mathfrak{q}) = \phi^{-1}\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}\right) = \phi^{-1}(r(\mathfrak{b})) = r(\mathfrak{b})^c = r(\mathfrak{b}^c).$$

Thus,

$$\overline{\phi^*(V(\mathfrak{b}))} = V\left(\bigcap_{\mathfrak{b} \subset \mathfrak{q}} \mathfrak{q}^c\right) = V(r(\mathfrak{b}^c)) = V(\mathfrak{b}^c).$$

□

(iv) Proof. Since ϕ is surjective, we have a ring isomorphism $A/(\ker\phi) \to B$. Thus, we have an inverse homomorphism $\psi : B \to A/(\ker\phi)$. From these maps, we obtain $\psi^* : \operatorname{Spec}(A/(\ker\phi)) \to \operatorname{Spec}(B)$ and $\phi^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A/(\ker\phi))$, which are inverse maps describing a bijection; by (i), both are continuous, so they describe a homeomorphism.

Now, we have to show that the map $\operatorname{Spec}(A/(\ker\phi)) \to V(\ker\phi)$, viewed as a closed subspace of $\operatorname{Spec}(A)$, is a homeomorphism.

But this follows easily from Proposition 1.1, considering the map $A \to A/(\ker\phi)$. More concretely, there is a one-to-one, order-preserving, primality-preserving correspondence between ideals of A containing $\ker\phi$ and ideals of $A/(\ker\phi)$. Thus, the map induced by projection $\operatorname{Spec}(A/\ker\phi) \to V(\ker\phi)$ is a bijection. Because the map preserves the containment relation, it is a closed map ($V(\mathfrak{a}) \to V(\mathfrak{a}^c)$). Since it is a bicontinuous bijection, it is a homeomorphism. □

Remark. *This shows that* $\operatorname{Spec}(A)$ *and* $\operatorname{Spec}(A/\mathfrak{N}_A)$ *are naturally homeomorphic.*

(v) Proof. Suppose $\ker\phi \subset \mathfrak{N}$. Since $Y = V(0)$, by (iii), we have

$$\overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(\phi^{-1}(0)).$$

Since $\ker\phi$ is in the nilradical, which is contained by every prime ideal, $V(\phi^{-1}(0)) = X$, so $\phi^*(Y)$ is dense in X.

Conversely, if $\phi^*(Y)$ is dense in X, then we have, by (iii),

$$X = \overline{\phi^*(Y)} = \overline{\phi^*(V(0))} = V(\phi^{-1}(0)),$$

so $\phi^{-1}(0) \subset \mathfrak{p}$ for all prime ideals, so $\ker\phi \subset \mathfrak{N}$. □

(vi) Proof. We have two ring homomorphisms $\phi : A \to B$ and $\psi : B \to C$. If $\mathfrak{r} \in \operatorname{Spec}(C)$, then we have a map $(\psi \circ \phi)^* : \operatorname{Spec}(C) \to \operatorname{Spec}(A)$. This takes $\mathfrak{r} \in \operatorname{Spec}(C)$ to $(\psi \circ \phi)^{-1}(\mathfrak{r}) \in \operatorname{Spec}(A)$. But, $(\psi \circ \phi)^{-1} = \phi^{-1} \circ \psi^{-1}$. In particular, $(\psi \circ \phi)^*$ decomposes into maps

$$\operatorname{Spec}(C) \xrightarrow{\psi^*} \operatorname{Spec}(B) \xrightarrow{\phi^*} \operatorname{Spec}(A),$$

so $(\psi \circ \phi)^* = \phi^* \circ \psi^*$. □

(vii) Proof. A is an integral domain with one non-zero prime ideal $\mathfrak{p}$, K is the field of fractions of A, and $B = (A/\mathfrak{p}) \times K$. We define $\phi : A \to B$ by $\phi(x) = (\overline{x}, x)$, where $\overline{x}$ is the image of x in $A/\mathfrak{p}$.

Since ϕ is a ring homomorphism, we have $\phi^* : \operatorname{Spec} B \to \operatorname{Spec} A$. A has only one non-zero prime ideal $\mathfrak{p}$, so $\operatorname{Spec} A = \{0, \mathfrak{p}\}$. For B, the only prime ideal of K is 0, and the only prime ideal of $A/\mathfrak{p}$ is 0. Thus, $\operatorname{Spec} B = \{0 \times K, (A/\mathfrak{p}) \times 0\}$. Note that 0 is not prime in B since $0 = (0 \times K)((A/\mathfrak{p}) \times 0)$. We have $\phi^{-1}(0 \times K) = \mathfrak{p}$ and $\phi^{-1}((A/\mathfrak{p}) \times 0) = 0$. Thus, ϕ^* is bijective.

However, we will show that it is not a homeomorphism. We only need to look at the image $\phi^*(\{(A/\mathfrak{p}) \times 0\}) = \{0\}$. Note that $\{(A/\mathfrak{p}) \times 0\} = V((A/\mathfrak{p}) \times 0)$, so this set is closed. However, $\{0\}$ is not closed since for any $\mathfrak{a}$, $0 \in V(\mathfrak{a})$ implies $\mathfrak{a} \subset 0 \subset \mathfrak{p}$, so $\mathfrak{p} \in \overline{\{0\}}$, i.e. ϕ^* maps a closed set to a non-closed set. Therefore, ϕ^* is not a homeomorphism. □

1.22 Exercise

First, we show that if $A = \prod_{i=1}^n$, then $\bigsqcup_{i=1}^n \operatorname{Spec}(A_j) \cong \operatorname{Spec} A$ as topological spaces.

Proof. Let $A = \prod_{i=1}^n$. We claim that all primes take the form

$$A_1 \times \cdots \times A_{j-1} \times \mathfrak{p}_j \times A_{j+1} \times \cdots \times A_n,$$

where $\mathfrak{p}_j \in \operatorname{Spec} A_j$. If $\pi_j : A \to A_j$ is the projection map, this is equivalent to saying all primes are of the form $\pi_j^{-1}(\mathfrak{p}_j)$ for some j and $\mathfrak{p}_j \in \operatorname{Spec}(A_j)$. First of all, it's clear that ideals of this form are prime since for $a, b \in A$ with $\pi_j(ab) \in \mathfrak{p}$, we have $\pi_j(a)\pi_j(b) \in \mathfrak{p}$, so either $\pi_j(a) \in \mathfrak{p}$ or $\pi_j(b) \in \mathfrak{p}$. Thus, either $a \in \pi_j^{-1}(\mathfrak{p})$ or $b \in \pi_j^{-1}(\mathfrak{p})$, so $\pi_j^{-1}(\mathfrak{p}) \in \operatorname{Spec}(A)$. In particular, we've shown

$$\coprod_{i=1}^n \operatorname{Spec}(A_j) \hookrightarrow \operatorname{Spec} A.$$

Conversely, suppose $\mathfrak{p}$ is a prime of A. Suppose $\pi_i(\mathfrak{p}) \neq A_i$ and $\pi_j(\mathfrak{p}) \neq A_j$ for distinct i, j. Then, let the components of $a \in A$ be all zeros except for the ith slot, where we choose $a_i \in A_i - \pi_i(\mathfrak{p})$. Similarly, we define $b \in A$ by letting the components of b be all zeros except for the jth slot, where we choose $b_j \in A_j - \pi_j(\mathfrak{p})$. Then, neither a nor b is in $\mathfrak{p}$, yet $ab = 0 \in \mathfrak{p}$, violating the primality of $\mathfrak{p}$. Thus, all projection maps of $\mathfrak{p}$ must be surjective except for one. So,

$$\mathfrak{p} = A_1 \times \cdots \times A_{j-1} \times \pi_j(\mathfrak{p}) \times A_{j+1} \times \cdots \times A_n.$$

Let $a_j, b_j \in A_j$. Lift a_j, b_j to $a, b \in A$ by letting all other components be zero. If $ab \in \mathfrak{p}$, then $a_j b_j \in \pi_j(\mathfrak{p})$. Since $\mathfrak{p}$ is prime, either a or b is in $\mathfrak{p}$. Without loss of generality, suppose $a \in \mathfrak{p}$. Then, $\pi_j(a) = a_j \in \pi_j(\mathfrak{p})$. Thus, $\pi_j(\mathfrak{p})$ is prime. Hence, we have a bijection

$$\coprod_{i=1}^n \operatorname{Spec}(A_j) \cong \operatorname{Spec} A.$$

To be more concrete, consider the projection $\pi_j : A \to A_j$. By Exercise 1.21(iv), π_j^* is a homeomorphism of $\operatorname{Spec}(A_j)$ onto the closed subset $V(\ker \pi_j)$ of $\operatorname{Spec} A$. Explicitly, the kernel is $A_1 \times \cdots \times A_{j-1} \times 0 \times A_{j+1} \times \cdots \times A_n$, which matches our constructed maps. We identified $V(A_1 \times \cdots \times A_{j-1} \times 0 \times A_{j+1} \times \cdots \times A_n)$ directly with $\operatorname{Spec} A_j$, which are homeomorphic. To be slightly pedantic, note that the closed subsets just mentioned are disjoint for distinct j by our earlier proof that a prime cannot have two non-surjective projections. □

Proof. By the previous proof, (ii) implies (i) since $A \cong A_1 \times A_2$ implies $\operatorname{Spec}(A) \cong \operatorname{Spec}(A_1) \sqcup \operatorname{Spec}(A_2)$, so $\operatorname{Spec}(A)$ is disconnected.

Now, we show (i) implies (iii). Suppose $\operatorname{Spec}(A)$ is disconnected. By Exercise 1.21(iv), $\operatorname{Spec}(A)$ is naturally homeomorphic to $\operatorname{Spec}(A/\mathfrak{N})$, so consider $A' = A/\mathfrak{N}$, which has no nilradical and is disconnected (by the homeomorphism). We can write $\operatorname{Spec}(A')$ as the disjoint union of two closed sets $V(\mathfrak{a})$ and $V(\mathfrak{b})$ for ideals $\mathfrak{a}, \mathfrak{b} \subset A'$. By Exercise 1.15(iv),

$$\operatorname{Spec}(A') = V(\mathfrak{a}) \cup V(\mathfrak{b}) = V(\mathfrak{a} \cap \mathfrak{b}),$$

so $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{N}_{A'} = 0$. Further, since $V(\mathfrak{a}) \cap V(\mathfrak{b}) = \emptyset$, $\mathfrak{a} + \mathfrak{b} = A'$. Consider the ring homomorphism $\psi : A' \to A'/\mathfrak{a} \times A'/\mathfrak{b}$. If $\psi(x) = (0, 0)$, then $x \in \mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{N}_{A'} = 0$, so ψ is injective. By Proposition 1.10(ii), since $\mathfrak{a} + \mathfrak{b} = (1)$, ψ is surjective, so we have a ring isomorphism. Thus,

$$A/\mathfrak{N} \cong A_1 \times A_2$$

for nonzero rings A_1 and A_2. Now, consider the idempotent $\overline{e} = (1, 0)$. Then, there is an $e \in A$ such that $e^2 - e \in \mathfrak{N}$. If $e^2 - e = 0$, then we have our idempotent; otherwise, there is a minimal $n > 0$ such that $(e^2 - e)^n = 0$. We can use the identity in $\mathbb{Z}[X]$ (the binomial theorem), which expands $1 = (X + (1 - X))^{2n-1}$ as

$$1 = \sum_{k=0}^{2n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k}.$$

Let $f = \sum_{k=0}^{n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k}$. Then, we have

$$f = f^2 + f \sum_{k=n}^{2n-1} \binom{2n-1}{k} e^k (1-e)^{2n-1-k},$$

but every term of f contains $(1 - e)$ of degree at least n, and all terms of the remainder contain e of degree at least n. Thus, the latter sum is annihilated, and we have $f = f^2$. Note that since $e(1-e) \in \mathfrak{N}$, $f \equiv (1-e)^{2n-1} \mod \mathfrak{N}$. Projecting down to $A_1 \times A_2$, we have $\overline{f} = (1-(1,0))^{2n-1} = (0,1)^{2n-1} = (0,1)$, which is not the image of either 0 or 1. Thus, as desired, f is a non-trival idempotent.

Now, we show (iii) implies (ii). Suppose e is an idempotent of A not equal to 0 or 1. Note that

$$(1-e)^2 = 1 - 2e + e^2 = 1 - 2e + e = 1 - e,$$

so $1 - e$ is also an idempotent. Further, note that $e(1 - e) = e - e^2 = e - e = 0$, so e and $1 - e$ are orthogonal idempotents. For any $x \in A$, we have $x = xe + x(1 - e)$. Then, we have the decomposition $\psi : A \cong Ae \times A(1 - e)$ by multiplying by e and $1 - e$, respectively, i.e. $\psi(x) = (xe, x(1 - e))$. Note that the right-hand side is a ring since e acts as the identity in Ae and $(1 - e)$ acts as the identity in $A(1 - e)$. Further, it's easily checked that the map is a ring homomorphism by using the idempotent relation and the orthogonality of the two idempotents. The only thing left to check is that the map is bijective. If $\psi(x) = (0, 0)$, then $xe = 0$ and $x(1 - e) = 0$, so their sum $xe + x(1 - e) = x = 0$, so injectivity holds. Conversely, if we have $(ae, b(1 - e))$ in the direct product, then $ae + b(1 - e) \in A$ maps to this element, so surjectivity holds. Thus, this is an isomorphism of rings. Finally, (iii) implies (ii), and now we see that all three properties are equivalent. □

Remark. *By Exercise 1.12, no local ring has an idempotent distinct from* 0 *and* 1*. We've shown this cannot happen if* $\mathrm{Spec}(A)$ *is disconnected, so the spectrum of a local ring is always connected.*

1.23 Exercise

(i) Proof. For any $f \in A$, consider X_f. Clearly, X_f is closed and open if and only if $V(f)$ is open and closed. Since $V(f)$ is closed by definition, we just have to show that $V(f)$ is an open set, i.e. there is some $g \in A$ such that $V(f) = V(g)^c$. Since A is a boolean ring, $f = f^2$ and $(1-f)^2 = 1-f$. Note that for any $x \in A$, we can write $x = xf + x(1-f)$. Consider the map $\psi : A \to Af \times A(1-f)$. We will show that this is a ring isomorphism. It's a homomorphism since Af and $A(1 - f)$ are rings with multiplicative identity f and $1 - f$; also, since $f(1 - f) = f - f^2 = 0$, multiplication is respected, i.e. $(af+b(1-f))(cf+d(1-f)) = (ac)f+(bd)(1-f)$. Now, we must show it is bijective. If $\psi(x) = (0, 0)$, then $xf = 0$ and $x(1 - f) = 0$, so $0 = xf + x(1 - f) = x$, hence ψ is injective.

Conversely, if we have $(af, b(1-f)) \in Af \times A(1-f)$, then $\psi(af + b(1-f)) = (af, b(1-f))$. Thus, ψ is surjective and hence bijective.

Since ψ realizes A as a direct product of rings, by Exercise 1.22, we know that

$$\operatorname{Spec}(A) \cong \operatorname{Spec}(Af) \sqcup \operatorname{Spec}(A(1-f)).$$

Now, considering the projections $\pi_f : A \to Af$ and $\pi_{1-f}A \to A(1-f)$, we have by Exercise 1.21(iv) that $\operatorname{Spec}(Af) \cong V(f)$ and $\operatorname{Spec}(A(1-f)) \cong V(1-f)$, where $\cong$ denotes homeomorphism. Thus, $\operatorname{Spec}(A)$ is a disjoint union of the closed sets $V(f)$ and $V(1-f)$, so as desired, $V(f) = V(1-f)^c$. □

(ii) Proof. Let $f_1, \ldots, f_n \in A$. We want to show that $X_{f_1} \cup \cdots \cup X_{f_n} = X_f$ for some $f \in A$. Taking complements, this is equivalent to the statement

$$V(f_1) \cap \cdots \cap V(f_n) = V(f).$$

Note that by Exercise 1.15(iii), we have

$$V(f_1) \cap \cdots \cap V(f_n) = V\left(\bigcup_{i=1}^{n}(f_i)\right) = V((f_1, \ldots, f_n)).$$

However, by Exercise 1.11(i), every finitely generated ideal of A is principal, so $(f_1, \ldots, f_n) = (f)$ for some $f \in A$, as desired. □

(iii) Proof. Let $Y \subset X$ be a closed and open set. Since Y is closed and X is quasi-compact by Exercise 1.17(v), we can easily show that Y is also quasi-compact. Let $\{U\}_U$ be an open cover of Y. Then, Y^c is open, so $\{Y^c\} \cup \{U\}_U$ is an open cover of X. Since X is quasi-compact, there is a finite subcover $\{V\}_V$, which obviously covers Y as well. Removing Y^c if necessary, which does not cover any points of Y, we have a finite subcover of Y that uses only open sets in $\{U\}_U$. Thus, Y is quasi-compact.

Since Y is also open, it is a union of basic open sets, and since it is quasi-compact, it is a union of finitely-many basic open sets. In other words, $Y = \bigcup_{i=1}^{n} X_{f_i}$. By (ii), there is some $f \in A$ such that $\bigcup_{i=1}^{n} X_{f_i} = X_f$, so $Y = X_f$, as desired. □

(iv) Proof. Let $\mathfrak{p}, \mathfrak{q}$ be distinct primes in $\operatorname{Spec} A$. Since they are distinct, suppose without loss of generality that $\mathfrak{p}$ contains some $f \in A$ that is not in $\mathfrak{q}$; otherwise, swap $\mathfrak{q}$ and $\mathfrak{p}$. Then, $(f) \subset \mathfrak{p}$, so $\mathfrak{p} \notin X_f$. But, since $\mathfrak{q} \notin V(f)$, we have $\mathfrak{q} \in X_f$. Since X_f is both open and closed by (i), we have two disjoint open sets X_f and X_f^c that contain $\mathfrak{q}$ and $\mathfrak{p}$, respectively. Thus, $\operatorname{Spec} A$ is Hausdorff. Since X is quasi-compact by Exercise 1.17(v), X is a compact Hausdorff space. □

1.24 Exercise

Proof. We start by showing $A(L)$ is a ring. We'll first prove the useful fact:

$$(a \wedge b)' = a' \vee b'.$$

Since the complement is unique, we just have to verify that it is the complement of $a \wedge b$. We have

$$\begin{aligned}(a \wedge b) \wedge (a' \vee b') &= a \wedge b \wedge (a' \vee b') \\ &= a \wedge ((b \wedge a') \vee (b \wedge b')) \\ &= a \wedge ((b \wedge a') \vee 0) \\ &= a \wedge (b \wedge a') \\ &= (a \wedge a') \wedge b \\ &= 0 \wedge b \\ &= 0.\end{aligned}$$

Then, we have

$$\begin{aligned}(a \wedge b) \vee (a' \vee b') &= (a \wedge b) \vee a' \vee b' \\ &= ((a \vee a') \wedge (b \vee a')) \vee b' \\ &= (1 \wedge (b \vee a')) \vee b' \\ &= (b \vee a') \vee b' \\ &= a' \vee (b \vee b') \\ &= a' \vee 1 \\ &= 1,\end{aligned}$$

so indeed, $(a \wedge b)' = a' \vee b'$. In fact, we can permute aa' and $b \mapsto b'$ and take the complement of both sides to obtain the complementary property:

$$a' \wedge b' = (a \vee b)'.$$

Now, we can begin proving that $A(L)$ is a ring, starting with the additive properties. Note that because the complements are unique, we have $0' = 1$ since $0 \wedge 1 = 0$ and $0 \vee 1 = 1$. Also note that since a and a' are symmetric in the definition of a', we have $(a')' = a$.

By the commutativity of $\vee$ and $\wedge$, $+$ is also commutative. We'll now show associativity.

$$\begin{aligned}(a + b) + c &= ((a \wedge b') \vee (a' \wedge b)) + c \\ &= (((a \wedge b') \vee (a' \wedge b)) \wedge c') \vee (((a \wedge b') \vee (a' \wedge b))' \wedge c) \\ &= ((a \wedge b' \wedge c') \vee (a' \wedge b \wedge c')) \vee (((a \wedge b') \vee (a' \wedge b))' \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (((a \wedge b')' \wedge (a' \wedge b)') \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee ((a' \vee b) \wedge (a \vee b') \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (a' \wedge a \wedge c) \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee (b \wedge b' \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (0 \wedge c) \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee (0 \wedge c) \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee 0 \vee (a' \wedge b' \wedge c) \vee (b \wedge a \wedge c) \vee 0 \\ &= (a \wedge b' \wedge c') \vee (a' \wedge b \wedge c') \vee (a' \wedge b' \wedge c) \vee (a \wedge b \wedge c).\end{aligned}$$

Note that the final line is completely symmetric under permutations of a, b, and c. Since we already have commutativity, this is equal to $c + (b + a)$. Since permuting a and c does not change the value of the final line (by aforementioned symmetry), we have $(a + b) + c = a + (b + c)$, so addition is associative.

For any $a \in A(L)$, we have

$$a + 0 = (a \wedge 0') \vee (a' \wedge 0) = (a \wedge 1) \vee (a' \wedge 0) = a \vee 0 = a,$$

so 0 is the additive identity. Further, for any $a \in A(L)$, we have

$$a + a' = (a \wedge a') \vee (a \wedge a') = a \wedge a' = 0,$$

by definition, so each a has an additive inverse. Thus, $(A(L), +)$ forms an abelian group.

Moving on, note that multiplication is commutative because $\wedge$ is commutative. Then, let $a, b \in A(L)$. We have $a \cdot 1 = a \wedge 1 = a$ since 1 is a global max. Thus, $A(L)$ has commutative multiplication with unit. Associativity follows easily since

$$a \cdot (b \cdot c) = a \wedge (b \wedge c) = (a \wedge b) \wedge c = (a \cdot b) \cdot c.$$

Lastly, we must show that multiplication distributes over addition. We have

$$\begin{aligned}
ab + ac &= (a \wedge b) + (a \wedge c) \\
&= ((a \wedge b) \wedge (a \wedge c)') \vee ((a \wedge b)' \wedge (a \wedge c)) \\
&= (a \wedge b \wedge (a' \vee c')) \vee ((a' \vee b') \wedge a \wedge c) \\
&= (a \wedge b \wedge a') \vee (a \wedge b \wedge c') \vee (a' \wedge a \wedge c) \vee (b' \wedge a \wedge c) \\
&= (0 \wedge b) \vee (a \wedge b \wedge c') \vee (0 \wedge c) \vee (b' \wedge a \wedge c) \\
&= 0 \vee (a \wedge b \wedge c') \vee 0 \vee (b' \wedge a \wedge c) \\
&= (a \wedge b \wedge c') \vee (a \wedge b' \wedge c) \\
&= a \wedge ((b \wedge c') \vee (b' \wedge c)) \\
&= a \wedge (b + c) \\
&= a(b + c),
\end{aligned}$$

so the distributive property holds, and $A(L)$ is a ring. It is now easy to see that it is a Boolean ring since for any $a \in A(L)$, $a^2 = a \wedge a = a$.

Conversely, suppose A is a Boolean ring. We define an ordering $a \leq b$ when $a = ab$. Thus, 0 is clearly a minimal element and 1 is clearly a maximal element, satisfying property (i) of a Boolean lattice. Define $a \wedge b = ab$ and $a \vee b = a + b + ab$. We now verify the two distributive properties:

$$\begin{aligned}
a \wedge (b \vee c) &= a(b + c + bc) \\
&= ab + ac + abc \\
&= ab + ac + a^2bc \\
&= ab + ac + (ab)(ac) \\
&= (ab) \vee (ac) \\
&= (a \wedge b) \vee (a \wedge c).
\end{aligned}$$

Also,

$$\begin{aligned}(a \vee b) \wedge (a \vee c) &= (a+b+ab)(a+c+ac) \\ &= a^2 + ac + a^2c + ab + bc + abc + a^2b + abc + a^2bc \\ &= a + ac + ac + ab + bc + abc + ab + abc + abc \\ &= a + 2ac + 2ab + bc + abc + 2abc \\ &= a + bc + abc \\ &= a \vee (bc) \\ &= a \vee (b \wedge c),\end{aligned}$$

so property (ii) is satisfied.

Lastly, we must show that there is a unique complement a' for each $a \in A$. We claim that this is satisfied by $1 - a$. We can prove existence and uniqueness at once by applying the definitions. We have

$$0 = a \wedge a' = aa' \quad \text{and} \quad 1 = a \vee a' = a + a' + aa'.$$

Since $aa' = 0$, we have that $1 = a + a' + 0$, so $a' = 1 - a$. This proves (iii). Thus, A defined by our $\wedge$ and $\vee$ is a Boolean lattice. □

1.25 Exercise

Proof. Here, we prove Stone's theorem. Let L be a Boolean lattice. By Exercise 1.24, L can be made into a Boolean ring, $A(L)$. We define $\mathcal{T}$ to be the open-and-closed subsets of $\mathrm{Spec}(A(L))$. By Exercise 1.23, the elements of $\mathcal{T}$ are exactly X_f, where $f \in A(L)$. By Exercise 1.23, $\mathrm{Spec}(A(L))$ is a compact Hausdorff space, so $\mathcal{T}$ is the set of open-and-closed subsets of a compact Hausdorff space. Thus, we aim to show that $L \to \mathcal{T}$ is an isomorphism of lattices.

We show that $\mathcal{T}$ is a Boolean lattice. We define $U \leq V$ when $U \subset V$. (i) X_1 is the maximal element; X_0 is the minimal element. (ii) We define $\wedge$ and $\vee$ by $\cap$ and $\cup$. These are closed operations (two open-and-closed sets yield an open-and-closed set), and they distribute over each other. (iii) We define the complement of X_f by $X_f^c = X_{f'}$; there is such an f' since X_f^c is open-and-closed. $X_f \wedge X_{f'} = X_f \cap X_{f'} = \emptyset = X_0$ and $X_f \vee X_{f'} = X_f \cup X_{f'} = X = X_1$. Thus, $\mathcal{T}$ is a lattice.

The map in view factors as $L \to A(L) \to \mathcal{T}$, where the first map is clearly a bijection. Suppose $f, g \in A(L)$. Then, if $F, G \in L$ yield $f, g \in A(L)$, then $fg = F \wedge G$ by our construction of the Boolean ring (Exercise 1.24). Thus, $F \wedge G \mapsto fg \mapsto X_{fg}$. However, $F \mapsto f \mapsto X_f$ and $G \mapsto g \mapsto X_g$. However, $X_f \wedge X_g = X_f \cap X_g = X_{fg}$ by Exercise 1.17, so the map preserves $\wedge$. If f, f' are complements in L, then in $A(L)$, they satisfy

$$f + f' = (f \wedge f) \vee (f' \wedge f') = f \vee f' = 1,$$

so $f' = 1 - f$. However, $(f) + (1 - f) = (1)$ and $(f) \cap (1 - f)$ contains no prime ideals. Thus, $X_f^c = X_{1-f}$, so the map preserves complements. Because the map preserves complements and $\wedge$, de Morgan's law here shows that it also preserves $\vee$. In particular, $L \to \mathcal{T}$ is a homomorphism of lattices.

Lastly, we show bijectivity. Note that $L \to A(L)$ is a bijection since the map preserves the underlying set. Every element of $\mathcal{T}$ has the form X_f for some $f \in A$, and the map is $f \mapsto X_f$, so clearly the map is surjective. To show injectivity, suppose $f, g \in X_f$ and $X_f = X_g$. Then, for every prime ideal $\mathfrak{p}$ of $A(L)$, $f \in \mathfrak{p} \iff g \in \mathfrak{p}$. By Exercise 1.11, $A/\mathfrak{p}$ is the field of two

elements, so for all $\mathfrak{p}$, $f \equiv g \mod \mathfrak{p}$. In particular, $f - g \in \mathfrak{p}$ for all $\mathfrak{p}$. Hence, $f - g \in \mathfrak{N}$. Since $a^2 = a$ for all $a \in A(L)$ (Boolean ring), $(f-g)^2 = f - g$. However, $f - g$ is nilpotent, so $(f-g)^n = 0$ for some $n > 0$. By a simple inductive argument using $(f-g)^2 = f - g$, we have $f - g = (f-g)^n = 0$, so $f = g$. Thus, the map is injective and therefore bijective. Hence, $L \to \mathcal{T}$ is a lattice isomorphism. □

1.26 Exercise

Let A be a ring. We define $\mathrm{Max}(A)$ to be the subspace of $\mathrm{Spec}(A)$ containing the maximal ideals, and we endow it with the induced topology. It is called the *maximal spectrum* of A. Let X be a compact Hausdorff space, and let $C(X)$ denote the ring of all real-valued continuous functions on X. For each $x \in X$, let $\mathfrak{m}_x$ be the set of all $f \in C(X)$ such that $f(x) = 0$. The ideal $\mathfrak{m}_x$ is maximal since it is the kernel of the map $C(X) \to \mathbb{R}$ via $f \mapsto f(x)$. Thus, we have a map $\mu : X \to \mathrm{Max}(C(X))$ via $x \mapsto \mathfrak{m}_x$. We aim to show that μ is a homeomorphism of X onto $\mathrm{Max}(C(X))$.

Proof. Let $\mathfrak{m} \in \mathrm{Max}(C(X))$, and define $V = V(\mathfrak{m})$ by

$$V = \{x \in X : f(x) = 0 \text{ for all } f \in \mathfrak{m}\}.$$

Suppose V is empty. Then, for each $x \in X$, there exists $f_x \in \mathfrak{m}$ such that $f_x(x) \neq 0$. Since f_x is continuous, there is an open neighborhood U_x of x in X on which f_x does not vanish. By compactness, a finite number of the neighborhoods, say $U_{x_1}, \ldots, U_{x_n}$ cover X. Let

$$f = f_{x_1}^2 + \cdots + f_{x_n}^2.$$

Then, f does not vanish anywhere on X, hence $1/f$ is a well-defined real-valued function on X. Open intervals are a basis of $\mathbb{R}$, so since 0 is not in the image of $1/f$, it is enough to consider intervals (a, b), where $0 < a < b$ or $a < b < 0$. Note that $(1/f)^{-1}((a,b)) = f^{-1}((1/b, 1/a))$, but this is open by the continuity of f, so $1/f$ is also continuous. Hence, $1/f \in C(X)$ and f is a unit, violating the maximality of $\mathfrak{m}$. Hence, V is non-empty, and there is some $x \in V$. Then, $\mathfrak{m} \subset \mathfrak{m}_x$, so by maximality, $\mathfrak{m} = \mathfrak{m}_x$. Hence, μ is surjective.

Since X is compact and Hausdorff, it is also normal. By the Hausdorff property, with distinct $x, y \in X$, we have disjoint open sets U_x and U_y such that $x \in U_x$ and $y \in U_y$. By normality, there are disjoint closed sets V_x and V_y containing U_x and U_y, respectively. By Urysohn's lemma (since X is normal), there exists a continuous function which is 0 on V_x and 1 on V_y, and vice-versa. Thus, $\mathfrak{m}_x$ contains an element that $\mathfrak{m}_y$ does not and vice-versa, so $\mu : x \mapsto \mathfrak{m}_x$ is injective. Compiling results, we've shown that μ is bijective.

Let $f \in C(X)$. Define

$$U_f = \{x \in X : f(x) \neq 0\} \quad \text{and} \quad \tilde{U}_f = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : f \notin \mathfrak{m}\}.$$

Consider any closed set C of X. If $y \notin C$, then since $\{y\}$ is closed, we have by Urysohn's lemma that there is some $f \in C(X)$ such that $f(y) = 1$ and f is 0 on C. Thus, C is the intersection of U_f^c for all $f \in C(X)$ such that f is zero on C. Thus, any open set is a union of U_f, so $\{U_f\}$ forms a basis of X. Also, for any $f \in C(X)$, X_f forms a basis of $\mathrm{Spec}(C(X))$ by Exercise 1.17. Restricted to $\mathrm{Max}(C(X))$, these open sets have the form

$$\{\mathfrak{m} \in \mathrm{Max}(C(X)) : (f) \subset \mathfrak{m}\}^c = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : (f) \not\subset \mathfrak{m}\} = \{\mathfrak{m} \in \mathrm{Max}(C(X)) : f \notin \mathfrak{m}\} = \tilde{U}_f,$$

so $\{\tilde{U}_f\}$ defines a basis for $\mathrm{Max}(C(X))$ since the topology is inherited from $\mathrm{Spec}(C(X))$.

Thus, if we can show $\mu(U_f) = \tilde{U}_f$, we will have shown that μ is an open map and thus a homeomorphism by its already established bijectivity. If $f(x) \neq 0$, then $f \notin \mathfrak{m}_x$. Conversely, if $f \notin \mathfrak{m}_x$, then by definition, $f(x) \neq 0$. Thus, $\mu(U_f) = \tilde{U}_f$, completing the proof. □

Remark. *This implies the topological space X can be reconstructed from the ring $C(X)$ via* $\mathrm{Max}(C(X))$.

1.27 Exercise

Let k be an algebraically closed field, and let

$$f_a(x_1, \ldots, x_n) = 0$$

be a set of polynomial equations in n variables with coefficients in k. The set X of all points $x = (x_1, \ldots, x_n) \in k^n$ which satisfy the equations is an *affine algebraic variety*. Consider the set of all polynomials $g \in k[x_1, \ldots, x_n]$ with the property that $g(x) = 0$ for all $x \in X$. This set is an ideal $I(X)$ in the polynomial ring, and is called the *ideal of the variety* X. The quotient ring

$$P(X) = k[x_1, \ldots, x_n]/I(X)$$

is the ring of polynomial functions on X because two polynomials g, h define the same polynomial on X if and only if $g - h$ vanishes at every point of X, i.e. if and only if $g - h \in I(X)$.

Let ξ_i be the image of t_i under the projection $k[x_1, \ldots, x_n] \to P(X)$. These are the *coordinate functions* on $P(X)$. If $x \in X$, then $\xi_i(x)$ is well-defined and gives the ith coordinate of x. Clearly, $P(X)$ is generated as a k-algebra by the coordinate functions, and is called the *coordinate ring* (or *affine algebra*) of X.

As in Exercise 1.26, for each $x \in X$, let $\mathfrak{m}_x$ be the ideal of all $f \in P(X)$ such that $f(x) = 0$; it is a maximal ideal of $P(X)$. Hence, we have a mapping $\mu : X \to \mathrm{Max}(P(X))$ by $x \mapsto \mathfrak{m}_x$. It is easy to show that μ is injective: if $x \neq y$, we have $\xi_i(x) \neq \xi_i(y)$ for some i, so $\xi_i - \xi_i(x) \in \mathfrak{m}_x$, but it is not in $\mathfrak{m}_y$. Therefore, $\mathfrak{m}_x \neq \mathfrak{m}_y$. We now show that μ is surjective; this creates a one-to-one correspondence between X and $\mathrm{Max}(P(X))$, one form of Hilbert's Nullstellensatz.

We'll start by showing that the maximal ideals of $k[x]$ are exactly $(x - a)$ for $a \in k$.

Proof. Clearly, $(x-a)$ is maximal as it is the kernel of the surjective ring homomorphism $k[x] \to k$ via $f \mapsto f(a)$ since k is algebraically closed. Let $\mathfrak{m}$ be a maximal ideal of $k[x]$. Let $f \in \mathfrak{m}$. Since the degree of a polynomial is a nonnegative integer, there is a minimal degree m of $\mathfrak{m}$ (it's clear that $\mathfrak{m} \neq 0$). Note that $m > 0$; otherwise $\mathfrak{m}$ contains a unit, violating maximality. Let f be a polynomial of degree m. Then, since k is algebraically closed, it factors as $a_0(x-a_1)\cdots(x-a_m)$ for $a_0 \neq 0$. Redefine f as $a_0^{-1}f$. Since $\mathfrak{m}$ is an ideal and k is a subring of $k[x]$, $f = (x-a_1)\cdots(x-a_m)$ is in $\mathfrak{m}$. If g is also a monic degree m polynomial of $\mathfrak{m}$, then if $f - g$ is nonzero, it has degree less than m, violating the minimality of m. Hence, $f = g$, so all degree m polynomials in $\mathfrak{m}$ have $a_1, \ldots, a_m$ as roots.

Let g be a polynomial in $\mathfrak{m}$ of minimal degree such that $g(a_1) \neq 0$. Then, it has degree $n > m$ since f is not proportional to g (values at a_1). Multiply by some element of k to make it monic; it's still in $\mathfrak{m}$. Since k is algebraically closed, $g(x) = (x-b_1)\cdots(x-b_n)$. Now, $f(x)(x-b_{m+1})\cdots(x-b_n)$ is in $\mathfrak{m}$ since $(x-b_{m+1})\cdots(x-b_n) \in k[x]$. Thus,

$$h(x) = f(x)(x-b_{m+1})\cdots(x-b_n) - g(x) = (f(x) - (x-b_1)\cdots(x-b_m))(x-b_{m+1})\cdots(x-b_n).$$

Note that $h(x) \in \mathfrak{m}$. Also, $f(x)-(x-b_1)\cdots(x-b_m) \neq 0$ since $f(a_1) = 0$, but $(x-b_1)\cdots(x-b_m) \neq 0$. However, both are monic of degree m, so their difference has degree less than m. Thus, $h(x)$ has degree less than n. However, $f(x)(x-b_{m+1})\cdots(x-b_n)$ evaluates to 0 at a_1, but $g(a_1) \neq 0$, so $h(a_1) \neq 0$. This violates the minimality of n. Thus, we have that no such g exists, meaning that $g(a_1) = 0$ for all $g \in \mathfrak{m}$. Hence, $\mathfrak{m} \subset (x - a_1)$, so by maximality, $\mathfrak{m} = (x - a_1)$. □

Now, we show that the maximal ideals of $k[x_1, \ldots, x_n]$ are exactly $(x_1 - a_1, \ldots, x_n - a_n)$ for $a_i \in k$.

Proof. Consider the ideal $(x_1 - a_1, \ldots, x_n - a_n)$ of $k[x_1, \ldots, x_n]$ for $a_i \in k$. It is the kernel of the surjective ring homomorphism $k[x_1, \ldots, x_n] \to k$ via $f \mapsto f(a_1, \ldots, a_n)$, so it is a maximal ideal.

Conversely, suppose $\mathfrak{m}$ is a maximal ideal of $k[x_1, \ldots, x_n]$. For each i, we have an injective ring homomorphism $k[x_i] \to k[x_1, \ldots, x_n]$. The contraction of $\mathfrak{m}$ is a prime ideal of $k[x_i]$. This is contained in some maximal ideal $(x_i - a_i)$ by the previous proof. Note that $\mathfrak{m} \subset (x_i - a_i) + \mathfrak{m}$; since the contraction of $\mathfrak{m}$ is contained in $(x_i - a_i)$, the contraction of the sum is also contained in $(x_i - a_i)$. Thus, the sum cannot be the whole ring, so by the maximality of $\mathfrak{m}$, we have $(x_i - a_i) \subset \mathfrak{m}$. Doing this for each i, we have that $(x_i - a_i)$ is a subset of $\mathfrak{m}$ for each i. Thus, $(x_1 - a_1, \ldots, x_n - a_n) \subset \mathfrak{m}$, so by the maximality of both sets, $\mathfrak{m} = (x_1 - a_1, \ldots, x_n - a_n)$. □

Now, we can prove the surjectivity of μ.

Proof. Suppose $\mathfrak{m} \in \mathrm{Max}(P(X))$. Using the ring homomorphism $\phi : k[x_1, \ldots, x_n] \to P(X)$, we have that $\phi^{-1}(\mathfrak{m})$ is a prime ideal of $k[x_1, \ldots, x_n]$. Let $\mathfrak{M}$ be a maximal ideal of $k[x_1, \ldots, x_n]$ containing $\phi^{-1}(\mathfrak{m})$. Since ϕ is surjective, $\phi(\mathfrak{M})$ is an ideal of $P(X)$ containing $\mathfrak{m}$. By maximality, $\phi(\mathfrak{M}) = \mathfrak{m}$ or $\phi(\mathfrak{M}) = P(X)$.

In the first case, $\mathfrak{M} \subset \phi^{-1}(\mathfrak{m})$, so $\mathfrak{M} = \phi^{-1}(\mathfrak{m})$, and $\phi^{-1}(\mathfrak{m})$ is a maximal ideal of $k[x_1, \ldots, x_n]$. In the second case, since $P(X) = k[x_1, \ldots, x_n]/I(X)$, we have that $I(X) + \mathfrak{M} = k[x_1, \ldots, x_2]$. In particular, since $\mathfrak{M}$ is maximal, $\mathfrak{M}$ does not contain $I(X)$; else, $k[x_1, \ldots, x_2] = I(X) + \mathfrak{M} = \mathfrak{M}$. But, $I(X) \subset \phi^{-1}(\mathfrak{m})$, violating the fact that $\mathfrak{M}$ contains $\phi^{-1}(\mathfrak{m})$. Hence, we have that $\phi^{-1}(\mathfrak{m})$ is a maximal ideal of $k[x_1, \ldots, x_n]$.

By our previous proof, we then have that $\phi^{-1}(\mathfrak{m}) = (x_1 - a_1, \ldots, x_n - a_n)$ for $a_i \in k$. Since $I(X) \subset \phi^{-1}(\mathfrak{m})$, we must have $a = (a_1, \ldots, a_n) \in X$. By the surjectivity of the projection map ϕ, $\phi((x_1 - a_1, \ldots, x_n - a_n)) = \mathfrak{m}$. Since for any $f \in \phi((x_1 - a_1, \ldots, x_n - a_n))$, $f(a) = 0$, we have that $\mathfrak{m}_a \subset \mathfrak{m}$. By maximality, $\mathfrak{m} = \mathfrak{m}_a$, proving the surjectivity of μ. □

1.28 Exercise

Let $f_1, \ldots, f_m$ be elements of $k[x_1, \ldots, x_n]$. They determine a *polynomial mapping* $\phi : k^n \to k^m$; if $x \in k^n$, the cooordinates of $\phi(x)$ are $f_1(x), \ldots, f_m(x)$. Let X, Y be affine algebraic varieties in k^n, k^m, respectively. A mapping $\phi : X \to Y$ is said to be *regular* if ϕ is the restriction to X of a polynomial mapping from k^n to k^m. If η is a polynomial function on Y, then $\eta \circ \phi$ is a polynomial function on X. Hence ϕ induces a k-algebra homomorphism $P(Y) \to P(X)$, namely $\eta \mapsto \eta \circ \phi$. We will show that this yields a one-to-one correspondence between the regular mappings $X \to Y$ and the k-algebra homomorphisms $P(Y) \to P(X)$.

Proof. Let $\phi : X \to Y$ be a regular mapping. We'll start by showing that the k-homomorphism $\eta \mapsto \eta \circ \phi$ is unique to ϕ. Suppose $\psi : X \to Y$ is a regular function and $\eta \circ \psi = \eta \circ \phi$ for all $\eta \in P(Y)$. Consider the coordinate functions $\xi_i \in P(Y)$. Then, $\xi_i \circ \psi = \xi_i \circ \phi$ for all i, so $\psi = \phi$.

Now, consider an arbitrary k-algebra homomorphism $\theta : P(Y) \to P(X)$, or

$$\theta : k[y_1, \ldots, y_m]/I(Y) \to k[x_1, \ldots, x_n]/I(X).$$

Since ξ_i generate $P(Y)$ as a k-algebra, θ is defined entirely by $\theta(\xi_i)$. Define θ_1 to be the map

$$\theta_1 : k[y_1, \ldots, y_m] \to k[x_1, \ldots, x_n]/I(X)$$

that sends $y_i \mapsto \theta(\xi_i)$. Then, θ_1 is a k-algebra homomorphism. Likewise, there are Ξ_i that generate $P(X)$. Hence, $\theta_1(y_i) = \sum_{\alpha \in I_i} c_{i,\alpha} \Xi^\alpha$ for $c_{i,\alpha} \in k$. Then, we can define ψ to be the unique k-algebra homomorphism

$$\psi : k[y_1, \ldots, y_m] \to k[x_1, \ldots, x_n]$$

defined by $\psi(y_i) = \sum_{\alpha \in I_i} c_{i,\alpha} x^\alpha$. Then, if the canonical projection maps defining the varieties are $\pi_Y : k[y_1, \ldots, y_m] \to k[y_1, \ldots, y_m]/I(Y)$ and $\pi_X : k[x_1, \ldots, x_n] \to k[x_1, \ldots, x_n]/I(X)$, we have $\pi_X \circ \psi = \theta \circ \pi_Y$ by our definition of ψ. In particular, $\pi_X(\psi(I(Y))) = \theta(0) = 0$, so $\psi : I(Y) \to I(X)$.

For each y_i, $\psi(y_i)$ is a polynomial of $k[x_1, \ldots, x_n]$. Hence, $(\psi(y_1)(x), \ldots, \psi(y_m)(x))$ defines a map $k[x_1, \ldots, x_n] \to k[y_1, \ldots, y_m]$. Since $\psi : I(Y) \to I(X)$, this defines a map $X \to Y$. This is regular as it is the restriction of a k-homomorphism $k[x_1, \ldots, x_n] \to k[y_1, \ldots, y_m]$.

The regular functions $X \to Y$ are one-to-one with the k-homomorphisms $P(Y) \to P(X)$. $\square$

2 Modules (Done)

2.1 Exercise

Proof. Let $x \otimes y \in (\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z})$. Since m, n are coprime, Bezout's lemma says that there are $a, b \in \mathbb{Z}$ such that $am + bn = 1$. Then,

$$x \otimes y = am(x \otimes y) + bn(x \otimes y) = (amx) \otimes y + x \otimes (bny) = 0 \otimes y + x \otimes 0 = 0,$$

so $(\mathbb{Z}/m\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/n\mathbb{Z}) = 0$. □

2.2 Exercise

Proof. Consider the exact sequence

$$0 \to \mathfrak{a} \to A \to A/\mathfrak{a} \to 0.$$

Tensoring with M, we have, by the right exactness of tensor products,

$$\mathfrak{a} \otimes_A M \to A \otimes_A M \to (A/\mathfrak{a}) \otimes_A M \to 0.$$

Under canonical isomorphism, we have $A \otimes_A M = M$. Through this canonical isomorphism the image of $\mathfrak{a} \otimes_A M$ is generated via $a \otimes m \mapsto am \in \mathfrak{a}M$. In fact, any element of $\mathfrak{a}M$ has the form $\sum_i a_i m_i$ for $a_i \in \mathfrak{a}$ and $m_i \in M$; this can be obtained through the map via the element $\sum_i (a_i, m_i) \in \mathfrak{a} \otimes_A M$. By exactness, we have

$$(A/\mathfrak{a}) \otimes_A M \cong (A \otimes_A M)/\operatorname{im}(\mathfrak{a} \otimes_A M) \cong M/\mathfrak{a}M,$$

completing the proof. □

2.3 Exercise

Proof. Suppose $M \otimes_A N = 0$ for a local ring A and finitely-generated A-modules M, N. If $k = A/\mathfrak{m}$ is the residue field, let $M_k = k \otimes_A M = (A/\mathfrak{m}) \otimes_A M \cong M/\mathfrak{m}M$ (Exercise 2.2). Since $M \otimes_A N = 0$, we have

$$0 = k \otimes_A (M \otimes_A N) \otimes_A k = M_k \otimes_A N_k.$$

By Exercise 2.2, we have

$$M_k \otimes_A N_k \cong (M/\mathfrak{m}M) \otimes_A (N/\mathfrak{m}N).$$

Note that the A-linearity of $(M/\mathfrak{m}M) \otimes_A (N/\mathfrak{m}N)$ factors through $k = A/\mathfrak{m}$, so

$$0 = M_k \otimes_A N_k \cong M_k \otimes_k N_k.$$

However, note that since M and N are finitely-generated, M_k and N_k are finite-dimensional vector spaces over k. Thus, since the tensor product vanishes, either M_k or N_k must vanish (dimensions multiply). By symmetry, assume $M_k = 0$ without loss of generality.

Since A is a local ring, $\mathfrak{m}$ is identical to the Jacobson radical. Since $M/\mathfrak{m}M \cong M_k = 0$, we have that $\mathfrak{m}M = M$. Since M is also finitely-generated, we have by Nakayama's lemma that $M = 0$, as desired. □

2.4 Exercise

Proof. Suppose M is flat and that we have an injective homomorphism of A-modules $j : N' \to N$. By flatness,

$$1_M \otimes j : M \otimes N' \to M \otimes N$$

is also injective. As $M \cong \bigoplus_i M_i$, we can define maps $1_{M_i} \otimes j : M_i \otimes N' \to M_i \otimes N$. Under canonical isomorphism, we have for N (and N')

$$\bigoplus_{i \in I} (M_i \otimes N) \cong \left(\bigoplus_{i \in I} M_i \right) \otimes N \cong M \otimes N.$$

Thus, $1_M \otimes j = \oplus_{i \in I}(1_{M_i} \otimes j)$. By the flatness of M, we have

$$0 = \ker(1_M \otimes j) = \ker\left(\oplus_{i \in I}(1_{M_i} \otimes j)\right) = \oplus_{i \in I} \ker(1_{M_i} \otimes j),$$

so each $1_{M_i} \otimes j$ is injective, ensuring that M_i is flat for all $i \in I$.

Conversely, suppose that M_i is flat for all $i \in I$. Then, any exact sequence of A-modules

$$0 \to N' \to N \to N'' \to 0$$

yields the exact sequence

$$0 \to M_i \otimes N' \to M_i \otimes N \to M_i \otimes N'' \to 0$$

for all $i \in I$. Exact sequence commute with direct sums, so we have

$$0 \to \bigoplus_{i \in I} (M_i \otimes N') \to \bigoplus_{i \in I} (M_i \otimes N) \to \bigoplus_{i \in I} (M_i \otimes N'') \to 0$$

$$0 \to \left(\bigoplus_{i \in I} M_i \right) \otimes N' \to \left(\bigoplus_{i \in I} M_i \right) \otimes N \to \left(\bigoplus_{i \in I} M_i \right) \otimes N'' \to 0$$

$$0 \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0,$$

so M is flat. □

2.5 Exercise

Proof. Consider $A[x]$ as an A-module. For any non-negative integers $n \geq 0$, we can view $M_n := Ax^n$ as an A-submodule since it is additively-closed, and A has a closed action on it. As a module, $M_n \cong A$ under the A-module homomorphism $a \mapsto ax^n$. Thus, for any A-module N, $M_n \otimes_A N \cong N$ under $M_n \cong A$ and the canonical isomorphism $A \otimes_A N \cong N \cong N$. Thus, tensoring a short exact sequence of A-modules with M_n yields an isomorphic short exact sequence, so by definition, M_n is a flat A-module.

Now, notice that any $f \in A[x]$ can be written as a finite and unique sum of elements in various M_n. In particular, $A[x] \cong \bigoplus_{n \geq 0} M_n$ as A-modules. Invoking Exercise 2.4, $A[x]$ is also flat. □

2.6 Exercise

Proof. If M is an A-module, then we can define an action of $A[x]$ on $M[x]$ in a natural way. More explicitly, the action of $ax^i \in A[x]$ on $mx^j \in M[x]$ is $(am)x^{i+j} \in M[x]$, exploiting the fact that $am \in M$. We extend the action A-linearly to all of $A[x]$ and $M[x]$, so $M[x]$ has an $A[x]$-action. Clearly, $M[x]$ is an abelian group additively. Thus, the only non-trivial condition to check on $M[x]$ being an $A[x]$-module is $(fg)m = f(gm)$. Thus, let $f = \sum_i f_i x^i$, $g = \sum_j g_j x^j$, and $m = \sum_k m_k x^k$, where $m \in M[x]$ and $f, g \in A[x]$. Then,

$$(fg)m = \left(\sum_i \sum_j f_i g_j x^{i+j}\right)\left(\sum_k m_k x^k\right) = \sum_i \sum_j \sum_k f_i g_j m_k x^{i+j+k}$$

and

$$f(gm) = \left(\sum_i f_i x^i\right)\left(\sum_j \sum_k g_j m_k x^{j+k}\right) = \sum_i \sum_j \sum_k f_i g_j m_k x^{i+j+k}$$

so $(fg)m = f(gm)$. Thus, $M[x]$ is an $A[x]$-module.

Now, we show that $M[x] \cong A[x] \otimes_A M$ as $A[x]$-modules. We use the direct product decomposition of Exercise 2.5, where as an A-module, $A[x] = \bigoplus_{i\geq 0} Ax^i$; we saw that $A \cong Ax^i$ as an A-module via $a \mapsto ax^i$. Thus, under canonical isomorphism, we have

$$A[x] \otimes_A M \cong \left(\bigoplus_{i\geq 0} Ax^i\right) \otimes_A M \cong \bigoplus_{i\geq 0} \left((Ax^i) \otimes_A M\right).$$

Since $Ax^i \cong A$ and $A \otimes_A M \cong M$, both as A-modules, we can compose the maps to obtain the isomorphism of A-modules $(Ax^i) \otimes_A M \cong Mx^i$. Thus, we have

$$A[x] \otimes_A M \cong \bigoplus_{i\geq 0} \left((Ax^i) \otimes_A M\right) \cong \bigoplus_{i\geq 0} Mx^i \cong M[x].$$

As A-modules then, $A[x] \otimes_A M \cong M[x]$. This establishes a bijection. To show this is truly an $A[x]$-module homomorphism, we'll show the action commutes with $(x^i, m) \mapsto mx^i$; terms of this form generate the modules A-linearly, so this will suffice. Let $f = \sum_j f_j x^j \in A[x]$. Then,

$$f(x).(x^i \otimes m) = (f(x) \otimes 1)(x^i \otimes m) = \left(\sum_j f_j x^{i+j}\right) \otimes m$$

$$\mapsto \sum_j f_j m x^{i+j} = \left(\sum_j f_j x^j\right)(mx^i) = f(x).(mx^i),$$

so we have the desired homomorphism. Thus, $M[x] \cong A[x] \otimes_A M$ as $A[x]$-modules. □

2.7 Exercise

Proof. Let $\mathfrak{p}$ be a prime ideal of A. Since $\mathfrak{p}$ is an A-module, by Exercise 2.6, $\mathfrak{p}[x]$ is an $A[x]$-module, so it is an ideal of $A[x]$. Suppose $f, g \in A[x]$ and $fg \in \mathfrak{p}[x]$. With the map $A \to A/\mathfrak{p}$, we have a surjective $A[x]$-module homomorphism $A[x] \to (A/\mathfrak{p})[x]$. Under this map, $\overline{fg} = 0$. Suppose that $\overline{f}$ is a zero-divisor. Then, by Exercise 1.2, there is a nonzero element $a \in A/\mathfrak{p}$ such that $a\overline{f} = 0$. Thus, $\sum_i a f_i x^i = 0$, but since $A/\mathfrak{p}$ is an integral domain, we must have $f_i = 0$ for all i, i.e. $\overline{f} = 0$. Hence, $f \in \mathfrak{p}[x]$, and $\mathfrak{p}[x]$ is a prime ideal of $A[x]$. □

Remark. *Note that if* $\mathfrak{m}$ *is a maximal ideal of* A*, then* $\mathfrak{m}[x]$ *is not maximal as an ideal of* $A[x]$*. Simply note that* $\mathfrak{m} \subset \mathfrak{m} + xA[x] \subset A[x]$ *is a proper chain of ideals.*

2.8 Exercise

(i) Proof. Suppose M and N are flat A-modules. Let $j : J \to J'$ be any injection of A-modules. Then, since N is flat,

$$(1 \otimes j) : N \otimes J \to N \otimes J'$$

is also injective. But M is also flat, so

$$1 \otimes (1 \otimes j) : M \otimes (N \otimes J) \to M \otimes (N \otimes J')$$

is injective. Recall that we have the natural isomorphism $M \otimes (N \otimes J) \cong (M \otimes N) \otimes J$ (same with J' in place of J). Thus, we have an injection

$$1 \otimes j : (M \otimes N) \otimes J \to (M \otimes N) \otimes J'.$$

Hence, $M \otimes_A N$ is a flat A-module. □

(ii) Proof. Suppose B is a flat A-algebra and N is a flat B-module. Let $j : J \to J'$ be any injection of A-modules. Then, since B is flat as an A-module,

$$(1 \otimes j) : B \otimes_A J \to B \otimes_A J'$$

is injective. Also, note that $B \otimes_A J$ and $B \otimes_A J'$ are naturally B-modules. Since N is flat as a B-module,

$$1 \otimes (1 \otimes j) : N \otimes_B (B \otimes_A J) \to N \otimes_B (B \otimes_A J')$$

is injective. Since B is an (A, B)-bimodule, under canonical isomorphism, we have an injection

$$1 \otimes j : (N \otimes_B B) \otimes_A J \to (N \otimes_B B) \otimes_A J'.$$

Yet again, under canonical isomorphism, we have an injection

$$1 \otimes j : N \otimes_A J \to N \otimes_A J',$$

so N is flat as an A-module. □

2.9 Exercise

Proof. Let

$$0 \to M' \xrightarrow{f} M \xrightarrow{g} M'' \to 0$$

be an exact sequence of A-modules, where M' and M'' are finitely-generated. By exactness, we have the isomorphism

$$M/f(M') \cong M''.$$

Since M'' is finitely-generated, $M/f(M')$ is generated by finitely-many elements $\{y_i\} \subset M$. Likewise, since f is injective and M' is finitely-generated, $f(M')$ is generated by finitely-many elements $\{x_j\} \subset M$. We claim $\{y_i\} \cup \{x_j\}$ generate M, so suppose $m \in M$. Then,

$$m + f(M') = \sum_i a_i y_i + f(M')$$

for $a_i \in A$, so $m - \sum_i a_i y_i \in f(M')$. Thus,

$$m - \sum_i a_i y_i = \sum_j b_j x_j$$

for $b_j \in A$, so $m = \sum_i a_i y_i + \sum_j b_j x_j$. Hence, M is finitely-generated. □

2.10 Exercise

Proof. By assumption, u induces a surjective homomorphism $M/\mathfrak{a}M \to N/\mathfrak{a}N$, so $N = \mathfrak{a}N + \operatorname{im} u$. Note that $\operatorname{im} u$ is an A-submodule of N. Thus, since N is a finitely-generated A-module and $\mathfrak{a}$ is in the Jacobson radical of A, we have by Corollary 2.7 that $N = \operatorname{im} u$, so u is surjective. □

2.11 Exercise

Proof. Let $A \neq 0$ be a ring, so it has a maximal ideal $\mathfrak{m}$. Suppose $\phi : A^m \to A^n$ is an A-module isomorphism. Then, if $k = A/\mathfrak{m}$ is the residue field, we have an isomorphism

$$1 \otimes \phi : k \otimes A^m \to k \otimes A^n.$$

Note that for $N > 0$, $A^N = \oplus_{i=1}^N A$, so $k \otimes A^N \cong \oplus_{i=1}^N (k \otimes A)$ by natural isomorphism, and $k \otimes_A A \cong k$ by natural isomorphism. Thus, our map $1 \otimes \phi$ is a map $k^m \to k^n$. This is an A-module isomorphism and thus a k-module (vector space) isomorphism. Hence, $m = n$. □

Proof. Now, suppose $A \neq 0$, and $\phi : A^m \to A^n$ is a surjective A-module homomorphism. Since A is nonzero, it has a maximal ideal $\mathfrak{m}$, so let $k = A/\mathfrak{m}$ be the residue field. Then, we have a surjection (right exactness)

$$1 \otimes \phi : k \otimes A^m \to k \otimes A^n.$$

Perfectly analogous to the last proof, $k \otimes A^N \cong k^N$, so we have a surjective homomorphism of A-modules $k^m \to k^n$. Thus, it is also a surjective k-module (vector space) homomorphism, so $m \geq n$. □

Proof. Again, assume $A \neq 0$ is a ring. Here, we prove that if $\phi : A^m \to A^n$ is injective, then $n \geq m$. We'll prove this by contradiction. Suppose we have an injective A-module homomorphism $\phi : A^m \to A^n$ with $m > n$. We can compose ϕ with the map $\psi : A^n \to A^n \oplus 0^{m-n} \subset A^m$ so that $\epsilon = \psi \circ \phi$ is an endomorphism of the A-module A^m. As the composition of two injective maps, it is itself injective. Note that $\epsilon(A^m) \subset A^n \oplus 0^{m-n}$. Since A^m is clearly finitely-generated as an A-module, by Proposition 2.4, ϵ satisfies an equation of the form:

$$\epsilon^k + a_1 \epsilon^{k-1} + \cdots + a_k \operatorname{id}_{A^m} = 0,$$

where $a_i \in A$ for all i. Choose an equation of this form such that k is minimal. Note that if $k = 0$, then $0 = \operatorname{id}_{A^m}$, but $A \neq 0$, so $k > 0$.

Consider the projection map $\pi : A^m \to A$ that projects onto the last direct summand. Since $m > n$ by assumption, $\pi \circ \epsilon = 0$. Thus, taking π on both sides, we have $\pi(a_k \operatorname{id}_{A^m}) = 0$. But,

$$\pi(a_k \operatorname{id}_{A^m}((0, \ldots, 0, 1))) = \pi(0, \ldots, 0, a_k) = a_k,$$

so $a_k = 0$. Thus, our equation is

$$\epsilon \circ (\epsilon^{k-1} + a_1 \epsilon^{k-2} + \cdots + a_{k-1} \operatorname{id}_{A^m}) = 0.$$

Since ϵ is injective,

$$\epsilon^{k-1} + a_1\epsilon^{k-2} + \cdots + a_{k-1}\,\mathrm{id}_{A^m} = 0,$$

violating the minimality of k and giving us the desired contradiction. □

2.12 Exercise

Proof. Let $\{e_1, \ldots, e_n\}$ be an A-basis of A^n, and choose $u_i \in M$ such that $\phi(u_i) = e_i$. This can be done since the given A-module homomorphism $M \to A^n$ is surjective. Let $m \in M$, and let $\sum_i a_i e_i$ be the unique decomposition of $\phi(m) \in A^n$, where $a_i \in A$. Then, $\phi\left(m - \sum_i a_i u_i\right) = 0$, so $m - \sum_i a_i u_i \in \ker\phi$. Also, if $\sum_i a_i u_i \in \ker\phi$, then $0 = \phi\left(\sum_i a_i u_i\right) = \sum_i a_i e_i$, but since $\{e_i\}$ is an A-basis, $a_i = 0$ for all i. If U is the A-submodule generated by $\{u_i\}$, we've shown that $\ker\phi \cap U = \{0\}$ and $M = \ker\phi + U$. Hence, $M = \ker\phi \oplus U$.

Since M is finitely-generated, we have finitely-many generators $\{v_j\}$ of M. Under our direct sum decomposition, we have a projection map $\pi : M \to \ker\phi$. Thus, $\{\pi(v_j)\}$ generate $\ker\phi$, so $\ker\phi$ is finitely-generated. □

2.13 Exercise

Proof. If $f : A \to B$ is a ring homomorphism and N is a B-module, we can restrict scalars to A, then extend scalars back to B. Thus, we have another B-module $N_B := B \otimes_A N$ ($b \in B$ acts via $b \otimes 1$) and a B-module homomorphism $g : N \to N_B$ via $y \mapsto 1 \otimes y$. We have a canonical "absorption" map $p : N_B \to N$ that maps $b \otimes y \mapsto by$. Let $1 \otimes y \in \operatorname{im} g$; $\operatorname{im} g$ is generated by elements of this form. Then, if $1 \otimes y \in \ker p$, $0 = p(1 \otimes y) = y$, so $1 \otimes y = 0$. Thus, $\operatorname{im} g \cap \ker p = \{0\}$. Consider $b \otimes y \in N_B$; elements of this form generate N_B. Note that $b \otimes y = (b \otimes y - 1 \otimes (by)) + 1 \otimes (by)$, and that $b \otimes y - 1 \otimes (by) \in \ker p$ and $1 \otimes (by) \in \operatorname{im} g$. Thus, $N_B = \operatorname{im} g \oplus \ker p$. □

2.14 Exercise

A partially ordered set I is said to be a *directed* set if for each pair $i, j \in I$ there exists $k \in I$ such that $i \leq k$ and $j \leq k$.

Let A be a ring, let I be a directed set, and let $(M_i)_{i \in I}$ be a family of A-modules indexed by I. For each pair $i, j \in I$ such that $i \leq j$, let $\mu_{ij} : M_i \to M_j$ be an A-homomorphism, and suppose that the following axioms are satisfied:

(1) μ_{ii} is the identity mapping of M_i for all $i \in I$;

(2) $\mu_{ik} = \mu_{jk} \circ \mu_{ij}$ whenever $i \leq j \leq k$.

Then, the modules M_i and homomorphisms μ_{ij} are said to form a *direct system* $\mathbf{M} = (M_i, \mu_{ij})$ over the directed set I.

We shall construct an A-module M called the *direct limit* of the direct system $\mathbf{M}$. Let C be the direct sum of the M_i and identity each module M_i with its canonical image in C. Let D be the submodule of C generated by all elements of the form $x_i - \mu_{ij}(x_i)$ where $i \leq j$ and $x_i \in M_i$. Let $M = C/D$, let $\mu : C \to M$ be the projection, and let μ_i be the restriction of μ to M_i.

The module M, or more correctly the pair consisting of M and the family of homomorphisms $\mu_i : M_i \to M$, is called the *direct limit* of the direct system $\mathbf{M}$, and is written $\varinjlim M_i$. For $x_i \in M_i$ and $i \leq j$, we have by construction that $x_i - \mu_{ij}(x_i) \in D = \ker\mu$. Thus, $\mu_i = \mu_j \circ \mu_{ij}$ since

$$\mu_i(x_i) = \mu(x_i) = \mu(\mu_{ij}(x_i)) = \mu_j(\mu_{ij}(x_i)).$$

2.15 Exercise

Proof. In the construction of M in Exercise 2.14, we saw that M is a quotient of the direct sum $C = \bigoplus_{i\in I} M_i$. Let $y \in M$, so there is some $x \in C$ such that $\mu(x) = y$. Then, $x = \sum_{j\in J} x_j$ for a finite subset $J \subset I$. Since I is a directed set, for any two elements $i, j \in I$, there exists a $k \in I$ such that $i \leq k$ and $j \leq k$. Since J is a finite subset, a routine inductive argument shows there is some $i \in I$ such that $j \leq i$ for all $j \in J$. Thus,

$$y = \mu(x) = \mu\left(\sum_j x_j\right) = \sum_j \mu(x_j) = \sum_j \mu_j(x_j) = \sum_j (\mu_i \circ \mu_{ji})(x_j) = \mu_i\left(\sum_j \mu_{ji}(x_j)\right),$$

where we invoked $\mu_j = \mu_i \circ \mu_{ji}$ whenever $j \leq i$ from Exercise 2.14. Thus, letting $x_i = \sum_j \mu_{ji}(x_j)$, we see that $y = \mu_i(x_i)$ for some $i \in I$ and $x_i \in M_i$, as desired. □

Proof. Let's first lay some groundwork. D, the kernel of μ, is generated by elements of the form $x_i - \mu_{ij}(x_i)$. In other words, the general element has the form

$$\sum_k a_k(x_{i_k} - \mu_{i_k j_k}(x_{i_k})).$$

But, since μ_{**} is an A-module homomorphism, we have $a_k(x_{i_k} - \mu_{i_k j_k}(x_{i_k})) = (a_k x_{i_k}) - \mu_{i_k j_k}(a_k x_{i_k})$, so we can always choose a representation such that $a_k = 1$ for all k. Further, if we have a repeated pair $(i, j) = (i_k, j_k)$, then $(x_1 - \mu_{ij}(x_1)) + (x_2 + \mu_{ij}(x_2)) = (x_1 + x_2) - \mu_{ij}(x_1 + x_2)$, so we can choose a representation such that pairs (i, j) occur only once.

Suppose $\mu_i(x_i) = 0$. Then, $x_i \in M_i \cap D$, so it has a representation

$$x_i = \sum_{(j,k)\in P} (y_j - \mu_{jk}(y_j)) = \sum_{j\in J} z_j$$

for a finite subset $J \subset I$. Since we are currently in the direct sum C and $x_i \in M_i$, the projection map onto M_j for $i \neq j$ applied to the equation vanishes. Since the right-hand side breaks the middle sum into M_j components, we have $z_j = 0$ if $j \neq i$ and $z_i = x_i$. Let $\ell \in I$ be such that $j \leq \ell$ for all $j \in J$; such an element can be found since I is a directed set and J is finite. Since $z_i = x_i$ and $z_j = 0$ for $j \neq i$, we have trivially that $\mu_{i\ell}(x_i) = \sum_{j\in J} \mu_{j\ell}(z_j)$. Thus,

$$\mu_{i\ell}(x_i) = \sum_{j\in J} \mu_{j\ell}(z_j) = \sum_{(j,k)\in P} (\mu_{j\ell}(y_j) - (\mu_{k\ell} \circ \mu_{jk})(y_j)) = 0$$

since $\mu_{j\ell} = \mu_{k\ell} \circ \mu_{jk}$, as desired. □

2.16 Exercise

Proof. Let's start by showing the direct limit M satisfies these properties. Let N be an A-module such that for each $i \in I$, $\alpha_i : M_i \to N$ is an A-module homomorphism such that $\alpha_i = \alpha_j \circ \mu_{ij}$ whenever $i \leq j$. Let $y \in M$. By Exercise 2.15, $y = \mu_i(x_i)$ for some $x_i \in M_i$. We'd like to define $\alpha : M \to N$ such that $\alpha(y) = \alpha_i(x_i)$, but we have to first show this is well-defined. Let $x_i' \in M_i$ be another element such that $y = \mu_i(x_i')$. Then, $\mu_i(x_i - x_i') = 0$. So, to show the map α is well-defined, we have to show that $\mu_i(y_i) = 0$ implies $\alpha_i(y_i) = 0$ (and thus that $\alpha_i(x_i) = \alpha_i(x_i')$). By Exercise 2.15, we know that $\mu_i(y_i) = 0$ implies there is some $j \geq i$ such that

$\mu_{ij}(y_i) = 0$. By assumption, $\alpha_i(y_i) = \alpha_j(\mu_{ij}(y_i)) = \alpha_j(0) = 0$, so α is well-defined. By definition, $\alpha_i = \alpha \circ \mu_i$. Also, this homomorphism is clearly unique since if $y \in M$, $y = \mu_i(x_i)$ for some $i \in I$ and $x_i \in M_i$, so $\alpha(y) = \alpha(\mu_i(x_i)) = \alpha_i(x_i)$. In particular, α is completely determined by the given data $\{(M_i, \alpha_i)\}_{i \in I}$.

We now show uniqueness. Let M' satisfy the given properties. Then, if M is the direct limit, M is an A-module with compatible maps $\mu_i : M_i \to M$ since $\mu_i = \mu_j \circ \mu_{ij}$ for $j \geq i$ by Exercise 2.14. Then, $\mu_i = \alpha' \circ \mu_i'$ for all i, where $\alpha' : M' \to M$ is an A-module homomorphism. Symmetrically, we have a map $\alpha : M \to M'$ such that $\mu_i' = \alpha \circ \mu_i$. Thus, we have $(\alpha' \circ \alpha)$ is the identity on the image of M_i in M for all i; every element of M has this form, so $\alpha' \circ \alpha = \mathrm{id}_M$. Similarly, $\alpha \circ \alpha' = \mathrm{id}_{M'}$, so $\alpha : M \to M'$ is an A-module isomorphism. □

2.17 Exercise

Proof. By construction, $\varinjlim M_i = \left(\bigoplus_{i \in I} M_i\right) / D$, where D is the A-module generated by $x_i - \mu_{ij}(x_i)$ for $i \leq j$. Consider the canonical homomorphism $\psi : \left(\bigoplus_{i \in I} M_i\right) \to \sum M_i$. In this case, μ_{ij} is the map that identifies M_i as a submodule of M_j. In particular, the submodule D of the direct sum is in $\ker \psi$ since $x_i - \mu_{ij}(x_i) = 0 \in \sum_k M_k$ for $i \leq j$. Thus, we have a well-defined A-module homomorphism

$$\phi : \left(\bigoplus_{i \in I} M_i\right) / D \to \sum M_i.$$

It is clearly surjective, since $\sum_i x_i \in \sum M_i$ satisfies $\psi\left(\sum_i x_i\right) = \sum_i x_i$. Now, suppose we have $\phi\left(D + \sum_i x_i\right) = 0$, i.e. $\sum_i x_i = 0$ in $\sum M_i$. Since there are finitely-many i indexing this sum, there is some k such that for each i, $x_i \in M_k$. Then,

$$D + \sum_i x_i = \left(D + \sum_i (\mu_{ik}(x_i) - x_i)\right) + \sum_i x_i = D + \sum_i \mu_{ik}(x_i).$$

In $\bigoplus_{i \in I} M_i$, $\sum_i \mu_{ik}(x_i)$ lies entirely in the direct summand M_k; since it is 0 in M_k, it is zero in the direct sum, so ϕ is injective. Thus, ϕ is an A-module isomorphism, so $\varinjlim M_i \cong \sum M_i$. Any element of $\sum M_i$ is a finite sum $\sum_i x_i$, where $x_i \in M_i$. Since I is directed, there is some M_k that contains each M_i, so $\sum M_i \subset \bigcup M_i$. The converse clearly holds, so $\varinjlim M_i \cong \sum M_i \cong \bigcup M_i$. □

2.18 Exercise

Proof. Consider the A-module homomorphisms $\alpha_i = \nu_i \circ \phi_i$ from M_i to N. If $i \leq j$, then

$$\alpha_j \circ \mu_{ij} = \nu_j \circ \phi_j \circ \mu_{ij} = \nu_j \circ \nu_{ij} \circ \phi_i = \nu_i \circ \phi_i = \alpha_i,$$

so by the universal mapping property of direct limits (Exercise 2.16), we have a unique homomorphism $\phi : M \to N$ such that $\alpha_i = \phi \circ \mu_i$ for all i. Unwinding the definition, $\phi \circ \mu_i = \nu_i \circ \phi_i$ for all $i \in I$, as desired. □

2.19 Exercise

Proof. Suppose we have directed systems of A-modules $\mathbf{M}$, $\mathbf{N}$, and $\mathbf{P}$ over the same directed set I such that

$$M_i \xrightarrow{f_i} N_i \xrightarrow{g_i} P_i$$

is exact for all $i \in I$. Consider the sequence of direct limits

$$M \xrightarrow{f} N \xrightarrow{g} P,$$

where $f = \varinjlim f_i$ and $g = \varinjlim g_i$ (Exercise 2.18). Let μ_{ij}, ν_{ij}, and π_{ij} be the corresponding compatibility maps. Suppose $m \in M$. Then, by Exercise 2.15, there is some $i \in I$ such that $m = \mu_i(x_i)$. Then,

$$(g \circ f)(m) = (g \circ f \circ \mu_i)(x_i) = (g \circ \nu_i \circ f_i)(x_i) = \pi_i((g_i \circ f_i)(x_i)) = \pi_i(0) = 0$$

since $g_i \circ f_i = 0$ by exactness. Thus, $g \circ f = 0$, so $\operatorname{im} f \subset \ker g$.

Now, let $n \in \ker g$. By Exercise 2.15, there is some $i \in I$ such that $n = \nu_i(x_i)$. Thus,

$$0 = g(n) = (g \circ \nu_i)(x_i) = (\pi_i \circ g_i)(x_i) = \pi_i(g_i(x_i)).$$

By Exercise 2.15, there is some $j \geq i$ such that $\pi_{ij}(g_i(x_i)) = 0$. So,

$$0 = (\pi_{ij} \circ g_i)(x_i) = (g_j \circ \nu_{ij})(x_i) = g_j(\nu_{ij}(x_i)) = g_j(x_j),$$

where we define $x_j = \nu_{ij}(x_i)$. Note that $n = \nu_i(x_i) = \nu_j(\nu_{ij}(x_i)) = \nu_j(x_j)$ and $x_j \in \ker g_j$. By exactness, $x_j \in \operatorname{im} f_j$, so there is some $y_j \in M_j$ such that $x_j = f_j(y_j)$. Let $y = \mu_j(y_j)$. Then,

$$f(y) = (f \circ \mu_j)(y_j) = (\nu_j \circ f_j)(y_j) = \nu_j(f_j(y_j)) = \nu_j(x_j) = n,$$

so $n \in \operatorname{im} f$. Thus, $\ker g \subset \operatorname{im} f$, so finally, $\ker g = \operatorname{im} f$, giving us the desired exactness. □

2.20 Exercise

Proof. Let N be any A-module. Then, $(M_i \otimes N, \mu_{ij} \otimes 1)$ is a direct system; let $P = \varinjlim(M_i \otimes N)$. For each $i \in I$ we have a homomorphism $\mu_i \otimes 1 : M_i \otimes N \to M \otimes N$, so by Exercise 2.16, we have a homomorphism $\psi : P \to M \otimes N$.

For each $i \in I$, define $g_i : M_i \times N \to M_i \otimes N$ to be the canonical A-bilinear map. For every $n \in N$, this defines an A-module homomorphism $g_i(-, n) \to M_i \otimes N$. Passing to the direct limit, we obtain $g_n = \varinjlim g_i(-, n)$, which satisfies $g_n \circ \mu_i = (\mu_i \otimes 1)(g_i(-, n))$. We define $g : M \times N \to P$ as $g(m, n) = g_n(m)$. This is A-linear in the first coordinate, so we must show it in the second.

$$(g_n + g_m) \circ \mu_i = g_n \circ \mu_i + g_m \circ \mu_i = (\mu_i \otimes 1)(g_i(-, n)) + (\mu_i \otimes 1)(g_i(-, m))$$

$$= (\mu_i \otimes 1)(g_i(-, n + m)) = g_{n+m} \circ \mu_i$$

for all i. By the uniqueness of the direct limit map (Exercise 2.18), $g_n + g_m = g_{n+m}$. Also,

$$a(g_n \circ \mu_i) = a(\mu_i \otimes 1)(g_i(-, n)) = (\mu_i \otimes 1)(g_i(-, an)) = g_{an} \circ \mu_i$$

for all i. By the uniqueness of the direct limit map again, $ag_n = g_{an}$, so $g : M \times N \to P$ is A-bilinear. Thus, g factors through $M \otimes N$ giving us an A-module homomorphism

$$\phi : M \otimes N \to P.$$

Consider the map $\psi \circ \phi$. Since it is an A-module endomorphism on $M \otimes N$, it is defined by its image on generators $m \otimes n$. Since $m \in M$, note that there is some $i \in I$ such that $\mu_i(x_i) = m$ for $x_i \in M_i$. Then,

$$(\psi \circ \phi)(m \otimes n) = \psi(g(m, n)) = \psi(g_n(m)) = \psi((g_n \circ \mu_i)(x_i)) = \psi(g_i(x_i, n))$$

$$= \psi(x_i \otimes n) = (\mu_i \otimes 1)(x_i \otimes n) = \mu_i(x_i) \otimes n = m \otimes n,$$

so $\psi \circ \phi = 1_{M \otimes N}$. Conversely, let $p \in P$. Then, by Exercise 2.15, there is some $i \in I$ and $\sum x_i \otimes n \in M_i \otimes N$ such that $p = (\mu_i \otimes 1)\left(\sum x_i \otimes n\right)$. Since we already know that $\phi \circ \psi$ is an A-module homomorphism, it is enough to show that $(\phi \circ \psi)((\mu_i \otimes 1)(x_i \otimes n)) = (\mu_i \otimes 1)(x_i \otimes n)$ to see that $\phi \circ \psi = \mathrm{id}_P$. So let's prove that:

$$(\phi \circ \psi)((\mu_i \otimes 1)(x_i \otimes n)) = \phi(\mu_i(x_i) \otimes n) = g(\mu_i(x_i), n)$$

$$= g_n(\mu_i(x_i)) = (\mu_i \otimes 1)(g_i(x_i, n)) = (\mu_i \otimes 1)(x_i \otimes n),$$

as desired. □

2.21 Exercise

Proof. Let $\{(A_i, \alpha_{ij})\}$ be a directed system of rings so that $\alpha_{ij} : A_i \to A_j$ is a ring homomorphism; trivially, they are $\mathbb{Z}$-modules and α_{ij} are $\mathbb{Z}$-module homomorphisms, so via the construction of Exercise 2.14, we have a direct limit $A = \varinjlim A_i$. A is thus a $\mathbb{Z}$-module, so $(A, +)$ is an abelian group.

Let $x, y \in A$. Then, by Exercise 2.15, there are $i, j \in I$, $x_i \in A_i$, and $y_j \in A_j$ such that $\alpha_i(x_i) = x$ and $\alpha_j(y_j) = y$. Since I is a directed set, there is some k such that $i \leq k$ and $j \leq k$. Consider then, $x_k = \alpha_{ik}(x_i)$ and $y_k = \alpha_{jk}(y_j)$. Note that

$$\alpha_k(x_k) = (\alpha_k \circ \alpha_{ik})(x_i) = \alpha_i(x_i) = x,$$

and similiarly, $\alpha_k(y_k) = y$. Thus, we define $x \cdot y = \alpha_k(x_k \cdot y_k)$, using the ring structure of A_k. To show that this is well-defined, we have to show that this is independent of both k and the representatives x_k and y_k. Suppose we have another representative x'_k of x. Then,

$$\alpha_k(x'_k \cdot y_k) - \alpha_k(x_k \cdot y_k) = \alpha_k((x'_k - x_k) \cdot y_k).$$

Note that since $\alpha_k(x'_k - x_k) = 0$, we have by Exercise 2.15, that there is some $\ell \geq k$ such that $\alpha_{k\ell}(x'_k - x_k) = 0$. Since $\alpha_{k\ell}$ is a ring homomorphism, we have

$$\alpha_k((x'_k - x_k) \cdot y_k) = (\alpha_\ell \circ \alpha_{k\ell})((x'_k - x_k) \cdot y_k) = \alpha_\ell(\alpha_{k\ell}(x'_k - x_k) \cdot \alpha_{k\ell}(y_k)) = \alpha_\ell(0 \cdot \alpha_{k\ell}(y_k)) = 0,$$

By exactly the same argument, or rather by symmetry, if y'_k is another representative of y, the same invariance holds: i.e.,

$$\alpha(x_k \cdot y_k) = \alpha(x'_k \cdot y_k) = \alpha(x'_k \cdot y'_k).$$

Thus, we've shown the multiplication does not depend on the particular representations of x and y.

Now, we must show that it does not depend on k. Let $i \in I$ be such that there are $x_i, y_i \in A_i$ satisfying $x = \alpha_i(x_i)$ and $y = \alpha_i(y_i)$. Then, since I is directed, there is some $j \in J$ such that $i \leq j$ and $k \leq j$. Define $x_{ij} = \alpha_{ij}(x_i)$, and define x_{kj}, y_{ij}, and y_{kj} similarly. Then,

$$\alpha_j(x_{ij}) = \alpha_j(\alpha_{ij}(x_i)) = \alpha_i(x_i) = x;$$

the same argument applies to x_{kj}, y_{ij}, and y_{kj}. In particular, x_{ij} and x_{kj} are two representations of x in A_j, and y_{ij} and y_{kj} are two representations of y in A_j. Since we've already shown the representations in the same ring produce the same multiplication, we have

$$\alpha_i(x_i \cdot y_i) = \alpha_j(\alpha_{ij}(x_i \cdot y_i)) = \alpha_j(x_{ij} \cdot y_{ij}) = \alpha_j(x_{kj} \cdot y_{kj}) = \alpha_j(\alpha_{kj}(x_k \cdot y_k)) = \alpha_k(x_k \cdot y_k),$$

so the multiplication depends neither on the particular ring A_i or on the particular representation of x and y, so $x \cdot y$ is well-defined.

Using this definition, we can easily verify other properties. Given $x, y, z \in A$, we can pass to a ring A_i that has a representative for each of x, y, and z. Then, the representatives satisfy the associativity $(x_i y_i)z_i = x_i(y_i z_i)$, commutativity $x_i y_i = y_i x_i$, and the distributive property $x_i(y_i + z_i) = x_i y_i = x_i z_i$, so x, y, and z satisfy these properties. Likewise, $1 \in A_i$ for all $i \in I$, and $\alpha_{ij}(1) = 1$ whenever $i \leq j$. Thus, $1 = \alpha_i(1) \in A$ for any $i \in I$. Thus, if x has a representative in A_i, $1 \cdot x = \alpha_i(1_i \cdot x_i) = \alpha_i(x_i) = x$, so finally, A is a ring.

Further, if $x, y \in A_i$, $\alpha_i(x + y) = \alpha_i(x) + \alpha_i(y)$, and by our multiplication definition, we have $\alpha_i(xy) = \alpha_i(x)\alpha_i(y)$ and $\alpha_i(1) = 1$. Thus, $\alpha_i : A_i \to A$ is a ring homomorphism for all $i \in I$. □

Proof. Suppose $A_i \neq 0$ for all $i \in I$, and suppose to the contrary that $A = \varinjlim A_i = 0$. Since $A = 0$, for any $i \in I$, we have $\alpha_i(1_i) = 0$. By Exercise 2.15, there is some j such that $\alpha_{ij}(1_i) = 0$. But $\alpha_{ij}(1_i) = 1_j \neq 0$ since $A_j \neq 0$. Hence, if $A = 0$, then there is some $i \in I$ such that $A_i = 0$. □

2.22 Exercise

Proof. First, we verify that given a directed system of ring $\{(A_i, \alpha_{ij})\}$, the nilradicals $\mathfrak{N}_i$ form a directed system $\{(\mathfrak{N}_i, \alpha_{ij})\}$. Suppose that $n_i \in \mathfrak{N}_i$ and $j \geq i$. By definition, there is some m such that $n_i^m = 0$. Thus, $\alpha_{ij}(n_i)^m = \alpha_{ij}(n_i^m) = \alpha_{ij}(0) = 0$, using that fact that α_{ij} is a ring homomorphism. Thus, the directed system is well-defined.

Also, note that for each i, since $\mathfrak{N}_i$ is an ideal of A_i, we have exact sequences

$$0 \to \mathfrak{N}_i \to A_i \to A_i/\mathfrak{N}_i \to 0.$$

By Exercise 2.19, we have an exact sequence of direct limits. In particular, $\varinjlim \mathfrak{N}_i \overset{\iota}{\hookrightarrow} A$, where $A = \varinjlim A_i$. The map $\iota = \alpha_i \circ \iota_i$, where $\iota_i : \mathfrak{N}_i \to A_i$ is just inclusion. In particular, the map $\mathfrak{N}_i \to \varinjlim \mathfrak{N}_j$ is just the restriction of α_i.

If $\mathfrak{N}_A$ is the nilradical of A, let $n \in \mathfrak{N}_A$. By Exercise 2.15, there is some $i \in I$ and $n_i \in A_i$ such that $\alpha_i(n_i) = n$. Since, per Exercise 2.21, α_i is a ring homomorphism, $\alpha_i(n_i^m) = n^m = 0$. By Exercise 2.15, there is some $j \geq i$ such that $0 = \alpha_{ij}(n_i^m)$. But α_{ij} is ring homomorphism, so $n_j = \alpha_{ij}(n_i)$ satisfies $n_j^m = 0$, so $n_j \in \mathfrak{N}_j$. Further, we have $\alpha_j(n_j) = \alpha_j(\alpha_{ij}(n_i)) = \alpha_i(n_i) = n$. Thus, $n \in \varinjlim \mathfrak{N}_i$.

Conversely, if $n \in \varinjlim \mathfrak{N}_i$, then there is some $i \in I$ such that $n = \alpha_i(n_i)$ for some $n_i \in \mathfrak{N}_i$. Since $n_i^m = 0$ for some m, we have $n^m = \alpha_i(n_i)^m = \alpha_i(n_i^m) = \alpha_i(0) = 0$, so $n \in \mathfrak{N}_A$. Thus, we have $\mathfrak{N}_A = \varinjlim \mathfrak{N}_i$. □

Proof. Let $A = \varinjlim A_i$, and suppose A is not an integral domain. So, there are nonzero $x, y \in A$ such that $xy = 0$. By Exercise 2.21, there is some $i \in I$ such that $x = \alpha_i(x_i)$ and $y = \alpha_i(y_i)$ for some $x_i, y_i \in A_i$. Then,

$$0 = xy = \alpha_i(x_i)\alpha_i(y_i) = \alpha_i(x_i y_i).$$

By Exercise 2.15, there is some $j \geq i$ such that $0 = \alpha_{ij}(x_i y_i) = \alpha_{ij}(x_i)\alpha_{ij}(y_i)$ since α_{ij} is a ring homomorphism. Suppose $\alpha_{ij}(x_i) = 0$. Then,

$$0 = \alpha_j(0) = \alpha_j(\alpha_{ij}(x_i)) = \alpha_i(x_i) = x,$$

but we assumed x is nonzero, so $\alpha_{ij}(x_i) \neq 0$. Similarly, since we assumed y is nonzero, we have $\alpha_{ij}(y_i) \neq 0$. Thus, A_j is not an integral domain. Therefore, if A_i are all integral domains, then A is also an integral domain. □

2.23 Exercise

Let $(B_\lambda)_{\lambda\in\Lambda}$ be a family of A-algebras. For each finite subset of $J \subset \Lambda$, let B_J denote the tensor product (over A) of the B_λ for $\lambda \in J$. If J' is another finite subset of Λ and $J \subset J'$, there is a canonical A-algebra homomorphism $\phi_{JJ'} : B_J \to B_{J'}$. Explicitly, write $B_{J'} = B_J \otimes B_K$, where $K = J' - J$. Then, $B_J \to B_J \otimes B_K$ via $x \mapsto x \otimes \left(\bigotimes_{k\in K} 1_k\right)$. Clearly, this is an A-algebra homomorphism. Note that if we have sets $I \subset J \subset K$, then the maps defined in this way satisfy $\psi_{JK} \circ \psi_{IJ} = \psi_{IK}$. Also, $\psi_{II} = \mathrm{id}_{B_I}$. In particular, $\{(B_I, B_{IJ})\}$ form a directed system.

Let B denote the direct limit of the rings B_J as J runs through all finite subsets of Λ. The ring B has a natural A-algebra structure for which the homomorphisms $B_J \to B$ are A-algebra homomorphisms (as in Exercise 2.14).

Proof. Our canonical maps $\mu_J : B_J \to B$ are compatible A-module homomorphisms by the construction in Exercise 2.14. By Exercise 2.21, $\mu_J : B_J \to B$ are ring homomorphisms. Thus, the μ_J are A-algebra homomorphisms. □

The A-algebra B is the *tensor product* of the family $(B_\lambda)_{\lambda\in\Lambda}$.

2.24 Exercise

Proof. (i) $\implies$ (ii). Suppose the A-module M is flat, and let N be an A-module. Then, we can take a free resolution of N:

$$\cdots \to F_2 \to F_1 \to N \to 0.$$

Since this sequence is exact and M is flat, tensoring with M yields another exact sequence. Thus, the homology of this sequence is trivial, so $\mathrm{Tor}_n^A(M, N) = 0$ for all $n > 0$ and all A-modules N.

(ii) $\implies$ (iii). Trivial.

(iii) $\implies$ (i). Let

$$0 \to N' \to N \to N'' \to 0$$

be any exact sequence of A-modules. Tensoring with M, we obtain the exact sequence

$$\mathrm{Tor}_1^A(M, N'') \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0.$$

By assumption, $\mathrm{Tor}_1^A(M, N'') = 0$, so the sequence is a short exact sequence. Thus, M is flat. □

2.25 Exercise

Proof. Let M be any A-module. Since N'' is flat, both $\mathrm{Tor}_1^A(M, N'')$ and $\mathrm{Tor}_2^A(M, N'')$ vanish by Exercise 2.24. Thus, tensoring

$$0 \to N' \to N \to N'' \to 0$$

with M, we have

$$0 \to \mathrm{Tor}_1^A(M, N') \to \mathrm{Tor}_1^A(M, N) \to 0$$

so that $\mathrm{Tor}_1^A(M, N') \cong \mathrm{Tor}_1^A(M, N)$. Thus, if one vanishes for all M, the other vanishes for all M. By Exercise 2.24, N is flat if and only if N' is flat. □

2.26 Exercise

Proof. The first direction is trivial. If N is flat, then by Exercise 2.24, we have $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated ideals $\mathfrak{a}$ in A simply because $\mathrm{Tor}_1(-, N) = 0$.

For the other direction, we'll start by showing N is flat if $\mathrm{Tor}_1(M, N) = 0$ for all finitely generated A-modules M. Suppose $M' \to M$ is an injection of finitely generated A-modules. Then,

$$\mathrm{Tor}_1(N, M/M') \to M' \otimes N \to M \otimes N$$

is exact. By the symmetry of Tor, we have $\mathrm{Tor}_1(N, M/M') \cong \mathrm{Tor}_1(M/M', N)$. As the projection of a finitely generated A-module, M/M' is also a finitely generated A-module, so by assumption, $\mathrm{Tor}_1(M/M', N) = 0$. Thus, $M' \otimes N \to M \otimes N$ is injective by exactness. Invoking Proposition 2.19, we see that N is flat.

Using this, we'll show that N is flat if $\mathrm{Tor}_1(M, N) = 0$ for all modules M generated by a single element $x \in M$. By the result of the previous paragraph, its enough to show that this implies any finitely generated module M satisfies $\mathrm{Tor}_1(M, N) = 0$. Let M be finitely generated, and let $x_1, \ldots, x_n$ be a collection of generators. Let M_i be the submodule generated by $x_1, \ldots, x_i$. By assumption, $\mathrm{Tor}_1(M_1, N) = 0$. Also note that the quotients M_i/M_{i-1} are generated by the projection of a single element x_i, so $\mathrm{Tor}_1(M_i/M_{i-1}) = 0$. We'll use induction to show that $\mathrm{Tor}_1(M_i, N) = 0$ for all i and thus that $\mathrm{Tor}_1(M, N) = 0$ since $M = M_n$. We'll shown the base case, so let's assume the inductive hypothesis. The exact sequence

$$0 \to M_{i-1} \to M_i \to M_i/M_{i-1} \to 0$$

gives rise to an exact sequence

$$\mathrm{Tor}_1(M_{i-1}, N) \to \mathrm{Tor}_1(M_i, N) \to \mathrm{Tor}_1(M_i/M_{i-1}, N).$$

Since both ends vanish, $\mathrm{Tor}_1(M_i, N) = 0$ by exactness. By induction, $\mathrm{Tor}_1(M, N) = 0$, so N is flat whenever $\mathrm{Tor}_1(M, N) = 0$ for all *cyclic* (generated by one element) modules M.

We're now ready to prove the other direction. Suppose $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated ideals $\mathfrak{a}$. Let M be any cyclic module, so $M = Ax$ for some $x \in M$. Consider the map $\phi : A \to Ax$ defined by $a \mapsto ax$. Since $M = Ax$, this map is surjective, so we have $M \cong A/\mathfrak{a}$, where $\mathfrak{a} = \ker \phi$ is an ideal of A. Since $M \cong A/\mathfrak{a}$, then if we had $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for arbitrary ideals $\mathfrak{a}$, we would have $\mathrm{Tor}_1(M, N) = 0$, implying by the previous paragraph that N is flat.

Thus, we just have to show that $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated ideals $\mathfrak{a}$ implies $\mathrm{Tor}_1(A/\mathfrak{b}, N) = 0$ for arbitary ideals $\mathfrak{b}$. Note that for an arbitrary ideal $\mathfrak{b}$, we have a direct system of finitely generated ideals $\mathfrak{a}$ contained in $\mathfrak{b}$, where the maps are inclusions. Since $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$ for all finitely generated $\mathfrak{a}$, we have exact sequences

$$0 \to \mathfrak{a} \otimes N \to A \otimes N.$$

By Exercise 2.19 (direct limits preserve exact sequences) and Exercise 2.20 (direct limit commutes with tensor product), we have that

$$0 \to \mathfrak{b} \otimes N \to A \otimes N$$

is exact, so $\mathfrak{b} \otimes N \to A \otimes N$ is injective. By this injectivity and the right exactness of the tensor product, we have

$$0 \to \mathfrak{b} \otimes N \to A \otimes N \to (A/\mathfrak{b}) \otimes N \to 0,$$

so $\mathrm{Tor}_1(A/\mathfrak{b}, N) = 0$ for any ideal $\mathfrak{b}$ of A, completing the proof. □

2.27 Exercise

Proof. (i) $\implies$ (ii). Suppose A is absolutely flat, and let $x \in A$. Since A is absolutely flat, (x) is flat, so

$$0 \to (x) \otimes (x) \to (x) \otimes A \to (x) \otimes A/(x) \to 0$$

is exact. By canonical isomorphism, this is

$$0 \to (x^2) \to (x) \xrightarrow{f} (x) \otimes A/(x) \to 0.$$

However, consider the commutative diagram

$$\begin{array}{ccc} (x) \otimes A & \xrightarrow{f} & (x) \otimes A/(x) \\ \downarrow{\scriptstyle \pi_1} & & \downarrow{\scriptstyle \pi_2} \\ A & \xrightarrow{f'} & A/(x) \end{array}$$

An arbitrary generating element of $(x) \otimes A$ is $ax \otimes b$. We have

$$f'(\pi_1(ax \otimes b)) = f'(abx) = 0$$

since $abx \in (x)$. Thus, $f' \circ \pi_1 = 0$ and thus $\pi_2 \circ f = 0$. Since A is absolutely flat, $A/(x)$ is flat, so

$$0 \to (x) \otimes A/(x) \to A \otimes A/(x)$$

is injective since $(x) \to A$ is injective. But, this is just $\pi_2 : (x) \otimes A/(x) \to A/(x) \cong A \otimes A/(x)$. Thus, since $\pi_2 \circ f = 0$ and π_2 is injective, we must have $\operatorname{im} f = 0$. By exactness, we have that $\ker((x^2) \to (x)) = 0$, implying $(x) = (x^2)$. Thus, every principal ideal is idempotent.

(ii) $\implies$ (iii). Suppose every principal ideal is idempotent. Then, for every $x \in A$, $(x) = (x^2)$. so $x = ax^2$ for some $a \in A$. Thus, $e = ax$ is idempotent. Note that $(e) = (x)$ since if $b \in (x)$, then $b = cx = cax^2 = cxe \in (e)$. Thus, every principal ideal is generated by a single idempotent.

Similarly, if we have a finitely generated ideal $(x_1, \ldots, x_n)$, then this is $(x_1) + \cdots + (x_n) = (e_1) + \cdots + (e_n)$, where e_i is the idempotent such that $(e_i) = (x_i)$. Thus, every finitely generated ideal is generated by idempotents. Suppose we have two idempotents $e, f \in A$. Then $(e, f) = (e + f - ef)$ since $e(e + f - ef) = e + ef - ef = e$ and $f(e + f - ef) = ef + f - ef = f$. But any principal ideal can be generated by an idempotent, so $(e, f) = (e + f - fg) = (g)$ for some idempotent g. By a routine inductive argument, every ideal generated by finitely many idempotents is generated by one idempotent. But we saw that all finitely generated ideals have this form, so every finitely generated ideal is (e) for some idempotent e.

Thus, to show that every finitely generated ideal is a direct summand of A, we just have to show that (e) is a direct summand of A for every idempotent e. We claim that $A = (e) \oplus (1 - e)$. If $x \in A$, then $x = xe + x(1 - e)$, so $(e) + (1 - e) = A$. Now suppose $x \in (e) \cap (1 - e)$, i.e. we have $x = ae = b(1 - e)$. Then, $(a + b)e = b$. Multiplying both sides by e, we obtain $(a + b)e = be$. Thus, $b = be$, so $x = b(1 - e) = b - be = 0$. Finally, $(e) \cap (1 - e)$ and $A = (e) \oplus (1 - e)$.

(iii) $\implies$ (i). Suppose every finitely generated ideal $\mathfrak{a}$ of A is a direct summand. Let $M_\mathfrak{a}$ denote the complement. Note that by injecting into the first coordinate and projecting onto the second, for any A-module N, we have the exact sequence

$$0 \to \mathfrak{a} \otimes N \to (\mathfrak{a} \otimes N) \oplus (M_\mathfrak{a} \otimes N) \to M_\mathfrak{a} \otimes N \to 0.$$

But the middle term is

$$(\mathfrak{a} \otimes N) \oplus (M_{\mathfrak{a}} \otimes N) \cong (\mathfrak{a} \oplus M_{\mathfrak{a}}) \otimes N \cong A \otimes N,$$

and $M_{\mathfrak{a}} \cong A/\mathfrak{a}$. Thus, we have an exact sequence

$$0 \to \mathfrak{a} \otimes N \to A \otimes N \to (A/\mathfrak{a}) \otimes N \to 0,$$

so $\mathrm{Tor}_1(A/\mathfrak{a}, N) = 0$. Therefore, N is flat by Exercise 2.26, and A is absolutely flat. □

2.28 Exercise

Every Boolean ring is absolutely flat.

Proof. Every $x \in A$ satisfies $x^2 = x$. For every principal ideal (x), we have $(x)^2 = (x^2) = (x)$, so by Exercise 2.27, A is absolutely flat. □

Let A be such that for any $x \in A$, there is some $n > 1$ such that $x^n = x$. A is absolutely flat.

Proof. Let (x) be an arbitrary principal ideal; x satisfies $x^n = x$ for some $n > 1$. Thus,

$$(x) = (x^n) \subset (x^{n-1}) \subset \cdots \subset (x^2) \subset (x),$$

so all are equal. Hence, $(x) = (x^2) = (x)^2$, so by Exercise 2.27, A is absolutely flat. □

Every homomorphic image of an absolutely flat ring is absolutely flat.

Proof. Let A be absolutely flat, and let $f : A \to B$ be a homomorphism. Then, $f : A \to f(A)$ is surjective. Let $(f(x)) \in f(A)$ be an arbitrary principal ideal. By surjectivity and the fact that f is a ring homomorphism, $(f(y)) = f((y))$ for any $y \in A$. By Exercise 2.27, $(x) = (x)^2$. Thus,

$$(f(x))^2 = f((x))^2 = f((x)^2) = f((x)) = (f(x)),$$

so by Exercise 2.27, $f(A)$ is absolutely flat. □

If a local ring is absolutely flat, then it is a field.

Proof. Suppose A is an absolutely flat local ring. Let $x \in A$ be nonzero. By Exercise 2.27, $(x) = (x)^2$ so that $x = ax^2$. Then, $e = ax$ is an idempotent. By Exercise 1.12, the only idempotents of a local ring are 0 and 1, so $ax = 1$ or $ax = 0$. If $ax = 0$, then $x = ax^2 = 0$. Since we chose x to be nonzero, we must have $ax = 1$, so x is a unit. All nonzero elements are units, so A is a field. □

If A is absolutely flat, every non-unit in A is a zero divisor.

Proof. Let $x \in A$ be a non-unit. By Exercise 2.27, $(x) = (x)^2$, so $x = ax^2$ for some $a \in A$. Thus, $x(1 - ax) = 0$. Since x is a non-unit, $1 - ax \neq 0$, so x is a zero divisor. □

3 Rings and Modules of Fractions (Done)

3.1 Exercise

Proof. By Proposition 3.14, since S is multiplicatively closed and M is finitely generated,

$$S^{-1}\operatorname{Ann}(M) = \operatorname{Ann}(S^{-1}M).$$

Thus, if $S^{-1}M = 0$, then $\operatorname{Ann}(S^{-1}M) = S^{-1}A$, so $S^{-1}\operatorname{Ann}(M) = S^{-1}A$. But $1 = 1/1 \in S^{-1}A$, so there is some $a \in \operatorname{Ann}(M)$ and $s \in S$ such that $a/s = 1/1$. In other words, there is some $t \in S$ such that $t(a - s) = 0$ or $st = at$. Since $a \in \operatorname{Ann}(M)$, $st = at \in \operatorname{Ann}(M)$. But $s, t \in S$, so $st \in S$, hence $S \cap \operatorname{Ann}(M)$ is non-empty.

Conversely, suppose there is some $s \in S$ such that $sM = 0$. Let m/t be an arbitrary element of $S^{-1}M$, where $m \in M$ and $t \in S$. Consider the element $0 = 0/s$. We have that $sm - 0t = 0 - 0 = 0$, so $m/t \equiv 0$ and thus $S^{-1}M = 0$. □

3.2 Exercise

Proof. The Jacobson radical of $S^{-1}A$ is the intersection of maximal ideals of $S^{-1}A$. Thus, if we can show that $S^{-1}\mathfrak{a}$ is in an arbitrary maximal ideal, we will have shown that it lies in the Jacobson radical. By Proposition 3.11(iv), since maximal ideals are prime, any maximal ideal of $S^{-1}A$ is of the form $S^{-1}\mathfrak{p}$ for some prime ideal $\mathfrak{p}$ of A that does not meet $S = (1 + \mathfrak{a})$, i.e. $(1 + \mathfrak{a}) \cap \mathfrak{p} = \emptyset$. By assumption, $S^{-1}\mathfrak{p}$ is maximal, either $S^{-1}\mathfrak{p} + S^{-1}\mathfrak{a} = (1)$ or $S^{-1}\mathfrak{p} + S^{-1}\mathfrak{a} = S^{-1}\mathfrak{p}$ (and hence $S^{-1}\mathfrak{a} \subset S^{-1}\mathfrak{p}$). In the former case, we have by Proposition 3.11(v),

$$(1) = S^{-1}\mathfrak{p} + S^{-1}\mathfrak{a} = S^{-1}(\mathfrak{p} + \mathfrak{a}).$$

This means we can write $1/1 \equiv (p + x)/(1 + y)$, where $p \in \mathfrak{p}$, $x, y \in \mathfrak{a}$. Then, there is some $z \in \mathfrak{a}$ such that

$$\begin{aligned} 0 &= (1 + z)((p + x) - (1 + y)) \\ 0 &= (1 + z)(p + x) - (1 + y)(1 + z) \\ 0 &= p + x + pz + xz - 1 - yz - y - z \\ p &= 1 + (yz + y + z - xz - pz - x). \end{aligned}$$

Since $yz + y + z - xz - pz - x \in \mathfrak{a}$ and $p \in \mathfrak{p}$, we contradict $(1 + \mathfrak{a}) \cap \mathfrak{p} = \emptyset$. Thus, as desired, we have $S^{-1}\mathfrak{a} \subset S^{-1}\mathfrak{p}$, so $S^{-1}\mathfrak{a}$ is contained in the Jacobson radical of $S^{-1}A$. □

Let M be a finitely generated A-module, and let $\mathfrak{a}$ be an ideal of A such that $\mathfrak{a}M = M$. Then, there exists $x \equiv 1 \mod \mathfrak{a}$ such that $xM = 0$.

Proof. Let $S = 1 + \mathfrak{a}$. Then, by assumption,

$$S^{-1}M = S^{-1}(\mathfrak{a}M) = S^{-1}(\mathfrak{a})S^{-1}(M).$$

By our previous proof, $S^{-1}\mathfrak{a}$ lies in the Jacobson radical of $S^{-1}A$. Further, note that by Proposition 3.5, $S^{-1}M \cong S^{-1}A \otimes_A M$, so since M is finitely generated as an A-module, $S^{-1}M$ is finitely generated as a $S^{-1}A$-module (map generators $m \mapsto 1 \otimes m$). Thus, we have everything we need to invoke Nakayama's lemma: $S^{-1}M = 0$. By Exercise 3.1, there is some $x \in S = 1 + \mathfrak{a}$ such that $xM = 0$. □

3.3 Exercise

Proof. To show that $(ST)^{-1}A \cong U^{-1}(S^{-1}A)$, we show that $U^{-1}(S^{-1}A)$ satisfies the universal property for the multiplicatively closed set ST. We invoke the universal property a few times, which is Proposition 3.1.

Consider

$$A \xrightarrow{j} S^{-1}A \xrightarrow{k} U^{-1}(S^{-1}A),$$

letting $g = k \circ j$. For any $s \in S$, $j(s)$ is a unit in $S^{-1}A$, so $g(s) = k(j(s))$ is a unit. For $t \in T$, $j(t) \in U$ by the definition of U, so $k(j(t))$ is a unit. Thus, $g(st) = g(s)g(t)$ is a product of units and thus a unit for any $st \in ST$.

Suppose we have a map $f : A \to B$ such that $f(ST) \subset B^\times$. Since $S \subset ST$, $f(S) \subset B^\times$. By the universal property of $S^{-1}A$, there is a unique map $h : S^{-1}A \to B$ such that $h \circ j = f$. Every element of U has the form $j(t)$ for some $t \in T$, but $h(j(t)) = f(t)$. Since $T \subset ST$, $f(t)$ is a unit. In particular, $h(U) \subset B^\times$. By the universal property of $U^{-1}(S^{-1}A)$, there is a unique map $\overline{h} : U^{-1}(S^{-1}A) \to B$ such that $\overline{h} \circ k = h$. Combining these, $f = h \circ j = \overline{h} \circ (k \circ j) = \overline{h} \circ g$. Because $\overline{h}$ is unique, $U^{-1}(S^{-1}A)$ satisfies the defining universal property of $(ST)^{-1}A$, so the two rings are uniquely isomorphic. □

3.4 Exercise

Proof. Since $f : A \to B$ is a ring homomorphism, and S is a multiplicatively closed set, $T = f(S)$ is a multiplicatively closed set of B. Note that B has a canonical A-module structure via f. On the one hand, we have the ring construction $T^{-1}B$, but on the other hand, we have the $S^{-1}A$-module construction $S^{-1}B$. Define a map

$$\psi : S^{-1}B \to T^{-1}B, \quad (b, s) \mapsto b/f(s).$$

Since every element of T has the form $f(s)$, this is clearly a surjective map. Further, suppose $(b, s), (b', s') \in S^{-1}B$ are indentical in the image: $b/f(s) \equiv b'/f(s')$. Then, there is $f(s'') \in f(S)$ such that $f(s'')(f(s')b - f(s)b') = 0$ in B. But this immediately implies $(b, s) \equiv (b', s')$ in $S^{-1}B$ since S acts on B via f. Thus, ψ is injective, so ψ is bijective.

Since B is an A-module, $S^{-1}B$ is automatically an $S^{-1}A$-module. For an arbitrary element $a/s \in S^{-1}A$ and $(b, s') \in S^{-1}B$. The action is defined by $(a/s)(b, s') = (f(a)b, ss')$. Likewise, since $f : A \to B$ and the natural map $B \to T^{-1}B$ are ring homomorphisms, we have a ring homomorphism $A \to T^{-1}B$ that sends S to units since $T = f(S)$. Thus, we have a ring homomorphism $S^{-1}A \to T^{-1}B$ that endows $T^{-1}B$ with an $S^{-1}A$-module structure. For an arbitrary, $b/f(s') \in T^{-1}B$ and $a/s \in S^{-1}A$, the action is defined by $(a/s)(b/f(s')) = f(a)b/f(ss')$. Thus, for an arbitrary $a/s \in S^{-1}A$ and $(b, s') \in S^{-1}B$, we have

$$\psi((a/s)(b, s')) = \psi((f(a)b, ss')) = f(a)b/f(ss') = (a/s)(b/f(s')) = (a/s)\psi((b, s')),$$

so ψ commutes with the $S^{-1}A$-action. For the sake of completion, we also have additivity:

$$\psi((b, s) + (b', s')) = \psi((f(s')b, ss') + (f(s)b', ss')) = \psi((f(s')b + f(s)b', ss'))$$

$$= (f(s')b + f(s)b')/f(ss') = b/f(s) + b'/f(s') = \psi((b, s)) + \psi((b', s')).$$

Thus, $\psi : S^{-1}B \to T^{-1}B$ is a $S^{-1}A$-module isomorphism. □

3.5 Exercise

Proof. Consider the nilradical $\mathfrak{N}$ of A, which has a natural A-module structure. By Corollary 3.12, for any multiplicatively closed set, $S^{-1}\mathfrak{N}$ is the nilradical of $S^{-1}A$. Consider the special case of localization as the prime ideals. Then, $\mathfrak{N}_\mathfrak{p}$ is the nilradical of $A_\mathfrak{p}$. If for all prime ideals $\mathfrak{p}$, the only nilpotent elements are 0, then $\mathfrak{N}_\mathfrak{p} = 0$ for all $\mathfrak{p}$. By Proposition 3.8, $\mathfrak{N} = 0$, so there are no nilpotent elements of A. □

Consider the ring $A = \mathbb{Z}/6\mathbb{Z}$, which is not an integral domain. It has prime ideals (2) and (3). Localizing at (2), $S = \{1, 3, 5\}$. Since $3 \in S$ and $2 \cdot 3 = 0$, $\frac{2}{1} = \frac{3\cdot 2}{3} = 0$. Thus, $0, 2, 4$ map to zero under $A \to A_{(2)}$. But then, $1 = 3 = 5$, so $A_{(2)} = \mathbb{Z}/2\mathbb{Z}$. Similarly, $A_{(3)} = \mathbb{Z}/3\mathbb{Z}$. Thus, $A_\mathfrak{p}$ is an integral domain for all prime ideals $\mathfrak{p}$, but A is not an integral domain, so being an integral domain is not a "local property".

3.6 Exercise

Proof. The multiplicatively closed subsets of A not containing 0 have a partial order under set inclusion; call this set Σ. Let $\mathcal{T}$ be a chain of Σ. Define $U = \bigcup_{S\in\mathcal{T}} S$. Clearly, U contains 1. It's also multiplicatively closed since $x, y \in U$ implies $x \in S$ and $y \in S'$ for some $S, S' in \mathcal{T}$. Since $\mathcal{T}$ is a chain, $S \subset S'$ or $S' \subset S$. Either way, the product is in $S \cup S'$, so U is multiplicatively closed. Since $0 \notin S$ for any $S \in \Sigma$, $0 \notin U$, so $U \in \Sigma$, serving as a clear upper bound for $\mathcal{T}$. By Zorn's lemma, Σ has maximal elements.

Now, we aim to show that $S \in \Sigma$ is maximal if and only if $A - S$ is a minimal prime ideal of A.

Suppose $S \in \Sigma$ is maximal. We can localize to $S^{-1}A$. Since $0 \notin S$, $S^{-1}A$ is not the zero ring, so $S^{-1}A$ has a maximal ideal $\mathfrak{m}$. Suppose $a/s \in S^{-1}A - \mathfrak{m}$. If a/s is not a unit, then $a \notin S$, so consider the multiplicatively closed set $\{a^n\}_{n\geq 0}S$. Since S is maximal in Σ, we must have that $\{a^n\}_{n\geq 0}S$ contains 0, so $a^n t = 0$ for some $t \in S$. Then, $(a/s)^n = a^n/s^n \equiv 0/t = 0$, so a/s is in the nilradical. But, $\mathfrak{m}$ is the Jacobson radical (since we're in a local ring), which contains the nilradical, so we have a contradiction. Thus, a/s is a unit. By Proposition 1.6, $S^{-1}A$ is a local ring with maximal ideal $\mathfrak{m}$.

By Proposition 3.11(iv), the prime ideals of $S^{-1}A$ are in one-to-one correspondence with the prime ideals of A not meeting A, via $\mathfrak{p} \leftrightarrow S^{-1}\mathfrak{p}$. Thus, there is exactly one prime ideal $\mathfrak{p}$ of A such that $S^{-1}\mathfrak{p} = \mathfrak{m}$. Further, $\mathfrak{p} \cap S = \emptyset$, so $S \subset A - \mathfrak{p}$. Suppose $x, y \in A - \mathfrak{p}$, then $x, y \notin \mathfrak{p}$, so since $\mathfrak{p}$ is prime, $xy \notin \mathfrak{p}$, so $xy \in A - \mathfrak{p}$. Thus, $A - \mathfrak{p} \in \Sigma$ since it contains 1 but not 0 as is multiplicatively closed. Thus, $S = A - \mathfrak{p}$ so that $\mathfrak{p} = A - S$.

If there is some prime ideal $\mathfrak{q}$ such that $\mathfrak{q} \subset \mathfrak{p}$, then $\mathfrak{q} \cap S = \emptyset$, so there is another prime ideal of $S^{-1}A$ by Proposition 3.11(iv). This violates the fact that $S^{-1}A$ is a local ring. Thus, if $S \in \Sigma$ is maximal, $A - S$ is a minimal prime ideal.

Conversely, suppose $A - S$ is a minimal prime ideal. Then, if $x, y \notin A - S$, $xy \notin A - S$, so S is multiplicatively closed. Further, $1 \in S$ and $0 \notin S$, so $S \in \Sigma$. Thus, there is a maximal $T \in \Sigma$ such that $S \subset T$. By the other direction, $A - T$ is a minimal prime ideal. Since $A - T \subset A - S$ and $A - S$ is a minimal prime ideal, we must have equality, so S is maximal. □

3.7 Exercise

(i) Proof. Let S be saturated. If $0 \in S$, then for any $x \in A$, $x \cdot 0 = 0 \in S$, so $x \in S$ and thus $A - S = \emptyset$, which is an empty union of prime ideals. Thus, we can commit to the case where $0 \notin S$. Let $a \in A - S$. Then for any $b \in A$, $ab \notin S$ since S is saturated. Thus, $aA \cap S = \emptyset$,

so $(a) \subset A - S$. Let Σ be the set of ideals containing (a) disjoint from S. We've shown this set is non-empty; further, it is ordered by inclusion. Any chain of Σ has an upper bound by taking the union over the chain; the union clearly contains (a), is disjoint from S, and it satisfies the properties of ideals since any pair of elements of the union exist together in some ideal in the chain. By Zorn's lemma, Σ contains maximal elements, so let $\mathfrak{p}$ be a maximal element of Σ; note that this implies $\mathfrak{p}$ contains (a), is disjoint from S, and is an ideal. Suppose $xy \in \mathfrak{p}$. Since $(x) + \mathfrak{p}$ and $(y) + \mathfrak{p}$ are ideals containing $\mathfrak{a}$ and $\mathfrak{p}$, the maximality of $\mathfrak{p}$ in Σ implies that either x or y is in $\mathfrak{p}$ or that both $(x) + \mathfrak{p}$ and $(y) + \mathfrak{p}$ intersect S. In the second case, since $(x) + \mathfrak{p}$ and $(y) + \mathfrak{p}$ intersect S and S is multiplicatively closed, their product must intersect S. However, their product is $(xy) + \mathfrak{p} = \mathfrak{p}$, which does not intersect S, giving us a contradiction. Thus, the first case must hold, wherein either x or y lies in $\mathfrak{p}$, showing that $\mathfrak{p}$ is a prime ideal containing a; let's call it $\mathfrak{p}_a$. Thus, $A - S \subset \bigcup_{a \in A-S} \mathfrak{p}_a$ since every $a \in A - S$ is accounted for on the right-hand side, but further, since $\mathfrak{p}_a$ is disjoint from S for all $a \in A - S$, we have $\bigcup_{a \in A-S} \mathfrak{p}_a \subset A - S$, giving us the desired equality.

Conversely, suppose $A - S$ is a union of prime ideals. Since S is multiplicatively closed, $x, y \in S \implies xy \in S$. To show the other direction, suppose one of x or y is not in S; without loss of generality, suppose $x \in A - S$. Then, $x \in \mathfrak{p}$ for some prime ideal, so $xy \in \mathfrak{p} \subset A - S$. Thus, $x, y \in S \iff xy \in S$. □

(ii) Proof. If T is any saturated multiplicatively closed set containing S, then $A - T \subset A - S$. By (i), $A - T$ is a union of prime ideals, so these prime ideals must be disjoint from S since $A - T \subset A - S$. Now, let $U = \bigcup_{\mathfrak{p}:\mathfrak{p} \cap S = \emptyset} \mathfrak{p}$. Then,

$$A - U = \bigcap_{\mathfrak{p}:\mathfrak{p} \cap S = \emptyset} (A - \mathfrak{p}).$$

As the intersection of multiplicatively closed sets, $A - U$ is a multiplicatively closed set $\overline{S}$. Further, since $A - \overline{S}$ is a union of prime ideals, it is saturated by (i). Since $A - T$ is a union of prime ideals disjoint from S, $A - T \subset U = A - \overline{S}$, so $\overline{S} \subset T$ for any multiplicatively closed saturated set T containing S. Finally, since $A - \overline{S} = U \subset A - S$, $S \subset \overline{S}$, so $\overline{S}$ is the unique, smallest multiplicatively closed saturated subset containing S. □

Remark. *$\overline{S}$ is called the "saturation" of S.*

Let $S = 1 + \mathfrak{a}$, where $\mathfrak{a}$ is an ideal of A. By (ii),

$$\overline{S} = A - \bigcup_{\mathfrak{p}:\mathfrak{p} \cap S = \emptyset} \mathfrak{p}.$$

Let $\mathfrak{p}$ be such that $\mathfrak{p} \cap S \neq \emptyset$, i.e. there is some $p \in \mathfrak{p}$, $a \in \mathfrak{a}$, such that $p = 1 + a$. Thus, $p - a = 1$ so that $\mathfrak{a} + \mathfrak{p} = (1)$. Thus, $\mathfrak{a}$ and $\mathfrak{p}$ are coprime ideals. Conversely, if $\mathfrak{a} + \mathfrak{p} = (1)$, then $\mathfrak{p} \cap (1 + \mathfrak{a}) \neq \emptyset$. In particular, $\mathfrak{p} \cap S = \emptyset$ if and only if $\mathfrak{a}$ and $\mathfrak{p}$ are not coprime. But this happens exactly when $\mathfrak{a} + \mathfrak{p}$ are contained in a common maximal ideal $\mathfrak{m}$, meaning $\mathfrak{p}$ is contained in some $\mathfrak{m}$ containing $\mathfrak{a}$. Thus,

$$\bigcup_{\mathfrak{p}:\mathfrak{p} \cap S = \emptyset} \mathfrak{p} = \bigcup_{\mathfrak{p}:\mathfrak{p} + \mathfrak{a} \neq (1)} \mathfrak{p} = \bigcup_{\mathfrak{m}:\mathfrak{a} \subset \mathfrak{m}} \mathfrak{m} = \bigcup_{\mathfrak{m} \in V(\mathfrak{a}) \cap \mathrm{Max}(A)} \mathfrak{m} = \bigcup_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p},$$

so

$$\overline{S} = A - \bigcup_{\mathfrak{p} \in V(\mathfrak{a})} \mathfrak{p}.$$

3.8 Exercise

Proof. (i) $\iff$ (ii). Suppose the ring homomorphism $\phi : S^{-1}A \to T^{-1}A$ is a bijection. Since $S \subset T$, $ST = T$. By Exercise 3.3, we have a ring isomorphism $\psi : T^{-1}A \to U^{-1}(S^{-1}A)$, where $U = \{t/1\}_{t\in T} \subset S^{-1}A$. Thus, we have a ring isomorphism $\psi \circ \phi : S^{-1}A \to U^{-1}(S^{-1}A)$. In particular, the set U must already be units in $S^{-1}A$, i.e. $t/1 \in S^{-1}A$ is a unit for all $t \in T$. Conversely, if $t/1 \in S^{-1}A$ is a unit for all $t \in T$, then the ring homomorphism $\psi \circ \phi : S^{-1}A \to U^{-1}(S^{-1}A)$ is a bijection, so ϕ is a bijection.

(ii) $\iff$ (iii). For any $t \in T$, $t/1$ is a unit in $S^{-1}A$, so there is some $a \in A, s \in S$ such that $(t/1)(a/s) = (ta/s) \equiv 1$. This means there is some $s' \in S$ such that $s'(ta - s) = 0$, i.e. $s'at = ss' \in S$. Thus, letting $x = s'a \in A$, we have $xt \in S$. Conversely, if $t \in T$ and there is some $x \in A$ such that $xt \in S$, then there is some $s \in S$ such that $xt = s$. Then, $xt - s = 0$ so that $(t/1)(x/s) = (tx/s) \equiv 1$, so $t/1$ is a unit.

(iii) $\implies$ (iv). For all $t \in T$, there exists $x \in A$ such that $xt \in S$. If $\mathfrak{p}$ is a prime ideal of A disjoint from S, then if there is some $t \in T \cap \mathfrak{p}$, $xt \in S$ so that $S \cap \mathfrak{p}$ is nonempty, contradicting disjointness. Since $S \subset T$, the converse is also true; i.e. a prime is disjoint from S if and only if it is disjoint from T. By Exercise 3.7, $T \subset \overline{T} = \overline{S}$.

(iv) $\implies$ (v). Suppose $T \subset \overline{S}$. Then, $\overline{T} \subset \overline{S}$ since $\overline{S}$ is a saturated, multiplicatively closed set containing T and $\overline{T}$ is the minimal such set. Conversely, $S \subset T$, so $\overline{S} \subset \overline{T}$, so $\overline{S} = \overline{T}$. Thus, every prime ideal meeting T meets S via Exercise 3.7.

(v) $\implies$ (iii). Suppose every prime ideal meeting T meets S. Suppose to the contrary that there is some $t \in T$ such that $t/1 \in S^{-1}A$ is not a unit. Then, $(t/1)$ is a non-trivial ideal of S^{-1} and is thus contained in some maximal ideal of $S^{-1}A$. By Proposition 3.11(iii), this prime ideal has the form $S^{-1}\mathfrak{p}$, where $\mathfrak{p}$ is a prime ideal of A not meeting S; by assumption, it does not meet T either, so $t \notin \mathfrak{p}$. Since $t/1 \in S^{-1}\mathfrak{p}$, there is some $p \in \mathfrak{p}$ and $s, s' \in S$ such that $t/1 \equiv p/s$ or $s'(ts - p) = 0$. Thus, $s'ts = ps'$. Since $S \subset T$, $s'ts \in T$, and since $\mathfrak{p}$ is an ideal, $ps' \in \mathfrak{p}$, so $T \cap \mathfrak{p} \neq \emptyset$, a contradiction. Thus, $t/1$ must be a unit in $S^{-1}A$ for all $t \in T$.

This concludes the proof that all five statements are equivalent. □

3.9 Exercise

Proof. Consider the set S_0 of all non-zero-divisors in A. If xy is a zero divisor, then there is some nonzero z such that $xyz = 0$. If $yz = 0$, then y is a zero divisor, and if $yz \neq 0$, then x is a zero divisor since $x(yz) = 0$. Thus, $x, y \in S_0$ implies that $xy \in S_0$. Further $1 \in S_0$, so S_0 is a multiplicatively closed set. If x or y is a zero divisor (with the element $z \neq 0$), then $xyz = 0$, so xy is a zero divisor. Thus, if $xy \in S_0$, both x and y must be in S_0. Thus, S_0 is saturated. Interestingly, in light of Exercise 3.7, the set of zero-divisors $(A - S_0)$ is the union of prime ideals, which we proved in Exercise 1.14.

Now, we show that every minimal prime ideal of A is contained in $D = A - S_0$, the set of zero-divisors of A. Let $\mathfrak{p}$ be a minimal prime. Then, $A - \mathfrak{p}$ is a multiplicative set. By Exercise 3.6, $A - \mathfrak{p}$ is a maximal element of Σ, the set of multiplicative sets not containing zero. Then, $S_0(A - \mathfrak{p})$ is another multiplicative set. Further, if it contains 0, then there is some $x \in S_0$ and $y \in A - \mathfrak{p}$ such that $xy = 0$. Since x is not a zero divisor, we have $y = 0$, but $0 \in \mathfrak{p}$, contradicting the fact that $y \in A - \mathfrak{p}$. Thus, $S_0(A - \mathfrak{p}) \in \Sigma$. However, $A - \mathfrak{p} \subset S_0(A - \mathfrak{p})$, so since $A - \mathfrak{p}$ is maximal in Σ, $S_0(A - \mathfrak{p}) = A - \mathfrak{p}$. Since $1 \in A - \mathfrak{p}$, this implies $S_0 = A - D \subset A - \mathfrak{p}$. Thus, $\mathfrak{p} \subset D$, as desired. □

Remark. *The ring $S_0^{-1}A$ is the "total ring of fractions" of A.*

(i) Proof. Consider the natural homomorphism $A \to S_0^{-1}A$ defined by $a \mapsto a/1$. If $a/1 \equiv a'/1$, then there is some $s \in S_0$ such that $s(a - a') = 0$. Since s is a non-zero-divisor, $a = a'$, so the homomorphism is injective. Suppose S properly contains S_0 so that S contains a zero divisor z. Thus, there is some non-zero $x \in A$ such that $xz = 0$. Then, in $S^{-1}A$, $x/1 \equiv 0/1$ since $z(x-0) = 0$. In particular, $A \to S^{-1}A$ is not injective. Thus, S_0 is the largest multiplicatively closed set such that the homomorphism $A \to S_0^{-1}A$ is injective. □

(ii) Proof. Suppose $a/s \in S_0^{-1}A$ is not a zero-divisor of $S_0^{-1}A$ for $a \in A$ and $s \in S_0$. If $a \in A$ is a zero divisor, there is some non-zero $b \in A$ such that $ab = 0$. Thus,

$$(a/s)(b/1) = (ab)/s = 0/s = 0,$$

so $b/1 \equiv 0$ since a/s is not a zero-divisor. By (i), $A \to S_0^{-1}A$ is injective, so $b/1 = 0$ implies $b = 0$, giving us a contradiction. Thus, a is not a zero divisor of A, so $a \in S_0$. Then, $s/a \in S_0^{-1}A$; because $(a/s)(s/a) = 1$, a/s is a unit. Thus, every element of $S_0^{-1}A$ is either a unit or a zero divisor. □

(iii) Proof. Suppose A is a ring such that all non-zero-divisors are units. Then, S_0 is the set of units of A. By (i), we have an injective ring homomorphism $\phi : A \to S_0^{-1}A$. Let a/s be an arbitrary element of $S_0^{-1}A$. Then, since $s \in S_0$ is a unit, there is some $s' \in A$ such that $ss' = 1$. Thus, $a/s \equiv (as')/(ss') = (as')/1$, so ϕ is surjective. Thus, ϕ is a ring isomorphism. □

3.10 Exercise

(i) Proof. Suppose A is absolutely flat and S is any multiplicative subset of A. Let N be a $S^{-1}A$-module. Under the canonical map $f : A \to S^{-1}A$, N is an A-module N_A. We wish to show that N is isomorphic to $S^{-1}N_A$ as $S^{-1}A$-modules. We define the map $\phi : N \to S^{-1}N_A$ by $n \mapsto (n, 1)$. Since N is an $S^{-1}A$-module, an arbitrary element $(n, s) \in S^{-1}N_A$ has the form $(n, s) \equiv ((1/s)n, 1)$, so ϕ is surjective. If $(n, 1) \equiv (n', 1)$, there is some $t \in S$ such that $t(n - n') = 0$, so $tn = tn'$ in N. But, N is a $S^{-1}A$-module, so we have

$$n = (1/t)tn = (1/t)tn' = n',$$

showing that ϕ is injective. Thus, ϕ is bijective. It's an $S^{-1}A$-module homomorphism since $\phi(n + n') = (n + n', 1) = (n, 1) + (n', 1) = \phi(n) + \phi(n')$ and for $a/s \in S^{-1}A$, we have

$$\phi((a/s)n) = ((a/s)n, 1) = (an, s) = (a/s)(n, 1) = (a/s)\phi(n).$$

Thus, for any $S^{-1}A$-module M, we have $M \cong S^{-1}M_A$ as $S^{-1}A$-modules.

Now, let

$$0 \to M' \to M$$

be an exact sequence of $S^{-1}A$-modules. Then,

$$0 \to M'_A \to M_A$$

is an exact sequence of A-modules. Since N_A is a flat A-module,

$$0 \to M'_A \otimes_A N_A \to M_A \otimes_A N_A$$

is an exact sequence of A-modules. Since localization preserves exactness,

$$0 \to S^{-1}(M'_A \otimes_A N_A) \to S^{-1}(M_A \otimes_A N_A)$$

is an exact sequence of $S^{-1}A$-modules. We have an exact sequence of $S^{-1}A$-modules

$$0 \to S^{-1}M'_A \otimes_{S^{-1}A} S^{-1}N_A \to S^{-1}M_A \otimes_{S^{-1}A} S^{-1}N_A$$

by Proposition 3.7. Using the result of the last paragraph, we have that

$$0 \to M' \otimes_{S^{-1}A} N \to M \otimes_{S^{-1}A} N$$

is exact, so N is a flat $S^{-1}A$-module. Thus, $S^{-1}A$ is absolutely flat. □

(ii) Proof. If A is absolutely flat, then $A_\mathfrak{m}$ is absolutely flat by (i). Thus, every principal ideal of $A_\mathfrak{m}$ is idempotent by Exercise 2.27. Let $x \in A_\mathfrak{m}$ be non-zero. Then, $(x) = (x^2)$, so $x = ax^2$. Thus, $ax = (ax)^2$, so ax is an idempotent. Since $A_\mathfrak{m}$ is a local ring, Exercise 1.12 guarantees that the only idempotents are 0 and 1. If $ax = 0$, then $x = ax^2 = 0$, so $ax = 1$. In particular, every non-zero element of $A_\mathfrak{m}$ is a unit, so $A_\mathfrak{m}$ is a field.

Conversely, suppose $A_\mathfrak{m}$ is a field for all maximal ideals $\mathfrak{m}$ of A. Then, the only ideals are (0) and (1), so all principal ideals are idempotent. By Exercise 2.27, $A_\mathfrak{m}$ is absolutely flat. Let N be an A-module, and let $j : M \to M'$ be an injection of A-modules. We have an A-module homomorphism

$$\phi = j \otimes 1 : M \otimes_A N \to M' \otimes_A N.$$

Localizing, we have

$$\phi_\mathfrak{m} : (M \otimes_A N)_\mathfrak{m} \to (M' \otimes_A N)_\mathfrak{m},$$

which by canonical isomorphism is just

$$\phi_\mathfrak{m} : M_\mathfrak{m} \otimes_{A_\mathfrak{m}} N_\mathfrak{m} \to M'_\mathfrak{m} \otimes_{A_\mathfrak{m}} N_\mathfrak{m}.$$

By Proposition 3.9, since $j : M \to M'$ is an injection, $j_\mathfrak{m} : M_\mathfrak{m} \to M'_\mathfrak{m}$ is injective. Since $A_\mathfrak{m}$ is absolutely flat, $N_\mathfrak{m}$ is a flat $A_\mathfrak{m}$-module, so $\phi_\mathfrak{m}$ is injective for all $\mathfrak{m}$. By Proposition 3.9, ϕ is injective, so N is a flat A-module. Therefore, A is absolutely flat. □

3.11 Exercise

Proof. (i) $\implies$ (ii). Let $A' = A/\mathfrak{N}$ be absolutely flat, and let $\mathfrak{p} \in \operatorname{Spec} A'$. By Exercise 3.10, $A'_\mathfrak{p}$ is absolutely flat. Further, it is a local ring. By Exercise 2.28, $A'_\mathfrak{p}$ is a field, so the only prime ideal of $A'_\mathfrak{p}$ is 0. By Corollary 3.13, the prime ideals of $A'_\mathfrak{p}$ are in one-to-one correspondence with the prime ideals contained in $\mathfrak{p}$. Thus, the only prime ideal that any $\mathfrak{p}$ contains is itself. In particular, if $\mathfrak{m}$ is a maximal ideal such that $\mathfrak{p} \subset \mathfrak{m}$, then since $\mathfrak{m}$ is prime, $\mathfrak{m} = \mathfrak{p}$. Thus, all prime ideals are maximal. Since $A' \to A/\mathfrak{N}$ is a surjective ring homomorphism inducing a one-to-one order preserving map of ideals (in light of Proposition 1.1), all prime ideals of A are maximal.

(ii) $\implies$ (i). Consider $A' = A/\mathfrak{N}$. In light of Proposition 1.1, $A \to A/\mathfrak{N} = A'$ is a one-to-one order-preserving map of ideals. Thus, A' has no nilpotent elements, and all prime ideals are maximal. The prime ideals of the local ring $A'_\mathfrak{m}$ are in one-to-one correspondence with the prime ideals $\mathfrak{p}$ contained in $\mathfrak{m}$ (Corollary 3.13). But since all prime ideals are maximal, the only prime ideal of the local ring is 0, so $A'_\mathfrak{m}$ is a field. By Exercise 3.10, A' is absolutely flat.

(ii) $\iff$ (iii). By Exercise 1.18, $\{x\}$ is closed if and only if $\mathfrak{p}_x$ is maximal. Thus, $\operatorname{Spec} A$ is T_1 if and only if all prime ideals are maximal.

(iv) $\implies$ (iii). Any topology that is Hausdorff (T_2) is T_1.

(i) $\implies$ (iv). We assume now that $A' = A/\mathfrak{N}$ is absolutely flat. Choose distinct $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec}(A')$. We've already shown above that all prime ideals of A' are maximal. Thus, $\mathfrak{p} + \mathfrak{q} = (1)$ since they are distinct maximal ideals. Thus, there is some $x \in \mathfrak{p}$ and $y \in \mathfrak{q}$ such that $(x) + (y) = (1)$. Since (x) and (y) are finitely generated ideals, by Exercise 2.27, we have idempotents e and f such that $(e) = (x)$, $(f) = (y)$, so $(e, f) = (1)$. Let $g = f(1 - e)$. Then, $g^2 = f^2(1 - e)^2 = f(1 - e) = g$ and $eg = f(e - e^2) = 0$. Further, $f = g + ef$, so $f \in (g, e)$ so that $(1) = (e, f) \subset (g, e)$, so $(g, e) = (1)$. In particular, since $y \notin (e)$, $y \in (g)$ since $eg = 0$. Thus, $(y) = (e)$ since $(e) + (y) = (1)$.

Since $y \notin (e)$, $\mathfrak{q} \notin V(e)$. Similarly, since $x \notin (g)$ (otherwise $(1) = (e, g) = (x, g) = (g)$), $\mathfrak{p} \notin V(g)$. In particular, $\mathfrak{p} \in X_g$ and $\mathfrak{q} \in X_e$. By Exercise 1.17, $X_e \cap X_g = X_{eg} = X_0 = \emptyset$, so $\mathfrak{p}$ and $\mathfrak{q}$ are separable by disjoint open sets; thus, $\operatorname{Spec} A'$ is Hausdorff. By Exercise 1.21, $\operatorname{Spec} A$ and $\operatorname{Spec} A'$ are homeomorphic, so $\operatorname{Spec} A$ is Hausdorff.

This concludes the proof of the equivalency of the four statements. □

Proof. Suppose the ring A satisfies the equivalent statements above. Since $\operatorname{Spec} A$ is always quasi-compact in light of Exercise 1.17, then since $\operatorname{Spec} A$ is Hausdorff here, it is also compact.

In the final part of the previous proof, for $\mathfrak{p}, \mathfrak{q} \in \operatorname{Spec} A$ distinct, we found basic open sets X_g and X_e that are disjoint. Further, since $(g, e) = (1)$ and $ge = 0$, for any maximal ideal $\mathfrak{m}$, we have $((g) + \mathfrak{m})((e) + \mathfrak{m}) = (ge) + \mathfrak{m} = \mathfrak{m}$. Since $\mathfrak{m}$ is maximal, $(e) + \mathfrak{m}$ is either (1) or $\mathfrak{m}$. The is true for g in place of e, as well. However, they cannot both be (1) since the product would be (1). Thus, one of (e) and (g) is contained by $\mathfrak{m}$. However, both cannot be in $\mathfrak{m}$ either since $(e) + (g) = (1)$. Thus, since all prime ideals are maximal here, $V(e)^c = V(g)$ so that $X_e = X_g^c$. In particular, any two distinct points of $\operatorname{Spec} A$ can be separated by a closed-and-open set, so $\operatorname{Spec} A$ is totally disconnected. □

3.12 Exercise

Proof. Let $T(M)$ denote the set of torsion elements of M. If $x, y \in T(M)$, then there is some $a_x, a_y \in A$ (non-zero) such that $a_x x = 0$ and $a_y y = 0$. Let $a, b \in A$. Since A is an integral domain, $a_x a_y \neq 0$. We have

$$a_x a_y(ax + by) = a a_y(a_x x) + b a_x(a_y y) = 0,$$

so $ax + by \in T(M)$ whenever $x, y \in T(M)$. Thus, $T(M)$ is an A-submodule of M. □

(i) Proof. Suppose $m + T(M)$ is a torsion element of $M/T(M)$. Then, there is some non-zero $a \in A$ such that $am = x$, where $x \in T(M)$. Let $a_x \in A$ be a non-zero annihilator of x. Then, $aa_x \neq 0$ since A is an integral domain, so $0 = a(a_x x) = (a_x a)m$, so $m \in T(M)$. Thus, $M/T(M)$ is torsion-free. □

(ii) Proof. Suppose $f : M \to N$ is an A-module homomorphism. Let $x \in T(M)$, and let $a_x \in A$ be a non-zero annihilator of x. Then, $a_x f(x) = f(a_x x) = f(0) = 0$, so $f(T(M)) \subset T(N)$. □

(iii) Proof. Suppose we have an exact sequence of A-modules

$$0 \to M' \xrightarrow{f} M \xrightarrow{g} M''.$$

By (ii), the sequence

$$0 \to T(M') \xrightarrow{f} T(M) \xrightarrow{g} T(M'')$$

is well-defined. Further, since $T(M')$ is a submodule of M', the restriction of f to $T(M')$ maintains its injectivity, so the sequence is exact at $T(M')$. Likewise, since $\operatorname{im} f \subset \ker g$ in the original

sequence, this clearly holds for the restriction to $T(M)$. Now, suppose $x \in T(M)$ satisfies $g(x) = 0$. Then, by exactness of the original sequence, there is some $x' \in M'$ such that $f(x') = x$. Since $x \in T(M)$, there is some non-zero $a_x \in A$ annihilating x. Thus,

$$0 = a_x x = a_x f(x') = f(a_x x'),$$

so by the injectivity of f, $a_x x' = 0$, showing that $x' \in T(M')$. Thus, $\ker g \subset \operatorname{im} f$ so that we have equality. In particular, the sequence is exact at $T(M)$ so that, finally, the sequence is exact. $\square$

(iv) Proof. K, the field of fractions of A, has the obvious A-module structure. Since A is an integral domain, $K = S^{-1}A$ with $S = A - \{0\}$. $T^{-1}A$, as T ranges over finitely generated multiplicatively closed subsets of A (excluding 0), forms a directed system, as there is the natural homomorphism $T_i^{-1}A \to T_j^{-1}A$ whenever $T_i \subset T_j$ ($a/t \mapsto a/t$). If $T_i \subset T_j \subset T_k$, we have

$$T_i^{-1}A \xrightarrow{\phi_{ij}} T_j^{-1}A \xrightarrow{\phi_{jk}} T_k^{-1}A.$$

By Exercise 3.3, we have a natural isomorphism $T_k^{-1}A \cong \phi_{ik}(T_k)^{-1}A$ sending $a/s \mapsto (a/s)/1$. The composition of these natural maps achieves the same result so that $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$. Thus, the maps of the $\{(T_i^{-1}A, \phi_{ij})\}$ satisfy the necessary compatibility requirements, as well as being injective.

Since A is an integral domain, $\iota_i : A \to T_i^{-1}A$ is injective by Exercise 3.9, and so is $\iota : A \to S^{-1}A$. By the same compatibility requirements just mentioned, we have a map $\phi_i : T_i^{-1}A \to S^{-1}A$ that satisfies $\iota = \phi_i \circ \iota_i$ for all i. Now, we wish to show that $K = \varinjlim T_i^{-1}A$. Let N be an A-module such that for all $i \in I$, we have a map $\alpha_i : T_i^{-1}A \to N$ that is an A-module homomorphism. Further, they must satisfy $\alpha_i = \alpha_j \circ \phi_{ij}$ whenever $i \leq j$. Now, we define $\alpha : S^{-1}A \to N$. If $a/s \in S^{-1}A$, then $T = \{s^n\}_{n \geq 0}$ is a finitely generated multiplicatively closed set not containing zero, so this is in our directed system; let's write $T_s = T$. We define $\alpha(a/s) = \alpha_s(\phi_s(a/s))$. Note that if $s \in T_i$ for any $i \in I$, then since T_i is multiplicatively closed, $T_s \subset T_i$, so we have $\alpha_s = \alpha_i \circ \phi_{si}$ by definition; then, $\alpha_s(a/s) = \alpha_i(\phi_{si}(a/s)) = \alpha_i(a/s)$. In particular, our map α is well-defined. Finally, suppose $a/s \in T_i^{-1}A$; then, $\alpha_i(a/s) = \alpha_s(a/s) = \alpha(\phi_i(a/s))$. Thus, K satisfies the universal property for direct limits, so by Exercise 2.16, $K = \varinjlim T_i^{-1}A$.

By definition, any T_i in the directed system is generated by finitely many elements $x_1, \ldots, x_n \in A$. We show now that $T_i^{-1}A \cong T^{-1}A$, where $T = \{\xi^n\}$ and $\xi = x_1 \cdots x_n$. Clearly, $T \subset T_i$, so we have an injection $T^{-1}A \to T_i^{-1}A$. Note that $T_i^{-1}A$ is generated as an A-algebra by $\frac{1}{x_1}, \ldots, \frac{1}{x_n}$, and $T^{-1}A$ by $\frac{1}{\xi}$. However, $\frac{1}{x_i} = \frac{x_1 \cdots \hat{x}_i \cdots x_n}{\xi}$, so we have a ring isomorphism.

Let $M_i' = \ker(M \to T_i^{-1}A \otimes_A M)$ defined by $x \mapsto 1 \otimes x$. Then, we have an exact sequence

$$0 \to M_i' \to M \to T_i^{-1}A \otimes_A M.$$

Since direct limits commute with exact sequences (Exercise 2.19) and since, by Exercise 2.20,

$$\varinjlim(T_i \otimes_A M) = \varinjlim(T_i) \otimes_A M = K \otimes_A M,$$

we have an exact sequence

$$0 \to M' \to M \to K \otimes_A M,$$

where $M' = \ker(M \to K \otimes_A M)$ and also $M' = \varinjlim \ker(M_i' \to T_i^{-1}A \otimes_A M)$. Thus, if $x \in M'$, then by Exercise 2.15, there is some $i \in I$ such that $x \in M_i'$. The converse is also clearly true since the maps $M_i' \to M'$ are A-module homomorphisms.

By definition, any T_i in the directed system is generated by finitely many elements $x_1, \ldots, x_n \in A$. We show now that $T_i^{-1}A \cong T^{-1}A$, where $T = \{\xi^n\}$ and $\xi = x_1 \cdots x_n$. Clearly, $T \subset T_i$, so we have an injection $T^{-1}A \to T_i^{-1}A$. Note that $T_i^{-1}A$ is generated as an A-algebra by $\frac{1}{x_1}, \ldots, \frac{1}{x_n}$, and $T^{-1}A$ by $\frac{1}{\xi}$. However, $\frac{1}{x_i} = \frac{x_1 \cdots \hat{x}_i \cdots x_n}{\xi}$, so we have a ring isomorphism.

If $x \in M'$, then $x \in \ker(M \to T^{-1}A \otimes_A M)$ for some $T = \{\xi^n\}$. Note that by Proposition 3.5, $T^{-1}A \otimes_A M \cong T^{-1}M$ under $a \otimes x \mapsto ax$. Thus, $x \in \ker(M \to \{\xi^n\}^{-1}M)$, so $(0, 1) \equiv (x, 1)$. Thus, for some n, $\xi^n x = 0$, so $x \in T(M)$. Conversely, if $x \in T(M)$, then there is some non-zero $\xi \in A$ such that $\xi x = 0$. Then, in $\{\xi^n\}^{-1}A \otimes_A M$, we have $1 \otimes x = \frac{\xi}{\xi} \otimes x = \frac{1}{\xi} \otimes (\xi x) = 0$, so $x \in M_i'$ and thus $x \in M'$. □

3.13 Exercise

Proof. We showed in Exercise 3.12 that

$$0 \to T(M) \to M \to K \otimes_A M$$

is exact. Localization preserves exact sequences, so

$$0 \to S^{-1}(TM) \to S^{-1}M \to S^{-1}(K \otimes_A M)$$

is exact. Further, $S^{-1}(K \otimes_A M) = S^{-1}K \otimes_{S^{-1}A} S^{-1}M$ by Proposition 3.7. However, $S^{-1}K = K$ since all elements of S are already units in K. Thus, our exact sequence is now

$$0 \to S^{-1}(TM) \to S^{-1}M \to K \otimes_{S^{-1}A} S^{-1}M.$$

Note that $S^{-1}A$ and A share the same field of fractions K, so by Exercise 3.12,

$$0 \to T(S^{-1}M) \to S^{-1}M \to K \otimes_{S^{-1}A} S^{-1}M$$

is exact. Since $T(S^{-1}M)$ and $S^{-1}(TM)$ are both exactly the kernel of $S^{-1}M \to K \otimes_{S^{-1}A} S^{-1}M$, they are isomorphic as $S^{-1}A$-modules. □

Proof. (i) $\implies$ (ii). If M is torsion-free, then $T(M) = 0$, so $T(S^{-1}M) = S^{-1}(T(M)) = 0$ by the previous proof. Thus, $M_\mathfrak{p}$ is torsion-free for all prime ideals $\mathfrak{p}$.

(ii) $\implies$ (iii). Trivial. All maximal ideals are prime.

(iii) $\implies$ (i). Suppose $M_\mathfrak{m}$ is torsion-free for all maximal ideals $\mathfrak{m}$. For any maximal ideal $\mathfrak{m}$, we have by assumption and the previous result that

$$0 = T(M_\mathfrak{m}) = T((A - \mathfrak{m})^{-1}M) = (A - \mathfrak{m})^{-1}T(M) = T(M)_\mathfrak{m}.$$

Since $T(M)_\mathfrak{m} = 0$ for all maximal ideals, we have by Proposition 3.8 that $T(M) = 0$. □

Remark. *In particular, "torsion-free" is a local property.*

3.14 Exercise

Proof. The quotient $M/\mathfrak{a}M$ is an $A/\mathfrak{a}$-module, and the maximal ideals of $A/\mathfrak{a}$ are precisely the maximal ideals containing $\mathfrak{a}$. By assumption, for any of these maximal ideals $\mathfrak{m}$, $M_\mathfrak{m} = 0$. Since localization commutes with quotients, we have $(M/\mathfrak{a}M)_\mathfrak{m} = M_\mathfrak{m}/(\mathfrak{a}M)_\mathfrak{m} = 0$. Since $(M/\mathfrak{a}M)_\mathfrak{m} = 0$ for all maximal ideals such that $\mathfrak{a} \subset \mathfrak{m}$, we invoke Proposition 3.8 to conclude that $M/\mathfrak{a}M = 0$; hence, $M = \mathfrak{a}M$. □

3.15 Exercise

Proof. Let $x_1, \ldots, x_n$ be a generating set of the A-module $F = A^n$. Let $e_1, \ldots, e_n$ be the canonical basis (e_i is the unit of the ith direct summand). Define $\phi : F \to F$ by $\phi(x_i) = e_i$, extending A-linearly so that ϕ is an A-module homomorphism. Since $\{e_i\}$ is a basis, ϕ is surjective. In light of Proposition 3.9, to show that ϕ is injective, it is enough to show that $\phi_{\mathfrak{m}} : F_{\mathfrak{m}} \to F_{\mathfrak{m}}$ is injective for each maximal ideal of A. Note that $F_{\mathfrak{m}} = A_{\mathfrak{m}}^n$. Since $\phi_{\mathfrak{m}}$ is surjective, if N is the kernel of $\phi_{\mathfrak{m}}$, then

$$0 \to N \to A_{\mathfrak{m}}^n \xrightarrow{\phi_{\mathfrak{m}}} A_{\mathfrak{m}}^n \to 0$$

is exact. Tensoring with $k = A/\mathfrak{m}$, we obtain the exact sequence

$$0 \to k \otimes_{A_{\mathfrak{m}}} N \to (k \otimes_{A_{\mathfrak{m}}} A_{\mathfrak{m}})^n \xrightarrow{\phi_{\mathfrak{m}}} (k \otimes_{A_{\mathfrak{m}}} A_{\mathfrak{m}})^n \to 0,$$

noting that since $A_{\mathfrak{m}}^n$ is a flat $A_{\mathfrak{m}}$-module, $\operatorname{Tor}_1^{A_{\mathfrak{m}}}(k, A_{\mathfrak{m}}^n) = 0$ by Exercise 2.24. But under natural isomorphism, we have $k \cong k \otimes_{A_{\mathfrak{m}}} A_{\mathfrak{m}}$, so we have the short exact sequence

$$0 \to k \otimes N \to k^n \to k^n \to 0.$$

Since $k^n \to k^n$ is a surjective endomorphism of k-vector spaces, it is an isomorphism, so $k \otimes N = 0$.

Note that $k \otimes N = (A_{\mathfrak{m}}/\mathfrak{m}) \otimes_{A_{\mathfrak{m}}} N = N/\mathfrak{m}N$ by Exercise 2.2. Thus, $N/\mathfrak{m}N = 0$ so that $N = \mathfrak{m}N$. Since $A_{\mathfrak{m}}$ is a local ring, $\mathfrak{m}$ is the Jacobson radical. By Exercise 2.12, N is finitely generated, so we can invoke Nakayama's lemma to conclude that $N = 0$. In particular, $\phi_{\mathfrak{m}}$ is injective for all maximal ideals $\mathfrak{m}$, so ϕ is injective by Proposition 3.9. Thus, ϕ is an isomorphism, and $x_1, \ldots, x_n$ is a basis.

In particular, suppose we can generate the A-module A^n by elements $x_1, \ldots, x_j$, where $j < n$. We can define $x_i = 0$ for $j < i \leq n$ to obtain a set of n elements that generate A^n. By our result, $\{x_i\}_{i=1}^n$ is a basis, but since $x_n = 0$, this is a clear contradiction. Thus, every set of generators must contain at least n elements. □

3.16 Exercise

Proof. (i) $\implies$ (ii). If $\mathfrak{a}^{ec} = \mathfrak{a}$ for all ideals $\mathfrak{a}$ of A, then certainly, $\mathfrak{p}^{ec} = \mathfrak{p}$ for any $\mathfrak{p} \in \operatorname{Spec}(A)$. By Proposition 3.16, every $\mathfrak{p}$ is the contraction of some $\mathfrak{q} \in \operatorname{Spec}(B)$. Thus, $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective.

(ii) $\implies$ (iii). Since $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is surjective, any maximal ideal $\mathfrak{m}$ satisfies $\mathfrak{m} = \mathfrak{q}^c$ for some $\mathfrak{q} \in \operatorname{Spec}(B)$. By Proposition 1.17(i), $\mathfrak{m}^e = \mathfrak{q}^{ce} \subset \mathfrak{q}$, so $\mathfrak{m}^e \neq (1)$ for any $\mathfrak{m} \in \operatorname{Max}(A)$.

(iii) $\implies$ (iv). Let M be a non-zero A-module, and let $x \in M$ be a non-zero element. Let $M' = Ax$. Since $M' \to M$ is injective, and B is a flat A-algebra, $M'_B \to M_B$ is injective. Thus, to show $M_B \neq 0$, it is sufficient to show that $M'_B \neq 0$. The map $\phi : A \to Ax$ is clearly surjective, so since $\ker \phi$ is an ideal $\mathfrak{a}$ of A, we have $M' = Ax \cong A/\mathfrak{a}$ as an A-module. Hence,

$$M'_B = B \otimes_A M' \cong B \otimes_A (A/\mathfrak{a}) \cong B/\mathfrak{a}B = B/\mathfrak{a}^e.$$

Since $\mathfrak{a}$ is not (1) ($1 \mapsto x \neq 0$), there is some maximal ideal $\mathfrak{m}$ such that $\mathfrak{a} \subset \mathfrak{m}$. By our assumption, $\mathfrak{a}^e \subset \mathfrak{m}^e \neq (1)$, so $\mathfrak{a}^e$ is a proper ideal of B, implying that $M'_B \cong B/\mathfrak{a}^e \neq 0$.

(iv) $\implies$ (v). Let M' be the kernel of $M \to M_B$. As B is flat over A,

$$0 \to M'_B \to M_B \to (M_B)_B$$

is exact. Since M_B is a B-module, by Exercise 2.13, $M_B \to (M_B)_B$ is injective. Thus, $M'_B = 0$. By assumption, $M' = 0$, so $M \to M_B$ $(m \mapsto 1 \otimes m)$ is injective.

(v) $\implies$ (i). If $\mathfrak{a}$ is any ideal of A, consider the A-module $A/\mathfrak{a}$. Then, by assumption,

$$A/\mathfrak{a} \to B \otimes_A A/\mathfrak{a} \cong B/\mathfrak{a}B \cong B/\mathfrak{a}^e$$

is injective. Thus, $f(a) \in \mathfrak{a}^e \iff a \in \mathfrak{a}$, so $\mathfrak{a} = \mathfrak{a}^{ec}$. □

Remark. *B is said to be "faithfully flat" over A.*

3.17 Exercise

Proof. Let

$$0 \to M' \to M \to M'' \to 0$$

be an exact sequence of A-modules. Then, since C is flat over A,

$$0 \to C \otimes_A M' \to C \otimes_A M \to C \otimes_A M'' \to 0$$

is also exact. For any A-module N, we have $C \otimes_A N = C \otimes_B (B \otimes_A N)$. Thus, the above exact sequence can be written as

$$0 \to C \otimes_B M'_B \to C \otimes_B M_B \to C \otimes_B M''_B \to 0.$$

By the right-exactness of tensors, we already have the exact sequence

$$M'_B \to M_B \to M''_B \to 0.$$

Since C is faithfully flat over B, $M'_B \to C \otimes_B M'_B$ and $M_B \to C \otimes_B M_B$ are injective. But the map $C \otimes_B M'_B \to C \otimes_B M_B$ is also injective (see above). Consider the canonical commutative diagram:

$$\begin{array}{ccc} M'_B & \xrightarrow{\phi} & M_B \\ \downarrow{\scriptstyle \iota'} & & \downarrow{\scriptstyle \iota} \\ C \otimes_B M'_B & \xrightarrow{\phi'} & C \otimes_B M_B \end{array}$$

Since ϕ' and ι' are injective, $\phi' \circ \iota' = \iota \circ \phi$ is injective. Thus, ϕ is injective, so the sequence

$$0 \to M'_B \to M_B \to M''_B \to 0$$

is exact. Finally, B is flat over A. □

3.18 Exercise

Proof. By Proposition 3.10, $B_\mathfrak{p}$ is flat over $A_\mathfrak{p}$. Since $\mathfrak{p} = \mathfrak{q}^c$, $\mathfrak{p}^{ec} = \mathfrak{q}^{cec} = \mathfrak{q}^c = \mathfrak{p}$. Then, by Exercise 3.4,

$$B_\mathfrak{p} = (A - \mathfrak{p})^{-1}B = f(A - \mathfrak{p})^{-1}B.$$

Since $\mathfrak{p} = f^{-1}(\mathfrak{q})$, $f(A - \mathfrak{p}) \subset B - \mathfrak{q}$; if $b = f(x)$, where $x \notin \mathfrak{p}$, then if $b \in \mathfrak{q}$, $x \in f^{-1}(\mathfrak{q})$, a contradiction. Thus, $B_\mathfrak{q} = (B - \mathfrak{q})^{-1}B_\mathfrak{p}$. By Corollary 3.6, $B_\mathfrak{q}$ is flat over $B_\mathfrak{p}$. Since $B_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-algebra, and $B_\mathfrak{q}$ is flat over $B_\mathfrak{p}$, we have by Exercise 2.8 that $B_\mathfrak{q}$ is flat over $A_\mathfrak{p}$.

Note that $A_\mathfrak{p}$ is a local ring with maximal ideal $S^{-1}\mathfrak{p}$. The map $f : A \to B$ becomes a map $A \to B \to B_\mathfrak{q}$. Since $f(A - \mathfrak{p}) \subset B - \mathfrak{q}$, it factors through a map (also called f) $f : A_\mathfrak{p} \to B_\mathfrak{q}$. Note that $\mathfrak{p}^e \neq (1)$ since $\mathfrak{p}^{ec} = \mathfrak{p}$, so it does not map to (1) in $B_\mathfrak{q}$. Since $\mathfrak{p}$ maps to $S^{-1}\mathfrak{p}$ in $A_\mathfrak{p}$, the unique maximal ideal, we have $(S^{-1}\mathfrak{p})^e \neq (1)$. Thus, we have satisfied condition 3 of Exercise 3.16, so $f : A_\mathfrak{p} \to B_\mathfrak{q}$ is faithfully flat and $f^* : \mathrm{Spec}(B_\mathfrak{q}) \to \mathrm{Spec}(A_\mathfrak{p})$ is surjective. □

3.19 Exercise

(i) Proof. $M \neq 0 \iff \operatorname{Supp}(M) \neq \emptyset$ follows immediately from Proposition 3.8. □

(ii) Proof. Consider the short exact sequence of A-modules

$$0 \to \mathfrak{a}A \to A \to A/\mathfrak{a} \to 0.$$

Localization is exact, so for any prime ideal $\mathfrak{p}$,

$$0 \to \mathfrak{a}^e A_\mathfrak{p} \to A_\mathfrak{p} \to (A/\mathfrak{a})_\mathfrak{p} \to 0$$

is exact. From the exact sequence, it is clear that $(A/\mathfrak{a})_\mathfrak{p} = 0$ if and only if $\mathfrak{a}^e = (1)$. But $\mathfrak{a}^e = (1)$ if and only if $(A - \mathfrak{p}) \cap \mathfrak{a} \neq \emptyset$ by Proposition 3.11. In particular, $\mathfrak{a}^e \neq (1)$ if and only if $\mathfrak{a} \subset \mathfrak{p}$. Thus, $V(\mathfrak{a}) = \operatorname{Supp}(A/\mathfrak{a})$, as desired. □

(iii) Proof. Suppose we have an exact sequence of A-modules

$$0 \to M' \to M \to M'' \to 0.$$

Localization is exact, so if $\mathfrak{p}$ is a prime ideal, we have

$$0 \to M'_\mathfrak{p} \to M_\mathfrak{p} \to M''_\mathfrak{p} \to 0.$$

If $M_\mathfrak{p} = 0$, then clearly $M'_\mathfrak{p} = 0$ and $M''_\mathfrak{p} = 0$. Conversely, if $M'_\mathfrak{p} = 0$ and $M''_\mathfrak{p} = 0$, $M_\mathfrak{p} = 0$. Thus, $\operatorname{Supp}(M)^c = \operatorname{Supp}(M')^c \cap \operatorname{Supp}(M'')^c$ so that $\operatorname{Supp}(M) = \operatorname{Supp}(M') \cup \operatorname{Supp}(M'')$. □

(iv) Proof. If $M = \sum M_i$, then for any prime ideal $\mathfrak{p}$, we have $M_\mathfrak{p} = \sum (M_i)_\mathfrak{p}$ (Proposition 3.11(v)). Thus, $M_\mathfrak{p} = 0$ if and only if all $(M_i)_\mathfrak{p} = 0$. Hence, $\operatorname{Supp}(M)^c = \bigcap \operatorname{Supp}(M_i)^c$. □

(v) Proof. If M is finitely generated, is generated by elements $x_1, \ldots, x_n$. Let $M_i = Ax_i$ so that $M = \sum M_i$. Note that $A \to Ax_i \cong M$ is surjective and the kernel is the ideal $\operatorname{Ann}(M_i)$. Thus, $M_i \cong A/\operatorname{Ann}(M_i)$. Thus, $M = \sum A/\operatorname{Ann}(M_i)$. By (iv), (ii), Exercise 1.15, and the intra-text Exercise 2.2,

$$\operatorname{Supp}(M) = \bigcup_i \operatorname{Supp}(A/\operatorname{Ann}(M_i)) = \bigcup_i V(\operatorname{Ann}(M_i))$$

$$= V\left(\bigcap_i \operatorname{Ann}(M_i)\right) = V\left(\operatorname{Ann}\left(\sum_i M_i\right)\right) = V(\operatorname{Ann}(M)).$$

In particular, $\operatorname{Supp}(M)$ is a closed subset of $\operatorname{Spec}(A)$. □

(vi) Proof. For any prime ideal $\mathfrak{p}$, we have

$$(M \otimes_A N)_\mathfrak{p} = M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p}$$

by Proposition 3.7. Clearly if $M_\mathfrak{p} = 0$ or $N_\mathfrak{p} = 0$, then the tensor product is 0. M and N are finitely generated, so $M_\mathfrak{p}$ and $N_\mathfrak{p}$ are finitely generated. Further, since $A_\mathfrak{p}$ is a local ring, $M_\mathfrak{p} \otimes_{A_\mathfrak{p}} N_\mathfrak{p} = 0$ implies $M_\mathfrak{p} = 0$ or $N_\mathfrak{p} = 0$ by Exercise 2.3. Thus, $\operatorname{Supp}(M \otimes_A N) = \operatorname{Supp}(M) \cap \operatorname{Supp}(N)$. □

(vii) Proof. Note that $M/\mathfrak{a}M \cong (A/\mathfrak{a}) \otimes_A M$. Since M is finitely generated as an A-module and $A/\mathfrak{a}$ is also, we have by (vi), (ii), (v), and Exercise 1.15 that

$$\operatorname{Supp}(M/\mathfrak{a}M) = \operatorname{Supp}(A/\mathfrak{a}) \cap \operatorname{Supp}(M) = V(\mathfrak{a}) \cap V(\operatorname{Ann}(M)) = V(\mathfrak{a} + \operatorname{Ann}(M)),$$

as desired. □

(viii) Proof. If $\mathfrak{q}$ is a prime ideal of B and $\mathfrak{p} = \mathfrak{q}^c$, then by Proposition 3.5,

$$(B \otimes_A M)_\mathfrak{q} \cong B_\mathfrak{q} \otimes_A M \cong (B_\mathfrak{q} \otimes_{A_\mathfrak{p}} A_\mathfrak{p}) \otimes_A M \cong B_\mathfrak{q} \otimes_{A_\mathfrak{p}} (A_\mathfrak{p} \otimes_A M) \cong B_\mathfrak{q} \otimes_{A_\mathfrak{p}} M_\mathfrak{p}.$$

If $M_\mathfrak{p} = 0$, then clearly, $(B \otimes_A M)_\mathfrak{q} = 0$. Thus, $\mathrm{Supp}(B \otimes_A M) \subset f^{*-1}(\mathrm{Supp}(M))$.

Conversely, suppose $0 = (B \otimes_A M)_\mathfrak{q} = B_\mathfrak{q} \otimes_{A_\mathfrak{p}} M_\mathfrak{p}$. Since M is finitely generated, $M_\mathfrak{p}$ is finitely generated by elements $x_1, \ldots, x_n$. Thus, $(1/1) \otimes x_i = 0$ in $B_\mathfrak{q} \otimes_{A_\mathfrak{p}} M_\mathfrak{p}$. By Corollary 2.13, there is a finitely generated $A_\mathfrak{p}$-submodule $N_i \subset B_\mathfrak{q}$ such that $(1/1) \otimes x_i = 0$. Thus, $N = \sum_i N \subset B_\mathfrak{q}$ is also finitely generated, and $(1/1) \otimes x = 0$ in $N \otimes_{A_\mathfrak{p}} M_\mathfrak{p}$ for all $x \in M_\mathfrak{p}$ so that $N \otimes_{A_\mathfrak{p}} M_\mathfrak{p} = 0$. Since both tensor factors are finitely generated and $A_\mathfrak{p}$ is a local ring, we have by Exercise 2.3 that $N = 0$ or $M_\mathfrak{p} = 0$. But, $1/1 \in N \subset B_\mathfrak{q}$, so $N \neq 0$ and $M_\mathfrak{p} = 0$. Hence, $\mathrm{Supp}(B \otimes_A M) = f^{*-1}(\mathrm{Supp}(M))$. □

3.20 Exercise

Consider $f^* : \mathrm{Spec}(B) \to \mathrm{Spec}(A)$.

(i) Proof. f^* is surjective $\iff$ for every $\mathfrak{p} \in \mathrm{Spec}(A)$, there is some $\mathfrak{q} \in \mathrm{Spec}(B)$ such that $\mathfrak{p} = f^*(\mathfrak{q}) = f^{-1}(\mathfrak{q}) = \mathfrak{q}^c$. □

(ii) Proof. Suppose every $\mathfrak{q} \in \mathrm{Spec}(B)$ is of the form $\mathfrak{a}^e$. Suppose now that $f^*(\mathfrak{q}_x) = f^*(\mathfrak{q}_y)$ so that $\mathfrak{q}_x^c = \mathfrak{q}_y^c$. Then, since $\mathfrak{q}_x = \mathfrak{a}_x^e$ and $\mathfrak{q}_y = \mathfrak{a}_y^e$, we have $\mathfrak{a}_x^{ec} = \mathfrak{a}_y^{ec}$. Extending again, we have by Proposition 1.17,

$$\mathfrak{q}_x = \mathfrak{a}_x^e = \mathfrak{a}_x^{ece} = \mathfrak{a}_y^{ece} = \mathfrak{a}_y^e = \mathfrak{q}_y,$$

so f^* is injective. □

We show now that the converse is false. For a ring A, we have $A[y] \cong A[x]/(x^2)$, where $y^2 = 0$. Then, $(y)^2 = 0$, so (y) is in the nilradical of $A[y]$. Thus, every prime ideal of $A[y]$ contains (y). We have the ring homomorphism $f : A[y] \to A$ that sends $a + by \mapsto a$; this is inherited from the natural augmentation map $A[x] \to A$. For any prime ideal $\mathfrak{p} \in A$, we have $f^{-1}(\mathfrak{p}) = \mathfrak{p}A + (y)$. Thus, $f^* : \mathrm{Spec}(A) \to \mathrm{Spec}(A[y])$ is clearly injective. For any ideal $\mathfrak{a}$ of A, $\mathfrak{a}^e = \mathfrak{a}A + \mathfrak{a}y$, which does not include y unless $\mathfrak{a} = A$. However, each prime $\mathfrak{p}A + (y)$ of $A[y]$ does contain y. Thus, f^* injective does not imply that all prime ideals of $\mathrm{Spec}(B)$ are extended ideals.

3.21 Exercise

(i) Proof. The canonical homomorphism $\phi : S^{-1}A \to A$ gives way to the map $\phi^* : \mathrm{Spec}(S^{-1}A) \to \mathrm{Spec}(A)$. To show that ϕ^* is a homeomorphism of $\mathrm{Spec}(S^{-1}A)$ onto its image, we have to show that ϕ^* is bijective and bicontinuous. We showed in Exercise 1.21 that ϕ^* is continuous in general, and ϕ^* is obviously surjective onto its image. So, we just have to show that ϕ^* is injective and that it is a closed map.

By Proposition 3.11, the prime ideals of $S^{-1}A$ are in one-to-one correspondence with the prime ideals of A which don't meet S. In particular, ϕ^* is injective and thus bijective onto its image. Writing $X = \mathrm{Spec}(A)$, we denote this image by $\phi^*(\mathrm{Spec}(S^{-1}A)) = S^{-1}X$.

To show that the map ϕ^* is closed, we invoke Proposition 3.11, which gives the aforementioned bijection via contraction and extension. In particular, for primes $\mathfrak{p} \in A - S$, $\mathfrak{p}^{ec} = \mathfrak{p}$. Thus, any closed set of $\mathrm{Spec}(S^{-1}A)$ has the form $V(\mathfrak{a}^e)$, where $\mathfrak{a}$ is an ideal of A. But if $\mathfrak{a} \subset \mathfrak{p} \subset A - S$, then $\mathfrak{a}^e \subset \mathfrak{p}^e \neq (1)$ so that $\mathfrak{a} \subset \mathfrak{a}^{ec} \subset \mathfrak{p}^{ec} = \mathfrak{p}$. Thus, $\phi^*(V(\mathfrak{a}^e)) = V(\mathfrak{a}) \cap S^{-1}X$. Thus, ϕ^* is a closed map and thus a homeomorphism, as desired.

In particular, consider the case of $A_f = \{f^n\}^{-1}A$. By Proposition 3.11 again, the prime ideals of A_f are in one-to-one correspondence with the prime ideals of A which don't meet $\{f^n\}$. Thus, $\phi^*(\operatorname{Spec}(A_f)) = \{\mathfrak{p} \in \operatorname{Spec}(A) : \{f^n\} \cap \mathfrak{p} = \emptyset\}$. If $\mathfrak{p} \cap \{f^n\} \neq \emptyset$, then there is some smallest n such that $f^n \in \mathfrak{p}$. However, if $n > 1$, then $f^n = f \cdot f^{n-1} \in \mathfrak{p}$, so either f or f^{n-1} is in the prime ideal, contradicting minimality. Furthermore, $1 \notin \mathfrak{p}$ since $\mathfrak{p}$ cannot be the whole ring. Thus, $\mathfrak{p} \notin \phi^*(A_f) \iff f \in \mathfrak{p}$. In particular, $\phi^*(\operatorname{Spec}(A_f)) = V((f))^c = X_f$, the basic open set of $\operatorname{Spec}(A)$. □

(ii) Proof. For a ring homomorphism $f : A \to B$, we have a homomorphism $S^{-1}f : S^{-1}A \to S^{-1}B$. By (i), we've shown that there is a canonical image $\operatorname{Spec}(S^{-1}A) \to S^{-1}\operatorname{Spec}(A) \subset \operatorname{Spec}(A)$. So, we have a square

$$\begin{array}{ccc} \operatorname{Spec}(S^{-1}B) & \xrightarrow{(S^{-1}f)^*} & \operatorname{Spec}(S^{-1}A) \\ \downarrow{\scriptstyle \phi_B^*} & & \downarrow{\scriptstyle \phi_A^*} \\ \operatorname{Spec}(B) & \xrightarrow{f^*} & \operatorname{Spec}(A) \end{array}$$

The goal is to show that $\phi_A^* \circ (S^{-1}f)^* = \phi_B^* \circ f^*$. In light of Exercise 1.21, it is sufficient to show that $(S^{-1}f) \circ \phi_A = \phi_B \circ f$, i.e. that we have the commutative diagram

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow{\scriptstyle \phi_A} & & \downarrow{\scriptstyle \phi_B} \\ S^{-1}A & \xrightarrow{S^{-1}f} & S^{-1}B \end{array}$$

So, taking $a \in A$, $(\phi_B \circ f)(a) = \phi_B(f(a)) = f(a)/1$ and $((S^{-1}f) \circ \phi_A)(a) = (S^{-1}f)(a/1) = f(a)/1$, so indeed, commutativity holds. Identifying $\operatorname{Spec}(S^{-1}B)$ with $S^{-1}\operatorname{Spec}(B)$ and $\operatorname{Spec}(S^{-1}A)$ with $S^{-1}\operatorname{Spec}(A)$, we see that $(S^{-1}f)^* = f^* \mid_{S^{-1}\operatorname{Spec}(B)}$. Thus, $(S^{-1}f)^* : \operatorname{Spec}(S^{-1}B) \to \operatorname{Spec}(S^{-1}A)$ is a homeomorphism by (i), so we have $S^{-1}\operatorname{Spec}(B) = f^{*-1}(S^{-1}\operatorname{Spec}(A))$. □

(iii) Proof. As before, we will start with a diagram:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ \downarrow{\scriptstyle \pi_A} & & \downarrow{\scriptstyle \pi_B} \\ A/\mathfrak{a} & \xrightarrow{\overline{f}} & B/\mathfrak{a}^e \end{array}$$

If $a \in A$, then we have $(\pi_B \circ f)(a) = \pi_B(f(a)) = f(a) + \mathfrak{a}^e$ and $(\overline{f} \circ \pi_A)(a) = \overline{f}(a + \mathfrak{a}) = f(a) + \mathfrak{a}^e$, so the diagram commutes. Thus, we have a commutative diagram (Exercise 1.21)

$$\begin{array}{ccc} \operatorname{Spec}(B/\mathfrak{a}^e) & \xrightarrow{\overline{f}^*} & \operatorname{Spec}(A/\mathfrak{a}) \\ \downarrow{\scriptstyle \pi_B^*} & & \downarrow{\scriptstyle \pi_A^*} \\ \operatorname{Spec}(B) & \xrightarrow{f^*} & \operatorname{Spec}(A) \end{array}$$

Now, if $\mathfrak{q} \in V(\mathfrak{a}^e)$, then $\mathfrak{a}^{ec} \subset \mathfrak{q}^c = f^*(\mathfrak{q})$, so $f^*(V(\mathfrak{a}^e)) \subset V(\mathfrak{a})$. Since the images of π_B^* and π_A^* are exactly $V(\mathfrak{a}^e)$ and $V(\mathfrak{a})$, and since $f^*(V(\mathfrak{a}^e)) \subset V(\mathfrak{a})$, it's clear from the diagram that $\overline{f}^*$ is just the restriction of f^* to $V(\mathfrak{a}^e)$. □

(iv) Proof. By the commutative diagrams of (ii) and (iii), we have

$$\begin{array}{ccccc} \mathrm{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}) & \longrightarrow & \mathrm{Spec}(B_\mathfrak{p}) & \longrightarrow & \mathrm{Spec}(B) \\ \downarrow{\scriptstyle (f_\mathfrak{p})^*} & & \downarrow{\scriptstyle (f_\mathfrak{p})^*} & & \downarrow{\scriptstyle f^*} \\ \mathrm{Spec}(A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}) & \longrightarrow & \mathrm{Spec}(A_\mathfrak{p}) & \longrightarrow & \mathrm{Spec}(A). \end{array}$$

Note that $k(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is a field, so $\mathrm{Spec}(A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}) = \{0\}$. By definition, $\mathrm{Spec}(A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p})$ is in one-to-one correspondence with the primes in $\mathrm{Spec}(A_\mathfrak{p})$ containing $\mathfrak{p}A_\mathfrak{p}$, which is itself in one-to-one correspondence with primes in $\mathrm{Spec}(A)$ that both contain $\mathfrak{p}$ and are contained in $\mathfrak{p}$ (so that they do not meet $S = A - \mathfrak{p}$). This is clearly $\{\mathfrak{p}\}$. Thus, $\mathrm{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}) = f^{*-1}(\mathfrak{p})$, as desired.

We also have, letting $U = A/\mathfrak{p} - \{0\}$,

$$k(\mathfrak{p}) \otimes_A B = (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}) \otimes_A B \cong (A/\mathfrak{p})_\mathfrak{p} \otimes_A B \cong U^{-1}(A/\mathfrak{p}) \otimes_A B$$

$$\cong U^{-1}(A/\mathfrak{p}) \otimes_A A/\mathfrak{p} \otimes_{A/\mathfrak{p}} B \cong U^{-1}(A/\mathfrak{p}) \otimes_{A/\mathfrak{p}} B/\mathfrak{p}B \cong U^{-1}(B/\mathfrak{p}B) \cong B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$$

so that $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\mathrm{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}) = \mathrm{Spec}(k(\mathfrak{p}) \otimes_A B)$. □

Remark. *$f^{*-1}(\mathfrak{p}) = \mathrm{Spec}(k(\mathfrak{p}) \otimes_A B)$ is called the "fiber" of f^* over $\mathfrak{p}$.*

3.22 Exercise

Proof. First, we show that the intersection of all open sets containing $\mathfrak{p}$ is equivalent to the intersection of all basic open sets containing $\mathfrak{p}$. Any open set U containing $\mathfrak{p}$ is the union of basic open sets, as they form a basis for the topology. Thus, there is $X_f \subset U$ that contains $\mathfrak{p}$ so that $f \in A - \mathfrak{p}$. However, X_f is also an open set containing $\mathfrak{p}$, so the intersection of open sets containing $\mathfrak{p}$ is contained in X_f. So, we can replace U with X_f without changing the resulting intersection.

Then, it is sufficient to show that the canonical image of $\mathrm{Spec}(A_\mathfrak{p})$ in $\mathrm{Spec}(A)$ is the intersection of all X_f containing $\mathfrak{p}$. For any $f \in A - \mathfrak{p}$, we have a canonical composition $A \to A_f \to A_\mathfrak{p}$. By Exercise 3.3, this map is exactly the canonical map $A \to A_\mathfrak{p}$. The map $\mathrm{Spec}(A_\mathfrak{p}) \to \mathrm{Spec}(A)$ then factors through $\mathrm{Spec}(A_f)$. By Exercise 3.21, the image of $\mathrm{Spec}(A_f)$ in $\mathrm{Spec}(A)$ is X_f. Thus, $\mathrm{Spec}(A_\mathfrak{p}) \subset X_f$ for any $f \notin \mathfrak{p}$ so that

$$\mathrm{Spec}(A_\mathfrak{p}) \subset \bigcap_{f \notin \mathfrak{p}} X_f.$$

To show the converse, we can take complements and show that $\mathrm{Spec}(A_\mathfrak{p})^c \subset \bigcup_{f \notin \mathfrak{p}} V(f)$. By Proposition 3.11, the prime ideals of $A_\mathfrak{p}$ are precisely the prime ideals of A that don't meet $A - \mathfrak{p}$. Thus, $\mathrm{Spec}(A_\mathfrak{p})^c$ is the set of prime ideals of A that meet $A - \mathfrak{p}$, i.e. the prime ideals $\mathfrak{q}$ such that $(A - \mathfrak{p}) \cap \mathfrak{q} \neq \emptyset$. For every such prime ideal $\mathfrak{q}$, there is an element $a_\mathfrak{q} \in A$ such that $a_\mathfrak{q} \in \mathfrak{q}$ and $a_\mathfrak{q} \in A - \mathfrak{p}$, or equivalently, $a_\mathfrak{q} \notin \mathfrak{p}$. So $\mathfrak{q} \in V(a_\mathfrak{q})$. In particular,

$$\mathrm{Spec}(A_\mathfrak{p})^c \subset \bigcup_{\mathfrak{q} \in \mathrm{Spec}(A_\mathfrak{p})^c} V(a_\mathfrak{q}) \subset \bigcup_{f \notin \mathfrak{p}} V(f)$$

so that, finally, $\mathrm{Spec}(A_\mathfrak{p}) = \bigcap_{f \notin \mathfrak{p}} X_f$. □

3.23 Exercise

(i) Proof. Suppose $X_f = X_g$. By Exercise 1.17, $r((f)) = r((g))$ so that $f^n \in (g)$ and $g^m \in (f)$. Thus, the image of f is a unit in A_g and the image of g is a unit in A_f. Thus, if S is the multiplicatively closed set generated by f and g, both A_f and A_g satisfy the universal property of localization for $S^{-1}A$, so there is a canonical isomorphism of rings $A_f \cong A_g$. In particular, $A(U) := A_f$ depends only on U. □

(ii) Proof. Suppose $U' \subset U$, where $U' = X_g$ for some $g \in A$ and $U = X_f$ for some $f \in A$. By (i), this choice is inconsequential. By Exercise 1.17, $X_g = X_f \cap X_g = X_{fg}$, so $r((fg)) = r((g))$. Since $r((g)) = r((fg)) \subset r((f))$, there is some $u \in A$ and $n > 0$ such that $g^n = uf$. Then, we have a ring homomorphism $\rho : A_f \to A_g$ defined by $a/f^m \mapsto au^m/g^{mn}$. Consider the canonical homomorphism $\psi_g : A \to A_g$. Since $uf = g^n$, $\psi_g(f)$ is a unit. Thus, by the universal property of localization (Proposition 3.1), there is a unique ring homomorphism $h : A_f \to A_g$ such that $\psi_g = h \circ \psi_f$. But h is just ρ since $\psi_g(a/1) = a/1$ and $\rho(\psi_f(a/1)) = \rho(a/1) = a/1$. So, ρ is canonical.

In (i), we showed that $A(U)$ and $A(U')$ depend only on U and U'. Thus, if $A \to A_f \to A_g$ is such a choice, we have $\psi_g = \rho \circ \psi_f$. □

(iii) Proof. By (ii), if we let X_f represent U, then $\psi_f = \mathrm{id}_{A_f} \circ \psi_f$. Since ρ is the unique homomorphism that satisfies $\psi_f = \rho \circ \psi_f$ by (ii), $\rho = \mathrm{id}_{A_f}$. □

(iv) Proof. If $U'' \subset U' \subset U$ are basic open sets, we have canonical maps $\psi_V : A \to A(V)$ that commute with restriction, by (ii). In particular, we have $\psi_{U'} = \rho_{UU'} \circ \psi_U$, $\psi_{U''} = \rho_{U'U''} \circ \psi_{U'}$, and $\psi_{U''} = \rho_{UU''} \circ \psi_U$. Thus, we can write: $\rho_{UU''} \circ \psi_U = \psi_{U''} = \rho_{U'U''} \circ \psi_{U'} = (\rho_{U'U''} \circ \rho_{UU'}) \circ \psi_U$. Since $\rho_{UU''}$ is the unique homomorphism satisfying the property $\psi_{U''} = \rho_{UU''} \circ \psi_U$, we must have $\rho_{UU''} = \rho_{U'U''} \circ \rho_{UU'}$, as desired. □

(v) Proof. We show that $A_\mathfrak{p}$ is the direct limit $\varinjlim A(U)$, where U ranges over those basis open sets X_f such that $f \notin \mathfrak{p}$ (U ranges over neighborhoods of $\mathfrak{p}$). The directed system is formed by restrictions, i.e. if $U' \subset U$, then $\rho_{UU'} : A(U) \to A(U')$ defines the homomorphisms. These maps are compatible by (ii) and (iii), so $\varinjlim A(U)$ is well-defined.

For every basic open set X_f, $\{f^n\} \subset A - \mathfrak{p}$, so there is a canonical map $\beta_U : A(U)(= A_f) \to A_\mathfrak{p}$ that commutes with the canonical maps $A \to A_f$ and $A \to A_\mathfrak{p}$. By Exercise 2.21, we have a ring homomorphism $\psi : \varinjlim A(U) \to A_\mathfrak{p}$. For any $a \in \varinjlim A(U)$, there is some U and $a_U \in A(U)$ such that $\alpha_U(a_U) = a$. Then, $\psi(a) = \beta_U(a_U)$.

To show this is an isomorphism, we need to show injectivity and surjectivity. If $\psi(a) = 0$, then there is some U and $a_U \in A(U)$ such that $\alpha_U(a_U) = a$ and $\beta_U(a_U) = 0$. Since β_U commutes with the canonical maps $\phi_U : A \to A(U)$ and $\phi : A \to A_\mathfrak{p}$, there is some $b \in \ker(A \to A_f)$ such that $\phi_U(b) = a_U$. In particular, there is some $f \in A - \mathfrak{p}$ such that $fb = 0$. But then, if $U = X_g$, then $U' = U \cap X_f = X_{fg}$ (Exercise 1.17) is a subset of both X_f and U. Thus, the restriction maps give us a commutative square with $A \to A(U) \to A(X_{fg})$ and $A \to A(X_f) \to A(X_{fg})$ as the two paths. Since $A \to A(X_f)$ maps b to 0, b is zero in the image $A(X_{fg})$. Thus, $a = \alpha_U(a_U) = \alpha_{X_{fg}}(0) = 0$, so injectivity holds.

Further, any element of $A_\mathfrak{p}$ has the form a/f, where $f \in A - \mathfrak{p}$. Then, a/f is the image of a/f in $A_f \to A_\mathfrak{p}$. Thus, $\psi(\beta_{X_f}(a/f)) = a/f$, so surjectivity holds and thus $A_\mathfrak{p} \cong \varinjlim A(U)$. □

Remark. *The assignment of the ring $A(U)$ to each basic open set U of X, and the restriction homomorphisms ρ, satisfying (iii) and (iv), constitutes a "presheaf of rings" on the basis of open sets $\{X_f\}_{f \in A}$. (v) says that the "stalk" of this presheaf at $x(= \mathfrak{p}) \in X$ is the local ring $A_\mathfrak{p}$.*

3.24 Exercise

Proof. Let $\{X_{f_i}\}_{i\in I}$ be a covering of X by basic open sets. For each $i \in I$, suppose there is an assignment $i \mapsto s_i \in A(X_{f_i})$ such that, for each pair $i, j \in I$, the restrictions of s_i and s_j to $A(X_{f_i} \cap X_{f_j}) = A(X_{f_i f_j})$ (Exercise 1.17) are identical.

By Exercise 1.17, we have a finite subcover $\{X_{g_i}\}_{i=1}^n$. For each, we can write $s_i = a_i/g_i^{m_i} \equiv a_i g_i^n / g_i^{m_i+n}$. In particular, since there are finitely many g_i, we can use a common exponent to obtain $s_i = a_i/g_i^M$ for all i. For any i, j, we are given that in $A_{g_i g_j}$, we have

$$\frac{a_i}{g_i^M} = \frac{a_j}{g_j^M}.$$

Thus, there is some m such that

$$(g_i g_j)^m (a_i g_j^M - a_j g_i^M) = 0$$

so that $a_i' g_j^{m+M} = a_j' g_i^{m+M}$ in A. Since there are finitely many g_i, we can adjust coefficients to find a common denominator N such that $a_i g_j^N = a_j g_i^N$ for all i, j and such that $s_i \equiv a_i/g_i^N$.

Since X is covered by $\{X_{g_i}\}$, we know that $(g_1, \ldots, g_n) = (1)$. Otherwise, there is a maximal ideal that contains it, but then $\bigcup X_{g_i}$ would not contain this maximal ideal. In particular, for every maximal ideal, there is some g_i not contained in it so that g_i^N is not contained in it either. Thus, $(g_1^N, \ldots, g_n^N) = (1)$, as well. We can find $h_i \in A$ such that $\sum_i h_i g_i^N = 1$. Let $s = \sum h_i a_i$. Then,

$$s g_j^N = \sum h_i (a_i g_j^N) = \sum h_i a_j g_i^N = a_j \sum h_i g_i^N = a_j.$$

Thus, under the canonical map $s \mapsto s/1$ in any $A(X_{g_i})$, we have $s/1 = s g_i^N / g_i^N = a_i/g_i^N = s_i$, giving us our desired global section.

Let's show uniqueness. Let s, s' both satisfy the desired properties, then $s - s' \mapsto 0$ in each $A(X_{g_i})$. Then, $g_i^{m_i}(s - s') = 0$ for some m_i. Thus, there is some common M such that $g_i^M(s - s') = 0$ for all i. As before, $(g_1^M, \ldots, g_n^M) = (1)$, so we can choose k_i such that $1 = \sum k_i g_i^M$. Then,

$$s - s' = \sum k_i g_i^M (s - s') = \sum k_i \cdot 0 = 0,$$

as desired.

Let X_f be any other basic open set. Then, $\{X_f \cap X_{g_i}\} = \{X_{f g_i}\}$ cover X_f. Using the global section s for the finite covering and transivity of restriction (Exercise 3.23), we have

$$\rho_{X_f, X_{f g_i}}(s_f) = \rho_{X_{g_i}, X_{f g_i}}(s_i) = \rho_{X, X_{f g_i}}(s).$$

Adding X_f to our collection $\{X_{g_i}\}$, we can invoke the uniqueness already shown to see that $s_f = \rho_{X, X_f}(s)$. Hence, the global section s exists and is unique. □

3.25 Exercise

Proof. If $\mathfrak{p} \in X$ and $k = k(\mathfrak{p})$ is its residue field, then by Exercise 3.21,

$$h^{*-1}(\mathfrak{p}) \cong \operatorname{Spec}((B \otimes_A C) \otimes_A k) \cong \operatorname{Spec}((B \otimes_A k) \otimes_k (C \otimes_A k)).$$

Thus, $\mathfrak{p} \in h^*(T)$ if and only if $(B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0$. Since the tensor factors are vector spaces, $(B \otimes_A k) \otimes_k (C \otimes_A k) \neq 0$ if and only if $(B \otimes_A k) \neq 0$ and $(C \otimes_A k) \neq 0$. By Exercise 3.21 again, this happens exactly when $\mathfrak{p} \in f^*(Y)$ and $\mathfrak{p} \in g^*(Z)$. Therefore, $\mathfrak{p} \in h^*(T)$ if and only if $\mathfrak{p} \in f^*(Y) \cap g^*(Z)$, so $h^*(T) = f^*(Y) \cap g^*(Z)$. □

3.26 Exercise

Proof. Let $\mathfrak{p} \in \operatorname{Spec}(A)$, and $k := k(\mathfrak{p})$. Then,

$$B \otimes_A k \cong (\varinjlim B_\alpha) \otimes_A k \cong \varinjlim (B_\alpha \otimes_A k)$$

by Exercise 2.20. Thus, $\mathfrak{p} \in f^*(\operatorname{Spec}(B))$ if and only if $B \otimes_A k \neq 0$ (Exercise 3.21) if and only if $\varinjlim(B_\alpha \otimes_A k) \neq 0$.

Suppose $B_\alpha \otimes_A k = 0$ for some α. Then, $f_\alpha = 0$ so that for any $\beta \geq \alpha$, $f_\beta = g_{\alpha\beta} \circ f_\alpha = 0$. Thus, $B \otimes_A k = 0$ since the maps are ring homomorphisms, implying that $0 = 1$ in $B \otimes_A k$. Conversely, since $(B_\alpha \otimes_A k)$ is a direct limit of rings and thus a ring by Exercise 2.21, we also have by Exercise 2.21, that $B \otimes_A k = 0$ implies that $B_\alpha \otimes_A k = 0$ for some α. Thus, $\varinjlim(B_\alpha \otimes_A k) \neq 0$ if and only if $B_\alpha \otimes_A k \neq 0$ for all α. But this happens, by Exercise 3.21, if and only if $\mathfrak{p} \in f_\alpha^*(\operatorname{Spec}(B_\alpha))$ for all α. Connecting the chain, we have $f^*(\operatorname{Spec}(B)) = \bigcap_\alpha f_\alpha^*(\operatorname{Spec}(B_\alpha))$. □

3.27 Exercise

(i) Proof. By Exercise 2.23, the tensor product is the direct limit of tensor products over finite subsets of $\{B_\alpha\}$ with the ordering given by inclusion. If $\{I\}_{I \in \Lambda}$ are the finite subsets of indices, then by Exercise 3.26, we have

$$f^*(\operatorname{Spec}(B)) = \bigcap_{I \in \Lambda} f_I^* \left(\operatorname{Spec} \left(\bigotimes_{\alpha \in I} B_\alpha \right) \right),$$

where $f_I : A \to \bigotimes_{\alpha \in I} B_\alpha$ is just $\bigotimes_{\alpha \in I} f_\alpha$. By Exercise 2.25 and a basic inductive argument, we have that for any $I \in \Lambda$,

$$f_I^* \left(\operatorname{Spec} \left(\bigotimes_{\alpha \in I} B_\alpha \right) \right) = \bigcap_{\alpha \in I} f_\alpha^*(\operatorname{Spec}(B_\alpha)).$$

Putting both together, we have $f^*(\operatorname{Spec}(B)) = \bigcap_\alpha f_\alpha^*(\operatorname{Spec}(B_\alpha))$, as desired. □

(ii) Proof. By Exercise 1.22, we know that $\operatorname{Spec}(B) \cong \bigsqcup_\alpha \operatorname{Spec}(B_\alpha)$. The map $f = (f_\alpha)$ respects this decomposition so that $f^*(\operatorname{Spec}(B)) = \bigcup_\alpha f_\alpha^*(\operatorname{Spec}(B_\alpha))$. □

(iii) Proof. Note that the ring homomorphism $f : A \to 0$ induces $f^*(\operatorname{Spec}(0)) = \emptyset$. Further, $\mathrm{id} : A \to A$ induces a map $\mathrm{id}^*(\operatorname{Spec}(A)) = \operatorname{Spec}(A)$. Along with the results of (i) and (ii), we see that subsets of $\operatorname{Spec}(A)$ of the form $f^*(\operatorname{Spec}(B))$ for a ring homomorphism $f : A \to B$ satisfy the properties for closed sets of a topology.

Every closed set of the Zariski topology is of the form $V(\mathfrak{a})$ for some ideal $\mathfrak{a}$. By Exercise 3.19, we know that the natural ring homomorphism $\pi : A \to A/\mathfrak{a}$ satisfies $V(\mathfrak{a}) = \pi^*(\operatorname{Spec}(A/\mathfrak{a}))$. Thus, every closed set of the Zariski topology is closed in this topology, so this topology is finer. □

Remark. *We refer to this topology as the "constructible topology" on* $\operatorname{Spec}(A)$.

(iv) Proof. Showing that X_C is quasi-compact means showing that any open covering has a finite subcovering. Equivalently, we can show that any family of closed sets having empty intersection has a closed subfamily with empty intersection.

Let $\{f_\alpha^*(\mathrm{Spec}(B_\alpha))\}$ be a closed family such that $\bigcap_\alpha f_\alpha^*(\mathrm{Spec}(B_\alpha)) = \emptyset$. Then, if B is the tensor product of (B_α), by (i), we have

$$f^*(\mathrm{Spec}(B)) = \bigcap_\alpha f_\alpha^*(\mathrm{Spec}(B_\alpha)) = \emptyset,$$

Hence, $\mathrm{Spec}(B)$ is empty and $B = 0$. Since $B = \varinjlim_I \bigotimes_{\alpha\in I} B_\alpha$, we must have $\bigotimes_{\alpha\in I} B_\alpha = 0$ for some I since this is a direct limit of A-algebras (Exercise 2.21). By (i) again, we have

$$\emptyset = f_I^*\left(\mathrm{Spec}\left(\bigotimes_{\alpha\in I} B_\alpha\right)\right) = \bigcap_{\alpha\in I} f_\alpha^*(\mathrm{Spec}(B_\alpha)),$$

so X_C is quasi-compact. □

3.28 Exercise

(i) Proof. Since X_C has a finer topology than the Zariski topology on $X = \mathrm{Spec}(A)$ (Exercise 3.27), X_g is open in X_C since it is open in the Zariski topology. Conversely, consider the canonical map $f : A \to A_g$. By Proposition 3.11, the primes of A_g are in one-to-one correspondence with the primes of A that do not meet $\{g^n\}$. If $\mathfrak{p} \in \mathrm{Spec}(A)$ intersects $\{g^n\}$, then it must contain g (otherwise, choose $n > 1$ minimal such that $g^n \in \mathfrak{p}$, then invoke primality on $g^n = g \cdot g^{n-1}$). Thus, $\mathfrak{p} \in f^*(\mathrm{Spec}(A_g))$ if and only if $g \notin \mathfrak{p}$. Thus, $f^*(\mathrm{Spec}(A_g)) = V(g)^c = X_g$, so X_g is also closed. □

Remark. *We've shown that for any $f \in A$,* $\mathrm{Spec}(A)$ *with the constructible topology is the disjoint union of* $\mathrm{Spec}(A/(f))$ *and* $\mathrm{Spec}(A_f)$*.*

(ii) Proof. Let C' be the coarsest topology on $X = \mathrm{Spec}(A)$ such that each X_g for $g \in A$ is both open and closed. Let $\mathfrak{p}$ and $\mathfrak{q}$ be distinct primes in X. Since they are not equal, we can assume that one contains an element that the other does not. Without loss of generality, suppose $f \in \mathfrak{p}$ and $f \notin \mathfrak{q}$. Then, $\mathfrak{q} \in X_f$ and $\mathfrak{p} \in X_f^c$. Since these are disjoint open sets, $X_{C'}$ is Hausdorff. □

(iii) Proof. Consider the identity map $\mathrm{id} : X_C \to X_{C'}$; clearly, this is a bijection. Since $X_{C'}$ is the coarsest topology such that X_f is open and closed for all $f \in A$ by (ii), and since X_C satisfies this property by (i), X_C is finer than $X_{C'}$. Therefore, any open set of $X_{C'}$ is open in X_C, so id is continuous.

For the converse, note that X_C is compact, as it is both Hausdorff (from (ii) and (i)) and quasi-compact (Exercise 3.27). Now, let K be an arbitrary closed set of X_C. As K is closed and X_C is compact, K is compact. Since id is continuous, $\mathrm{id}(K)$ is compact. As a compact subset of a Hausdorff space (from (ii)), $\mathrm{id}(K)$ is closed. Therefore, id is a closed mapping and finally a homeomorphism. □

(iv) Proof. By (i) and (ii), X_C is Hausdorff. Further, X_C is quasi-compact by Exercise 3.27, so X_C is compact. To show that X_C is totally disconnected, let S be some subset of X_C. Suppose we have distinct elements $\mathfrak{p}, \mathfrak{q} \in S$. We showed in (i) that there exists some $f \in A$ such that $\mathfrak{p} \in X_f$ and $\mathfrak{q} \in X_f^c$ (switching $\mathfrak{p}$ and $\mathfrak{q}$ if necessary). Since both X_f and X_f^c are open by (i), $S \cap X_f$ and $S \cap X_f^c$ are open (in the subspace topology), disjoint, non-empty, and their union gives S. Thus, any subset with more than one element is disconnected, as desired. □

3.29 Exercise

Proof. By Exercise 3.28, $\mathrm{Spec}(A)_C$ is generated as a topology by letting the sets of the form X_g and $V(g)$ for all $g \in A$ be open. By Exercise 1.21, $f^{*-1}(X_g) = Y_{f(g)}$ and $f^{*-1}(V(g)) = V(f(g))$. In particular, these generating open sets are mapped under f^{*-1} to the generating open sets of $\mathrm{Spec}(B)_C$. Thus, f^* is continuous.

Now, note that an arbitrary closed set of $\mathrm{Spec}(B)_C$ has the form $h^*(\mathrm{Spec}(R))$ for some ring homeomorphism $h : B \to R$. Note that $(h \circ f) : A \to R$ is also a ring homomorphism, so $(h \circ f)^*(\mathrm{Spec}(R))$ is closed in $\mathrm{Spec}(A)_C$. By Exercise 1.21, $f^*(h^*(\mathrm{Spec}(R))) = (h \circ f)^*(\mathrm{Spec}(R))$, so f^* is a closed mapping. □

3.30 Exercise

Proof. If the Zariski topology and constructible topology are equivalent on $\mathrm{Spec}(A)$, then since $\mathrm{Spec}(A)_C$ is Hausdorff (Exercise 3.28), $\mathrm{Spec}(A)$ must be Hausdorff in the Zariski topology, so by Exercise 3.11, $A/\mathfrak{N}$ is absolutely flat.

Conversely, if $A/\mathfrak{N}$ is absolutely flat, then by Exercise 3.11, $\mathrm{Spec}(A)$ is Hausdorff and compact in the Zariski topology. The identity mapping $\mathrm{id} : \mathrm{Spec}(A)_C \to \mathrm{Spec}(A)$ is bijective and continuous since $\mathrm{Spec}(A)_C$ is finer than $\mathrm{Spec}(A)$ by Exercise 3.27. Now, let K be a closed set of $\mathrm{Spec}(A)_C$. Because K is closed and $\mathrm{Spec}(A)_C$ is compact (Exericse 3.28), K is compact. Then, since id is continuous, $\mathrm{id}(K)$ is also compact. This implies that $\mathrm{id}(K)$ is closed since $\mathrm{Spec}(A)$ is Hausdorff. Finally, id is a homeomorphism, as desired. □

4 Primary Decomposition (Done)

4.1 Exercise

Proof. Since $\mathfrak{a}$ is decomposable, we have by Proposition 4.6 that the minimal elements in the set of all prime ideals containing $\mathfrak{a}$, or equivalently the minimal prime ideals of $A/\mathfrak{a}$ (Proposition 1.1), are exactly the minimal prime ideals belonging to $\mathfrak{a}$. Again, since $\mathfrak{a}$ is decomposable, there are only finitely many prime ideals belonging to $\mathfrak{a}$, so $A/\mathfrak{a}$ has finitely many minimal prime ideals. By Exercise 1.20, the irreducible components of $\operatorname{Spec}(A/\mathfrak{a})$ are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime of $A/\mathfrak{a}$. Therefore, $\operatorname{Spec}(A/\mathfrak{a})$ has finitely many irreducible components. □

4.2 Exercise

Proof. Since $\mathfrak{a} = r(\mathfrak{a})$, $\mathfrak{a}$ is the intersection of the prime ideals containing $\mathfrak{a}$. By Proposition 4.6, each prime containing $\mathfrak{a}$ contains a minimal prime also containing $\mathfrak{a}$. Then, replacing the intersection with the intersection over only minimal primes yields the same intersection. Since $\mathfrak{a}$ is decomposable, we invoke Exercise 4.1 to see that there are finitely many such primes, so the intersection of minimal prime ideals containing $\mathfrak{a}$ is a primary decomposition. Since $r(\mathfrak{p}_i) = \mathfrak{p}_i$ for all primes, the primes belonging to $\mathfrak{a}$ must all be minimal, so $\mathfrak{a}$ has no embedded prime ideals. □

4.3 Exercise

Proof. Let $\mathfrak{q}$ be primary so that, by Proposition 4.1, $r(\mathfrak{q})$ is prime. Now, suppose $x \in r(\mathfrak{q})$ so that $x^n \in \mathfrak{q}$ for some $n > 0$. Since A is absolutely flat, we know by Exercise 2.27 that $(x) = (x^n)$, so $x \in (x^n) \subset \mathfrak{q}$. Therefore, $\mathfrak{q} = r(\mathfrak{q})$ so that $\mathfrak{q}$ is a prime ideal. By Exercise 3.11, every prime ideal of A is maximal. Hence, $\mathfrak{q}$ is maximal. □

4.4 Exercise

If I is an ideal properly containing $(2,t)$ in $\mathbb{Z}[t]$, then it has an element $f \notin (2,t)$. But $f = a + g$, where $a \in \mathbb{Z}$ and $g(0) = 0$. Then, $g \in (2,t)$, so $a \in I$. By assumption, $a \notin (2)$ (otherwise $f = a + g \in (2,t)$). Thus, $a = 2n + 1$, where $n \in \mathbb{Z}$. But, $2n \in (2,t) \subset I$, so $a - 2n = 1 \in I$ and thus $I = (1)$. In particular, $(2,t)$ is maximal.

Note that $(2,t)^2 = (4, 2t, t^2) \subset (4,t)$, so $(2,t) \subset r((4,t))$. Since $r((4,t))$ is an ideal and $(2,t)$ is maximal, $r((4,t))$ is either (1) or $(2,t)$. However, $r(\mathfrak{a}) = (1) \iff \mathfrak{a} = 1$ by the intra-text Exercise 1.13, so since $(4,t) \neq (1)$, $r((4,t)) = (2,t)$. Thus, $(4,t)$ is $(2,t)$-primary by Proposition 4.2.

However, we showed that $(4, 2t, t^2) = (2,t)^2 \subset (4,t) \subset (2,t)$. Since $t \in (4,t) - (4, 2t, t^2)$ and $2 \in (2,t) - (4,t)$, the inclusions are proper. Thus, $(4,t)$ is primary with respect to a maximal ideal, and yet, is not a power of the maximal ideal.

4.5 Exercise

$K[x,y,z]/(x,y) \cong K[z]$ is an integral domain; if f,g are non-zero, $\deg(f)$ and $\deg(g)$ are finite, so $\deg(fg) = \deg(f)\deg(g)$ is finite and thus fg is non-zero. Therefore, (x,y) is prime. By the same argument, (x,z) is prime. (x,y,z) is maximal since $K[x,y,z]/(x,y,z) \cong K$, a field.

Let $\mathfrak{a} = (x,y)(x,z) = (x^2, xy, xz, yz)$. Note that $(x,y,z)^2 = (x^2, xy, xz, y^2, yz, z^2)$ and also $(x,y) \cap (x,z) = (x, xy, xz, yz) = (x, yz)$ so that

$$(x,y) \cap (x,z) \cap (x,y,z)^2 = (x,yz) \cap (x^2, xy, xz, y^2, yz, z^2) = (x^2, xy, xz, yz) = \mathfrak{a}.$$

Since (x,y) and (x,z) are prime, they are primary; also, since (x,y,z) is maximal, $(x,y,z)^2$ is primary, so we have a primary decomposition of $\mathfrak{a}$. Clearly, $r((x,y)) = (x,y)$, $r((x,z)) = (x,z)$, and since $r((x,y,z)^2)$ contains the maximal ideal (x,y,z) and is not the whole ring, we must have $r((x,y,z)^2) = (x,y,z)$. Thus, the associated primes of the decomposition are distinct. Moreover,

$$(x,z) \not\subset (x,y,z) \cap (x,y) = (x,y), \quad (x,y) \not\subset (x,y,z) \cap (x,z) = (x,z),$$

$$\text{and } (x,y,z) \not\subset (x,y) \cap (x,z) = (x, xy, xz, yz),$$

so the primary decomposition is reduced.

The prime ideals belonging to $\mathfrak{a}$ are thus (x,y,z), (x,y), and (x,z). Since (x,y) and (x,z) each do not contain any of the other primes, they are isolated. Since $(x,y) \subset (x,y,z)$, (x,y,z) is embedded.

4.6 Exercise

We show that if X is an infinite, compact Hausdorff space, then in the ring of real-valued continuous functions on X, $C(X)$, 0 is not decomposable.

Proof. We proceed by contradiction. Let

$$0 = \bigcap_{i=1}^{n} \mathfrak{q}_i$$

be a reduced, primary decomposition of 0. By Exercise 1.26, $\operatorname{Max}(C(X)) = \{\mathfrak{m}_x : x \in X\}$, where $\mathfrak{m}_x$ is the maximal ideal of functions $f \in C(X)$ satisfying $f(x) = 0$.

Suppose there are two distinct maximal ideals $\mathfrak{m}_x$, $\mathfrak{m}_y$ containing $\mathfrak{q}_i$. Since X is Hausdorff and compact, it is normal, so we can make use of Urysohn's lemma. X is Hausdorff, so the sets $\{x\}$ and $\{y\}$ are closed. Further, they have disjoint open sets $x \in U_x$ and $y \in U_y$. By Urysohn's lemma, there is a real-valued continuous function f_x such that $f_x(x) = 1$ and $f_x(U_x^c) = 0$. We define f_y symmetrically. Then, since $U_x \cap U_y = \emptyset$, $U_x^c \cup U_y^c = X$ so that $f_x f_y = 0$. Note that $f_x \in \mathfrak{m}_y - \mathfrak{m}_x$ and $f_y \in \mathfrak{m}_x - \mathfrak{m}_y$, so neither f_x nor f_y is in $\mathfrak{q}_i$ since we assume that $\mathfrak{q}_i \subset \mathfrak{m}_x \cap \mathfrak{m}_y$. However, note that $f_x f_y = 0 \in \mathfrak{q}_i$. Since $f_x \notin \mathfrak{q}_i$, we must have $f_y^n \in \mathfrak{q}_i \subset \mathfrak{m}_y$ for some $n > 0$ since $\mathfrak{q}_i$ is primary. But since $f_y(y) = 1$, $f_y^n(y) = 1$, so $f_y^n \notin \mathfrak{m}_y$ and thus not in $\mathfrak{q}_i$, a contradiction. Thus, for all $\mathfrak{q}_i$ in the decomposition, $\mathfrak{q}_i$ has a unique maximal ideal $\mathfrak{m}_{x_i}$ containing it for some $x_i \in X$.

Let $\{x_i\}$ be the points defining these finitely many maximal ideals. Since X is infinite, by assumption, there is some $y \in X - \{x_i\}$. By the previous paragraph, $\mathfrak{m}_y$ does not contain any of the $\mathfrak{q}_i$. Thus, for each i, there is some $f_i \in \mathfrak{q}_i$ such that $f_i(y) \neq 0$. Note that the finite product $f_1 \cdots f_n \in \bigcap_{i=1}^n \mathfrak{q}_i$, but $(f_1 \cdots f_n)(y) \neq 0$, violating our assumption that $0 = \bigcap_{i=1}^n \mathfrak{q}_i$. □

4.7 Exercise

(i) Proof. Consider the injective ring homomorphism $A \to A[x]$, $a \mapsto a$. Let $\mathfrak{a}$ be an ideal of A. Note that $\mathfrak{a}[x]$ is additively generated by elements of the form ax^n, where $n \geq 0$ and $a \in \mathfrak{a}$. Since

$x^n \in A[x]$ and $a \in \mathfrak{a}$, $ax^n \in \mathfrak{a}A[x]$, so $\mathfrak{a}[x] \subset \mathfrak{a}A[x]$. Conversely, if $f = f_0 + f_1x^1 + \cdots + f_nx^n \in A[x]$, then for any $a \in \mathfrak{a}$ and any $f_i \in A$, $af_i \in \mathfrak{a}$, so $af = (af_0) + (af_1)x^1 + \cdots + (af_n)x^n \in \mathfrak{a}[x]$; finally, $\mathfrak{a}^e = \mathfrak{a}A[x] = \mathfrak{a}[x]$. □

(ii) Proof. Consider the ring homomorphism $A[x] \to (A/\mathfrak{p})[x]$. The kernel of this map is clearly $\mathfrak{p}[x]$. Suppose $f, g \in A[x] - \mathfrak{p}[x]$. Then, $\overline{f}$ and $\overline{g}$ are non-zero, so they have highest degree terms, ax^m and bx^n. Then, $\overline{f} \cdot \overline{g}$ has highest degree term $(ab)x^{m+n}$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain, so $ab \neq 0$ and thus $\overline{fg} = \overline{f} \cdot \overline{g} \neq 0$. In particular, $fg \notin \mathfrak{p}[x]$, so $\mathfrak{p}[x]$ is a prime ideal. □

(iii) Proof. Suppose $\mathfrak{q}$ is primary. Then, $A/\mathfrak{q} \neq 0$ and all its zero divisors are nilpotent. Now, consider $A[x]/\mathfrak{q}[x] \cong (A/\mathfrak{q})[x]$. Since $A/\mathfrak{q} \neq 0$, $(A/\mathfrak{q})[x] \neq 0$. Now, suppose $f = \sum f_ix^i \in (A/\mathfrak{q})[x]$ is a zero-divisor. By Exercise 1.2, there is some non-zero $a \in A/\mathfrak{q}$ such that $0 = af = \sum(af_i)x^i$. Thus, each f_i is a zero divisor of $A/\mathfrak{q}$ and thus nilpotent. Again by Exercise 1.2, f is itself nilpotent, so $\mathfrak{q}[x]$ is primary.

Now, we just have to show that $r(\mathfrak{q}[x]) = \mathfrak{p}[x]$. We start by showing $\mathfrak{p}[x] \subset r(\mathfrak{q}[x])$. Since $r(\mathfrak{q}[x])$ is additively closed, it is enough to show that the additive generators ax^n of $\mathfrak{p}[x]$ (i.e., $a \in \mathfrak{p}$) are in $r(\mathfrak{q}[x])$. Since $\mathfrak{q}$ is $\mathfrak{p}$-primary, $a \in r(\mathfrak{q})$. Thus, there is some $m > 0$ such that $a^m \in \mathfrak{q}$. Thus, $(ax^n)^m = (a^m)x^{nm} \in \mathfrak{q}[x]$, as desired. Thus, $\mathfrak{p}[x] \subset r(\mathfrak{q}[x])$. Conversely, since $\mathfrak{q} \subset \mathfrak{p}$, $\mathfrak{q}[x] \subset \mathfrak{p}[x]$, so $r(\mathfrak{q}[x]) \subset r(\mathfrak{p}[x]) = \mathfrak{p}[x]$ since $\mathfrak{p}[x]$ is prime by (ii). Hence, we have equality: $\mathfrak{p}[x] = r(\mathfrak{q}[x])$. □

(iv) Proof. Suppose $\bigcap_i \mathfrak{q}_i$ is a minimal primary decomposition of $\mathfrak{a}$. Then clearly,

$$\bigcap_i (\mathfrak{q}_i[x]) = \mathfrak{a}[x];$$

$f \in \mathfrak{a}[x] \iff f_j \in \mathfrak{a}$ for all $j \iff f_j \in \mathfrak{q}_i$ for all $i, j \iff f \in \mathfrak{q}_i[x]$ for all $i \iff f \in \bigcap \mathfrak{q}_i[x]$.

By (iii), $\mathfrak{q}_i[x]$ is primary for each i, so we have a primary decomposition. By (iii), if $r(\mathfrak{q}_i) = \mathfrak{p}_i$, then $r(\mathfrak{q}_i[x]) = \mathfrak{p}_i[x]$. Since the $\mathfrak{p}_i$ are distinct, the $\mathfrak{p}_i[x]$ are distinct. Also, since for all i

$$\bigcap_{i \neq j} \mathfrak{q}_j \not\subset \mathfrak{q}_i,$$

we have

$$\bigcap_{i \neq j} (\mathfrak{q}_j[x]) \not\subset \mathfrak{q}_i[x];$$

simply choose some $a \in \mathfrak{q}_i - \bigcap_{i \neq j} \mathfrak{q}_j$ and consider a as a constant polynomial. Thus, $\bigcap_i (\mathfrak{q}_i[x])$ is a minimal primary decomposition of $\mathfrak{a}[x]$. □

(v) Proof. By (iv), if $\{\mathfrak{p}_i\}$ are the primes belonging to $\mathfrak{a}$, $\{\mathfrak{p}_i[x]\}$ are the primes belonging to $\mathfrak{a}[x]$. Further, if $\mathfrak{p}_i \subset \mathfrak{p}_j$, then $\mathfrak{p}_i[x] \subset \mathfrak{p}_j[x]$. In particular, if $\mathfrak{p}$ is a minimal prime belonging to $\mathfrak{a}$, then $\mathfrak{p}[x]$ is a minimal prime belonging to $\mathfrak{a}[x]$. □

4.8 Exercise

Proof. By Exercise 1.2, if $f \in k[x_1, \ldots, x_n]$ is a zero-divisor, then there is some non-zero $a \in k$ such that $af = 0$. Then, $f = a^{-1}af = 0$, so $k[x_1, \ldots, x_n]$ is an integral domain for all $n > 0$. Consider the ring homomorphism $k[x_1, \ldots, x_n] \to k[x_{i+1}, \ldots, x_n]$ that sends $x_j \mapsto 0$ when $1 \leq j \leq i$. Clearly, this map is surjective, and the kernel is the ideal $(x_1, \ldots, x_i)$. Thus,

$$k[x_1, \ldots, x_n]/(x_1, \ldots, x_i) \cong k[x_{i+1}, \ldots, x_n].$$

But as before, $k[x_{i+1}, \ldots, x_n]$ is an integral domain, so $(x_1, \ldots, x_i)$ is a prime ideal.

Now, consider the power $(x_1)^N$ of $k[x_1, \ldots, x_n]$. Let $f \in k[x_1, \ldots, x_n]$ represent a zero-divisor of $k[x_1, \ldots, x_n]/(x_1)^N$. I.e., there is some $g \in k[x_1, \ldots, x_n]$ non-zero in the quotient such that

$$fg \in (x_1)^N.$$

There is some maximal m such that $g = x_1^m h$, where $h \in k[x_1, \ldots, x_n]$. Since g is non-zero in the quotient, $m < N$. Then, we have

$$fhx_1^m = \ell x_1^N,$$

where $\ell \in k[x_1, \ldots, x_n]$. Then,

$$fh = \ell x_1^{N-m}.$$

Consider again the ring homomorphism $k[x_1, \ldots, x_n] \to k[x_2, \ldots, x_n]$ that sends $x_1 \mapsto 0$. Since we chose m minimal, $h(0, x_2, \ldots, x_n)$ is not identically zero. Otherwise, we could find some h' such that $h = x_1 h'$, violating the minimality of m. Then, $\overline{f} \cdot \overline{h} = \ell \cdot 0 = 0$. Since $k[x_2, \ldots, x_n]$ is an integral domain, $\overline{f} = 0$ so that $f \in (x_1)$ and $f^N \in (x_1)^N$. Thus, $(x_1)^N$ is primary in $k[x_1, \ldots, x_n]$ for all $n > 0$, $N > 0$.

We wish to show now that $(x_1, \ldots, x_i)$ is primary in $k[x_1, \ldots, x_n]$ for all $1 \leq i \leq n$. Since the result of the previous paragraph holds for any $n > 0$, (x_1) is primary in $k[x_1, x_{i+1}, \ldots, x_n]$. For the inductive step, assume $(x_1, \ldots, x_{j-1})$ is primary in $k[x_1, \ldots, x_{j-1}, x_{i+1}, \ldots, x_n]$ for $j \leq i$. Then, by Exercise 4.7, $(x_1, \ldots, x_{j-1})[x_j] = (x_1, \ldots, x_j)$ is primary in $k[x_1, \ldots, x_{j-1}, x_{i+1}, \ldots, x_n][x_j] \cong k[x_1, \ldots, x_j, x_{i+1}, \ldots, x_n]$. By induction, $(x_1, \ldots, x_i)$ is primary in $k[x_1, \ldots, x_i, x_{i+1}, \ldots, x_n] = k[x_1, \ldots, x_n]$, as desired. □

4.9 Exercise

Proof. If $x \in A$ is a zero divisor, there is some non-zero $y \in A$ such that $xy = 0$. Then, $x \in (0 : y)$. Since $y \neq 0$, $1 \notin (0 : y)$, so there are primes containing $(0 : y)$. Now, consider $A \to A/(0 : y)$. By Proposition 1.1, the primes of $A/(0 : y)$ are in a one-to-one correspondence with primes of A containing $(0 : y)$, and the map is order-preserving. By Exercise 1.8, $A/(0 : y)$ has minimal ideals, so there is a minimal prime ideal $\mathfrak{p}$ of A containing $(0 : y)$. By definition, $\mathfrak{p} \in D(A)$, so $x \in (0 : y) \subset \mathfrak{p}$.

Conversely, suppose $x \in \mathfrak{p}$ for some $\mathfrak{p} \in D(A)$. Then, there is some $y \in A$ such that $\mathfrak{p}$ is minimal in the set of prime ideals containing $(0 : y)$. So, consider $\pi : A \to A/(0 : y)$. By Proposition 1.1, $\pi(\mathfrak{p})$ is a minimal prime of $A/(0 : y)$. Then, by Exercise 3.6, $S = A/(0 : y) - \pi(\mathfrak{p})$ is maximal in the set of multiplicatively closed sets of $A/(0 : y)$ not containing 0. Since $x \in \mathfrak{p}$, $\pi(x) \notin S$, so let S' be the multiplicatively closed set generated by S and $\pi(x)$. Then clearly, $S \subset S'$, so S' must contain 0. Thus, there is some $s \in S$ and some minimal $n > 0$ such that $s\pi(x^n) = 0$. Since $s \in A/(0 : y)$ is non-zero, there is some non-zero $z \in A$ such that $\pi(s) = z$, so $0 = \pi(z)\pi(x^n) = \pi(zx^n)$, so $zx^n \in (0 : y)$. Then, $zyx^n = 0$, so $x(zyx^{n-1}) = 0$. If $zyx^{n-1} = 0$, this violates the minimality of n, so x is a zero-divisor. □

Proof. Suppose $\mathfrak{p} \in D(A) \cap f^* \operatorname{Spec}(S^{-1}A)$. Then, $\mathfrak{p} \cap S = \emptyset$ by Proposition 3.11 and $\mathfrak{p}$ is a minimal prime containing some $(0 : a)$ for some $a \in A$. Since (a) is clearly finitely generated, we have by Proposition 3.15 that $S^{-1}(0 : a) = (S^{-1}0 : S^{-1}a) = (0 : a/1)$. Since S^{-1} preserves ordering and $S^{-1}\mathfrak{p}$ is prime since $\mathfrak{p} \cap S = \emptyset$ (Proposition 3.11), $S^{-1}\mathfrak{p}$ is a minimal prime containing $(0 : a/1)$. Thus, $\mathfrak{p} \in f^* D(S^{-1}A)$.

Conversely, if $S^{-1}\mathfrak{p} \in D(S^{-1}A)$ (all primes of $S^{-1}A$ have this form via Proposition 3.11), then clearly, $S^{-1}\mathfrak{p} \in \operatorname{Spec}(S^{-1}A)$ so that $\mathfrak{p} \in f^* \operatorname{Spec}(S^{-1}A)$. Further, $S^{-1}\mathfrak{p}$ is a minimal prime containing $(0 : a/s)$ for some $a/s \in S^{-1}A$.

We split the remainder into two cases. The first assumes $a \in S$. If $b \in (0 : a/s)$, we have $0 = (s/a) \cdot (b \cdot a/s) = b \cdot 1 = b$, so $(0 : a/s) = 0$. Since $S^{-1}\mathfrak{p}$ is the minimal prime containing $(0 : a/s) = 0$, it must be the nilradical of $S^{-1}A$; since all primes contain 0, all primes contain $S^{-1}\mathfrak{p}$, so since $S^{-1}\mathfrak{p}$ is prime, it is also the intersection of all primes, the nilradical. Thus, $\mathfrak{p}$ is the nilradical of A by Corollary 3.12. Then, since $\mathfrak{p}$ is prime, it is the minimal prime containing $0 = (0 : 1)$.

Finally, we have the case $a \notin S$. Let $b \in (0 : a/s)$. Then, $b \cdot a/s = 0$, so $0 = s \cdot (b \cdot a/s) = b \cdot a/1$; i.e., $(0 : a/s) \subset (0 : a/1)$. Conversely, if $b \cdot a/1 = 0$, then $0 = (1/s) \cdot b \cdot a/1 = b \cdot a/s$, so $(0 : a/s) = (0 : a/1)$. Thus, as before, we have $S^{-1}(0 : a) = (S^{-1}0 : S^{-1}a) = (0 : a/1) = (0 : a/s)$ (Proposition 3.15), so $S^{-1}\mathfrak{p}$ is a minimal prime containing $S^{-1}(0 : a)$. Since $\mathfrak{p} \mapsto S^{-1}\mathfrak{p}$ is a one-to-one, order-preserving relationship between primes of A not containing S and primes of $S^{-1}A$ (Proposition 3.11), $\mathfrak{p}$ is a minimal prime containing $(0 : a)$. Finally, $f^*D(S^{-1}A) = D(A) \cap f^* \operatorname{Spec}(S^{-1}A)$. □

Proof. Suppose 0 is decomposable. By Theorem 4.5, Ass(0) is precisely the set of prime ideals of the form $r(0 : x)$ for some $x \in A$.

By definition, $\mathfrak{p} \in D(A)$ when there is some $x \in A$ such that $\mathfrak{p}$ is minimal among the primes containing $(0 : x)$. Since $r(0 : x)$ is the intersection of prime ideals containing $(0 : x)$, $\mathfrak{p} = r(0 : x)$. Thus, $D(A) \subset \operatorname{Ass}(0)$. Conversely, if $\mathfrak{p} \in \operatorname{Ass}(0)$, then $\mathfrak{p} = r(0 : x)$ for some $x \in A$. $r(0 : x)$ is the intersection of prime ideals containing $(0 : x)$, so since $(0 : x) \subset r(0 : x) = \mathfrak{p}$, $\mathfrak{p}$ is minimal among the primes containing $(0 : x)$. Finally, $\operatorname{Ass}(0) = D(A)$. □

4.10 Exercise

(i) Proof. Let $a \in S_{\mathfrak{p}}(0)$. Then, $0 \equiv a/1 \in A_{\mathfrak{p}}$, so there is some $s \in A - \mathfrak{p}$ such that $sa = 0$. Since $0 \in \mathfrak{p}$ and $\mathfrak{p}$ is prime, we must have $s \in \mathfrak{p}$ or $a \in \mathfrak{p}$. But, $s \in A - \mathfrak{p}$, so we must have $a \in \mathfrak{p}$. □

(ii) Proof. Here, we use Exercise 2.6: Let Σ be the set of multiplicatively closed sets of A not containing 0. $S \in \Sigma$ is maximal if and only if $A - S$ is a minimal prime.

If $\mathfrak{p}$ is a minimal prime, then $S = A - \mathfrak{p}$ is a maximal mutliplicatively closed set. Let $x \in \mathfrak{p}$. Then, $\{x\} \cup S$ generates a multiplicatively closed set strictly bigger than S, so by maximality, this set contains 0. Therefore, there is some $n > 0$ and $s \in A - \mathfrak{p}$ such that $sx^n = 0$. But then, $x^n/1 = 0$ in $A_{\mathfrak{p}}$, so $x \in S_{\mathfrak{p}}(0)$ and thus $\mathfrak{p} \subset r(S_{\mathfrak{p}})(0)$. By (i), $S_{\mathfrak{p}}(0) \subset \mathfrak{p}$, so $r(S_{\mathfrak{p}}) \subset r(\mathfrak{p}) = \mathfrak{p}$, so actually, $\mathfrak{p} = r(S_{\mathfrak{p}})(0)$.

Conversely, suppose $\mathfrak{p} = r(S_{\mathfrak{p}})(0)$. Then, for any $x \in \mathfrak{p}$, there is some $n > 0$ such that $x^n/1 = 0$ in $A_{\mathfrak{p}}$. Thus, $A - \mathfrak{p}$ is maximal in the set of multiplicatively closed sets not containing 0, so $\mathfrak{p}$ is a minimal prime. □

(iii) Proof. Suppose $\mathfrak{p} \subset \mathfrak{q}$, where $\mathfrak{p}$ and $\mathfrak{q}$ are both prime ideals. Then, $A - \mathfrak{q} \subset A - \mathfrak{p}$ so that the canonical homomorphism $A \to A_{\mathfrak{p}}$ factors through the canonical maps $A \to A_{\mathfrak{q}} \to (A - \mathfrak{p})^{-1}A_{\mathfrak{q}} \cong A_{\mathfrak{p}}$ (Exercise 3.3). Then clearly, $\ker(A \to A_{\mathfrak{q}}) \subset \ker(A \to A_{\mathfrak{p}})$, so $S_{\mathfrak{q}}(0) \subset S_{\mathfrak{p}}(0)$. □

(iv) Proof. Suppose $x \in \bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0)$; suppose to the contrary that $x \neq 0$ so that $(0 : x) \neq A$. Then, there is some maximal ideal $\mathfrak{m} \in A$ such that $(0 : x) \subset \mathfrak{m}$. The ring $A/(0 : x)$ has minimal prime ideals by Exercise 1.8. Then, there is some minimal prime $\mathfrak{p}$ over $(0 : x)$ since by Proposition 1.1, the primes of $A/(0 : x)$ are in an order-preserving, one-to-one correspondence with primes of

A containing $(0:x)$. In particular, $\mathfrak{p} \in D(A)$. By assumption, $x \in S_{\mathfrak{p}}(0)$ (since $x \in S_{\mathfrak{q}}(0)$ for all $\mathfrak{q} \in D(A)$), so there is some $y \in A - \mathfrak{p}$ such that $xy = 0$. However, this would imply that $y \in (0:x) \subset \mathfrak{p}$, contradicting the fact that $y \in A - \mathfrak{p}$. Hence, $x = 0$ and $\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = 0$. □

4.11 Exercise

Proof. First, we show that $S_{\mathfrak{p}}(0)$ is a $\mathfrak{p}$-primary ideal. Suppose $ab \in S_{\mathfrak{p}}(0)$ and $a \notin S_{\mathfrak{p}}(0)$. By Exercise 4.10, $S_{\mathfrak{p}}(0) \subset \mathfrak{p}$, so $a \in A - \mathfrak{p}$. By definition, $(ab)/1 = 0$ in $A_{\mathfrak{p}}$, so there is some $s \in A - \mathfrak{p}$ such that $sab = 0$. Since a and s are both outside of $\mathfrak{p}$, $sa \notin \mathfrak{p}$ since $\mathfrak{p}$ is prime. Then, if we project the relation $(sa)b = 0$ onto $A/\mathfrak{p}$, then $\overline{sa} \neq 0$. Since $A/\mathfrak{p}$ is an integral domain, $b \in \mathfrak{p}$. We invoke Exercise 4.10 and the fact that $\mathfrak{p}$ is a minimal prime to see that $\mathfrak{p} = r(S_{\mathfrak{p}}(0))$, so $b^n \in S_{\mathfrak{p}}(0)$ for some $n > 0$. Then, since the radical of $S_{\mathfrak{p}}(0)$ is $\mathfrak{p}$, it is $\mathfrak{p}$-primary.

Now, suppose $\mathfrak{q}$ is a $\mathfrak{p}$-primary ideal. By Proposition 4.8, $\mathfrak{q}^{ec} = \mathfrak{q}$ with regard to the homomorphism $A \to A_{\mathfrak{p}}$. Clearly, $0 \subset \mathfrak{q}^e$, so $S_{\mathfrak{p}}(0) = 0^c \subset \mathfrak{q}^{ec} = \mathfrak{q}$. Thus, $S_{\mathfrak{p}}(0)$ is the smallest $\mathfrak{p}$-primary ideal. □

Proof. By Exercise 4.10, if $S_{\mathfrak{p}}(0) \subset \mathfrak{p}$ for all primes $\mathfrak{p}$. Thus, if $\mathfrak{a}$ is the intersection of $S_{\mathfrak{p}}(0)$ as $\mathfrak{p}$ ranges over all minimal primes, then

$$\mathfrak{a} \subset \bigcap_{\mathfrak{p} \text{ minimal}} \mathfrak{p} = \bigcap_{\mathfrak{p} \in \operatorname{Spec}(A)} \mathfrak{p} = \mathfrak{N},$$

as desired. The first equality is justified by Exercise 1.8: every prime contains a minimal prime. □

Proof. Since we're assuming 0 has a primary decomposition, we have $D(A) = \operatorname{Ass}(0)$ by Exercise 4.9. In particular, since 0 is decomposable, $D(A)$ is finite. By Exercise 4.10,

$$0 = \bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0).$$

Now, suppose $\bigcap_{\mathfrak{p} \text{ minimal}} S_{\mathfrak{p}}(0) = 0$. By Exercise 4.10, $D(A)$ is the set of prime ideals minimal over $(0:x)$ for some $x \in A$. Clearly, the minimal primes of A are in this set since they are minimal primes containing $(0:1) = 0$. In particular, there are finitely many minimal prime ideals since $D(A)$ is a finite set here. Thus,

$$0 = \bigcap_{\mathfrak{p} \text{ minimal}} S_{\mathfrak{p}}(0)$$

is a primary decomposition of 0. Hence, there is some restriction of the indexing set such that we obtain a minimal primary decomposition. Then, all the associated primes of 0 are minimal in A and thus isolated.

Conversely, suppose the primes of $\operatorname{Ass}(0)$ are all isolated. Then, we have a minimal primary decomposition of 0 as

$$0 = \bigcap_{\mathfrak{p} \in \operatorname{Ass}(0)} \mathfrak{q}_{\mathfrak{p}},$$

where $\mathfrak{q}_{\mathfrak{p}}$ is a $\mathfrak{p}$-primary ideal. We've shown above that $S_{\mathfrak{p}}(0)$ is the minimal $\mathfrak{p}$-primary ideal in A, so $S_{\mathfrak{p}}(0) \subset \mathfrak{q}_{\mathfrak{p}}$ for all $\mathfrak{p}$. Further, from Exercise 4.10, we have $D(A) = \operatorname{Ass}(0)$, so

$$\bigcap_{\mathfrak{p} \in D(A)} S_{\mathfrak{p}}(0) = \bigcap_{\mathfrak{p} \in \operatorname{Ass}(0)} S_{\mathfrak{p}}(0) \subset \bigcap_{\mathfrak{p} \in \operatorname{Ass}(0)} \mathfrak{q}_{\mathfrak{p}} = 0.$$

We showed above that all minimal primes of A are in $D(A)$. By Exercise 1.8, $\mathfrak{p} \in D(A)$ contains some minimal prime $\mathfrak{q}$ of A. But then, $\mathfrak{q} \in D(A)$. Since we're supposing all primes are isolated in $\mathrm{Ass}(0) = D(A)$, $\mathfrak{p} = \mathfrak{q}$, so $D(A)$ is exactly the minimal primes of A. Thus,

$$\bigcap_{\mathfrak{p} \text{ minimal}} S_{\mathfrak{p}}(0) = 0,$$

as desired. □

4.12 Exercise

(i) Proof. By Proposition 3.11(v), $S^{-1}(\mathfrak{a}) \cap S^{-1}(\mathfrak{b}) = S^{-1}(\mathfrak{a} \cap \mathfrak{b})$. By the intra-text Exercise 1.18, we then have

$$S(\mathfrak{a} \cap \mathfrak{b}) = (S^{-1}(\mathfrak{a} \cap \mathfrak{b}))^c = (S^{-1}(\mathfrak{a}) \cap S^{-1}(\mathfrak{b}))^c = (S^{-1}(\mathfrak{a}))^c \cap (S^{-1}(\mathfrak{b}))^c = S(\mathfrak{a}) \cap S(\mathfrak{b}),$$

as desired. □

(ii) Proof. By Proposition 3.11(v), we have $S^{-1}(r(\mathfrak{a})) = r(S^{-1}(\mathfrak{a}))$. By the intra-text Exercise 1.18, we then have

$$r(S(\mathfrak{a})) = r((S^{-1}(\mathfrak{a}))^c) = r(S^{-1}(\mathfrak{a}))^c = (S^{-1}(r(\mathfrak{a})))^c = S(r(\mathfrak{a})),$$

as desired. □

(iii) Proof. By Proposition 3.11(ii), $\mathfrak{a} \cap S \neq \emptyset \iff S^{-1}(\mathfrak{a}) = (1)$.

If $S^{-1}(\mathfrak{a}) = (1)$, then $(S^{-1}(\mathfrak{a}))^c = (1)^c$, i.e. $S(\mathfrak{a}) = (1)$. Conversely, if $S(\mathfrak{a}) = (1)$, then $1 \in (S^{-1}(\mathfrak{a}))^c = \mathfrak{a}^{ec}$. Therefore, $1 \in \mathfrak{a}^{ece} = \mathfrak{a}^e = S^{-1}(\mathfrak{a})$ (Proposition 1.17), so $S^{-1}(\mathfrak{a}) = (1)$. □

(iv) Proof. By Exercise 3.3, $S_1^{-1}(S_2^{-1}(A)) \cong (S_1S_2)^{-1}(A)$ canonically so that the composite homomorphism $A \to S_2^{-1}(A) \to S_1^{-1}(S_2^{-1}(A))$ commutes with the isomorphism and the direct canonical homomorphism $A \to (S_1S_2)^{-1}(A)$. Thus, $(S_1S_2)^{-1}(\mathfrak{a}) = S_1^{-1}(S_2^{-1}(\mathfrak{a}))$. In fact,

$$(S_1S_2)(\mathfrak{a}) = ((S_1S_2)^{-1}(\mathfrak{a}))^c = (S_1^{-1}(S_2^{-1}(\mathfrak{a})))^c = S_1((S_2^{-1}(\mathfrak{a}))^c) = S_1(S_2(\mathfrak{a})),$$

where, of course, $-^c$ means contraction back to A. □

Proof. We can write the primary decomposition of $\mathfrak{a}$ as

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i,$$

where $\mathfrak{q}_i$ is primary for all i. By (i),

$$S(\mathfrak{a}) = \bigcap_{i=1}^{n} S(\mathfrak{q}_i).$$

Therefore, to see that there are only finitely many ideals $S(\mathfrak{a})$ as S ranges over all multiplicatively closed sets, it will suffice to replace $\mathfrak{a}$ with a $\mathfrak{p}$-primary ideal $\mathfrak{q}$.

Suppose $S \cap \mathfrak{p} \neq \emptyset$. By Proposition 4.8(i), $S^{-1}\mathfrak{q} = S^{-1}A$, so $S(\mathfrak{q}) = (1)$. If, on the other hand, $S \cap \mathfrak{p} = \emptyset$, then by Proposition 4.8(ii), $\mathfrak{q} = (S^{-1}(\mathfrak{q}))^c = S(\mathfrak{q})$. Thus, for any multiplicatively closed subset of A, we must have that $S(\mathfrak{q})$ is (1) or $\mathfrak{q}$, as desired. □

4.13 Exercise

(i) Proof. Suppose $xy \in \mathfrak{p}^{(n)}$ and $x \notin \mathfrak{p}^{(n)}$. Note that, by Proposition 3.11, we have

$$r(S_{\mathfrak{p}}^{-1}(\mathfrak{p}^n)) = S_{\mathfrak{p}}^{-1}(r(\mathfrak{p}^n)) = S_{\mathfrak{p}}^{-1}(\mathfrak{p}).$$

Since $S_{\mathfrak{p}}^{-1}(\mathfrak{p})$ is maximal in $A_{\mathfrak{p}}$, we have by Proposition 4.2 that $S_{\mathfrak{p}}^{-1}(\mathfrak{p}^n)$ is primary. Then, since $(xy)/1 \in S_{\mathfrak{p}}^{-1}(\mathfrak{p}^n)$ and $x/1 \notin S_{\mathfrak{p}}^{-1}(\mathfrak{p}^n)$ by assumption, we must have $(y/1)^k \in S_{\mathfrak{p}}^{-1}(\mathfrak{p}^n)$ for some $k > 0$. Then, since $(y/1)^k = y^k/1$, we have $y^k \in \mathfrak{p}^{(n)}$, so $\mathfrak{p}^{(n)}$ is primary.

By Exercise 4.12, we have

$$r(S_{\mathfrak{p}}(\mathfrak{p}^n)) = S_{\mathfrak{p}}(r(\mathfrak{p}^n)) = S_{\mathfrak{p}}(\mathfrak{p}) = \mathfrak{p},$$

the latter equality following from Proposition 4.8. Finally, $\mathfrak{p}^{(n)}$ is $\mathfrak{p}$-primary. □

(ii) Proof. Suppose $\mathfrak{p}^n$ has a primary decomposition so that we have a minimal primary decomposition

$$\mathfrak{p}^n = \bigcap_{i=1}^{m} \mathfrak{q}_i,$$

where $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary for each i. Now, by Exercise 4.12, we have

$$\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{p}^n) = \bigcap_{i=1}^{m} S_{\mathfrak{p}}(\mathfrak{q}_i).$$

By Proposition 4.9, this is a primary decomposition of $\mathfrak{p}^{(n)}$. However, we already showed in (i) that $\mathfrak{p}^{(n)}$ is $\mathfrak{p}$-primary, so it is already a primary decomposition. Thus, since the decomposition is minimal, we have $\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{q})$ for some $\mathfrak{q} \in \{\mathfrak{q}_i\}$.

By Exercise 4.12, we have

$$r(S_{\mathfrak{p}}(\mathfrak{q})) = r(S_{\mathfrak{p}}(\mathfrak{p}^n)) = S_{\mathfrak{p}}(r(\mathfrak{p}^n)) = S_{\mathfrak{p}}(\mathfrak{p}) = \mathfrak{p},$$

where the last equality follows by Proposition 4.8. Since $S_{\mathfrak{p}}(\mathfrak{q}) \neq (1)$, we have again by Proposition 4.8 that $\mathfrak{p}'$ (the prime associated to the primary ideal $\mathfrak{q}$) is contained in $\mathfrak{p}$. Thus, by Proposition 4.8 and Exercise 4.12, we have

$$r(S_{\mathfrak{p}}(\mathfrak{q})) = S_{\mathfrak{p}}(r(\mathfrak{q})) = S_{\mathfrak{p}}(\mathfrak{p}') = \mathfrak{p}',$$

so $\mathfrak{p}' = \mathfrak{p}$. Finally, by Proposition 4.8, we have

$$\mathfrak{p}^{(n)} = S_{\mathfrak{p}}(\mathfrak{q}) = \mathfrak{q},$$

as desired. In particular, $\mathfrak{p}^{(n)}$ is the $\mathfrak{p}$-primary component of $\mathfrak{p}^n$. □

(iii) Proof. Suppose $\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}$ is decomposable, so it has a minimal primary decomposition

$$\mathfrak{p}^{(m)}\mathfrak{p}^{(n)} = \bigcap_{i=1}^{N} \mathfrak{q}_i.$$

By Proposition 4.9, we have a minimal decomposition

$$S_{\mathfrak{p}}^{-1}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = \bigcap_{j=1}^{M} S_{\mathfrak{p}}^{-1}(\mathfrak{q}_j),$$

where we reindex to exclude the $\mathfrak{q}_j$ that meet $S_\mathfrak{p} = A - \mathfrak{p}$. By Exercise 3.11(v), we have

$$S_\mathfrak{p}^{-1}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = S_\mathfrak{p}^{-1}(S_\mathfrak{p}(\mathfrak{p}^m))S_\mathfrak{p}^{-1}(S_\mathfrak{p}(\mathfrak{p}^n)) = S_\mathfrak{p}^{-1}(\mathfrak{p}^m)S_\mathfrak{p}^{-1}(\mathfrak{p}^n) = S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n}).$$

However, note that by Exercise 4.12, we have

$$r(S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n})) = S_\mathfrak{p}^{-1}(r(\mathfrak{p}^{m+n})) = S_\mathfrak{p}^{-1}(\mathfrak{p}).$$

Since $S_\mathfrak{p}^{-1}(\mathfrak{p})$ is a maximal ideal in $A_\mathfrak{p}$, we conclude by Proposition 4.2 that $S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n})$ is $S_\mathfrak{p}^{-1}(\mathfrak{p})$-primary. Then, since $S_\mathfrak{p}^{-1}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n})$ is primary, we must have $M = 1$ in the minimal primary decomposition, so $S_\mathfrak{p}^{-1}(\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}) = S_\mathfrak{p}^{-1}(\mathfrak{q})$ for some $\mathfrak{q} \in \{\mathfrak{q}_j\}$.

Suppose $\mathfrak{q}$ is $\mathfrak{p}'$-primary. Then, by Proposition 3.11, we have

$$r(S_\mathfrak{p}^{-1}(\mathfrak{q})) = S_\mathfrak{p}^{-1}(r(\mathfrak{q})) = S_\mathfrak{p}^{-1}(\mathfrak{p}').$$

Also, from the preceding paragraph, we have

$$r(S_\mathfrak{p}^{-1}(\mathfrak{q})) = r(S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n})) = S_\mathfrak{p}^{-1}(r(\mathfrak{p}^{m+n})) = S_\mathfrak{p}^{-1}(\mathfrak{p}).$$

Since $\mathfrak{p}$ and $\mathfrak{p}'$ both do not meet $A - \mathfrak{p}$ (otherwise $S_\mathfrak{p}^{-1}\mathfrak{q} = (1)$), we have by the one-to-one correspondence of Proposition 3.11(iv) that $S_\mathfrak{p}^{-1}(\mathfrak{p}) = S_\mathfrak{p}^{-1}(\mathfrak{p}')$ implies $\mathfrak{p} = \mathfrak{p}'$. Therefore, $\mathfrak{q}$ is $\mathfrak{p}$-primary.

Since $\mathfrak{q}$ is $\mathfrak{p}$-primary and $S_\mathfrak{p} \cap \mathfrak{q} = \emptyset$, we have by Proposition 4.8 that $(S_\mathfrak{p}^{-1}(\mathfrak{q}))^c = \mathfrak{q}$. Thus,

$$\mathfrak{q} = (S_\mathfrak{p}^{-1}(\mathfrak{q}))^c = (S_\mathfrak{p}^{-1}(\mathfrak{p}^{m+n}))^c = \mathfrak{p}^{(m+n)}.$$

In particular, $\mathfrak{p}^{(m+n)}$ is the $\mathfrak{p}$-primary component of $\mathfrak{p}^{(m)}\mathfrak{p}^{(n)}$. □

(iv) Proof. If $\mathfrak{p}^{(n)} = \mathfrak{p}^n$, then since $\mathfrak{p}^{(n)}$ is $\mathfrak{p}$-primary by (i), $\mathfrak{p}^n$ is $\mathfrak{p}$-primary. Conversely, suppose $\mathfrak{p}^n$ is $\mathfrak{p}$-primary. Then, $\mathfrak{p}^n$ is trivially decomposable. By (ii), $\mathfrak{p}^{(n)}$ is the $\mathfrak{p}$-primary component of the decomposition. However, $\mathfrak{p}^n$ is the $\mathfrak{p}$-primary component, so $\mathfrak{p}^n = \mathfrak{p}^{(n)}$. □

4.14 Exercise

Proof. To start, let $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal, and let $x \in A$ be such that $(\mathfrak{q} : x)$ is maximal in the set of all ideals of this form, excluding (1) of course. Then, by Lemma 4.4, $x \notin \mathfrak{q}$. Now, suppose $y \in A - (\mathfrak{q} : x)$. Then, by intra-text Exercise 1.12,

$$(\mathfrak{q} : x) \subset ((\mathfrak{q} : x) : y) = (\mathfrak{q} : xy).$$

As $(\mathfrak{q} : x)$ is $\mathfrak{p}$-primary by Lemma 4.4 and $y \notin (\mathfrak{q} : x)$, $(\mathfrak{q} : xy) = ((\mathfrak{q} : x) : y) \neq (1)$ (again by Lemma 4.4). By the maximality of $(\mathfrak{q} : x)$, we must have $(\mathfrak{q} : x) = (\mathfrak{q} : xy)$, so for all $a \in A$, $axy \in \mathfrak{q} \implies ax \in \mathfrak{q}$. Letting $a = y^n$, we have that $xy^{n+1} \in \mathfrak{q}$ implies that $xy^n \in \mathfrak{q}$. Continuing along this chain, $xy^n \in \mathfrak{q} \implies x \in \mathfrak{q}$. Then, $y^n \notin \mathfrak{q}$ for any $n > 0$; otherwise, letting $x = 1$, we have $xy^n \in \mathfrak{q}$, implying $1 \in \mathfrak{q}$, a clear contradiction. Thus, $y \notin r(\mathfrak{q}) = \mathfrak{p}$. In particular, we've shown $y \notin (\mathfrak{q} : x)$ implies that $y \notin \mathfrak{p}$, so $\mathfrak{p} \subset (\mathfrak{q} : x)$. But, $(\mathfrak{q} : x) \subset r(\mathfrak{q} : x) = \mathfrak{p}$, so we have $(\mathfrak{q} : x) = \mathfrak{p}$.

Now, since $\mathfrak{a}$ is a decomposable ideal, we have a minimal primary decomposition

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i,$$

where $\mathfrak{q}_i$ is a $\mathfrak{p}_i$-primary ideal. Since the decomposition is minimal, for any $\mathfrak{q}_j$, there is some $y \in \bigcap_{i \neq j} \mathfrak{q}_i - \mathfrak{q}_j$. Then, by Lemma 4.4 and intra-text Exercise 1.12,

$$(\mathfrak{a} : y) = \bigcap_{i=1}^{n} (\mathfrak{q}_i : y) = (\mathfrak{q}_j : y).$$

If we choose y such that $(\mathfrak{q}_i : y)$ is maximal for $y \notin \mathfrak{q}_i$, then by the preceding paragraph, $(\mathfrak{q}_i : y) = \mathfrak{p}_i$. Thus, the maximal elements of the set of ideals of the form $(\mathfrak{a} : x)$ are exactly $\mathrm{Ass}(\mathfrak{a})$. □

4.15 Exercise

Proof. Since $\mathfrak{a}$ is decomposable, we have a minimal primary decomposition

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i = \left(\bigcap_{i:\mathfrak{p}_i \in \Sigma} \mathfrak{q}_i \right) \cap \left(\bigcap_{i:\mathfrak{p}_i \notin \Sigma} \mathfrak{q}_i \right),$$

where $\mathfrak{q}_i$ is a $\mathfrak{p}_i$-primary ideal. We have $f \in A$ such that $f \in \mathfrak{p} \iff \mathfrak{p} \notin \Sigma$. Since $\mathfrak{p}$ is prime, for any $m > 0$, we actually have $f^m \in \mathfrak{p} \iff \mathfrak{p} \notin \Sigma$. Note that $r(\mathfrak{q}) = \mathfrak{p}$, so for any i with $\mathfrak{p}_i \notin \Sigma$, there is some $n_i > 0$ such that $f^{n_i} \in \mathfrak{q}_i$. Since $\mathfrak{a}$ is decomposable, there are finitely many such i so let N be the max over $\{n_i\}$. Then, for all $\mathfrak{q}$ with $r(\mathfrak{q}) = \mathfrak{p} \notin \Sigma$, $f^N \in \mathfrak{q}$. On the other hand, if $r(\mathfrak{q}) = \mathfrak{p} \in \Sigma$, then $f^N \notin \mathfrak{p}$ and thus $f^N \notin \mathfrak{q}$. Therefore, for all $n \geq N$, $f^n \notin \mathfrak{p}$ for $\mathfrak{p} \in \Sigma$, and $f^n \in \mathfrak{q}$ if $r(\mathfrak{q}) = \mathfrak{p} \notin \Sigma$.

For $n \geq N$, we have by Lemma 4.4 that if $\mathfrak{p} \in \Sigma$, $(\mathfrak{q} : f^n) = \mathfrak{q}$ since $f^n \notin \mathfrak{p}$ ($f \notin \mathfrak{p} \implies f^n \notin \mathfrak{p}$ since $\mathfrak{p}$ is prime); on the other hand, $f^n \in \mathfrak{q}$ if $r(\mathfrak{q}) = \mathfrak{p} \notin \Sigma$, so $(\mathfrak{q} : f^n) = (1)$. Thus, by intra-text Exercise 1.12,

$$(\mathfrak{a} : f^n) = \left(\bigcap_{i:\mathfrak{p}_i \in \Sigma} (\mathfrak{q}_i : f^n) \right) \cap \left(\bigcap_{i:\mathfrak{p}_i \notin \Sigma} (\mathfrak{q}_i : f^n) \right) = \left(\bigcap_{i:\mathfrak{p}_i \in \Sigma} \mathfrak{q}_i \right) \cap \left(\bigcap_{i:\mathfrak{p}_i \notin \Sigma} (1) \right) = \mathfrak{q}_\Sigma,$$

as desired.

If $S_f = \{f^n\}_{n \geq 0}$, then $\mathfrak{p} \cap S_f = \emptyset \iff \mathfrak{p} \in \Sigma$ by the first paragraph. By Proposition 4.9, we have

$$S_f(\mathfrak{a}) = \bigcap_{i:\mathfrak{p}_i \in \Sigma} \mathfrak{q}_i = \mathfrak{q}_\Sigma,$$

as desired. □

4.16 Exercise

Proof. By Proposition 3.11(i), every ideal of $S^{-1}A$ is an extended ideal from A. So, if $\phi : A \to S^{-1}A$ is the canonical homomorphism, every ideal of $S^{-1}A$ has the form $S^{-1}\mathfrak{a}$, where $\mathfrak{a}$ is an ideal of A. Since every ideal of A is decomposable, we have a minimal primary decomposition of $\mathfrak{a}$:

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i,$$

so by Proposition 4.9, if we reindex so that $\mathfrak{q}_i \cap S = \emptyset \iff i \leq m$, we have a minimal primary decomposition

$$S^{-1}\mathfrak{a} = \bigcap_{i=1}^{m} S^{-1}\mathfrak{q}_i,$$

so every ideal of $S^{-1}A$ is decomposable. □

4.17 Exercise

Proof. We start by generalizing Exercise 4.11: if $\mathfrak{a} \neq (1)$ and $\mathfrak{p}$ is a prime ideal minimal over $\mathfrak{a}$, then $S_{\mathfrak{p}}(\mathfrak{a})$ is $\mathfrak{p}$-primary. Since $\mathfrak{p}$ is minimal among prime ideals over $\mathfrak{a}$, $\overline{\mathfrak{p}}$ is a minimal prime ideal of $A/\mathfrak{a}$. By Exercise 4.11, $\overline{\mathfrak{q}} = S_{\overline{\mathfrak{p}}}(0)$ is $\overline{\mathfrak{p}}$-primary. If $sa \in \mathfrak{a}$ for $s \in S_{\mathfrak{p}}$, then $\overline{sa} = 0$, where $\overline{s} \in S_{\overline{\mathfrak{p}}}$. Thus, $a \in S_{\overline{\mathfrak{p}}}(0)$. Conversely, if $\overline{sa} = 0$, where $\overline{s} \in S_{\overline{\mathfrak{p}}}$, then $sa \in \mathfrak{a}$. Thus, $\overline{S_{\mathfrak{p}}(\mathfrak{a})} = S_{\overline{\mathfrak{p}}(0)}$. In particular, $S_{\mathfrak{p}}(\mathfrak{a})$ is the contraction of a primary ideal, so it is itself primary. Note that $r(S_{\overline{\mathfrak{p}}}(0)) = \overline{\mathfrak{p}}$ by Exercise 4.11, so contracting along $A \to A/\mathfrak{a}$, we have

$$r(S_{\mathfrak{p}}(\mathfrak{a})) = r(S_{\overline{\mathfrak{p}}}(0)^c) = (r(S_{\overline{\mathfrak{p}}}(0)))^c = (\overline{\mathfrak{p}})^c = \mathfrak{p},$$

so $S_{\mathfrak{p}}(\mathfrak{a})$ is $\mathfrak{p}$-primary.

Let $\mathfrak{a} \neq (1)$ be an ideal of A. Let $\mathfrak{p}_0$ be a prime ideal minimal over $\mathfrak{a}_0 = \mathfrak{a}$. As we just showed, $\mathfrak{q}_0 = S_{\mathfrak{p}_0}(\mathfrak{a}_0)$ is $\mathfrak{p}_0$-primary. By (L1), there is some $x_0 \notin \mathfrak{p}_0$ such that $\mathfrak{q}_0 = (\mathfrak{a}_0 : x_0)$. We wish to show now that $\mathfrak{a}_0 = \mathfrak{q}_0 \cap [\mathfrak{a}_0 + (x_0)]$. The first inclusion follows easily since $\mathfrak{a}_0 \subset \mathfrak{q}_0$ and $\mathfrak{a}_0 \subset \mathfrak{a}_0 + (x_0)$. Conversely, choose some $a_0 + b_0 x_0 \in \mathfrak{a}_0 + (x_0)$ that is also in $\mathfrak{q}_0 = (\mathfrak{a}_0 : x)$. Then, $x_0(a_0 + b_0 x_0) = a_0 x_0 + b_0 x_0^2 \in \mathfrak{a}_0$ so that $b_0 x_0^2 \in \mathfrak{a}_0 \subset \mathfrak{q}_0$. Since $x_0 \notin \mathfrak{p}_0$, $(x_0^2)^k = x_0^{2k} \notin \mathfrak{p}_0 = r(\mathfrak{q}_0)$ for any $k > 0$. Since $b_0 x_0^2 \in \mathfrak{q}_0$, $\mathfrak{q}_0$ is primary, and $(x_0^2)^k \notin \mathfrak{q}_0$ for any $k > 0$, we must have $b_0 \in \mathfrak{q}_0 = (a_0 : x_0)$. Thus, $b_0 x_0 \in \mathfrak{a}_0$ so that $a_0 + b_0 x_0 \in \mathfrak{a}_0$, giving us the second inclusion. Therefore, $\mathfrak{a}_0 = \mathfrak{q}_0 \cap [\mathfrak{a}_0 + (x_0)]$.

In particular, the set of ideals $\mathfrak{b}$ containing $\mathfrak{a}_0 + (x_0)$ such that $\mathfrak{q}_0 \cap \mathfrak{b} = \mathfrak{a}_0$ is non-empty; let therefore $\mathfrak{a}_1$ be a maximal element of this set. If $\mathfrak{a}_1 = (1)$, then $\mathfrak{q}_0 = \mathfrak{a}_0$, so we have a primary decomposition of $\mathfrak{a}_0$. Thus, suppose $\mathfrak{a}_1 \neq (1)$. Then, there is a prime ideal $\mathfrak{p}_1$ minimal among those containing $\mathfrak{a}_1$, and $\mathfrak{q}_1 = S_{\mathfrak{p}_1}(\mathfrak{a}_1)$ is $\mathfrak{p}_1$-primary by the first paragraph. By (L1), $\mathfrak{q}_1 = (\mathfrak{a}_1 : x_1)$ for some $x_1 \notin \mathfrak{p}_1$. As before, we have $\mathfrak{a}_1 = \mathfrak{q}_1 \cap (\mathfrak{a}_1 + (x_1))$. Then, $\mathfrak{a}_0 = \mathfrak{q}_0 \cap \mathfrak{a}_1 = \mathfrak{q}_0 \cap \mathfrak{q}_1 \cap (\mathfrak{a}_1 + (x_1))$.

We use transfinite induction to pull the pieces together. Suppose $\mathfrak{a}_0 = [\mathfrak{a}_\alpha + (x_\alpha)] \cap \bigcap_{\beta \leq \alpha} \mathfrak{q}_\beta$ for some ordinal α and primary ideals $\mathfrak{q}_\beta$ such that $\mathfrak{a}_\alpha$ is not contained in $r(\mathfrak{q}_\beta)$ for any $\beta < \alpha$. If $\mathfrak{a}_\alpha \neq (1)$, then as before, there is some ideal $\mathfrak{a}_{\alpha+1}$ containing $\mathfrak{a}_\alpha + (x_\alpha)$, maximal under the contraint $\mathfrak{a}_0 = \mathfrak{a}_{\alpha+1} \cap \bigcap_{\beta \leq \alpha} \mathfrak{q}_\beta$. If $\mathfrak{a}_{\alpha+1} = (1)$, we're finished, so suppose $\mathfrak{a}_{\alpha+1} \neq (1)$ so that there is some prime $\mathfrak{p}_{\alpha+1}$ minimal over $\mathfrak{a}_{\alpha+1}$ so that $\mathfrak{q}_{\alpha+1} = S_{\mathfrak{p}_{\alpha+1}}(\mathfrak{a}_{\alpha+1})$ is $\mathfrak{p}_{\alpha+1}$-primary. By (L1), there is some $x_{\alpha+1} \notin \mathfrak{p}_{\alpha+1}$ such that $\mathfrak{q}_{\alpha+1} = (\mathfrak{a}_{\alpha+1} : x_{\alpha+1})$. As before, we have $\mathfrak{a}_{\alpha+1} = \mathfrak{q}_{\alpha+1} \cap [\mathfrak{a}_{\alpha+1} + (x_{\alpha+1})]$ so that

$$\mathfrak{a}_0 = \mathfrak{a}_{\alpha+1} \cap \bigcap_{\beta \leq \alpha} \mathfrak{q}_\beta = [\mathfrak{a}_{\alpha+1} + (x_{\alpha+1})] \cap \bigcap_{\beta \leq \alpha+1} \mathfrak{q}_\beta.$$

So, we've proven the zero case and the successor step. For the limit step, suppose that for all $\alpha < \beta$, we have $\mathfrak{a}_0 = \mathfrak{a}_{\alpha+1} \cap \bigcap_{\gamma \leq \alpha} \mathfrak{q}_\gamma$ with $\mathfrak{a}_\gamma$ properly containing $\mathfrak{a}_\alpha$ for all $\alpha < \gamma$. Define $\mathfrak{a}_\beta = \bigcup_{\alpha < \beta} \mathfrak{a}_\alpha$. Then, $\mathfrak{a}_\beta$ properly contains $\mathfrak{a}_\alpha$ for each $\alpha < \beta$ so that

$$\mathfrak{a}_\beta \cap \bigcap_{\alpha<\beta} \mathfrak{q}_\alpha = \bigcup_{\gamma<\beta} \left(\mathfrak{a}_\gamma \cap \bigcap_{\alpha<\beta} \mathfrak{q}_\alpha \right) = \bigcup_{\gamma+1<\beta} \left(\mathfrak{a}_{\gamma+1} \cap \bigcap_{\alpha \leq \gamma} \mathfrak{q}_\alpha \cap \bigcap_{\gamma<\alpha<\beta} \mathfrak{q}_\alpha \right)$$

$$= \bigcup_{\gamma+1<\beta} \left(\mathfrak{a}_0 \cap \bigcap_{\alpha<\beta} \mathfrak{q}_\alpha \right) = \bigcup_{\gamma+1<\beta} \mathfrak{a}_0 = \mathfrak{a}_0,$$

as desired.

If for any α, we obtain $\mathfrak{a}_\alpha = (1)$, we've shown $\mathfrak{a}_0$ is the intersection of primary ideals. For $\alpha < \beta$, $\mathfrak{a}_\beta$ strictly contains $\mathfrak{a}_\alpha$. The cardinality of A imposes a bound on the number of distinct $\mathfrak{a}_\alpha$ we can have, so we have by transfinite induction that $\mathfrak{a} = \bigcap_{\alpha<\beta} \mathfrak{q}_\alpha$ for some β of cardinality less than A. Thus, $\mathfrak{a}$ is the intersection of (possibly infinitely many) primary ideals. □

4.18 Exercise

Proof. (i) $\implies$ (L1). Let $\mathfrak{a} \neq (1)$ be an ideal of A. Then, it has a minimal primary decomposition

$$\mathfrak{a} = \bigcap_{i=1}^{n} \mathfrak{q}_i.$$

Let Σ be the set of indices i such that $\mathfrak{q}_i \subset \mathfrak{p}$. Then, if $i \notin \Sigma$, $\mathfrak{q}_i \not\subset \mathfrak{p}$ so that $\mathfrak{q}_i \cap S_\mathfrak{p} \neq \emptyset$. By Proposition 4.9, we have $S_\mathfrak{p} = \bigcap_{i\in\Sigma} \mathfrak{q}_i$. On the other hand, $(\mathfrak{a} : x) = \bigcap_{i=1}^{n} (\mathfrak{q}_i : x)$; $(\mathfrak{q}_i : x) = (1)$ when $x \in \mathfrak{q}_i$ (Lemma 4.4) and $(\mathfrak{q}_i : x) = \mathfrak{q}_i$ when $x \notin \mathfrak{p}_i$. Thus, we are looking for some $x \in \bigcap_{i:i\notin\Sigma} \mathfrak{q}_i - \bigcup_{i:i\in\Sigma} \mathfrak{p}_i$. By Proposition 4.1, since $\mathfrak{q}_i \subset \mathfrak{p}$ for all $i \in \Sigma$, $\mathfrak{p}_i \subset \mathfrak{p}$ so that $\bigcup_{i:i\in\Sigma} \mathfrak{p}_i \subset \mathfrak{p}$. Therefore,

$$\bigcap_{i:i\notin\Sigma} (\mathfrak{q}_i - \mathfrak{p}) = \bigcap_{i:i\notin\Sigma} \mathfrak{q}_i - \mathfrak{p} \subset \bigcap_{i:i\notin\Sigma} \mathfrak{q}_i - \bigcup_{i:i\in\Sigma} \mathfrak{p}_i,$$

so we just have to show that $\bigcap_{i:i\notin\Sigma}(\mathfrak{q}_i - \mathfrak{p})$. For each $i \notin \Sigma$, since $\mathfrak{q}_i \not\subset \mathfrak{p}$, there is some $x_i \in \mathfrak{q}_i - \mathfrak{p}$. Multiplying them all together, we clearly have $x = \prod_{i\notin\Sigma} x_i \subset \bigcap_{i:i\notin\Sigma} \mathfrak{q}_i$. But since $\mathfrak{p}$ is prime, a product of finitely many elements not in $\mathfrak{p}$ is also not in $\mathfrak{p}$. Finally, we have $x \in \bigcap_{i:i\notin\Sigma}(\mathfrak{q}_i - \mathfrak{p}) \subset \bigcap_{i:i\notin\Sigma} \mathfrak{q}_i - \bigcup_{i:i\in\Sigma} \mathfrak{p}_i$ so that $S_\mathfrak{p}(\mathfrak{a}) = (\mathfrak{a} : x)$.

(i) $\implies$ (L2). Suppose we have a descending chain $S_1 \supset S_2 \supset \cdots$ of multiplicatively closed subsets of A. This leads to a descending chain $S_1(\mathfrak{a}) \supset S_2(\mathfrak{a}) \supset \cdots$. By Exercise 4.12, there are only finitely many sets of the form $S(\mathfrak{a})$, so the chain stabilizes.

(ii) $\implies$ (i). Let $\mathfrak{a} \neq (1)$ be an ideal of A. Since we have (L1), we have by Exercise 4.17 that there exist primary $\mathfrak{q}_\alpha$ for $\alpha < \beta$ such that $\mathfrak{a} = \bigcap_\alpha \mathfrak{q}_\alpha$. Let $mathfrakp_\alpha = r(\mathfrak{q}_\alpha)$ and $S_\beta = A - \bigcup_{\alpha<\beta} \mathfrak{p}_\alpha$. By Exercise 3.7, these are saturated multiplicatively closed sets of A and $S_\alpha \supset S_\beta$ for $\alpha < \beta$. In Exercise 4.17, we defined α for finite ordinals. I.e., there is some ideal $\mathfrak{a}_{n+1}$ such that

$$\mathfrak{a} = \mathfrak{a}_{n+1} \cap \mathfrak{q}_n \cap \cdots \cap \mathfrak{q}_0$$

and $\mathfrak{a}_{n+1} \not\subset \mathfrak{p}_m$ for $m < n+1$. Then, $\mathfrak{a}_{n+1} \cap S_n \neq \emptyset$, so Exercise 4.12 gives us $S_n(\mathfrak{a}_{n+1}) = (1)$, so

$$S_n(\mathfrak{a}) = \mathfrak{q}_n \cap \cdots \cap \mathfrak{q}_0.$$

(L2) tells us that the sequence $S_n(\mathfrak{a})$ stabilizes so that for some m, we have, for all $\alpha > m$, $S_\alpha(\mathfrak{a}) = \mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m$.

Let $\alpha > n$ for all finite ordinals n. We can write $\mathfrak{a} = S_n(\mathfrak{a}) \cap \mathfrak{q}_\alpha \cap \bigcap_{\gamma\notin 0,\ldots,n,\alpha} \mathfrak{q}_\gamma$. Taking S_α and using Exercise 4.12, we have (since $S_\alpha S_n = S_n$)

$$S_\alpha(\mathfrak{a}) = S_n(\mathfrak{a}) \cap S_\alpha(\mathfrak{q}_\alpha) \cap \bigcap_{\gamma\notin 0,\ldots,n,\alpha} S_\alpha(\mathfrak{q}_\gamma).$$

Note that $\mathfrak{q}_\alpha$ does not meet $A - \mathfrak{p}_\alpha \supset S_\alpha$, so if $x \in S_\alpha(\mathfrak{q}_\alpha)$, there is some $s \in S_\alpha$ such that $sx \in \mathfrak{q}_\alpha$. Since $\mathfrak{q}_\alpha$ is primary, there is some n such that $s^n \in \mathfrak{q}_\alpha$ (which cannot happen since S_α is saturated) or $x \in \mathfrak{q}_\alpha$. Thus, it must be the latter condition, forcing $S_\alpha(\mathfrak{q}_\alpha) = \mathfrak{q}_\alpha$. Thus, we have

$$S_\alpha(\mathfrak{a}) = S_n(\mathfrak{a}) \cap S_\alpha(\mathfrak{q}_\alpha) \cap \bigcap_{\gamma\notin 0,\ldots,n,\alpha} S_\alpha(\mathfrak{q}_\gamma) = \mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m \cap \mathfrak{q}_\alpha \cap \bigcap_{\gamma\notin 0,\ldots,n,\alpha} S_\alpha(\mathfrak{q}_\gamma)$$

$$\mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m = \mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m \cap \mathfrak{q}_\alpha \cap \cdots \subset \mathfrak{q}_\alpha,$$

so $\mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m \subset \mathfrak{q}_\alpha$ for any ordinal $\alpha > m$. Thus,

$$\mathfrak{a} = \bigcap_{\alpha<\beta} \mathfrak{q}_\alpha = \mathfrak{q}_0 \cap \cdots \cap \mathfrak{q}_m,$$

so $\mathfrak{a}$ has a primary decomposition. □

4.19 Exercise

Proof. Every $\mathfrak{p}$-primary ideal contains $S_{\mathfrak{p}}(0) = \ker(A \to A_{\mathfrak{p}})$ by Exercise 4.11.

We prove that if $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ are distinct prime ideals, none of which is minimal in A, then there exists an ideal $\mathfrak{a}$ with $\operatorname{Ass}(\mathfrak{a}) = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$. We do this by induction on n. The base case is trivial; if the set of primes is $\{\mathfrak{p}\}$, then $\operatorname{Ass}(\mathfrak{p}) = \mathfrak{p}$.

Now, suppose the property holds for all sets of distinct primes (none of which is minimal in A) with cardinality $< n$. Now, suppose we have a set $\{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$ of distinct primes, none of which are minimal in A. Reindex so that $\mathfrak{p}_n$ is maximal in the set. By the inductive hypothesis, there is some ideal $\mathfrak{b}$ with a minimal primary decomposition $\mathfrak{b} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_{n-1}$, where $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary. If $\mathfrak{b} \subset S_{\mathfrak{p}_n}(0)$, let $\mathfrak{p}$ be a minimal prime of A contained in $\mathfrak{p}_n$. Then, by Exercise 4.10, we have $S_{\mathfrak{p}_n}(0) = S_{\mathfrak{p}}(0)$ so that $\mathfrak{b} \subset S_{\mathfrak{p}}(0)$. Now,

$$\mathfrak{p} \supset \mathfrak{b} = \bigcap_{i=1}^{n-1} \mathfrak{q}_i,$$

so taking radicals, we have

$$\mathfrak{p} = r(\mathfrak{p}) \supset \bigcap_{i=1}^{n-1} r(\mathfrak{q}_i) = \bigcap_{i=1}^{n-1} \mathfrak{p}_i.$$

Hence, there is some $\mathfrak{p}_i$ such that $\mathfrak{p}_i \subset \mathfrak{p}$, but since we chose $\mathfrak{p}$ to be minimal in A, we have $\mathfrak{p} = \mathfrak{p}_i$. But by assumption, no prime ideal in our set can be minimal in A, giving us a contradiction. Therefore, $\mathfrak{b} \not\subset S_{\mathfrak{p}_n}(0)$. Let $\mathfrak{q}_n = S_{\mathfrak{p}_n}(0)$, which is a $\mathfrak{p}_n$-primary ideal by Exercise 4.10.

Now, defining $\mathfrak{a} = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$, we clearly have a primary decomposition. We must show that it is minimal. First of all, the $r(\mathfrak{q}_i)$ are all distinct, as required. Further, $\mathfrak{q}_n \not\supset \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_{n-1}$ by the preceding paragraph. Finally, we must show that for $i < n$

$$\mathfrak{q}_i \not\supset \mathfrak{q}_1 \cap \cdots \cap \hat{\mathfrak{q}}_i \cap \cdots \cap \mathfrak{q}_n.$$

We'll prove this by contradiction; suppose $\mathfrak{q}_i$ does contain the above intersection for a fixed $i < n$. Then, localizing at $\mathfrak{p}_i$, we obtain ($S = A - \mathfrak{p}_i$)

$$S^{-1}\mathfrak{q}_i \supset S^{-1}\mathfrak{q}_1 \cap \cdots \cap \widehat{S^{-1}\mathfrak{q}_i} \cap \cdots \cap S^{-1}\mathfrak{q}_n.$$

Since we chose $\mathfrak{p}_n$ to be maximal, we have $\mathfrak{p}_n \not\subset \mathfrak{p}_i$, so by Proposition 3.11, $S^{-1}\mathfrak{q}_n = A_{\mathfrak{p}_i}$. Thus,

$$S^{-1}\mathfrak{q}_i \supset S^{-1}\mathfrak{q}_1 \cap \cdots \cap \widehat{S^{-1}\mathfrak{q}_i} \cap \cdots \cap S^{-1}\mathfrak{q}_{n-1}.$$

Note that $\mathfrak{p}_i \supset \mathfrak{q}_i$, so $\mathfrak{q}_i \cap S = \emptyset$. Let Σ be the set of indices j satisfying $\mathfrak{q}_j \cap S = \emptyset$, excluding i. For $j \notin \Sigma$, we have by Proposition 4.8 that $S^{-1}\mathfrak{q}_j = A_{\mathfrak{p}_i}$. Then, by Proposition 4.8, we have

$$S^{-1}\mathfrak{q}_i \supset S^{-1}\mathfrak{q}_1 \cap \cdots \cap \widehat{S^{-1}\mathfrak{q}_i} \cap \cdots \cap S^{-1}\mathfrak{q}_{n-1} = \bigcap_{j \notin \Sigma} A_{\mathfrak{p}_i} \cap \bigcap_{j \in \Sigma} S^{-1}\mathfrak{q}_j = \bigcap_{j \in \Sigma} S^{-1}\mathfrak{q}_j.$$

By Proposition 4.8 again, after contracting, we have

$$\mathfrak{q}_i \supset \bigcap_{j \in \Sigma} \mathfrak{q}_j \supset \bigcap_{j \neq i,n} \mathfrak{q}_j,$$

violating the minimality of the primary decomposition for $\mathfrak{b}$ (inductive step). Thus,

$$\mathfrak{q}_i \not\supset \mathfrak{q}_1 \cap \cdots \cap \hat{\mathfrak{q}}_i \cap \cdots \cap \mathfrak{q}_n$$

for all i, so indeed, $\operatorname{Ass}(\mathfrak{a}) = \{\mathfrak{p}_1, \dots, \mathfrak{p}_n\}$. By induction, such an ideal can be found for any n. $\square$

4.20 Exercise

Proof. Suppose $x \in r(N:M)$. Then, for some $q > 0$, $x^q \in (N:M)$, so $x^q M \subset N$; i.e., $x \in r_M(N)$. Conversely, if $x \in r_M(N)$, then there is some $q > 0$ such that $x^q M \subset N$. Thus, $x^q \in (N:M)$, so $x \in r(N:M)$; hence, $r_M(N) = r(N:M)$. To see that $r(N:M) = r(\mathrm{Ann}(M/N))$, we simply observe that

$$x \in (N:M) \iff xM \subset N \iff x \in \mathrm{Ann}(M/N),$$

so $(N:M) = \mathrm{Ann}(M/N)$. Thus, $r_M(N) = r(N:M) = r(\mathrm{Ann}(M/N))$. As $r_M(N)$ takes the form of a radical, it is then clearly an ideal.

(i) $N \subset P \implies r_M(N) \subset r_M(P)$. This is clear: $x^q M \subset N \subset P$.

(ii) $r(r_M(N)) = r_M(N)$. We prove the non-obvious inclusion. Suppose $x \in r(r_M(N))$. Thus, there is some $m > 0$ such that $x^m \in r_M(N)$. Then, there is some $q > 0$ such that $x^{mq}M = (x^m)^q M \subset N$, so $x \in r_M(N)$.

(iii) $r_M(N \cap P) = r_M(N) \cap r_M(P)$. If $x \in r_M(N \cap P)$, there is some $q > 0$ such that $x^q M \subset N \cap P$, but $N \cap P$ is contained in N and P. Thus, x is in $r_M(N)$ and $r_M(P)$ so that $x \in r_M(N) \cap r_M(P)$. Conversely, there are $q, r > 0$ such that $x^q M \subset N$ and $x^r M \subset P$. Letting n be larger than q and r, we have $x^n M \subset x^{n-q}(x^q M) \subset x^{n-q} N \subset$; similarly, $x^n M \subset P$ so that $x^n M \subset N \cap P$. Then, $x \in r_M(N \cap P)$.

(iv) $r_M(N) = (1) \iff M = N$. First of all, we always have $N \subset M$ by assumption. Now, if $1 \in r_M(N)$, then $M = 1^q M \subset N$, so $M = N$. Conversely, if $M = N$, then for all $x \in A$, $xM \subset M = N$, so $x \in r_M(N)$, i.e. $r_M(N) = (1)$.

(v) $r_M(N+P) \supset r(r_M(N) + r_M(P))$. By (i), we have $r_M(N), r_M(P) \subset r_M(N+P)$. Thus, $r_M(N) + r_M(P) \subset r_M(N+P)$. Taking radicals, $r(r_M(N) + r_M(P)) \subset r(r_M(N+P)) = r_M(N+P)$ by (ii). □

4.21 Exercise

Proof. Suppose Q is a primary submodule of M. Note that M/Q is a $A/\mathrm{Ann}(M/Q)$-module. Let $x \in A/\mathrm{Ann}(M/Q)$ be a zero-divisor. I.e., there is some non-zero $y \in A/\mathrm{Ann}(M/Q)$ such that $xy = 0$. Let ϕ_x, ϕ_y, and $\phi_{xy} = 0$ denote the endomorphisms on M/Q. Since y is non-zero, there is some non-zero $m \in M/Q$ such that $\phi_y m \neq 0$. Then, $0 = (\phi_x \circ \phi_y)(m) = \phi_x(\phi_y(m))$ so that ϕ_x is not injective. Since Q is primary in M, ϕ_x must be nilpotent, so $\phi_x^q = 0$ for some $q > 0$. Thus, $x^q \in \mathrm{Ann}(M/Q)$, so x is nilpotent in $A/\mathrm{Ann}(M/Q)$. Thus, $\mathrm{Ann}(M/Q)$ is a primary ideal of A.

By Exercise 4.20, $\mathrm{Ann}(M/Q) = (Q:M)$. Further, $r(\mathrm{Ann}(M/Q)) = r_M(Q)$, so $r_M(Q)$ is prime and $\mathrm{Ann}(M/Q) = (Q:M)$ is $r_M(Q)$-primary. □

(4.3) If Q_i are $\mathfrak{p}$-primary, then $Q = \bigcap_{i=1}^n Q_i$ is $\mathfrak{p}$-primary.

(4.3) Proof. By Exercise 4.20, we have

$$r_M(Q) = \bigcap_{i=1}^n r_M(Q_i) = \bigcap_{i=1}^n \mathfrak{p} = \mathfrak{p},$$

so we just have to show that Q is primary.

Suppose $x \in A$ is a zero-divisor on M/Q. Then, there is some $m \in M - Q$ such that $xm \in Q \subset Q_i$ for all i. Since $m \notin Q$, there is some j such that $m \notin Q_j$. Since Q_j is primary in M and x is a zero-divisor on M/Q_j, it is nilpotent on M/Q_j. Thus, $x \in r(\mathrm{Ann}(M/Q_j)) = \mathfrak{p} = r_M(Q)$, so x is nilpotent on M/Q. Thus, x is nilpotent as an endomorphism of M/Q. Finally, Q is a $\mathfrak{p}$-primary submodule of M. □

(4.4) (i) Let $N \subset M$ be A-modules and $m \in N$. Then, $(N : m) = (1)$; (ii) Let $Q \subset M$ be a $\mathfrak{p}$-primary submodule, and $m \in M$. If $m \notin Q$, then $(Q : m)$ is $\mathfrak{p}$-primary; (iii) Let $Q \subset M$ be a $\mathfrak{p}$-primary submodule, and $x \in A$. If $x \notin \mathfrak{p}$, then $(Q : x) := \{m \in M : xm \in Q\} = Q$.

(4.4) Proof. (i) If $m \in N$, then since N is an A-module, we have $Am \subset N$, so $(N : m) = (1)$.

(ii) Note that $(Q : m)$ is an ideal of A. Suppose $x, y \in A$ and $xy \in (Q : m)$. Then, $(xy)m \in Q$, so xy is a zero-divisor on M/Q. Since Q is primary in M, $(xy)^q \in \mathrm{Ann}(M/Q)$ for some $q > 0$. Thus, $xy \in r(\mathrm{Ann}(M/Q)) = r_M(Q) = \mathfrak{p}$ by Exercise 4.20. Since $\mathfrak{p}$ is prime, $x \in \mathfrak{p}$ or $y \in \mathfrak{p}$. Without loss of generality, suppose $y \in \mathfrak{p}$. Then, since $\mathfrak{p} = r_M(Q)$, there is some $q > 0$ such that $y^q M \subset Q$, so $y^q m \in Q$. In other words, $y^q \in (Q : m)$, so $(Q : m)$ is primary.

Note that $\mathrm{Ann}(M/Q) \subset (Q : m)$, so $r(\mathrm{Ann}(M/Q)) \subset r(Q : m)$. Suppose $y \in r(Q : m)$ so that $y^q \in (Q : m)$ for some $q > 0$. Then, $y^q m \in Q$, so y^q is a zero-divisor on M/Q. Since Q is primary in M, there is some $n > 0$ such that $(y^q)^n = y^{qn} \in \mathrm{Ann}(M/Q)$, so $y \in r(\mathrm{Ann}(M/Q))$. Thus, $r(Q : m) = r(\mathrm{Ann}(M/Q)) = \mathfrak{p}$, so $(Q : m)$ is a $\mathfrak{p}$-primary ideal.

(iii) Suppose $x \notin \mathfrak{p}$. First of all, it's clear that $Q \subset (Q : x)$. Conversely, suppose to the contrary that there exists some $m \in (Q : x) - Q$. Then, $xm \subset Q$ while $m \notin Q$, so x is a zero-divisor on M/Q. Since Q is primary in M, $x^q \in \mathrm{Ann}(M/Q)$ for some $q > 0$. Thus, $x \in r(\mathrm{Ann}(M/Q)) = \mathfrak{p}$, a contradiction. Hence, $(Q : x) = Q$. □

4.22 Exercise

(4.5) Proof. Let $N = Q_1 \cap \cdots \cap Q_n$ be a minimal primary decomposition of N, and let $\mathfrak{p}_i = r_M(Q_i)$. For any $m \in M$, we have

$$r(N : m) = r\left(\bigcap_{i=1}^{n} Q_i : m\right) = \bigcap_{i=1}^{n} r(Q_i : m) = \bigcap_{i : m \notin Q_i} \mathfrak{p}_i.$$

If $\mathfrak{p} = r(M : n)$ is prime, then $\mathfrak{p} = \bigcap_{i:m\notin Q_i} \mathfrak{p}_i$ implies by Proposition 1.11 that $\mathfrak{p} = \mathfrak{p}_j$ for some j. Conversely, since we have a minimal decomposition, we can choose $m_j \in \bigcap_{i \neq j} Q_i - Q_j$. Then, by the calculation above, $r(N : m) = \bigcap_{i:m\notin Q_i} \mathfrak{p}_i = \mathfrak{p}_j$. Thus, the primes belonging to N in M are exactly the prime ideals of the form $r(N : m)$, $m \in M$. In particular, they depend only on N and M, not on the specific decomposition.

Again, let $N = Q_1 \cap \cdots \cap Q_n$ be a minimal primary decomposition of N. Then, since $N \subset Q_i$, we have

$$0 = N/N = (Q_1 \cap \cdots \cap Q_n)/N = (Q_1/N) \cap \cdots \cap (Q_n/N).$$

Since $M/Q_i \cong (M/N)/(Q_i/N)$, Q_i/N is a primary submodule of M/N, so this is a primary decomposition of 0. Now, note that

$$r_{M/N}(Q_i/N) = r(\mathrm{Ann}((M/N)/(Q_i/N))) = r(\mathrm{Ann}(M/Q_i)) = \mathfrak{p}_i.$$

Thus, the primes belonging to 0 in M/N are also primes belonging to N in M. Conversely, if $\mathfrak{p}$ is a prime belonging to N in M, then as above, there is some $m \in M$ such that $\mathfrak{p} = r(N : m)$. But

$$x \in r(N : m) \iff \exists q > 0, x^q m \in N \iff \exists q > 0, x^q m = 0 \in M/N \iff x \in r(0 : m + N),$$

so we have $\mathfrak{p} = r(N : m) = r(0 : m + N)$. By the preceding paragraph, $\mathfrak{p}$ belongs to 0 in M/N. □

4.23 Exercise

(4.6) Let N be a decomposable submodule of M. Then any prime ideal $\mathfrak{p} \supset r_M(N)$ contains a minimal prime ideal belonging to N. Thus, the minimal prime ideals of N are precisely the minimal elements in the set of all prime ideals containing $r_M(N)$.

(4.6) Proof. Suppose $N = \bigcap_i Q_i$ is a minimal primary decomposition. By Exercise 4.20, $r_M(N) = \bigcap_i r_M(Q_i) = \bigcap_i \mathfrak{p}_i$. By Proposition 1.11, since $\mathfrak{p} \supset r_M(N) = \bigcap_i \mathfrak{p}_i$, we have $\mathfrak{p} \supset \mathfrak{p}_i$ for some i. Clearly, $\mathfrak{p}_i$ contains some minimal element $\mathfrak{p}_j$ in the set of associated primes (possibly itself), so $\mathfrak{p} \supset \mathfrak{p}_i \supset \mathfrak{p}_j$. Thus, $\mathfrak{p}$ contains a minimal prime belonging to N in M. In particular, the minimal elements in the set of prime ideals over $r_M(N)$ are exactly the minimal primes of $\mathrm{Ass}(M/N)$. □

(4.7) Let N be a decomposable submodule of M so that we have a minimal primary decomposition $N = \bigcap_{i=1}^n Q_i$. Letting $\mathfrak{p}_i = r_M(Q_i)$, we have

$$\bigcup_{i=1}^{n} \mathfrak{p}_i = \{x \in A : (N : x) \neq N\}.$$

In particular, if $0 \subset M$ is decomposable, the set D of zero-divisors of M is the union of prime ideals belonging to 0.

(4.7) Proof. By Exercise 4.22, the primes belonging to N in M is the same as the primes belonging to 0 in M/N; further, for $x \in A$, $(N : x) \neq N \iff (0 : x) \neq 0 \subset M/N$. Thus, we've reduced the proof to the case $N = 0$.

Since we're in the case $N = 0$, we have

$$\{x \in A : (0 : x) \neq 0\} = D,$$

the set of zero-divisors in A on M, i.e. $x \in D$ when there exists some $m \neq 0$ such that $xm = 0$. If $x \in r(D)$, then there exists some non-zero $m \in M$ and some minimal $n > 0$ such that $x^n m = x(x^{n-1}m) = 0$; since n was chosen to be minimal, $x^{n-1}m \neq 0$, so $x \in D$. Thus,

$$D = r(D) = r\left(\bigcup_{m \neq 0} (0 : m)\right) = \bigcup_{m \neq 0} r(0 : m).$$

But by Exercise 4.22, each $r(0 : m)$ is the intersection of primes associated to 0 in M. Since by Exercise 4.22 all such primes have the form $r(0 : m)$ for some $m \in M$, we see that

$$D = \bigcup_{m \neq 0} r(0 : m) = \bigcup_{\mathfrak{p} \in \mathrm{Ass}(M)} \mathfrak{p},$$

as desired. □

(4.8) Let S be a multiplicatively closed subset of A, and let Q be a $\mathfrak{p}$-primary submodule of M. (i) If $S \cap \mathfrak{p} \neq \emptyset$, then $S^{-1}Q = S^{-1}M$. (ii) If $S \cap \mathfrak{p} = \emptyset$, then $S^{-1}Q$ is a $S^{-1}\mathfrak{p}$-primary submodule of $S^{-1}M$, and its preimage under the canonical map $M \to S^{-1}M$ is Q. Hence, primary $S^{-1}A$-submodules of $S^{-1}M$ correspond to primary A-submodules of M.

(4.8) Proof. (i) Let $s \in S \cap \mathfrak{p}$. Since $\mathfrak{p} = r_M(Q)$, there is some $n > 0$ such that $s^n M \subset Q$. Thus, for any $m/t \in S^{-1}M$, we have $m/t = s^n m/s^n t \in S^{-1}Q$, so $S^{-1}Q = S^{-1}M$.

(ii) Let $x/s \in S^{-1}A$ be a zero-divisor in $S^{-1}M/S^{-1}Q$ so that there is some non-zero $\overline{m/t} \in S^{-1}M/S^{-1}Q$ such that $(x/s)\overline{m/t} = \overline{xm/st} = 0$. Then, $xm/st \in S^{-1}Q$, so there is some $u \in S$ such that $uxm \in Q$. Since $\overline{m/t} \neq 0$, we have $m \notin Q$, so ux is a zero-divisor on M/Q. Since Q is primary in M, there is some $q > 0$ such that $(ux)^q M \subset Q$. Thus, $x^q S^{-1}M = x^q u^q S^{-1}M \subset S^{-1}Q$, so x is nilpotent in $S^{-1}M/S^{-1}Q$. Thus, $S^{-1}Q$ is primary in $S^{-1}M$.

Now, suppose $x \in \mathfrak{p} = r_M(Q)$, so there is some $n > 0$ such that $x^n M \subset Q$. For any $s \in S$, we then have $(x/s)^n S^{-1}M \subset S^{-1}Q$, so $x/s \in r(S^{-1}Q : S^{-1}M) = r_{S^{-1}M}(S^{-1}Q)$. Conversely, if $x/s \in r_{S^{-1}M}(S^{-1}Q)$, then $(x/1)^n S^{-1}M = (x/s)^n S^{-1}M \subset S^{-1}Q$ for some $n > 0$. Then, for every $m/t \in S^{-1}M$, we have $x^n m/t \in S^{-1}Q$. Thus, there is some $u \in S$ such that $ux^n m \in Q$ for all $m \in M$, so $ux^n \in \mathrm{Ann}(M/Q) \subset r(\mathrm{Ann}(M/Q)) = \mathfrak{p}$. Since $\mathfrak{p}$ is prime, $S \cap \mathfrak{p} = \emptyset$, and $u \in S$, we have $u \notin S$ so that $x^n \in \mathfrak{p}$. Then, $x \in \mathfrak{p}$ by a standard inductive argument, so $x/s \in S^{-1}\mathfrak{p}$. Finally, $r_{S^{-1}M}(S^{-1}Q) = S^{-1}\mathfrak{p}$, so $S^{-1}Q$ is a $S^{-1}\mathfrak{p}$-primary submodule of $S^{-1}M$.

If $m \in M$ such that $m/1 \in S^{-1}Q$, then there is some $s \in S$ such that $sm \in Q$. By (4.7), $\mathfrak{p} = \{x \in A : (Q : x) \neq Q\}$. Since $S \cap \mathfrak{p} = \emptyset$ and $s \in S$, $(Q : s) = Q$, so $m \in Q$.

To see that every $S^{-1}A$-submodule N' of $S^{-1}M$ is an extended module, let $N = \{n \in M : n/1 \in N'\}$. If $n/s \in N'$, then $n/1 = s(n/s) \in N'$, so $n \in N$ and hence $n/s \in S^{-1}N$. Conversely, if $n/s \in S^{-1}N$, then $n \in N$ by definition, so $n/1 \in N'$. Then, $n/s = (1/s)(n/1) \in N'$, so $N' = S^{-1}N$. □

(4.9) Let S be a multiplicatively closed subset of A and let N be a decomposable submodule of M. Let $N = \bigcap_{i=1}^n Q_i$ be a minimal primary decomposition of N. Let $\mathfrak{p}_i = r_M(Q_i)$, and suppose Q_i is numbered such that S meets $\mathfrak{p}_{j+1}, \ldots, \mathfrak{p}_n$ but none of $\mathfrak{p}_1, \ldots, \mathfrak{p}_j$. Then,

$$S^{-1}N = \bigcap_{i=1}^{j} S^{-1}Q_i, \qquad S(N) = \bigcap_{i=1}^{j} Q_i,$$

and these are minimal primary decompositions. Note that $S(N) = \{m \in M : m/1 \in S^{-1}N\}$.

(4.9) Proof. By Proposition 3.4, we have $S^{-1}N = \bigcap_{i=1}^n S^{-1}Q_i$. By (4.8), $S^{-1}Q_i = S^{-1}M$ for $i > j$, so $S^{-1}N = \bigcap_{i=1}^j S^{-1}Q_i$. Thus, we just have to show that this is a minimal primary decomposition. By (4.8), $S^{-1}Q_i$ is $S^{-1}\mathfrak{p}_i$-primary for $i \leq j$. By Proposition 3.11, since $\mathfrak{p}_i$ are distinct for $i \leq j$ and $\mathfrak{p}_i \cap S = \emptyset$, we have that $S^{-1}\mathfrak{p}_i$ are all distinct for $i \leq j$. Now, suppose for some $k \leq j$, $S^{-1}Q_k \supset \bigcap_{i=1:i\neq k}^j S^{-1}Q_i$, then taking the inverse image of $f : M \to S^{-1}M$, we have by (4.8) that $Q_k \supset \bigcap_{i=1:i\neq k}^j Q_i \supset \bigcap_{i=1:i\neq k}^n Q_i$, violating the minimality of our original decomposition. Thus, $S^{-1}N = \bigcap_{i=1}^j S^{-1}Q_i$ is a minimal primary decomposition.

Taking inverse images of $f : M \to S^{-1}M$ again and using (4.8) twice, we have

$$S(N) = f^{-1}(S^{-1}N) = f^{-1}\left(\bigcap_{i=1}^{j} S^{-1}Q_i\right) = \bigcap_{i=1}^{j} f^{-1}(S^{-1}Q_i) = \bigcap_{i=1}^{j} Q_i.$$

This is clearly a minimal primary decomposition since $N = \bigcap_{i=1}^n Q_i$ is one. □

(4.10) Let N be a decomposable submodule of M, and let $N = \bigcap_{i=1}^n Q_i$ be a minimal primary decomposition of N, where $\mathfrak{p}_i = r_M(Q_i)$. Let Σ be a subset of the isolated primes of N. Then, $\bigcap_{\mathfrak{p}\in\Sigma} Q_i$ is independent of the decomposition.

(4.10) Proof. Let $S = A - \bigcup_{\mathfrak{p}\in\Sigma} \mathfrak{p}$, which is a multiplicatively closed subset of A. If $\mathfrak{p} \in \Sigma$, then clearly, $\mathfrak{p} \cap S = \emptyset$. Conversely, suppose $\mathfrak{p}' \subset \bigcup_{\mathfrak{p}\in\Sigma} \mathfrak{p}$. By Proposition 1.11, $\mathfrak{p}' \subset \mathfrak{p}$ for some $\mathfrak{p} \in \Sigma$. Since every $\mathfrak{p} \in \Sigma$ is isolated, we must have $\mathfrak{p}' = \mathfrak{p}$; hence, $\mathfrak{p} \in \Sigma \iff \mathfrak{p} \subset \bigcup_{\mathfrak{p}\in\Sigma} \mathfrak{p} \iff \mathfrak{p} \cap S = \emptyset$. Then, by (4.9), $S(N) = \bigcap_{\mathfrak{p}\in\Sigma} Q_i$; in particular, $\bigcap_{\mathfrak{p}\in\Sigma} Q_i$ is independent of the decomposition. □

(4.11) The isolated primary components of N are uniquely determined by N.

(4.11) Proof. Let $\mathfrak{p}_i$ be any isolated prime; $\Sigma = \{\mathfrak{p}_i\}$ satisfies the conditions of (4.10), so letting $S = A - \mathfrak{p}_i$, if Q_i is the $\mathfrak{p}_i$-primary component of a minimal primary decomposition of N, we obtain $S(N) = Q_i$, so Q_i is uniquely determined by N and $\mathfrak{p}_i$. □

5 Integral Dependence and Valuations (Done)

5.1 Exercise

Proof. Note that by Exercise 1.21, since f is surjective onto its image, $\mathrm{Spec}(A)$ is homeomorphic to $\mathrm{Spec}(f(A))$. Since f is an integral homomorphism, B is an $f(A)$-algebra and is integral over $f(A)$. To see that $f^* : \mathrm{Spec}(A) \to \mathrm{Spec}(B)$ is a closed mapping, it suffices then to consider f as an inclusion $A \hookrightarrow B$.

A closed set of $\mathrm{Spec}(B)$ has the form $V(\mathfrak{b})$ for some ideal $\mathfrak{b}$ of B. By Exercise 1.21(iii), $\overline{f^*(V(\mathfrak{b}))} = V(\mathfrak{b}^c)$ so that $f^*(V(\mathfrak{b})) \subset V(\mathfrak{b}^c)$. Note also that $\mathfrak{b}^c = \mathfrak{b} \cap A$ since $A \subset B$. By Proposition 5.6, since B is integral over A, $B/\mathfrak{b}$ is integral over $A/\mathfrak{b}^c$. Now, suppose that $\mathfrak{p} \in V(\mathfrak{b}^c)$. Then, $\overline{\mathfrak{p}} \in \mathrm{Spec}(A/\mathfrak{b}^c)$, so by Theorem 5.10, there is some $\overline{\mathfrak{q}} \in \mathrm{Spec}(B/\mathfrak{b})$ such that $\overline{\mathfrak{q}} \cap A/\mathfrak{b}^c = \overline{\mathfrak{p}}$. In particular, $\mathfrak{q} \supset \mathfrak{b}$ and $\mathfrak{q} \cap A = \mathfrak{p}$. Thus, $\mathfrak{p} = f^{-1}(\mathfrak{q}) \in f^*(V(\mathfrak{b}))$. Hence, $f^*(V(\mathfrak{b})) = V(\mathfrak{b}^c)$ so that f^* is a closed mapping. □

5.2 Exercise

Proof. Since Ω is an integral domain, $f(A) \cong A/\ker(f)$ is also an integral domain. Thus, $\mathfrak{p} := \ker(f)$ is a prime ideal. By Theorem 5.10, there is a prime ideal $\mathfrak{q}$ of B such that $\mathfrak{q} \cap A = \mathfrak{p}$. By Proposition 5.6, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. In particular, it suffices to prove the result when $A \subset B$ are integral domains with B integral over A and $f : A \to \Omega$ an injection.

Let Σ be the set of pairs (C, σ), where $A \subset C \subset B$ and $\sigma : C \to \Omega$ is an embedding such that σ restricts to f on A. We can put a partial ordering on Σ by letting $(C, \sigma) \leq (C', \sigma')$ if $C \subset C'$ and σ' restricts to σ on C. Then, Σ is non-empty since it contains (A, f). If we have a chain (C_i, σ_i) of Σ, then $C = \bigcup_i C_i \subset B$ and σ is defined on C by letting $\sigma(c) = \sigma_i(c)$ when $c \in C_i$; $c \in C$ implies $c \in C_i$ for some i, and this is well-defined by the compatibility requirement on restriction. Note that σ is injective; if $\sigma(a) = \sigma(b)$, then since we're in a chain, they are identical on some common i, implying $a = b$ by the injectivity of each σ_i. Thus, $(C, \sigma) \in \Sigma$, so by Zorn's lemma, Σ has maximal elements. If we have a maximal element of the form (B, σ), we are done, so suppose we have a maximal element (C, σ), where $C \neq B$.

Suppose to the contrary that there is some $b \in B - C$; by assumption, b is integral over A and hence over C. Let b satisfy $p(x) = \sum_i c_i x^i \in C[x]$. Write $(\sigma p)(x) = \sum_i \sigma(c_i) x^i \in \Omega[x]$. Then, the composition

$$C[x] \to \Omega[x] \to \Omega[x]/((\sigma p)(x)) \cong \Omega$$

has kernel $(p(x))$, so we have an injection $C[x]/(p(x)) \to \Omega$ restricting to σ on C. But $C[x]/(p(x)) \cong C[b]$, and by assumption $C \subset C[b] \subset B$, where the first inclusion is proper. Thus, the existence of the induced map $C[b] \to \Omega$ violates the maximality of σ. □

5.3 Exercise

Proof. Consider simple elements of $B' \otimes_A C$, i.e. elements of the form $b' \otimes c$, where $b' \in B'$ and $c \in C$. Since B' is integral over $f(B)$ by assumption, there is some $p(x) = x^n + f(b_{n-1})x^{n-1} + \cdots + f(b_0)$

with $b_i \in B$ such that $p(b') = 0$. Thus,

$$\begin{aligned}
0 &= p(b') \\
0 &= p(b') \otimes c^n \\
0 &= (b')^n \otimes c^n + f(b_{n-1})(b')^{n-1} \otimes c^n + \cdots + f(b_0) \otimes c^n \\
0 &= (b' \otimes c)^n + (f(b_{n-1}) \otimes c^1)(b' \otimes c)^{n-1} + \cdots + f(b_0) \otimes c^n.
\end{aligned}$$

Thus, $b' \otimes c$ is integral over $f(B) \otimes C$, so every simple element of $B' \otimes C$ is integral over $f(B) \otimes C$.

In general, an element $x \in B' \otimes C$ is a finite sum of simple elements of $B' \otimes C$. By Corollary 5.3, the elements of $B' \otimes C$ integral over $f(B) \otimes C$ form a subring of $B' \otimes C$. Since x is a finite sum of simple elements, each of which is integral over $f(B) \otimes C$, x is itself integral over $f(B) \otimes C$. Thus, $f \otimes 1 : B \otimes_A C \to B' \otimes_A C$ is integral whenever f is integral. □

5.4 Exercise

No, if B is integral over a subring A, $\mathfrak{n}$ is a maximal ideal of B, and $\mathfrak{m} = \mathfrak{n} \cap A$ is maximal in A, $B_\mathfrak{n}$ is not necessarily integral over $A_\mathfrak{m}$.

Let $A = k[x^2 - 1]$, $B = k[x]$, where k is a characteristic 0 field. Then, $f(t) = t^2 - ((x^2 - 1) + 1)$ shows that $x \in k[x]$ is integral over $k[x^2 - 1]$. The integral elements form a subring, so since everything in $k[x]$ is finitely generated by k and x through ring operations, $k[x]$ is integral over $k[x^2 - 1]$.

Now, let $\mathfrak{n} = (x - 1)$ in B. We have a k-algebra surjective homomorphism $f : k[x] \to k$ by $x \mapsto 1$, and $(x - 1)$ is the kernel, so $k[x]/(x - 1) \cong k$, a field. Thus, $\mathfrak{n}$ is maximal. Note that $(x^2 - 1) \subset k[x^2 - 1] \cap (x - 1)$, but this ideal is maximal by the same argument, so $\mathfrak{m} = (x^2 - 1) = (x - 1) \cap k[x^2 - 1]$ is maximal.

Note that $x + 1 \in B - (x - 1)$, so $\frac{1}{x+1} \in B_\mathfrak{n}$. Suppose $\frac{1}{x+1}$ is integral over $A_\mathfrak{m}$. Then,

$$0 = \left(\frac{1}{x+1}\right)^n + a_{n-1}\left(\frac{1}{x+1}\right)^{n-1} + \cdots + a_1\frac{1}{x+1} + a_0$$

for some $a_i \in A_\mathfrak{m}$. Then, multiplying by $(x + 1)^n$, we obtain

$$0 = 1 + a_{n-1}(x+1)^1 + \cdots + a_1(x+1)^{n-1} + a_0(x+1)^n.$$

Since $a_i \in A_\mathfrak{m}$ for each i, we can write $a_i = g_i(x)/h_i(x)$, where $g_i(x) \in k[x^2 - 1]$ and $h_i(x) \in k[x^2-1]-(x^2-1)$. In particular, $h_i(-1)$ is the constant term of $h_i(x)$ since $(-1)^2-1 = 0$. However, $h_i(x) \notin (x^2 - 1)$, so the constant term cannot be zero. Thus, for all i, we have a well-defined k-algebra homomorphism $A_\mathfrak{m} \to k$ that sends $x \mapsto -1$; since $h_i(-1) \neq 0$, $g_i(-1)/h_i(-1) \in k$. Under this k-algebra homomorphism, we obtain

$$0 = 1 + \frac{g_{n-1}(-1)}{h_{n-1}(-1)}(-1+1)^1 + \cdots + \frac{g_1(-1)}{h_1(-1)}(-1+1)^{n-1} + \frac{g_0(-1)}{h_0(-1)}(-1+1)^n,$$

implying that $0 = 1$, a clear contradiction. Thus, $\frac{1}{x+1} \in B_\mathfrak{n}$ is not integral over $A_\mathfrak{m}$.

5.5 Exercise

(i) Proof. Since $x \in A$ is a unit in B, there is some $b \in B$ such that $xb = 1$. B is integral over A, so there is some polynomial of minimal degree $n > 0$ such that

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1 b + a_0 = 0,$$

where $a_i \in A$. Assume to the contrary that $n > 1$. Multiplying by x, we obtain

$$(xb)b^{n-1} + a_{n-1}(xb)b^{n-2} + \cdots + a_1(xb) + a_0x = 0,$$

but since $xb = 1$,

$$b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_2b + (a_1 + a_0x) = 0.$$

Since $x \in A$, $a_1 + a_0x \in A$, so this is a monic polynomial of b (since $n > 1$) with coefficients in A. Since we chose $n > 1$ minimal, we clearly have a contradiction, so $n = 1$. In particular, b satisfies a monic polynomial of the form $f(t) = t - a = 0$, where $a \in A$, so $b = a \in A$, as desired. □

(ii) Proof. Let $\mathfrak{R}_B$ and $\mathfrak{R}_A$ be the Jacobson radicals of B and A respectively. We wish to show that $A \cap \mathfrak{R}_B = \mathfrak{R}_A$. We will make use of Proposition 1.9, that $x \in \mathfrak{R} \iff 1 - xy$ is a unit in R for all $y \in R$.

If $x \in \mathfrak{R}_B \cap A$, then $x \in A$ and $1 - xy$ is a unit in B for all $y \in B$. Since $A \subset B$, $1 - xy$ is a unit in B for all $y \in A$. In this case, $1 - xy \in A$, so by (i), since $1 - xy$ is a unit in B, it is also a unit in A. Thus, $1 - xy$ is a unit in A for all $y \in A$, so $x \in \mathfrak{R}_A$.

Conversely, suppose $x \notin \mathfrak{R}_B \cap A$. If $x \notin A$, then clearly $x \notin \mathfrak{R}_A$, as desired. So, assume $x \in A$ so that $x \notin \mathfrak{R}_B$. As $\mathfrak{R}_B$ is the intersection of maximal ideals of B, there is some maximal ideal $\mathfrak{m}$ of B such that $x \notin \mathfrak{m}$. Since $\mathfrak{m}$ is prime, $\mathfrak{m}^c$ is also a prime ideal, so there is some maximal ideal $\mathfrak{m}_A$ of A such that $\mathfrak{m}_A \supset \mathfrak{m}_B^c$. By Theorem 5.11 ("Going-Up"), there is a prime ideal $\mathfrak{q}$ such that $\mathfrak{q} \supset \mathfrak{m}$ and $\mathfrak{q} \cap A = \mathfrak{m}_A$. Since $\mathfrak{m}$ is maximal, $\mathfrak{q} = \mathfrak{m}$, so $\mathfrak{m}_A = \mathfrak{m}^c$. Thus, $\mathfrak{m}^c$ is maximal. Because $\mathfrak{m}^c = A \cap \mathfrak{m}$, $x \notin \mathfrak{m}$ implies $x \notin \mathfrak{m}^c$. Then, since $\mathfrak{m}^c$ is maximal, $x \notin \mathfrak{R}_A$. □

5.6 Exercise

Proof. Since each B_i is an integral A-algebra, each B_i is integral over $f_i(A)$. Clearly, $f = (f_1, \ldots, f_n)$ makes $B := \prod_{i=1}^n B_i$ into an A-algebra. So, we must show that B is integral over $(\prod_{i=1}^n f_i)(A) = f(A)$. Let $b \in B$, and let b_i be the component in B_i under the canonical projection map. Since b_i is integral over A, b_i satisfies some monic polynomial

$$b_i^m + f_i(a_{m-1})b_i^{m-1} + \cdots + f_i(a_1)b_i + f_i(a_0).$$

We can then define a polynomial $g_i(x)$ on $f(A)[x]$ as

$$g_i(x) = x^m + f(a_{m-1})x^{m-1} + \cdots + f(a_1)x + f(a_0).$$

If $\pi_i : B \to B_i$ is the canonical projection map,

$$\pi_i(g_i(b)) = \pi_i(b^m + f(a_{m-1})b^{m-1} + \cdots + f(a_0)) = b_i^m + f_i(a_{m-1})b_i^{m-1} + \cdots + f_i(a_0) = 0.$$

Thus, for each i, we have some monic polynomial $g_i \in f(A)[x]$ such that $\pi_i(g_i(b)) = 0$. Let $g = g_1 \cdots g_n \in f(A)[x]$; note that g is monic since each factor is monic. Then, for all i,

$$\pi_i(g(b)) = \pi_i(g_1(b)) \cdots \pi_i(g_i(b)) \cdots \pi_i(g_n(b)) = \pi_i(g_1(b)) \cdots 0 \cdots \pi_i(g_n(b)) = 0,$$

so $g(b) = 0$. Hence, b is integral over $f(A)$, so B is an integral A-algebra. □

5.7 Exercise

Proof. Suppose that $b \in B - A$ is integral over A. Then, it satisfies a monic polynomial in $A[x]$, so let's choose a monic polynomial of minimal degree $n > 0$ such that

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1 b + a_0 = 0.$$

Since $b \notin A$ by assumption, $n > 1$ since if $b + a_0 = 0$, $b = -a_0 \in A$. Rearranging terms, we have

$$b(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1) = -a_0 \in A$$

Since $b \in B - A$ and $B - A$ is multiplicatively closed, we must have $b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1 \in A$; otherwise, both factors are in $B - A$, so $-a_0 \in B - A$, a contradiction. In particular, we have that $b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1 = a \in A$, so

$$b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_2 b + (a_1 - a) = 0,$$

where $a_1 - a \in A$. This violates the minimality of n, so $b \in B - A$ cannot be integral. Thus, A is integrally closed in B. □

5.8 Exercise

(i) Proof. Let $f, g \in B[x]$ be monic polynomials such that $fg \in C[x]$. Since C is the integral closure of A in B, C is integrally closed in B (Corollary 5.5). Since B is an integral domain, it is a subring of its field of fractions, K (Exercise 3.9). Thus, $C \subset B \subset K$, so $fg \in K[x]$. Let L be the splitting field of fg over K; since $K \subset L$, $fg \in L[x]$ and splits into linear factors over $L[x]$. Thus, $f(x) = \prod_{i=1}^{n}(x - \xi_i) \in L[x]$ and $g(x) = \prod_{j=1}^{m}(x - \eta_j) \in L[x]$. Since $\{\xi_i, \eta_j\}_{i,j}$ are roots of the polynomial $fg \in C[x]$, then $\{\xi_i, \eta_j\}_{i,j} \subset L$ are all integral over C.

Let E be the integral closure of C in L. Then, $\{\xi_i, \eta_j\}_{i,j} \subset E$. Since E is a subring of L (Corollary 5.3), the coefficients of $f(x) = \prod_{i=1}^{n}(x - \xi_i)$ and $g(x) = \prod_{j=1}^{m}(x - \eta_j)$ are all in E. Thus, $f(x), g(x) \in B[x] \cap E[x]$ (B and E viewed as subrings of L). Since C is integrally closed in B, $C = B \cap E$, so $B[x] \cap E[x] = C[x]$. As desired, $f(x), g(x) \in C[x]$. □

(ii) Proof. Let $f, g \in B[x]$ be monic polynomials such that $fg \in C[x]$.

First, we will show that there is a ring B' containing B such that f and g split into linear factors. Let $\deg(f) = n$. Let's extend B by first creating the ring $D = B[x]/(f(x))$. In D, f has the root $\overline{x}$, which we will call α. Now, consider the canonical map $D[y] \to D[y]/(y - \alpha)$. Since $B \subset D$, $f \in D[y]$. Note that $f(y) \mapsto \overline{f(y)} = \overline{f(\alpha)} = 0$, so $f(y) \in (y - \alpha)$. Thus, there is some polynomial $f_1 \in D[y]$ such that $f(y) = f_1(y)(y - \alpha)$. Suppose by^m is the term of highest degree of $f_1(y)$. Then, the term of highest degree of the right-hand product is by^{m+1}. The term of highest degree on the left-hand side is y^n, so $m = n - 1$ and $b = 1$. Thus, f_1 is a monic polynomial of degree $n - 1$. Repeating this process finitely, we obtain a ring D' containing B such that $f(x)$ splits into linear factors in $D[x]$. Repeat the process again for g, starting with the ring D. Then, we obtain a ring B' containing B such that f and g both split into linear factors in $B'[x]$.

In particular, $f(x) = \prod_{i=1}^{n}(x - \xi_i) \in B'[x]$ and $g(x) = \prod_{j=1}^{m}(x - \eta_j) \in B'[x]$. Note that $\{\xi_i, \eta_j\}_{i,j}$ are all roots of fg in B', so each of these elements is a member of the integral closure C' of C in B'. Since C' is a subring of B' (Corollary 5.3), the coefficients of the expansion of f and g are also in C'. Recall that by assumption the coefficients of f and g are in B. Thus, the coefficients are in B and are integral over C. However, C is integrally closed in B, so the coefficients of f and g are in C. Thus, $f, g \in C[x]$. □

Remark. *Note that we've already shown $x, y \in C$ implies $xy \in C$ and $x+y \in C$. Here, we obtain a converse. Consider the polynomial $(t-x)(t-y) \in C[t]$ since $xy, x+y \in C$. By this result, $t-x, t-y \in C[t]$, so $x, y \in C$.*

5.9 Exercise

Proof. Suppose $f \in B[x]$ is integral over $A[x]$ so that

$$f^n + g_{n-1}f^{n-1} + \cdots + g_0 = 0,$$

where $g_i \in A[x]$. Choose $r > \max\{n, \deg(g_i)\}_i$, and let $p = f - x^r$. Then,

$$(p + x^r)^n + g_{n-1}(p + x^r)^{n-1} + \cdots + g_0 = 0.$$

After expanding, we obtain a polynomial in p:

$$p^n + h_{n-1}p^{n-1} + \cdots + h_0 = 0,$$

where $h_0(x) = (x^r)^n + g_{n-1}(x^r)^{n-1} + \cdots + g_0 \in A[x]$ (since $g_i \in A[x]$ for all i by assumption). Since $p^n + h_{n-1}p^{n-1} + \cdots + h_1p^1 = -h_0 \in A[x] \subset C[x]$, we have

$$p^n + h_{n-1}p^{n-1} + \cdots + h_1p^1 = p(p^{n-1} + h_{n-1}p^{n-2} + \cdots + h_1) \in C[x].$$

By Exericse 5.8, $p \in C[x]$. Since $x^r \in C[x]$ and $f = p + x^r$, $f \in C[x]$. Thus, the integral closure of $A[x]$ in $B[x]$ is contained in $C[x]$.

Conversely, to show that any element of $C[x]$ is integral over $A[x]$, it is enough to consider monomials in $C[x]$ since any element of $C[x]$ is a sum of monomials (and the integral closure of $A[x]$ in $B[x]$ forms a ring). The monomials of $C[x]$ have the form cx^n, where $c \in C$ and $n \geq 0$. Note that c is integral over $A[x]$ since it is integral over A via the definition of C. Further, x^n is integral over $A[x]$ since $x^n \in A[x]$. Thus, the product cx^n is integral since the integral elements of $B[x]$ over $A[x]$ form a ring. Thus, $C[x]$ is the integral closure of $A[x]$ in $B[x]$. □

5.10 Exercise

(i) Proof. $(a) \implies (b)$. By induction, it is sufficient to prove the going-up property with primes $\mathfrak{p}_1 \subset \mathfrak{p}_2$ of $f(A)$ and the prime $\mathfrak{q}_1 = \mathfrak{p}_1 \cap f(A)$ of B. Since f^* is closed, we know by Exercise 1.21 that $f^*(V(\mathfrak{q}_1)) = \overline{f^*(V(\mathfrak{q}_1))} = V(\mathfrak{q}_1^c) = V(\mathfrak{p}_1)$. Hence, $\mathfrak{p}_2 \in V(\mathfrak{p}_1)$, so there is some $\mathfrak{q}_2 \in V(\mathfrak{q}_1)$ such that $\mathfrak{q}_2^c = f^*(\mathfrak{q}_2) = \mathfrak{p}_2$. Thus, $\mathfrak{q}_2 \supset \mathfrak{q}_1$ and $\mathfrak{p}_2 = \mathfrak{q}_2^c = \mathfrak{q}_2 \cap f(A)$, as desired.

$(b) \implies (c)$. Now, suppose f has the going-up property; let $\mathfrak{q}$ be a prime ideal of B and let $\mathfrak{p} = \mathfrak{q}^c$. By Exercise 3.21, the map $f^* : \operatorname{Spec}(B/\mathfrak{q}) \to \operatorname{Spec}(A/\mathfrak{p})$ is equivalent to the map $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ restricted to $f^* : V(\mathfrak{q}) \to V(\mathfrak{p})$. Thus, to show surjectivity, we just have to show that $\mathfrak{p}' \supset \mathfrak{p}$ implies that there is some $\mathfrak{q}' \supset \mathfrak{q}$ such that $\mathfrak{p}' = f^*(\mathfrak{q}') = \mathfrak{q}'^c$. But this is exactly the going-up property, so f^* is indeed surjective.

$(c) \implies (b)$. Conversely, to show the going-up property, it's sufficient (by induction) to prove it for primes $\mathfrak{p}' \supset \mathfrak{p}$ of A and $\mathfrak{q}$ of B such that $\mathfrak{p} = \mathfrak{q}^c$. Since we're assuming $f^* : \operatorname{Spec}(B/\mathfrak{q}) \to \operatorname{Spec}(A/\mathfrak{p})$ is surjective, we have $f^*(V(\mathfrak{q})) = V(\mathfrak{p})$ by Exercise 3.21. Since $\mathfrak{p}' \in V(\mathfrak{p})$, there is some $\mathfrak{q}' \in V(\mathfrak{q})$ such that $\mathfrak{q}'^c = f^*(\mathfrak{q}) = \mathfrak{p}'$, so the going-up property holds. □

(ii) Proof. $(a') \implies (c')$. Let $\mathfrak{q} \in \operatorname{Spec}(B)$, and let $\mathfrak{q}^c = \mathfrak{p} \in \operatorname{Spec}(A)$. Note that $B_\mathfrak{q}$ is the direct limit of B_t as t ranges over $B - \mathfrak{q}$. Thus, by Exercise 3.26, we have

$$f^*(\operatorname{Spec}(B_\mathfrak{q})) = \bigcap_t f^*(\operatorname{Spec}(B_t)) = \bigcap_t f^*(Y_t),$$

where Y_t are open neighborhoods of $\operatorname{Spec}(B)$. Since f^* is an open mapping, $f^*(Y_t)$ are open neighborhoods of $\mathfrak{p}$ in $\operatorname{Spec}(A)$, so they each contain $\operatorname{Spec}(A_\mathfrak{p})$. Thus, $f^* : \operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$ is surjective.

$(c') \implies (b')$. By a simple inductive argument, it is enough to show the going-down property for primes $\mathfrak{p}' \subset \mathfrak{p}$ of A and for $\mathfrak{q} \in \operatorname{Spec}(B)$ such that $\mathfrak{p} = \mathfrak{q}^c$. Note that both $\overline{\mathfrak{p}}', \overline{\mathfrak{p}}$ are in $\operatorname{Spec}(A_\mathfrak{p})$ by Proposition 3.11. Also, $\overline{\mathfrak{q}} \in \operatorname{Spec}(B_\mathfrak{q})$. Since $f^* : \operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$ is surjective, there is some $\overline{\mathfrak{q}}' \in \operatorname{Spec}(B_\mathfrak{q})$ such that $f^*(\overline{\mathfrak{q}}') = \overline{\mathfrak{p}}'$. Again by Proposition 3.11, there is some $\mathfrak{q}' \in \operatorname{Spec}(B)$ with projection $\overline{\mathfrak{q}}'$. By the same proposition, $\mathfrak{q}'$ does not meet $B - \mathfrak{q}$, so $\mathfrak{q}' \subset \mathfrak{q}$. Since

$$\begin{array}{ccc} \operatorname{Spec}(B) & \longrightarrow & \operatorname{Spec}(A) \\ \downarrow & & \downarrow \\ \operatorname{Spec}(B_\mathfrak{q}) & \longrightarrow & \operatorname{Spec}(A_\mathfrak{p}) \end{array}$$

is commutative and $\mathfrak{q}' \mapsto \overline{\mathfrak{p}}'$, $f^*(\mathfrak{q}') \in \operatorname{Spec}(A)$ must be precisely $\mathfrak{p}'$ via Proposition 3.11. Thus, the going-down property holds.

$(b') \implies (c')$. Let $\mathfrak{q}$ be an ideal of B, and let $\mathfrak{p} = \mathfrak{q}^c \in \operatorname{Spec}(A)$. Now, consider $f^* : \operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$. Select some arbitrary $\overline{\mathfrak{p}}' \in \operatorname{Spec}(A_\mathfrak{p})$. By Proposition 3.11, there exists some prime ideal $\mathfrak{p}'$ of A that maps to $\overline{\mathfrak{p}}'$, justifying the notation. Again by Proposition 3.11, $\mathfrak{p}'$ does not meet $A - \mathfrak{p}$, so $\mathfrak{p}' \subset \mathfrak{p}$. By the going-down property, there is some $\mathfrak{q}' \subset \mathfrak{q}$ such that $\mathfrak{p}' = \mathfrak{q}'^c$. By the same commutative diagram from the previous paragraph, $f^*(\overline{\mathfrak{q}}') = \overline{\mathfrak{p}}'$, so f^* is surjective. □

5.11 Exercise

Proof. Since $f : A \to B$ is a flat homomorphism of rings, Exercise 3.18 gives us the surjectivity of $\operatorname{Spec}(B_\mathfrak{q}) \to \operatorname{Spec}(A_\mathfrak{p})$. Then, by Exercise 5.10, f has the going-down property. □

5.12 Exercise

Proof. Note that we can extend the automorphism group G of A to $A[x]$ by defining $\sigma(x) = x$. Then clearly, $(A[x])^G = A^G[x]$ since for an arbitary element of $A[x]$, we have

$$\sigma\left(a_n x^n + \cdots + a_1 x + a_0\right) = \sigma(a_n)x^n + \cdots + \sigma(a_1)x + \sigma(a_0).$$

For $a \in A$, consider the element $p_a(x) = \prod_{\sigma \in G}(x - \sigma(a))$. Since at least one linear term is $(x - a)$ (G is a group, so it contains the trivial automorphism), $p_a(a) = 0$. Note that for any $\tau \in G$,

$$\tau(p_a(x)) = \tau\left(\prod_{\sigma \in G}(x - \sigma(a))\right) = \prod_{\sigma \in G}\tau(x - \sigma(a)) = \prod_{\sigma \in G}(x - (\tau\sigma)(a)) = \prod_{\sigma \in G}\tau(x - \sigma(a)) = p_a(x),$$

so $p_a(x) \in (A[x])^G = A^G[x]$. Lastly, $p_a(x)$ is clearly monic, so a is integral over A^G; hence, A is integral over A^G. □

Proof. Let S be a multiplicatively closed subset of A such that $\sigma(S) \subset S$ for all $\sigma \in G$, and let $S^G = S \cap A^G$. We wish to show that the action of G on A extends to an action on $S^{-1}A$. Since $S \subset A$, the obvious definition will be thus: for arbitrary $a/s \in S^{-1}A$, where $a \in A$ and $s \in S$, $\sigma(a/s) := \sigma(a)/\sigma(s)$. Since $\sigma(S) \subset S$, this action is closed. Now, we must show that σ is a ring homomorphism for all $\sigma \in G$. Note that $\sigma(1) = \sigma(1/1) = \sigma(1)/\sigma(1) = 1/1 = 1$, and $\sigma(0) = \sigma(0/1) = \sigma(0)/\sigma(1) = 0/1 = 0$. If $a/s, a'/s' \in S^{-1}A$, then

$$\sigma\left(\frac{a}{s}\frac{a'}{s'}\right) = \sigma\left(\frac{aa'}{ss'}\right) = \frac{\sigma(aa')}{\sigma(ss')} = \frac{\sigma(a)\sigma(a')}{\sigma(s)\sigma(s')} = \frac{\sigma(a)}{\sigma(s)}\frac{\sigma(a')}{\sigma(s')} = \sigma\left(\frac{a}{s}\right)\sigma\left(\frac{a'}{s'}\right)$$

and

$$\sigma\left(\frac{a}{s} + \frac{a'}{s'}\right) = \sigma\left(\frac{as' + a's}{ss'}\right) = \frac{\sigma(as' + a's)}{\sigma(ss')} = \frac{\sigma(a)\sigma(s') + \sigma(a')\sigma(s)}{\sigma(s)\sigma(s')}$$
$$= \frac{\sigma(a)}{\sigma(s)} + \frac{\sigma(a')}{\sigma(s')} = \sigma\left(\frac{a}{s}\right) + \sigma\left(\frac{a'}{s'}\right),$$

so indeed the elements of G are ring homomorphisms of $S^{-1}A$ onto itself. Compatibility is also easy to show:

$$\sigma\left(\tau\left(\frac{a}{s}\right)\right) = \sigma\left(\frac{\tau(a)}{\tau(s)}\right) = \frac{\sigma(\tau(a))}{\sigma(\tau(s))} = \frac{(\sigma\tau)(a)}{(\sigma\tau)(s)} = (\sigma\tau)\left(\frac{a}{s}\right).$$

Thus, G is a group of ring homomorphisms on $S^{-1}A$. Since G is a group, all elements are invertible, so the elements are actually ring automorphisms. □

Proof. Finally, we wish to show that $(S^G)^{-1}A^G \cong (S^{-1}A)^G$. For any $a \in A^G$, $\sigma(a/1) = \sigma(a)/1 = a/1$, so the sequence of canonical maps $A^G \to A \to S^{-1}A$ factors as $A^G \to (S^{-1}A)^G \to S^{-1}A$. If $s \in S^G \subset S$, then s becomes a unit in $S^{-1}A$, so it is a unit in $(S^{-1}A)^G$ since $1/s$ is G-fixed (as s is). In particular, we can invoke the universal property of localizations (Proposition 3.1) to obtain a unique homomorphism $\psi : (S^G)^{-1}A^G \to (S^{-1}A)^G$ that takes $a/s \mapsto a/s$.

To show that ψ is bijective, we start by showing it is injective. Suppose $a \in A^G, s \in S^G$, and $\psi(a/s) = 0$. Then, $a/s = 0/1$ in $(S^{-1}A)^G$, so there is some $t \in S$ such that $ta = 0$. Since S is G-stable and multiplicatively closed, and since G is finite, $t' = \prod_{\sigma\in G}\sigma(t) \in S^G$, and $t'a = 0$. Since $t' \in S^G$, we have in $(S^G)^{-1}A^G$ that $a/s = (at')/(st') = 0/(st') = 0$, so ψ is injective.

To show surjectivity, suppose $a/s \in (S^{-1}A)^G$. Let $s' = \prod_{\sigma\in G:\sigma\neq\mathrm{id}}\sigma(s)$ so that $ss' = \prod_{\sigma\in G}\sigma(s)$. By assumption, a/s is G-fixed, and clearly, $ss'/1$ is fixed since the action of G on the product ss' simply permutes the factors. Thus, their product, $(a/s)(ss'/1) = (as')/1$ is G-fixed. Thus, for all $\sigma \in G$, $as'/1 = \sigma(as'/1) = \sigma(as')/1$, so there is some $t_\sigma \in S$ such that $t_\sigma(as' - \sigma(as')) = 0$. Let $t = \prod_{\tau\in G}\tau\left(\prod_{\sigma\in G} t_\sigma\right) \in S^G$. Since t_σ is a factor of t, we have $\sigma(tas') = t\sigma(as') = tas'$. Thus, $a/s = (tas')/(ts's)$, where $tas' \in A^G$, and $ts's \in S^G$ since $t \in S^G$ and $s's \in S^G$. Thus, ψ is surjective. □

5.13 Exercise

Proof. Let $\mathfrak{p}_1, \mathfrak{p}_2 \in P$, the set of prime ideals of A whose contraction in A^G is $\mathfrak{p}$. Let $x \in \mathfrak{p}_1$. Then, $\prod_{\sigma\in G}\sigma(x) \in \mathfrak{p}_1 \cap A^G = \mathfrak{p} \subset \mathfrak{p}_2$ (note that the product is in $\mathfrak{p}_1$ simply because it contains x). Since $\mathfrak{p}_2$ is prime, we see by induction that one of the factors of the product lies in $\mathfrak{p}_2$, i.e. there is some $\sigma \in G$ such that $\sigma(x) \in \mathfrak{p}_2$. Thus, $\mathfrak{p}_1 \subset \bigcup_{\sigma\in G}\sigma(\mathfrak{p}_2)$. Since each element of G is a ring automorphism, $\sigma(\mathfrak{p}_2)$ is prime for all $\sigma \in G$. Since G is finite, the right-hand side is a finite union

of prime ideals, so since it contains the prime ideal $\mathfrak{p}_1$, we have by Proposition 1.11, that there is some $\sigma \in G$ such that $\mathfrak{p}_1 \subset \sigma(\mathfrak{p}_2)$. Note that

$$A^G \cap \sigma(\mathfrak{p}_2) = \sigma^{-1}(A^G \cap \sigma(\mathfrak{p}_2)) = A^G \cap (\sigma^{-1}\sigma)(\mathfrak{p}_2) = A^G \cap \mathfrak{p}_2 = \mathfrak{p},$$

so both $\mathfrak{p}_1$ and $\sigma(\mathfrak{p}_2)$ are primes having contraction $\mathfrak{p}$. Because $\mathfrak{p}_1 \subset \sigma(\mathfrak{p}_2)$, we have by Corollary 5.9 that $\mathfrak{p}_1 = \sigma(\mathfrak{p}_2)$. Note that we used the integrality of A over A^G established in Exercise 5.12. Hence, G acts transitively on P; in particular, P is finite since G is finite. □

5.14 Exercise

Proof. Let $\sigma \in G$, and let $b \in B$. B is the integral closure of A in L, so there is a monic polynomial $f \in A[x]$ such that $f(b) = 0$. Since $A \subset K$, $\sigma(A) = A$, so $0 = \sigma(f(b)) = f(\sigma(b))$. Thus, $B \subset \sigma(B)$. Since we chose σ arbitrarily, we also have $B \subset \sigma^{-1}(B)$, so $\sigma(B) \subset \sigma(\sigma^{-1}B) = B$; thus, $B = \sigma(B)$.

Since L/K is a finite, separable, normal extension, we have by elementary Galois theory that $L^G = K$. Thus, $B^G = L^G \cap B = K \cap B = A$ since A is integrally closed in K and B is the ring of integral elements of L over A. □

5.15 Exercise

Proof. A is an integrally closed domain, K is its field of fractions, L is a finite extension field of K, and B is the integral closure of A in L. Let L_{sep} be the maximal separable subfield of K in L, and let C be the integral closure of A in L_{sep}. Note that L/L_{sep} is purely inseparable, and L_{sep}/K is separable. To show that the fibers of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ are finite, it is clearly sufficient to show that the fibers of $\mathrm{Spec}(B) \to \mathrm{Spec}(C)$ and $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ are each finite.

Since L_{sep}/K is a separable extension, we can embed L_{sep} in a finite normal, separable extension E of K. Let G be the Galois group of E/K, and let D be the integral closure of E in A (note that $C \subset D$). By Exercise 5.14, $A = D^G$, so by Exercise 5.13, $\mathrm{Spec}(D) \to \mathrm{Spec}(D^G) = \mathrm{Spec}(A)$ has finite fibers. Since we can factor this as $\mathrm{Spec}(D) \to \mathrm{Spec}(C) \to \mathrm{Spec}(A)$, $\mathrm{Spec}(C) \to \mathrm{Spec}(A)$ has finite fibers.

Now, we must consider the purely inseparable extension L/L_{sep}. If the characteristic of L_{sep} is 0, $L_{sep} = L$ so that $C = B$. Clearly then, $\mathrm{Spec}(B) \to \mathrm{Spec}(C)$ has finite fibers. So, suppose the characteristic of L_{sep} is $p > 0$. Let $\mathfrak{p} \in \mathrm{Spec}(C)$, and let $\mathfrak{q} \in \mathrm{Spec}(B)$ be such that $\mathfrak{q} \cap C = \mathfrak{p}$. In a finite purely inseparable extension, every $\ell \in L$ has a power p^n ($n \geq 0$) such that $\ell^{p^n} \in L_{sep}$. Let $\ell \in \mathfrak{q} \subset B \subset L$. Then, for some $n \geq 0$, $\ell^{p^n} \in L_{sep}$. Since ℓ is integral over A, ℓ^{p^n} is integral over A (integral elements form a subring), so $\ell^{p^n} \in B \cap L_{sep} = C$. But $\ell \in \mathfrak{q}$, so $\ell^{p^n} \in \mathfrak{q} \cap C = \mathfrak{p}$. Since $\mathfrak{p}$ is prime, we have (by a simple inductive argument) that $\ell \in \mathfrak{p}$. Thus, $\mathfrak{q} = \mathfrak{p}$, so the fibers of $\mathrm{Spec}(B) \to \mathrm{Spec}(C)$ are clearly finite. □

5.16 Exercise

Let k be a field, and let $A \neq 0$ be a finitely generated k-algebra. Then there exist elements $y_1, \ldots, y_r \in A$ which are algebraically independent over k and such that A is integral over $k[y_1, \ldots, y_r]$. (We will assume that k is infinite, even though it remains true when k is finite).

Proof. A is finitely generated, so let $x_1, \ldots, x_n$ generate A as a k-algebra. We can renumber the x_i so that $x_1, \ldots, x_r$ are algebraically independent over k and each of $x_{r+1}, \ldots, x_n$ is algebraic over $k[x_1, \ldots, x_r]$. Now we proceed by induction on n. If $n = r$ there is nothing to do; this

is the base case: $k[x_1, \ldots, x_r]$ is clearly integral over itself. So suppose $j > r$ and assume the result is true for $j-1$ generators, i.e. $k[z_1, \ldots, z_{j-1}]$ is integral over $k[y_1, \ldots, y_r]$ for generators $z_1, \ldots, z_{j-1}$, where $y_1, \ldots, y_r$ are a collection of algebraically independent elements over k. The generator x_j is algebraic over $k[x_1, \ldots, x_{j-1}]$, so there is a non-zero polynomial $f \in k[x_1, \ldots, x_j]$ such that $f(x_1, \ldots, x_{j-1}, x_j) = 0$. Let F be the homogeneous part of highest degree in f. Since k is infinite, there exist $\lambda_1, \ldots, \lambda_{j-1} \in k$ such that $F(\lambda_1, \ldots, \lambda_{j-1}, 1) \neq 0$. Let $x_i' = x_i - \lambda_i x_j$ so that $x_i = \lambda_i x_j + x_i'$.

Now, we'll show that x_j is integral over $k[x_1', \ldots, x_{j-1}']$. Let $a = F(\lambda_1, \ldots, \lambda_{j-1}, 1) \neq 0$.

$$F(x_1, \ldots, x_{j-1}, x_j) = F(\lambda_1 x_j + x_1', \ldots, \lambda_{j-1} x_j + x_{j-1}', x_j) \in k[x_1', \ldots, x_{j-1}'][x_j].$$

Since $F \in k[x_1, \ldots, x_j]$ is homogeneous, it has some homogeneous degree $m > 0$. Since we replace the arguments $x_1, \ldots, x_j$ with homogeneous polynomials of degree 1, the polynomial remains homogeneous of degree m. Collecting terms in $k[x_1', \ldots, x_{j-1}'][x_j]$, we have

$$F(\lambda_1 x_j + x_1', \ldots, \lambda_{j-1} x_j + x_{j-1}', x_j) = F_0(x_1', \ldots, x_{j-1}') x_j^m + \cdots + F_m(x_1', \ldots, x_{j-1}'),$$

where $F_i \in k[x_1', \ldots, x_{j-1}']$ is homogeneous of degree i. Finally, observe that the term of highest degree satisfies

$$F_0 x_j^m = F(\lambda_1 x_j, \ldots, \lambda_{j-1} x_j, x_j) = F(\lambda_1, \ldots, \lambda_{j-1}, 1) x_j^m = a x_j^m \neq 0,$$

as desired. Then, the term of highest degree in x_j is $a^{-1} F(\lambda_1 x_j + x_1', \ldots, \lambda_{j-1} x_j + x_{j-1}', x_j)$ is x_j^m.

Remember that $f(x_1, \ldots, x_{j-1}, x_j) = 0$, so $a^{-1} f(x_1, \ldots, x_{j-1}, x_j) = 0$. Then, $0 = a^{-1} f(\lambda_1 x_j + x_1, \ldots, \lambda_{j-1} x_j + x_{j-1}', x_j) \in k[x_1', \ldots, x_{j-1}'][x_j]$. Finally, note that since we replaced arguments with degree 1 homogeneous polynomials, the highest degree terms are still m. As we just saw, the degree m terms are $a^{-1} F(\lambda_1 x_j + x_1', \ldots, \lambda_{j-1} x_j + x_{j-1}', x_j)$, so when we collect terms in $k[x_1', \ldots, x_{j-1}'][x_j]$, $a^{-1} f(x_1, \ldots, x_{j-1}, x_j)$ is monic in x_j, so x_j is integral over $k[x_1', \ldots, x_{j-1}']$.

In particular, $k[x_1, \ldots, x_j]$ is integral over $k[x_1', \ldots, x_{j-1}']$. By the inductive hypothesis, we have that $k[x_1', \ldots, x_{j-1}']$ is integral over $k[y_1, \ldots, y_r]$. By the transitivity of integrality, A is integral over $k[y_1, \ldots, y_r]$. By induction, A is integral over $k[y_1, \ldots, y_r]$ for algebraically independent elements $y_1, \ldots, y_r$. □

Remark. *This is "Noether's Normalization Theorem".*

There is a geometric interpretation if k is algebraically closed: If X is an affine algebraic variety in k^n with coordinate ring $A \neq 0$, then there exists a linear subspace L of dimension r in k^n and a linear mapping of k^n onto L which maps X onto L.

Proof. The coordinate ring is given by $P(X) = k[t_1, \ldots, t_n]/I(X)$, which is clearly finitely generated as a k-algebra. Indeed, $P(X) = k[x_1, \ldots, x_n]$, where x_i is the coordinate function associated to t_i. By the previous proof, the collection of $y_1, \ldots, y_r$ we constructed was a sequence of linear combinations of $x_1, \ldots, x_n$ and thus is itself a linear combination of the $x_1, \ldots, x_n$; here, we use the fact that k is infinite (it is algebraically closed).

For each i then, $y_i = \sum_{j=1}^n c_{ij} x_j$. Define $\varphi : k[y_1, \ldots, y_r] \to k[t_1, \ldots, t_n]$ to be the k-algebra homomorphism generated by $\varphi(y_i) = \sum_{j=1}^n c_{ij} t_j$. Let $y_i' = \varphi(y_i)$. Since $\{y_i\}$ are algebraically independent, $\{y_i'\}$ are also algebraically independent. Since y_i' are each linear combinations of $\{t_j\}$, we can find and addend $y_{r+1}', \ldots, y_n' \in k[t_1, \ldots, t_n]$ to $\{y_i'\}_{i=1}^r$ so that $k[t_1, \ldots, t_n] \cong k[y_1', \ldots, y_n']$ as k-algebras.

For every point $a = (a_1, \ldots, a_n) \in k^n$, we have a k-algebra homomorphism $\ell : k^n \to k^r$ given by $(y'_1(a), \ldots, y'_r(a))$. This is a k-linear map since each y'_i is a k-linear combination of t_j, so $y'_i(a) = \sum_{j=1}^n c_{ij}a_j$. Let L be the r-dimensional subvariety of k^n defined by $(y'_{r+1} = 0, \ldots, y'_n = 0)$. This is a linear subspace, as $y'_i = 0$ represents a linear equation in terms of $\{t_j\}$.

Consider the canonical k-algebra homomorphism $\pi : k[y'_1, \ldots, y'_n] \to k[y'_1, \ldots, y'_n]/I(X) \cong P(X)$. Note that $k[y_1, \ldots, y_r] \cong k[y'_1, \ldots, y'_n]/(y'_{r+1} = 0, \ldots, y'_n = 0) = P(L)$, so $\tau : P(L) \hookrightarrow P(X)$. In light of Exercise 1.28, we obtain a regular map $\psi : X \to L$.

Now, let $a = (a_1, \ldots, a_n) \in L$. Then, we have a k-algebra homomorphism $\theta_a : P(L) \to k$ defined by evaluating functions at a. Since $P(X) \cong k[x_1, \ldots, x_n]$ is integral over $P(L) \cong k[y_1, \ldots, y_r]$ by the previous result, and since k is algebraically closed, we have by Exercise 5.2 that θ_a can be extended to a k-algebra homomorphism $\hat{\theta}_a : P(X) \to k$. A k-algebra homomorphism from $P(X)$ to k corresponds to a point $b \in X$ such that $\psi(b) = a$ (Exercise 1.28). Thus, ψ is surjective. □

5.17 Exercise

Proof. Since $I(X) \neq (1)$, the coordinate ring $P(X) = k[t_1, \ldots, t_n]/I(X) \neq 0$. By Exercise 5.16, there is a linear subspace L of dimension $r \geq 0$ and a k-linear map such that $X \to L$ is surjective. Hence, X is non-empty since L is non-empty. □

Remark. *This fact, that an affine algebraic variety X over an algebraically closed field k is non-empty if $I(X) \neq (1)$, is a weak form of the Nullstellensatz.*

Every maximal ideal of $k[t_1, \ldots, t_n]$ has the form $(t_1 - a_1, \ldots, t_n - a_n)$ for some $a \in k^n$.

Proof. Let $\mathfrak{m}$ be a maximal ideal of $k[t_1, \ldots, t_n]$. Note that $P := k[t_1, \ldots, t_n]/\mathfrak{m}$ is a finitely generated k-algebra, so by Exercise 3.16, there is some collection of algebraically indepedent $y_i \in P$ such that P is integral over $k[y_1, \ldots, y_r]$. Note that since $\mathfrak{m}$ is maximal, P is a field, so by Proposition 5.7, $k[y_1, \ldots, y_r]$ is a field so that $r = 0$; simply note that $y_1 \in k[y_1, \ldots, y_r]$ $(r > 0)$ does not have an inverse. Thus, the composition of k-algebra homomorphisms $k \to k[t_1, \ldots, t_n] \to P$ is an isomorphism. In particular, $t_i \mapsto a_i$ for some $a_i \in k$ for each i. The kernel of the map $k[t_1, \ldots, t_n] \to k[t_1, \ldots, t_n]/\mathfrak{m} \cong k$ is then $(t_1 - a_1, \ldots, t_n - a_n)$. But the kernel is also clearly $\mathfrak{m}$, so $\mathfrak{m} = (t_1 - a_1, \ldots, t_n - a_n)$, as desired. □

5.18 Exercise

Proof. Let B be a finitely generated k-algebra, where k is a field. Assume B is itself a field. Let $x_1, \ldots, x_n$ generate B as a k-algebra. We will prove that B is a finite algebraic extension of k by induction. If $n = 1$, then $B = k[x_1]$. Since B is a field, $x_1^{-1} \in B$, so we can write $x_1^{-1} = a_0 + a_1x_1 \cdots + a_nx_1^n$, where $a_i \in k$ and $a_n \neq 0$. Then,

$$a_nx_1^{n+1} + \cdots + a_1x_1^2 + a_0x_1 - 1 = 0,$$

so $B = k[x_1]$ is a finite algebraic extension of k. This proves the base case.

Now, assume $n > 1$ and the inductive hypothesis holds for $m < n$, and assume B has n generators $x_1, \ldots, x_n$ as a k-algebra. Let $A = k[x_1]$ and $K = k(x_1)$, the field of fractions of A. B is a finitely generated K-algebra with generators $x_2, \ldots, x_n$, so by the inductive hypothesis, B is a finite algebraic extension of K. Thus, $x_2, \ldots, x_n$ each satisfy a monic polynomial equation with coefficients in K, i.e. coefficients of the form a/b, where $a, b \in A$. Let $f \in A$ be the product of all denominators over each monic polynomial. Then, $x_2, \ldots, x_n$ are integral over A_f, as these same

monic polynomials map canonically to $A_f[x]$. Since B is generated as a ring by $x_2, \ldots, x_n$, B is integral over A_f. Since $K \subset B$, K is also integral over A_f.

If x_1 is algebraic over k, then K is a finite algebraic extension of k. Since B is a finite algebraic extension of K, B is a finite algebraic extension of k by transitivity, as desired. Then, the proof follows by induction. So, we only have to prove that x_1 is algebraic over k. Assume to the contrary that x_1 is transcendental over k. Then, $A = k[x_1]$ is a unique factorization domain, so A is integrally closed. By Proposition 5.12, since A_f has the same field of fractions K as A, which is integrally closed, A_f is itself integrally closed. But, we saw that K is integral over A_f, so we must have $A_f = K$; thus, $k(x_1) = k[x_1]$, showing that $x_1^{-1} \in k[x_1]$. By our proof of the base case above, this shows that x_1 is algebraic over k, giving the desired contradiction. □

Remark. *This is known as "Zariski's Lemma".*

5.19 Exercise

First, we prove that all maximal ideals of $k[t_1, \ldots, t_n]$ are of the form $(t_1 - a_1, \ldots, t_n - a_n)$, where $a \in k^n$.

Proof. Let $\mathfrak{m}$ be a maximal ideal of $k[t_1, \ldots, t_n]$ so that $B = k[t_1, \ldots, t_n]/\mathfrak{m}$ is both a field and a finitely generated k-algebra. By Exercise 5.18, B is a finite algebraic extension of k, so since k is algebraically closed, we have a k-algebra homomorphism $k[t_1, \ldots, t_n] \to k[t_1, \ldots, t_n]/\mathfrak{m} \longrightarrow k$. Thus, for each t_i, we have $t_i \mapsto a_i \in k$, so the kernel of this map is $(t_1 - a_1, \ldots, t_n - a_n)$. But also, the kernel is clearly $\mathfrak{m}$, so $\mathfrak{m} = (t_1 - a_1, \ldots, t_n - a_n)$. □

We use this to show that if X is an affine algebraic variety such that $I(X) \neq (1)$, then X is non-empty.

Proof. If $I(X) \neq (1)$, then there is some maximal ideal $\mathfrak{m}$ of $k[t_1, \ldots, t_n]$ such that $\mathfrak{m} \supset I(X)$. We've just shown that all maximal ideals are of the form $(t_1 - a_1, \ldots, t_n - a_n)$ for some $a \in k^n$, so $\mathfrak{m} = \mathfrak{m}_a$. But, $\mathfrak{m}_a = I(\{a\})$, where the point $\{a\}$ is the affine algebraic variety associated to $\{t_i - a_i\}$. Since $I(\{a\}) \supset I(X)$, we have the natural k-algebra homomorphism $P(X) \hookrightarrow P(\{a\})$. By Exercise 1.27, there is some regular function $f : \{a\} \to X$, so X is non-empty as $f(a) \in X$. □

5.20 Exercise

Proof. Let $S = A - \{0\}$, which is multiplicatively closed since A is an integral domain. This gives the field of fractions $K = S^{-1}A$. Since B is a ring that's finitely generated over A, $S^{-1}B$ is finitely generated over $S^{-1}A$ and is thus a finitely generated K-algebra. By Exercise 5.16, there are elements $x_1, \ldots, x_n$ in $S^{-1}B$, algebraically independent over K and such that $S^{-1}B$ is integral over $K[x_1, \ldots, x_n]$.

Let $z_1, \ldots, z_m$ generate B as an A-algebra. Then, for each i, $z_i = z_i/1 \in S^{-1}B$ is integral over $K[x_1, \ldots, x_n]$. Then, for each i, we have an equation of integral dependence:

$$z_i^k + f_{i,k-1} z_i^{k-1} + \cdots + f_{i,0} = 0,$$

where $f_{i,j} \in K[x_1, \ldots, x_n]$ for each j. Each $f_{i,j}$ is the sum of finitely many monomials of the form $c_{ij,\nu} x_1^{\nu_1} \cdots x_n^{\nu_n}$, where $c_{ij,\nu} \in K$, so $c_{ij,\nu} = a_{ij,\nu}/s_{ij,\nu}$, where $a_{ij,\nu} \in A$, $s_{ij,\nu} \in S$. Define $s_{ij} = \prod_\nu s_{ij,\nu}$.

$s_{ij} \in S$ since ν runs over the finitely many monomials that sum to $f_{i,j}$. Now, let $s_i = \prod_j s_{ij} \in S$. We have

$$0 = s_i^k(z_i^k + f_{i,k-1}z_i^{k-1} + \cdots + f_{i,0}) = (s_i z_i)^k + s_i f_{i,k-1}(z_i s_i)^{k-1} + \cdots + f_{i,0}s_i^k.$$

Since s_i was defined to clear denominators, this is the image in $K[x_1, \ldots, x_n]$ of $(s_i z_i)^k + s_i f_{i,k-1}(z_i s_i)^{k-1} + \cdots + f_{i,0}s_i^k \in A[x_1, \ldots, x_n]$. Note that since $A[x_1, \ldots, x_n]$ is an integral domain, the map $A[x_1, \ldots, x_n] \to K[x_1, \ldots, x_n]$ is injective, so since it is 0 in $K[x_1, \ldots, x_n]$, $s_i z_i$ is integral over $A[x_1, \ldots, x_n]$ for each i. If $t = \prod_i s_i$, then tz_i is integral over $A[x_1, \ldots, x_n]$ for each i; letting $t_i = \prod_{j \neq i} s_j$, $tz_i = t_i s_i z_i$, so since $t_i \in A$ and $s_i z$ is integral over $A[x_1, \ldots, x_n]$, tz_i is integral over $A[x_1, \ldots, x_n]$ for each i.

Now, for each x_i, note that $x_i \in S^{-1}B$, so $x_i = b_i/q_i$, where $b_i \in B$ and $q_i \in S$. Let $q = \prod_i q_i$, and let $s = tq$. Then, since tz_i is integral over $A[x_1, \ldots, x_n]$, $sz_i = qtz_i$ is integral over $A[x_1, \ldots, x_n]$. Further, since $s = q_i \cdot k_i$, where $k_i \in B$, we have $x_i = (b_i k_i)/(q_i k_i) = y_i/s$, where $y_i = b_i k_i \in B$.

Now, we confirm we have the desired s. Since x_i are algebraically independent over A, $y_i = sx_i$ are algebraically independent over A. Let $B' = A[y_1, \ldots, y_n]$. Then, $B'_s = A_s[y_1, \ldots, y_n] = A_s[x_1, \ldots, x_n]$ since s is a unit in A_s and $x_i = y_i/s$. Likewise, recall that B is generated as an A-algebra by $z_1, \ldots, z_m$. Thus, B_s is generated as an A_s-algebra by $z_1, \ldots, z_m$. Since $sz_i \in B$ is integral over $A[x_1, \ldots, x_n]$, $sz_i \in B_s$ is integral over $A_s[x_1, \ldots, x_n] = B'_s$. Since s is invertible and $s^{-1} \in A_s$, $z_i = s^{-1}(sz_i) \in B_s$ is integral over B'_s. Since the elements that generate B_s are integral over B'_s, B_s is integral over B'_s. □

5.21 Exercise

Proof. A is a subring of an integral domain B such that B is finitely generated over A. By Exercise 5.20, there exists some $s \neq 0$ and algebraically independent elements $y_1, \ldots, y_n$ over A in B such that B_s is integral over B'_s, where $B' = A[y_1, \ldots, y_n]$. We're given a homomorphism $f : A \to \Omega$ such that $f(s) \neq 0$.

Note that f is easily extended to B' by simply assigning $y_i \mapsto 0$; this is well-defined since the y_i are algebraically independent. We then extend $f : B' \to \Omega$ to $f : B'_s \to \Omega$; an element of B'_s has the form b/s^k, where $k \geq 0$ and $b \in B'$. Since $f(s) \neq 0$, $f(b/s^k) = f(b)/f(s)^k$. B_s is integral over B'_s, so we can extend to $f : B_s \to \Omega$ by Exercise 5.2. Finally, since B is an integral domain, B injects into B_s, so we have a ring homomorphism $f : B \to \Omega$ that restricts to the original $f : A \to \Omega$. □

5.22 Exercise

Proof. We have to show that for every non-zero $v \in B$, there is some maximal ideal of B not containing v. Since B is finitely generated over A as a ring, B_v is finitely generated over A as a ring. Further, since B is an integral domain, $B \hookrightarrow B_v$, so we can view A as a subring of B_v. By Exercise 5.21, there exists $s \neq 0$ in A such that whenever $f : A \to \Omega$ is a homomorphism (Ω algebraically closed) for which $f(s) \neq 0$, then f can be extended to a homomorphism $f : B_v \to \Omega$. Since the Jacobson radical of A is 0, there is some maximal ideal $\mathfrak{m}$ of A that does not contain s. Then, we have the canonical map $\psi : A \to A/\mathfrak{m} = k$, where k is a field. Let Ω_k be an algebraic closure of k so that $\psi : A \to k \subset \Omega_k$; note that $\psi(s) \neq 0$ since $s \notin \mathfrak{m}$. Thus, ψ can be extended to a homomorphism $\psi : B_v \to \Omega_k$. Note that $1 = \psi(v \cdot 1/v) = \psi(v) \cdot \psi(1/v)$. Thus, $\psi(v) \neq 0$, as required.

Let us describe this map more explicitly. In light of Exercise 5.20, s is such that $(B_v)_s$ is integral over $A_s[y_1, \ldots, y_n]$, where $y_1, \ldots, y_n$ are a collection of algebraically independent elements over A (and thus A_s). By our construction in Exercise 5.21, $\psi : A_s[y_1, \ldots, y_n] \to \Omega_k$ sends $y_i \mapsto 0$. Thus,

$A_s[y_1, \ldots, y_n] \cap \ker \psi \subset A_s$. Since $s \notin \mathfrak{m}$, $\mathfrak{m}A_s$ is a maximal ideal of A_s in light of Proposition 3.11. However, since $\mathfrak{m} \subset \ker \psi$ by construction, $\mathfrak{m}A_s \subset A_s[y_1, \ldots, y_n] \cap \ker \psi$. Since $A_s[y_1, \ldots, y_n] \cap \ker \psi$ is an ideal of A_s containing the maximal ideal $\mathfrak{m}A_s$, its either all of A_s or $\mathfrak{m}A_s$. But $s \in A_s$ and $\psi(s) \neq 0$, so $A_s[y_1, \ldots, y_n] \cap \ker \psi = \mathfrak{m}A_s$.

Now, we wish to show that $B \cap \ker \psi$ is a maximal ideal of B. Let $b \in B$. Since B is an integral domain, it injects into $(B_v)_s$, which is integral over $A_s[y_1, \ldots, y_n]$. Thus, b satisfies an integral equation

$$b^k + a_{k-1}b^{k-1} + \cdots + a_0 = 0,$$

where $a_i \in A_s[y_1, \ldots, y_n]$. Since $\psi : y_i \mapsto 0$, then $\psi(a_i) = \psi(c_i)$ for some $c_i \in A_s$. Then,

$$\begin{aligned} 0 &= \psi(b)^k + \psi(a_{k-1})\psi(b)^{k-1} + \cdots + \psi(a_0) \\ &= \psi(b)^k + \psi(c_{k-1})\psi(b)^{k-1} + \cdots + \psi(c_0) \\ &= \psi(b^k + c_{k-1}b^{k-1} + \cdots + c_0). \end{aligned}$$

Then, $b^k + c_{k-1}b^{k-1} + \cdots + c_0 \in \ker \psi$. Modulo $\ker \psi$,

$$\overline{b}^k + \overline{c_{k-1}}\overline{b}^{k-1} + \cdots + \overline{c_0} = 0,$$

so $B/(B \cap \ker \psi)$ is integral over $A_s/(A_s \cap \ker \psi)$.

If $f, g \in B$ and $fg \in B \cap \ker \psi$, then $\psi(fg) = \psi(f)\psi(g) = 0$. Since Ω_k is an integral domain, $\psi(f) = 0$ or $\psi(g) = 0$. Thus, $f \in B \cap \ker \psi$ or $g \in B \cap \ker \psi$, so $B \cap \ker \psi$ is a prime ideal. Thus, $B/(B \cap \ker \psi)$ is an integral domain. Further, $A_s/(A_s \cap \ker \psi)$ is a field since $A_s \cap \ker \psi$ is a maximal ideal. Also, $A_s/(A_s \cap \ker \psi) \hookrightarrow B/(B \cap \ker \psi)$, so by Proposition 5.7, $B/(B \cap \ker \psi)$ is a field, so $B \cap \ker \psi$ is a maximal ideal of B. Finally, since $\psi(v) \neq 0$, this maximal ideal does not contain v. Thus, the Jacobson radical of B is 0 since no non-zero element of B is contained in every maximal ideal. □

5.23 Exercise

Proof. (ii) $\implies$ (i). In every homomorphic image of A, the nilradical is equal to the Jacobson radical. Let $\mathfrak{p} \in \operatorname{Spec}(A)$. Then, we have a canonical homomorphism $A \to A/\mathfrak{p}$. Since $\mathfrak{p}$ is prime, $A/\mathfrak{p}$ is an integral domain, so the nilradical is zero. Thus, the Jacobson radical is zero, so the intersection of maximal ideals of $A/\mathfrak{p}$ is zero. By Proposition 1.1, the ideals of $A/\mathfrak{p}$ are in a one-to-one order-preserving correspondence with ideals of A containing $\mathfrak{p}$. Thus, the maximal ideals of A containing $\mathfrak{p}$ are precisely the maximal ideals of $A/\mathfrak{p}$. The intersection of these ideals is 0 in $A/\mathfrak{p}$, so $\mathfrak{p}$ contains the intersection. Conversely, since each maximal ideal of the intersection contains $\mathfrak{p}$, the intersection contains $\mathfrak{p}$, so $\mathfrak{p}$ is the intersection of the maximal ideals containing it, as desired.

(i) $\implies$ (ii). Suppose every prime in A is the intersection of the maximal ideals containing it. Now, let $f : A \to B$ be a homomorphism of rings. Then, $f(A)$ is itself a ring, and $f(A) \cong A/\ker f$. So, it is sufficient to show that for any ideal $\mathfrak{a}$, $\mathfrak{N}_{A/\mathfrak{a}} = \mathfrak{R}_{A/\mathfrak{a}}$. Note that the ideals of $A/\mathfrak{a}$ are precisely those of A that contain $\mathfrak{a}$, and this correspondence is order-preserving (Proposition 1.1). Let $\overline{\mathfrak{p}}$ be a prime ideal of $A/\mathfrak{a}$, so its contraction is a prime $\mathfrak{p}$ of A containing $\mathfrak{a}$. By assumption, $\mathfrak{p}$ is the intersection of the maximal ideals containing it; note these maximal ideals must also contain $\mathfrak{a}$. Further, since the ideal correspondence is order-preserving, these maximal ideals remain maximal in $A/\mathfrak{a}$, so $\overline{\mathfrak{p}}$ is an intersection of maximal ideals. Thus, the nilradical of $A/\mathfrak{a}$ is equal to the Jacobson radical, as desired.

(i) $\implies$ (iii). Suppose every prime ideal in A is the intersection of maximal ideals, and suppose $\mathfrak{p} \in \text{Spec}(A)$ is not maximal. Then, the prime ideals strictly containing $\mathfrak{p}$ includes all of the maximal ideals containing $\mathfrak{p}$. Thus, $\mathfrak{p}$ is a subset of the intersection of primes strictly containing $\mathfrak{p}$, which in turn, is a subset of the maximal ideals containing $\mathfrak{p}$. Since this last intersection is $\mathfrak{p}$ by assumption, we have equality throughout, so $\mathfrak{p}$ is the intersection of primes strictly containing $\mathfrak{p}$.

(iii) $\implies$ (ii). Assume (ii) is false, so (i) is false (we proved (i) $\implies$ (ii)). Therefore, there is some prime $\mathfrak{p}$ that is not the intersection of the maximal ideals containing it. Thus, $B = A/\mathfrak{p}$ is an integral domain with non-zero Jacobson radical $\mathfrak{R}$, so let $f \in \mathfrak{R}$ be a non-zero element. Since B is an integral domain, $B \to B_f$ is injective, so $B_f \neq 0$. Thus, B_f has some maximal ideal whose contraction in B is a prime ideal $\mathfrak{q}$. Note that $f \notin \mathfrak{q}$ (otherwise $(1/f)f = 1 \in \mathfrak{q}$). Any prime ideal in B that does not contain f is in B_f (Proposition 3.11), so since $\mathfrak{q}$ has this property and is maximal in B_f, it is maximal in the set of primes of B not containing f. Since f is in every maximal ideal of B, $\mathfrak{q}$ is not maximal. Further, any prime of B containing $\mathfrak{q}$ strictly must contain f since $\mathfrak{q}$ is maximal among primes that do not contain f. Thus, the intersection of primes strictly containing $\mathfrak{q}$ contains f, so the intersection does not equal $\mathfrak{q}$.

Finally, since $B = A/\mathfrak{p}$, $\mathfrak{q}$ lifts to a prime $\mathfrak{q}'$ of A. First, note that $\mathfrak{q}'$ is not maximal since $\mathfrak{q}$ is not maximal. Explicitly, there is some $\mathfrak{m}$ of B such that $\mathfrak{m}$ strictly contains $\mathfrak{q}$, and this lifts to a maximal ideal $\mathfrak{m}'$ that strictly contains $\mathfrak{q}'$; this is all in light of Proposition 1.1, which guarantees an order-preserving one-to-one correspondence between ideals of A containing $\mathfrak{p}$ and ideals of $A/\mathfrak{p}$. Finally, the intersection of primes of A strictly containing $\mathfrak{q}'$ is an ideal that maps to the intersection of primes of $A/\mathfrak{p}$ that strictly contain $\mathfrak{q}$. Since this latter intersection is not $\mathfrak{q}$, the former intersection is not $\mathfrak{q}'$ (since the correspondence of ideals is one-to-one). Thus, (iii) is false, as desired. □

Remark. *A ring with these three equivalent properties is called a "Jacobson ring".*

5.24 Exercise

(i) Proof. A is a Jacobson ring, and B is an A-algebra that is integral over A. Let $\mathfrak{q}$ be a prime ideal of B, and let $\mathfrak{p} = \mathfrak{q}^c$. By Proposition 5.6, $B/\mathfrak{q}$ is integral over $A/\mathfrak{p}$. Since A is a Jacobson ring, we have by Exercise 5.23 that $\mathfrak{p}$ is an intersection of maximal ideals. Therefore, $\mathfrak{R}(A/\mathfrak{p}) = 0$. By Exercise 5.5, $\mathfrak{R}(B/\mathfrak{q})^c = \mathfrak{R}(A/\mathfrak{p}) = 0$.

Now, suppose $b \in \mathfrak{R}(B/\mathfrak{q})$, so it satisfies an equation of integral dependence over $A/\mathfrak{p}$. We choose $n > 0$ to be minimal. Then,

$$b^n + a_{n-1}b^{n-1} + \cdots + a_1 b + a_0 = 0,$$

where $a_i \in A/\mathfrak{p}$, so

$$a_0 = -b(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1).$$

Since b is in every maximal ideal of $B/\mathfrak{q}$, a_0 is in every maximal ideal of $B/\mathfrak{q}$. Thus,

$$a_0 \in \mathfrak{R}(B/\mathfrak{q}) \cap A/\mathfrak{p} = \mathfrak{R}(B/\mathfrak{q})^c = 0.$$

Therefore, we have $b(b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1) = 0$. Since $\mathfrak{q}$ is prime, $B/\mathfrak{q}$ is an integral domain. If $b \neq 0$, then $b^{n-1} + a_{n-1}b^{n-2} + \cdots + a_1 = 0$, violating the minimality of n. Thus, $b = 0$ so that $\mathfrak{R}(B/\mathfrak{q}) = 0$. In particular, $\mathfrak{q}$ is the intersection of the maximal ideals of B containing it. By Exercise 5.23, B is a Jacobson ring. □

(ii) Proof. A is a Jacobson ring, and B is a finitely-generated A-algebra. First, we show that $f(A)$ ($f : A \to B$ the algebra-defining homomorphism) is a Jacobson ring so that it will be sufficient to assume A is a subring of B. Let C be a homomorphic image of $f(A)$; then, it is a homomorphic image of A, so the nilradical of C is equal to the Jacobson radical by Exercise 5.23. In particular, $f(A)$ is a Jacobson ring.

Therefore, we can assume A is a subring of B. Let $\mathfrak{q}$ be a prime ideal of B, so $\mathfrak{p} = \mathfrak{q}^c = A \cap \mathfrak{q}$ is a prime ideal of A. Then, $A/\mathfrak{p} \hookrightarrow B/\mathfrak{q}$; because B is a finitely-generated A-algebra, $B/\mathfrak{q}$ is a finitely-generated $A/\mathfrak{p}$-algebra. Since A is a Jacobson ring, we have by Exercise 5.23 that $\mathfrak{p}$ is the intersection of the maximal ideals containing it. Thus, the Jacobson radical of $A/\mathfrak{p}$ is 0.

Note that $B/\mathfrak{q}$ is an integral domain since $\mathfrak{q}$ is prime. Now, we can invoke Exercise 5.22: the Jacobson radical of $B/\mathfrak{q}$ is 0 since the Jacobson radical of $A/\mathfrak{p}$ is 0. Therefore, $\mathfrak{q}$ is the intersection of the maximal ideals in B containing $\mathfrak{q}$. By Exercise 5.23, B is a Jacobson ring, as desired. □

Proof. Let A be a finitely-generated ring. Then, it is a finitely-generated $\mathbb{Z}$-algebra. Note that every non-zero prime of $\mathbb{Z}$ is maximal. The only non-zero prime of $\mathbb{Z}$ is 0, but $0 = \bigcap_{(p)\in \mathrm{Spec}(\mathbb{Z})}(p)$, so it is the intersection of maximal ideals. Thus, it is a Jacobson ring by Exercise 5.23. In light on (ii), A is a Jacobson ring. □

Proof. Let A be a finitely-generated k-algebra, where k is a field. The only prime ideal of k is 0, which is maximal, so by Exercise 5.23, k is a Jacobson ring. By (ii), A is a Jacobson ring. □

5.25 Exercise

Proof. (i) $\implies$ (ii). A is a Jacobson ring; let B be a finitely-generated A-algebra that is also a field. First, we show that $f(A)$ ($f : A \to B$ the algebra-defining homomorphism) is a Jacobson ring so that it will be sufficient to assume A is a subring of B. Let C be a homomorphic image of $f(A)$; then, it is a homomorphic image of A, so the nilradical of C is equal to the Jacobson radical by Exercise 5.23. In particular, $f(A)$ is a Jacobson ring.

Since B is a field, it is an integral domain, so we can apply Exercise 5.21: there exists some non-zero $s \in A$ such that, if Ω is algebraically closed, $f : A \to \Omega$ is a ring homomorphism, and $f(s) \neq 0$, then f can be extended to B. By Exercise 5.20, there exists a non-zero $s \in A$ such that B_s is integral over $(A[y_1, \ldots, y_n])_s$, where $y_1, \ldots, y_n$ are algebraically independent.

If s is a unit in A, then $(A[y_1, \ldots, y_n])_s \cong A[y_1, \ldots, y_n]$ and $B_s \cong B$, so B is integral over $A[y_1, \ldots, y_n]$. Since B is a field, we have by Proposition 5.7 that $A[y_1, \ldots, y_n]$ is a field; in particular, $y_i^{-1} \in A[y_1, \ldots, y_n]$ if $1 \leq i \leq n$. Then, $y_1^{-1} = P(y_1, \ldots, y_n)$, so $P(y_1, \ldots, y_n)y_1 - 1 = 0$, violating the algebraic independence of $\{y_i\}$. Thus, $n = 0$, so A is a field, and B is a finitely generated A-algebra. Thus, B is a finite algebraic extension over A (since B is integral over A), as desired.

On the other hand, suppose s is not a unit in A. Since A is an integral domain, the nilradical is 0; since A is also Jacobson, the Jacobson radical must also be 0 since these ideals are equal in homomorphic images of A, by Exercise 5.23. Thus, s is not in the Jacobson radical, so there is some maximal ideal $\mathfrak{m}$ of A such that $s \notin \mathfrak{m}$. Define $f : A \to A/\mathfrak{m}$; let $k = A/\mathfrak{m}$ be the residue field, and let Ω_k be the algebraic closure of k. Then, we have $f : A \to \Omega_k$. Further, since $s \notin \mathfrak{m}$, $f(s) \neq 0$, so we can extend the homomorphism to B. Suppose there is some non-zero $b \in B$ such that $f(b) = 0$. Then, $1 = f(bb^{-1}) = f(b)f(b^{-1}) = 0$, a contradiction. Thus, f is injective on B. Hence, $B \cong f(B)$ is a finitely generated k-algebra (since $f(A) = k$), so it is a finite algebraic extension of k by Exercise 5.18 (Zariski's lemma).

(ii) $\implies$ (i). Suppose that every finitely generated A-algebra B which is a field is finite over A. Let $\mathfrak{p}$ be a non-maximal prime of A. Let $B = A/\mathfrak{p}$ (note that B is an integral domain), and let

$f \in B$ be some non-zero element. Then, B_f is a finitely generated A-algebra since B is clearly a finitely generated A-algebra. If B_f is a field, then it is finite over A (by assumption), so it is finite over B. Thus, it is integral over B (no transcendental elements), so B is a field by Proposition 5.7. This is a contradiction since we chose $\mathfrak{p}$ to be non-maximal, so B_f is not a field. Therefore, it has some non-zero maximal (prime) ideal whose contraction in B is a non-zero ideal $\mathfrak{p}'$ such that $f \notin \mathfrak{p}'$. Note that the non-zero primes of B correspond exactly to primes of A strictly containing $\mathfrak{p}$. For every non-zero element f of $A/\mathfrak{p}$, we saw that there is some prime that does not contain it, so the intersection of non-zero primes of $A/\mathfrak{p}$ is 0. Contracting back to A, the intersection of primes strictly containing $\mathfrak{p}$ is $\mathfrak{p}$. By Exercise 5.23, A is a Jacobson ring. □

5.26 Exercise

Proof. (1) $\implies$ (2). Suppose every non-empty locally closed subset of X meets X_0. Now, let E be a closed subset of X. Then, E is locally closed since $E = E \cap X$, E being closed and X being open. Therefore, $E \cap X_0 \neq \emptyset$. Since E is closed and $E \cap X_0 \subset E$, $C = \overline{E \cap X_0} \subset E$. Let $Y = E \cap C^c$; since C is closed, C^c is open, so Y is locally closed since E is closed. Therefore, if Y is non-empty, then $Y \cap X_0 \neq \emptyset$ by assumption. However, note that $C^c = \overline{E \cap X_0} \subset (E \cap X_0)^c$. Thus,

$$Y \cap X_0 = X_0 \cap E \cap C^c \subset (E \cap X_0) \cap (E \cap X_0)^c = \emptyset.$$

Thus, since $Y \cap X_0$ is empty, Y must also be empty (by assumption), so in fact, $\overline{E \cap X_0} = E$.

(2) $\implies$ (3). The map $U \mapsto U \cap X_0$ that maps open sets of X to open sets of X_0 is clearly surjective, as this is what defines open sets of X_0. Now, suppose we have two open sets U, V such that $U \cap X_0 = V \cap X_0$. Therefore,

$$U^c \cap X_0 = X_0 - (U \cap X_0) = X_0 - (V \cap X_0) = V^c \cap X_0.$$

By assumption, every closed set E of X satisfies $\overline{E \cap X_0} = E$. Thus, we have $V^c = \overline{V^c \cap X_0} = \overline{U^c \cap X_0} = U^c$, so $U = V$. Hence, the map is bijective.

(3) $\implies$ (1). Finally, assume the map $U \mapsto U \cap X_0$ from open sets of X onto open sets of X_0 is bijective. Let Y be a locally closed subset of X, so $Y = U \cap C$, where U is open and C is closed. We will prove the contrapositive, so assume that $Y \cap X_0 = \emptyset$. Then, $U \cap C \cap X_0 = \emptyset$. Since C is closed, C^c is an open set. Then,

$$U \cap X_0 = (U \cap C \cap X_0) \cup (U \cap C^c \cap X_0) = \emptyset \cup (U \cap C^c \cap X_0) = (U \cap C^c) \cap X_0.$$

By the bijectivity of the map (since $U \cap C^c$ is open), $U = U \cap C^c$, so $U \subset C^c$. Thus, $Y = U \cap C = \emptyset$. So, if a locally closed set does not meet X_0, it is empty. □

Remark. *A subset X_0 satisfying these properties is said to be "very dense" in X.*

Proof. (i) $\implies$ (ii). Suppose A is a Jacobson ring, so by Exercise 5.23, every prime ideal is an intersection of maximal ideals. Now, let $V(\mathfrak{a})$ be an arbitrary closed set of $\operatorname{Spec}(A)$, where $\mathfrak{a}$ is some ideal of A. If $\mathfrak{a} = (1)$ so that $V(\mathfrak{a}) = \emptyset$, then trivially, $\overline{V(\mathfrak{a}) \cap \operatorname{Max}(A)} = V(\mathfrak{a})$. So, assume $\mathfrak{a} \neq (1)$. Since $V(\mathfrak{a})$ is closed, we clearly have $\overline{V(\mathfrak{a}) \cap \operatorname{Max}(A)} \subset V(\mathfrak{a})$. To show the reverse, suppose $\mathfrak{p} \in V(\mathfrak{a})$. By assumption, there is a collection of maximal ideals $\{\mathfrak{m}_\alpha\}$ such that $\mathfrak{p} = \bigcap_\alpha \mathfrak{m}_\alpha$. Note that $\mathfrak{m}_\alpha \in V(\mathfrak{a})$ and thus is in $V(\mathfrak{a}) \cap \operatorname{Max}(A) \subset \overline{V(\mathfrak{a}) \cap \operatorname{Max}(A)}$. Finally, note that since $\overline{V(\mathfrak{a}) \cap \operatorname{Max}(A)}$ is closed, it is equal to some $V(\mathfrak{b})$, where $\mathfrak{b}$ is some ideal. Then, $\mathfrak{m}_\alpha \supset \mathfrak{b}$ for all $\mathfrak{m}_\alpha$, so $\mathfrak{p} = \bigcap_\alpha \mathfrak{m}_\alpha \supset \mathfrak{b}$. Thus, $\mathfrak{p} \in V(\mathfrak{b})$, so $V(\mathfrak{a}) = \overline{V(\mathfrak{a}) \cap \operatorname{Max}(A)}$ for all closed sets $V(\mathfrak{a})$ of $\operatorname{Spec}(A)$. By the previous results, $\operatorname{Max}(A)$ is very dense in $\operatorname{Spec}(A)$.

(ii) $\implies$ (iii). Since $\mathrm{Max}(A)$ is very dense in X, we have from the equivalent conditions proved above that any locally closed subset of $\mathrm{Spec}(A)$ must meet $\mathrm{Max}(A)$. Hence, any locally closed singleton set must contain a maximal ideal. For any maximal ideal $\mathfrak{m}$, we have $\{\mathfrak{m}\} = V(\mathfrak{m})$, by the definition of maximality, so every locally closed subset of $\mathrm{Spec}(A)$ consisting of a single point is closed.

(iii) $\implies$ (i). Let $\mathfrak{p} \in \mathrm{Spec}(A)$ be a non-maximal prime, and let $\{\mathfrak{p}_\alpha\} \subset \mathrm{Spec}(A)$ be the set of primes that contain $\mathfrak{p}$ strictly. Then, $\mathfrak{p} \subset \bigcap_\alpha \mathfrak{p}_\alpha$. Suppose for the sake of contradiction that $\mathfrak{p} \not\supset \bigcap_\alpha \mathfrak{p}_\alpha$. Then, $V\left(\bigcap_\alpha \mathfrak{p}_\alpha\right)^c \cap V(\mathfrak{p}) \neq \emptyset$ since this set contains $\mathfrak{p}$. Since it is the intersection of an open and closed set, it is a locally closed set. Now, we'll show that this set is the singleton set containing $\mathfrak{p}$. Suppose $\mathfrak{q} \in V(\mathfrak{p})$ and is not equal to $\mathfrak{p}$. Then, $\mathfrak{q} \supset \mathfrak{p}$, so by definition, $\mathfrak{q} \in \{\mathfrak{p}_\alpha\}$. Hence, $\mathfrak{q} \supset \bigcap_\alpha \mathfrak{p}_\alpha$, so $\mathfrak{q} \notin V\left(\bigcap_\alpha \mathfrak{p}_\alpha\right)^c$. Thus, $\{\mathfrak{p}\} = V\left(\bigcap_\alpha \mathfrak{p}_\alpha\right)^c \cap V(\mathfrak{p})$ is a locally closed set containing only one point, so by assumption, $\{\mathfrak{p}\}$ is a closed set. Thus, $\{\mathfrak{p}\} = V(\mathfrak{a})$ for some ideal $\mathfrak{a}$, so $\mathfrak{p} \supset \mathfrak{a}$. However, since $\mathfrak{p}$ is not maximal, there is some maximal ideal $\mathfrak{m}$ containing $\mathfrak{p}$ and thus also $\mathfrak{a}$. Therefore, $\mathfrak{m} \in V(\mathfrak{a})$, a contradiction. Hence, we must have $\mathfrak{p} = \bigcap_\alpha \mathfrak{p}_\alpha$, so by Exercise 5.23, A is a Jacobson ring. □

5.27 Exercise

Proof. Let K be a field and let Σ be the set of local subrings of K. We can apply a partial order on Σ using domination, i.e. B dominates A if $A \subset B$ and $\mathfrak{m}_A = \mathfrak{m}_B \cap A$. Now, let $\mathcal{T}$ be a chain of elements of Σ. Define $R = \bigcup_{A \in \mathcal{T}} A$ and $\mathfrak{m} = \bigcup_{A \in \mathcal{T}} \mathfrak{m}_A$. We aim to prove that this is an upper bound of $\mathcal{T}$ in Σ.

First, note that R is a subring of K since it is a union of subrings of K. We want to show that $\mathfrak{m}$ is a maximal ideal of R, so we need to show that it is an ideal. If $x, y \in \mathfrak{m}$, then there are rings $A_x, A_y \in \mathcal{T}$ such that $x \in \mathfrak{m}_{A_x}$ and $y \in \mathfrak{m}_{A_y}$. Since $\mathcal{T}$ is a chain, either A_x is a subring of A_y or vice-versa. Assume without loss of generality that $A_x \subset A_y$. Then, $\mathfrak{m}_{A_x} = \mathfrak{m}_{A_y} \cap A_x$, so $x \in \mathfrak{m}_{A_y}$ as well. Thus, $x + y \in \mathfrak{m}_{A_y} \subset \mathfrak{m}$. Finally, if $r \in R$, then $r \in A_1$ for some $A_1 \in \mathcal{T}$. If $x \in \mathfrak{m}$, it is in $\mathfrak{m}_{A_2}$ for some ring $A_2 \in \mathcal{T}$. Since $\mathcal{T}$ is a chain, either A_1 dominates A_2 or vice-versa. If $A_1 \subset A_2$, then $r \in A_2$, so $rx \in \mathfrak{m}_{A_2} \subset \mathfrak{m}$, so $\mathfrak{m}$ is R-stable. If $A_2 \subset A_1$, then $x \in \mathfrak{m}_{A_2} \subset \mathfrak{m}_{A_1}$ so that $rx \in \mathfrak{m}_{A_1} \subset \mathfrak{m}$. Thus, $\mathfrak{m}$ is an ideal of R.

We wish to show that for all $A \in \mathcal{T}$, $\mathfrak{m}_A = \mathfrak{m} \cap A$. We have

$$A \cap \mathfrak{m} = A \cap \left(\bigcup_{B \in \mathcal{T}} \mathfrak{m}_B\right) = A \cap \left(\bigcup_{B \leq A} \mathfrak{m}_B \cup \bigcup_{B > A} \mathfrak{m}_B\right)$$

$$= \left(\bigcup_{B \leq A} A \cap \mathfrak{m}_B\right) \cup \left(\bigcup_{B > A} A \cap \mathfrak{m}_B\right) = \left(\bigcup_{B \leq A} \mathfrak{m}_B\right) \cup \left(\bigcup_{B > A} \mathfrak{m}_A\right) = \mathfrak{m}_A \cup \mathfrak{m}_A = \mathfrak{m}_A,$$

as desired. Now, suppose I is an ideal of R that properly contains the ideal $\mathfrak{m}$. Then, there is some $x \in I - \mathfrak{m} \subset R$, so there is some $A \in \mathcal{T}$ such that $x \in A$. Then, $I \cap A \supset \mathfrak{m} \cap A = \mathfrak{m}_A$ properly. Since $\mathfrak{m}_A$ is a maximal ideal of A, $I \cap A$ must be A. Thus, $1 \in I$, so $I = R$. In particular, $\mathfrak{m}$ is a maximal ideal of R.

We must show that R is a local ring. Let $\mathfrak{m}'$ be an arbitrary maximal ideal of R. For any $A \in \mathcal{T}$, since A has a unique maximal ideal $\mathfrak{m}_A$, either $A \cap \mathfrak{m}' = (1)$ or $\mathfrak{m}_A \supset \mathfrak{m}' \cap A$. If there is any A for which $A \cap \mathfrak{m}' = (1)$, then $1 \in \mathfrak{m}'$, so $\mathfrak{m}' = R$, violating maximality. Therefore, for all A, we have $\mathfrak{m}_A \supset \mathfrak{m}' \cap A$. Taking the union over all A, we obtain $\mathfrak{m} \supset \mathfrak{m}' \cap R = \mathfrak{m}'$. Since $\mathfrak{m}'$ is maximal, $\mathfrak{m}' = \mathfrak{m}$, so R is indeed a local ring with maximal ideal $\mathfrak{m}$. Thus, $R \in \Sigma$. Since $A \subset R$

and $\mathfrak{m}_A = \mathfrak{m} \cap A$ for all $A \in \mathcal{T}$, R is an upper bound of $\mathcal{T}$ in Σ. By Zorn's lemma, Σ has maximal elements.

Finally, we wish to show that $A \in \Sigma$ is maximal if and only if A is a valuation ring of K. Let R be a maximal element of Σ. Consider the set of pairs (A, f), where A is a subring of K and $f : A \to \Omega$ (an algebraically closed field) is a homomorphism. Then, the set Σ' can be ordered such that $(A, f) \leq (B, g)$ if $A \subset B$ and g restricts to f on A. Prior to Lemma 5.19, the text proves that Σ' satisifes Zorn's lemma and thus has maximal elements. Let $\mathfrak{m}$ be the unique maximal ideal of R, and define $f : R \mapsto R/\mathfrak{m} \hookrightarrow \Omega$, where Ω is an algebraic closure of the residue field. Since $(R, f) \in \Sigma'$, there is some maximal element (B, g) of Σ' such that $(R, f) \leq (B, g)$. By Lemma 5.19, B is a local ring and its maximal ideal is $\mathfrak{m}_B = \ker g$. Note that $R \cap \mathfrak{m}_B = R \cap \ker g = \ker f = \mathfrak{m}$. Thus, B dominates R, so we have $R \leq B$ in Σ. Since R is a maximal element of Σ, we have $(R, f) = (B, g)$. By Theorem 5.21, B is a valuation ring of K, so R is a valuation ring of K.

Conversely, suppose A is a valuation ring of K. Then, it is a local subring of K, so $A \in \Sigma$. Thus, there is some maximal element $B \in \Sigma$ so that $A \leq B$. Since $A \subset B \subset K$, B is also a valuation ring by Proposition 5.18. Since B dominates A, $\mathfrak{m}_A = A \cap \mathfrak{m}_B$. Note that $\mathfrak{m}_A$ and $\mathfrak{m}_B$ are the set of non-units of A and B, respectively (Proposition 1.6). Let $b \in \mathfrak{m}_B$ (non-zero), so $b^{-1} \notin B$. Thus, $b^{-1} \notin A$, so since A is a valuation ring of K, $b \in A$. Thus, $\mathfrak{m}_B \subset A$, so $\mathfrak{m}_A = A \cap \mathfrak{m}_B = \mathfrak{m}_B$. Finally, suppose $b \in B - \mathfrak{m}_B$. Then, there is an inverse $b^{-1} \in B$. Since b^{-1} has its inverse b in B, we must have $b^{-1} \in B - \mathfrak{m}_B$, as well. Now, since A is a valuation ring of K, $b \in A$ or $b^{-1} \in A$. Swapping b and b^{-1} if necessary, we can assume $b \in A$. Since $\mathfrak{m}_A = \mathfrak{m}_B$ and $b \notin \mathfrak{m}_B$, $b \in A - \mathfrak{m}_A$. Thus, b is a unit in A, so $b^{-1} \in A$, and as before, $b^{-1} \in A - \mathfrak{m}_A$ since b^{-1} is a unit in A. Thus, $B - \mathfrak{m}_B \subset A - \mathfrak{m}_A$, so that by $\mathfrak{m}_A = \mathfrak{m}_B$, $B \subset A$. Finally, $A = B$ so that A is a maximal element of Σ, as desired. □

5.28 Exercise

Proof. (i) $\implies$ (ii). Assume A is a valuation ring in its field of fractions K; let $\mathfrak{a}$, $\mathfrak{b}$ be two ideals of A. If $\mathfrak{a} - \mathfrak{b}$ is empty, then $\mathfrak{a} \subset \mathfrak{b}$. So, assume there is some $a \in \mathfrak{a} - \mathfrak{b}$; clearly, $a \neq 0$. If $\mathfrak{b} = 0$, then $\mathfrak{b} \subset \mathfrak{a}$, so assume $\mathfrak{b} \neq 0$; let $b \in \mathfrak{b}$ be non-zero. Then, $a/b, b/a \in K^\times$, so $a/b \in A$ or $b/a \in A$. If $a/b \in A$, then $a = (a/b)b \in \mathfrak{b}$, a contradiction. Thus, $b/a \in A$, so $b = (b/a)a \in \mathfrak{a}$. Since $b \in \mathfrak{b}$ was chosen arbitrarily, $\mathfrak{b} \subset \mathfrak{a}$.

(ii) $\implies$ (i). Let K be the field of fractions of A, which contains A since A is an integral domain. Choose an arbitrary non-zero element $a/b \in K^\times$, so $a, b \in A$ are both non-zero. Then, we have the ideals (a) and (b) in A. By assumption, either $(a) \subset (b)$ or $(b) \subset (a)$. If $(a) \subset (b)$, then $a = xb$ for some $x \in A$. But $x = a/b$, so $a/b \in A$. On the other hand, if $(b) \subset (a)$, then $b = xa$ for some $x \in A$. But $x = b/a = (a/b)^{-1}$. Thus, either b/a or $(b/a)^{-1}$ lies in A, so A is a valuation ring of K. □

Proof. Assume A is a valuation ring of a field K and $\mathfrak{p}$ is a prime ideal of A. Since A is an integral domain as well, the ideals form a chain of A. Since the ordering of ideals of $A_\mathfrak{p}$ and $A/\mathfrak{p}$ is inherited from the ordering of ideals of A, the set of ideals of these two rings both form a chain. By the result, since $A/\mathfrak{p}$ and $A_\mathfrak{p}$ are also both integral domains, $A/\mathfrak{p}$ and $A_\mathfrak{p}$ are both valuation rings. □

5.29 Exercise

Proof. Let B be a subring of K that contains A, a valuation ring of K. By Proposition 5.18, B is a valuation ring of K and thus a local ring with maximal ideal $\mathfrak{m}_B$. Since $A \subset B$, $\mathfrak{m}_B^c = \mathfrak{m}_B \cap A$

is a prime ideal $\mathfrak{p}$ of A, so $\mathfrak{m}_A \supset \mathfrak{p}$ since $\mathfrak{m}_A$ is the unique maximal ideal of A.

Now, consider $A_\mathfrak{p}$. Since A is a valuation ring of K, we have by Exercise 5.28 that $A_\mathfrak{p}$ is a valuation ring of K. Suppose $s \in A - \mathfrak{p}$, so $s \in B$ and $s \notin \mathfrak{m}_B$. Then, since B is a valuation ring, s is a unit in B since $\mathfrak{m}_B$ is the set of non-units of B. Thus, we have a map $A_\mathfrak{p} \to B$ by mapping $a/s \mapsto as^{-1}$ (for $a \in A$ and $s \in A - \mathfrak{p}$). This is injective since $as^{-1} = 0 \implies a = 0 \implies a/s = 0$, so we have $A_\mathfrak{p} \subset B \subset K$.

Finally, note that $A_\mathfrak{p}$ is a local ring of K and has unique maximal ideal $\mathfrak{p}$ such that $\mathfrak{p} = A_\mathfrak{p} \cap \mathfrak{m}_B$. Thus, B dominates $A_\mathfrak{p}$. By Exercise 5.27, since $A_\mathfrak{p}$ is a valuation ring, it is maximal with respect to domination in the set of local rings of K, so $B = A_\mathfrak{p}$. Thus, B is a local ring of A. □

5.30 Exercise

Proof. Let A be a valuation ring of a field K. $U = A^\times$ is subgroup of $K^\times$, so let $\Gamma = K^\times/U$. Let $x, y \in K^\times$, and suppose $x \geq y$ so that $xy^{-1} \in A$. Now, let $u, v \in U = A^\times$. Then, $(ux)(vy)^{-1} = (uv^{-1})xy^{-1} \in A$ since $xy^{-1} \in A$ and $uv^{-1} \in U \subset A$, so $ux \geq vy$. In particular, the ordering $(K^\times, \geq)$ lifts to an ordering of $(\Gamma, \geq)$.

Let $\xi, \eta \in \Gamma$, and let $x, y \in K^\times$ be a choice of representatives for ξ and η, respectively. Since A is a valuation ring of K, $xy^{-1} \in A$ or $(xy^{-1})^{-1} = yx^{-1} \in A$. Thus, $\xi \geq \eta$ or $\eta \geq \xi$, so $(\Gamma, \geq)$ is a total ordering.

Now, suppose $\xi \geq \eta$ and $\omega \in \Gamma$. Let $x, y, z \in K^\times$ be representatives of ξ, η, ω, respectively. Then, $(xz)(yz)^{-1} = zz^{-1}xy^{-1} = xy^{-1} \in A$, so $\xi\omega \geq \eta\omega$. Thus, $\geq$ is compatible with the group, so Γ is a totally ordered abelian group.

Let $v : K^\times \to \Gamma$ be the canonical homomorphism, and let $x, y \in K^\times$ such that $x + y \neq 0$. Note that either $x/y \in A$ or $y/x \in A$. Relabelling if necessary, we can assume $x/y \in A$. Since $x/y \in A$, $x/y + 1 = (x + y)/y \in A$, so $x + y \geq y$; hence, $v(x + y) \geq v(y) \geq \min(v(x), v(y))$, as desired. □

Remark. Γ *is called the "value group" of* A.

5.31 Exercise

Proof. Let $A = \{x \in K \mid v(x) \geq 0\}$. Note that $v(1) = 0$ since v is a group homomorphism $K^\times \to \Gamma$, so $1 \in A$. Since $0/1 \in A$, $v(0) \geq v(1) = 0$, so $0 \in A$. If $x, y \in A$, then $v(x), v(y) \geq 0$, so $v(x + y) \geq \min(v(x), v(y)) \geq 0$ and thus $x + y \in A$. Further, $v(xy) = v(x) + v(y) \geq 0$, so $xy \in A$. Further, note that

$$0 = v(1) = v(-1 \cdot -1) = 2v(-1).$$

If $v(-1) > 0$, then $2v(-1) > 2 \cdot 0 = 0$, and if $v(-1) < 0$, then $2v(-1) < 2 \cdot 0 = 0$; both are contradictions. Thus, $v(-1) = 0$, so $-1 \in A$. Thus, for any $x \in A$, $-1 \cdot x = -x \in A$. Hence, A is a subring of K.

Let $x \in K^\times$. Then,

$$0 = v(1) = v(x \cdot x^{-1}) = v(x) + v(x^{-1}).$$

If $v(x) < 0$ and $v(x^{-1}) < 0$, then $v(x) + v(x^{-1}) < 0$, a contradiction. Thus, $v(x) \geq 0$ or $v(x^{-1}) \geq 0$, so $x \in A$ or $x^{-1} \in A$. Hence, A is a valuation ring of K. □

Proof. For the sake of completion, we want to show that valuations and valuation rings are essentially equivalent. We've shown that valuation rings have a valuation (Exercise 5.30), and we've just shown that a valuation gives rise to a valuation ring (Exercise 5.31). We want to show that these processes are inverse.

Let v be the valuation gives above, and let A be the valuation ring of K coming from v, i.e. $A = \{x \in K \mid v(x) \geq 0\}$. The valuation coming from A is $\pi : K^\times \to K^\times/A^\times$, so $\ker \pi \cong A^\times$. But if x is a unit in A, then $0 = v(1) = v(x) + v(x^{-1})$, so since $v(x) \geq 0$ and $v(x^{-1}) \geq 0$, we must have $v(x) = 0$. Thus, $\ker \pi \subset \ker v$. Conversely, if $x \in \ker v$, then $0 = v(1) = v(x) + v(x^{-1}) = v(x^{-1})$, so $x^{-1} \in A$. Thus, x is a unit so that $\ker v \subset U = \ker \pi$; then, $\ker \pi = \ker v$.

To show finally that π and v are equivalent, it is enough to show that $\pi(x) \geq \pi(y) \iff v(x) \geq v(y)$ for $x, y \in K^\times$. So, suppose $\pi(x) \geq \pi(y)$. Then, $x \geq y$ so that $xy^{-1} \in A$. Therefore, $v(xy^{-1}) \geq 0$; since $v(xy^{-1}) = v(x) - v(y)$, we have $v(x) \geq v(y)$. Conversely, if $v(x) \geq v(y)$, then $v(xy^{-1}) \geq 0$, so $xy^{-1} \in A$. Thus, $x \geq y$, so $\pi(x) \geq \pi(y)$. Thus, v and π produce the same ordering. □

Remark. Γ *is the "valuation group" of* v.

5.32 Exercise

Proof. A is a valuation ring of K. Define $\Delta' = v(A - \mathfrak{p}) \subset \Gamma$. If $x, y \in A - \mathfrak{p}$, then $xy \in A - \mathfrak{p}$, so $v(x) + v(y) = v(xy) \in \Delta'$. Also, $1 \in A - \mathfrak{p}$, so $0 = v(1) \in \Delta'$. Therefore, Δ' is a submonoid of Γ. Thus, $-\Delta'$ is also a submonoid of Γ. Clearly, $\Delta = \Delta' \cup (-\Delta')$ has the identity element and each element has an inverse. To show Δ is a subgroup, we just need to show closure. The only non-trivial case is when $v(x) \in \Delta'$ and $-v(y) \in -\Delta'$, where $x, y \in A - \mathfrak{p}$. Then, $v(x) + (-v(y)) = v(x) + v(y^{-1}) = v(xy^{-1})$. Since A is a valuation ring, either $xy^{-1} \in A$ or $(xy^{-1})^{-1} = yx^{-1} \in A$. Suppose $xy^{-1} \in A$. If $xy^{-1} \in \mathfrak{p}$, then since $y \in A$, $x = y(xy^{-1}) \in \mathfrak{p}$, a contradiction. Thus, $xy^{-1} \in A - \mathfrak{p}$, so $v(xy^{-1}) \in \Delta' \subset \Delta$. On the other hand, if $yx^{-1} \in A$, then by the same logic above, $yx^{-1} \in A - \mathfrak{p}$, so $v(yx^{-1}) \in \Delta'$. Then, $v(xy^{-1}) = -v(yx^{-1}) \in -\Delta' \subset \Delta$, so Δ is a subgroup of Γ. Further, for any $\delta \in \Delta$, if $\delta \geq 0$, then $\delta \in \Delta' = v(A - \mathfrak{p})$.

Now, suppose $\beta \in \Gamma$ and $0 \leq \beta \leq \alpha$, where $\alpha \in \Delta$. Then, $\alpha \geq 0$, so $\alpha = v(a)$ for some $a \in A - \mathfrak{p}$. Let $x \in K^\times$ such that $v(x) = \beta \geq 0$. Since $v(x) \geq 0$, $x \in A$. Since $v(a) \geq v(x)$, $ax^{-1} \in A$. Suppose that $x \in \mathfrak{p}$. Then, $a = (ax^{-1})x \in \mathfrak{p}$, a contradiction, so $x \in A - \mathfrak{p}$. Thus, $\beta = v(x) \in \Delta$, so Δ is an isolated subgroup of Γ.

Finally, let's show that the construction $\mathfrak{p} \mapsto \Delta_\mathfrak{p}$ is a bijection from $\mathrm{Spec}(A)$ into the set of isolated subgroups of Γ. First, we show injectivity. Suppose $\Delta_\mathfrak{p} = \Delta_\mathfrak{q}$ for $\mathfrak{p}, \mathfrak{q} \in \mathrm{Spec}(A)$. By Exercise 5.28, since A is a valuation ring of K, $\mathfrak{p} \subset \mathfrak{q}$ or $\mathfrak{q} \subset \mathfrak{p}$. Relabelling if necessary, we can assume that $\mathfrak{q} \subset \mathfrak{p}$. Then, $A - \mathfrak{p} \subset A - \mathfrak{q}$. Let $x \in A - \mathfrak{q}$. Since $\Delta_\mathfrak{p} = \Delta_\mathfrak{q}$, the subset of elements greater than 0 are identical; thus, $v(A - \mathfrak{p}) = v(A - \mathfrak{q})$. Therefore, there is some $y \in A - \mathfrak{p}$ such that $v(x) = v(y)$. Then, $xy^{-1} \in A$ and $yx^{-1} \in A$. If $x \in \mathfrak{p}$, then since $yx^{-1} \in A$, $y = x(yx^{-1}) \in \mathfrak{p}$, a contradiction. Thus, $x \in A - \mathfrak{p}$; hence, $A - \mathfrak{p} = A - \mathfrak{q}$ so that $\mathfrak{p} = \mathfrak{q}$.

To show surjectivity, let Δ be an isolated subgroup of Γ. Let Δ' be the subset of $\delta \in \Delta$ such that $\delta \geq 0$; let $S = v^{-1}(\Delta') \subset A$. Since Δ' is a submonoid of Δ, S is a multiplicatively closed subset of A. Now, we aim to show that $A - S$ is a prime ideal. Let $x, y \in A - S$. Then, $v(x + y) \geq \min(v(x), v(y)) \geq 0$. If $v(x + y) \in \Delta$, then since Δ is an isolated subgroup, $\min(v(x), v(y)) \in \Delta$. Then, either $x \in S$ or $y \in S$, a contradiction. Thus, $v(x + y) \notin \Delta$, so $x + y \in A - S$. Thus, $A - S$ is additively closed. Suppose $x \in A - S$ and $a \in A$. Then, $v(ax) = v(a) + v(x) \geq v(x) \geq 0$ since $v(a) \geq 0$. If $v(ax) \in \Delta$, then since Δ is an isolated subgroup, we must have $v(x) \in \Delta$, a contradiction. Thus, $ax \in A - S$, so $A - S$ is an ideal of A. Finally, suppose $x, y \in S$. Since S is multiplicatively closed, $xy \in S$. Thus, $A - S \in \mathrm{Spec}(A)$, so the map is surjective. □

Now, let $\mathfrak{p}$ be a prime of A. By Exercise 5.28, $A/\mathfrak{p}$ and $A_\mathfrak{p}$ are both valuation rings since A is a valuation ring. We prove now that $\Gamma(A/\mathfrak{p}) \cong \Delta_\mathfrak{p}$ and $\Gamma(A_\mathfrak{p}) \cong \Gamma/\Delta_\mathfrak{p}$.

Proof. Note that we have $A \subset A_\mathfrak{p} \subset K$. Since $A^\times \subset A_\mathfrak{p}^\times$, we have a homomorphism

$$\Gamma = K^\times/A^\times \to K^\times/A_\mathfrak{p}^\times = \Gamma(A_\mathfrak{p}).$$

This map is clearly surjective. Let Δ be the kernel of this map, which is a subgroup of Γ. Suppose $x \in K^\times$, $v(x) \in \Delta$, and $v(x) \geq 0$. Then, $x \in A$. Since $v(x) \in \Delta$, $x \in A_\mathfrak{p}^\times$. Thus, $x \in A \cap A_\mathfrak{p}^\times = A - \mathfrak{p}$. Conversely, if $x \in A - \mathfrak{p}$, then $x \in A_\mathfrak{p}^\times$ so that $x \in \Delta$. Thus, the positive part of Δ and $\Delta_\mathfrak{p}$ correspond. Then, since Γ is totally ordered, their negative parts also match (since they are both groups), so $\Delta = \Delta_\mathfrak{p}$. Thus, $\Gamma(A_\mathfrak{p}) \cong \Gamma/\Delta_\mathfrak{p}$.

Now, note that the field of fractions of $A/\mathfrak{p}$ is canonically isomorphic to $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$. Then, we have $\Gamma(A/\mathfrak{p}) \cong (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p})^\times/(A/\mathfrak{p})^\times$. Note that we have a commutative diagram

$$\begin{array}{ccc} A & \longrightarrow & A_\mathfrak{p} \\ \downarrow & & \downarrow \\ A/\mathfrak{p} & \longrightarrow & A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}, \end{array}$$

where the horizontal maps are injective homomorphisms. Thus, we have a commutative diagram

$$\begin{array}{ccc} A^\times & \longrightarrow & A_\mathfrak{p}^\times \\ \downarrow & & \downarrow \\ (A/\mathfrak{p})^\times & \longrightarrow & (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p})^\times, \end{array}$$

where the horizontal maps are injective group homomorphisms. Equating kernels, we have

$$A_\mathfrak{p}^\times/A^\times \cong (A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p})^\times/(A/\mathfrak{p})^\times \cong \Gamma(A/\mathfrak{p}).$$

From above, the kernel of $\Gamma \to \Gamma(A_\mathfrak{p})$ is

$$\Gamma/\Gamma(A_\mathfrak{p}) \cong (K^\times/A^\times)/(K^\times/A_\mathfrak{p}^\times) \cong A_\mathfrak{p}^\times/A^\times \cong \Gamma(A/\mathfrak{p}),$$

noting that $A^\times \subset A_\mathfrak{p}^\times$. But we showed above that this kernel is exactly $\Delta_\mathfrak{p}$, so $\Gamma(A/\mathfrak{p}) \cong \Delta_\mathfrak{p}$. □

5.33 Exercise

Let Γ be a totally ordered abelian group. We will show how to construct a field K and a valuation v of K with Γ as the value group. Let k be any field, and let $A = k[\Gamma]$ be the group algebra of Γ over k. By definition, A is freely generated as a k-vector space by elements x_α ($\alpha \in \Gamma$) such that $x_\alpha x_\beta = x_{\alpha+\beta}$. First, we show that A is an integral domain.

Proof. Arbitrary elements of A are of the form $a = \sum_\alpha a_\alpha x_\alpha$ and $b = \sum_\beta b_\beta x_\beta$, where each sum has finitely many terms; assume that each coefficient in the sum is non-zero and that a and b are non-zero. Let $S_a \subset \Gamma$ be the set of $\gamma \in \Gamma$ such that $a_{gamma} \neq 0$; define S_b similarly. Then, since there are finitely many non-zero terms and Γ is totally ordered, each set has a strict maximum. Let $\alpha' \in S_a$ and $\beta' \in S_b$ be these strictly maximal elements. Note that

$$ab = \sum_{\alpha,\beta} a_\alpha b_\beta x_\alpha x_\beta = \sum_{\alpha,\beta} a_\alpha b_\beta x_{\alpha+\beta}.$$

Thus, the terms of ab are indexed over $\{\alpha+\beta\}$. If either $\alpha<\alpha'$ or $\beta<\beta'$, then $\alpha+\beta<\alpha'+\beta'$. Therefore, the coefficient for $\alpha'+\beta'$ of ab is $a_{\alpha'}b_{\beta'}$. Since $a_{\alpha'}\neq 0$ and $b_{\beta'}\neq 0$, $a_{\alpha'}b_{\beta'}\neq 0$ since k is a field. Finally, since A is freely generated as a k-vector space by elements x_γ ($\gamma\in\Gamma$), $abneq0$ since the coefficient for $\alpha'+\beta'$ is non-zero. Hence, A is an integral domain. □

If $u=\lambda_1 x_{\alpha_1}+\cdots+\lambda_n x_{\alpha_n}$ is any non-zero element of A, where $\lambda_i\neq 0$ and $\alpha_1<\cdots<\alpha_n$, define $v_0(u)=\alpha_1$. We will show that $v_0: A-\{0\}$ satisfies the valuation properties.

Proof. (1) Let $u=\lambda_1 x_{\alpha_1}+\cdots+\lambda_n x_{\alpha_n}$ and $v=\gamma_1 x_{\beta_1}+\cdots+\gamma_m x_{\beta_m}$, where each coefficient is non-zero and $\alpha_i<\alpha_j$ and $\beta_i<\beta_j$ whenever $i<j$. Then, $v_0(u)=\alpha_1$ and $v_0(v)=\beta_1$. The product uv is indexed over $\alpha_i+\beta_j$. Note that if $i>1$ or $j>1$, then $\alpha_1+\beta_1<\alpha_i+\beta_j$. Thus, the term of minimal term of uv is $\lambda_1\gamma_1 x_{\alpha_1+\beta_1}$. Since $\lambda_1\neq 0$ and $\gamma_1\neq 0$, the product is also non-zero, so $v_0(uv)=\alpha_1+\beta_1=v_0(u)+v_0(v)$.

(2) Let u, v be as before. If $\alpha_1<\beta_1$, then $v_0(u+v)=\alpha_1=v_0(u)\geq\min(v_0(u),v_0(v))$. If $\alpha_1>\beta_1$, then $v_0(u+v)=\beta_1=v_0(v)\geq\min(v_0(u),v_0(v))$. If $\alpha_1=\beta_1$, then it's possible that $\lambda_1=-\gamma_1$ so that the α_1 term of $u+v$ vanishes. Nonetheless, the minimal non-zero index will be larger than α_1 (if the sum is 0, we define $v_0(0)=\infty$, which is by definition larger than every other element). Thus, $v_0(u+v)\geq\alpha_1=v_0(u)\geq\min(v_0(u),v_0(v))$. Thus, in all cases, $v(u+v)\geq\min(v_0(u),v_0(v))$. □

Since we showed that A is an integral domain, A is a subring of its field of fractions K. We now show that v_0 extends uniquely to a valuation v of K and that the value group of v is Γ.

Proof. First of all, for any $u\in A$, $v_0(u)=v_0(u\cdot 1)=v_0(u)+v_0(1)$, so $v_0(1)=0$. Suppose $u\in A-\{0\}$ so that $u^{-1}\in K^\times$. If v is an extension of v_0 to $K^\times$, then

$$0=v_0(1)=v(1)=v(u^{-1}u)=v(u^{-1})+v(u)=v(u^{-1})+v_0(u).$$

Thus, $v(u^{-1})=-v_0(u)$. Any element of $K^\times$ has the form $a/b=ab^{-1}$, where $a,b\in A-\{0\}$. Then,

$$v(ab^{-1})=v(a)+v(b^{-1})=v_0(a)-v_0(b).$$

If v exists, it is clearly unique since it is defined on an arbitrary element of $K^\times$ in terms of v_0.

Now, let's show v exists. First, v is well-defined since for any $u,a,b\in A-\{0\}$, we have

$$v(ua/ub)=v_0(ua)-v_0(ub)=v_0(u)+v_0(a)-v_0(u)-v_0(b)=v_0(a)-v_0(b)=v(a/b),$$

so this definition is well-defined on $K^\times$. Further, if $a,b,c,d\in A-\{0\}$, then

$$v((a/b)(c/d))=v(ac/bd)=v_0(ac)-v_0(bd)=v_0(a)+v_0(c)-v_0(b)-v_0(d)=v(a/b)+v(c/d),$$

so the first property of a valuation holds. For the second, we have

$$v((a/b)+(c/d))=v((ad+bc)/bd)=v_0(ad+bc)-v_0(bd)\geq\min(v_0(ad),v_0(bc))-v_0(bd)$$

$$=\min(v_0(ad)-v_0(bd),v_0(bc)-v_0(bd))=\min(v(ad/bd),v(bc/bd))=\min(v(a/b),v(c/d)),$$

so indeed, v is a valuation of K. As desired then, v_0 extends uniquely to a valuation v of K.

Note that v_0 takes values in Γ, by definition. Since the extension of v to $K^\times$ is defined by $v(a/b)=v_0(a)-v_0(b)$, $v(K^\times)$ is a subgroup of Γ. If $\alpha\in\Gamma$, then $x_\alpha\in A$ and $v(x_\alpha)=v_0(x_\alpha)=\alpha$, so $v(K^\times)=\Gamma$, as desired. □

5.34 Exercise

Proof. A is a valuation ring in its field of fractions K. $f : A \to B$ is a ring homomorphism such that $f^* : \text{Spec}(B) \to \text{Spec}(A)$ is a closed mapping. Let $g : B \to K$ be any A-algebra homomorphism, and let $C = g(B)$. Then, $g \circ f : A \to K$ is an A-algebra homomorphism. Thus, for any $a \in A$, we have $(g \circ f)(a) = a \cdot (g \circ f)(1) = a \cdot 1$, so $A \subset C$. Let $\mathfrak{n}$ be a maximal ideal of C. Then, $\mathfrak{n}_B = \mathfrak{n}^c \in B$ is a prime ideal. Since $\mathfrak{n}_B$ is a non-trivial ideal, there is some maximal ideal $\mathfrak{m}$ of B such that $\mathfrak{n}_B \subset \mathfrak{m}$. Then, $g(\mathfrak{m}) \supset g(\mathfrak{n}_B) = \mathfrak{n}$. Note that $g : B \to C$ is surjective, so the image of an ideal is an ideal. Since $\mathfrak{n}$ is maximal, either $g(\mathfrak{m}) = \mathfrak{n}$ so that $\mathfrak{m} = \mathfrak{n}_B$ or $g(\mathfrak{m}) = (1)$. In the latter case, $1 \in g(\mathfrak{m})$ so that $1 \in \mathfrak{m}$, a violation of maximality. Thus, $\mathfrak{n}_B$ is maximal in B.

Since $f^* : \text{Spec}(B) \to \text{Spec}(A)$ is a closed mapping, we have by Exercise 5.10 that f has the going-up property. Thus, $\mathfrak{m} = \mathfrak{n}_B^c$ is a maximal ideal of A. Since A is a valuation ring, it is a local ring, so $\mathfrak{m}$ is the unique maximal ideal of A. Thus, $A = A_\mathfrak{m} \subset K$. Note that $C_\mathfrak{n}$ is also a local ring of K and that $A_\mathfrak{m} = A \subset C \subset C_\mathfrak{n}$. Further,

$$\mathfrak{n} \cap A = \mathfrak{n} \cap B \cap A = \mathfrak{n}_B \cap A = \mathfrak{m}.$$

Therefore, $C_\mathfrak{n}$ dominates $A_\mathfrak{m}$ in the set of local rings of K. By Exercise 5.27, the maximal elements of this set with respect to domination are precisely the valuation rings of K. Since A is a valuation ring and $C_\mathfrak{n}$ is a local ring of K that dominates A, we must have $A = C_\mathfrak{n}$. Thus, $C \subset C_\mathfrak{n} = A$, so $A = C$, as desired. □

5.35 Exercise

Proof. Let $f : A \to B$ be integral, and let C be an A-algebra. By Exercise 5.3, $f \otimes 1 : A \otimes_A C \to B \otimes_A C$ is integral. We have a canonical isomorphism $A \otimes_A C \cong C$ so that $f \otimes 1 : C \to B \otimes_A C$ is integral. Now, by Exercise 5.1, $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \to \text{Spec}(C)$ is a closed mapping.

Conversely, suppose that $f : A \to B$ is a ring homomorphism and that $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \to \text{Spec}(C)$ is a closed mapping for any A-algebra C; further, assume that B is an integral domain. Note that we can factor f as $A \to f(A) \to B$. Clearly, $A \to f(A)$ is an integral map, so we just have to show that $\varphi : f(A) \to B$ is an integral map. Let D be any $f(A)$-algebra. Then, we have a map

$$(\varphi \otimes 1)^* : \text{Spec}(B \otimes_{f(A)} D) \to \text{Spec}(D)$$

Since B and D are both $f(A)$-algebras, they are both A-algebras via f. Under this action, $B \otimes_{f(A)} D \cong B \otimes_A D$, noting that the A-algebra action of B defined by $f(A)$ is precisely the given A-algebra structure of B. Thus, $(\varphi \otimes 1)^*$ is a closed mapping by the given assumption. Thus, we've reduced to the case when $A \subset B$ and f is injective.

So, suppose $f : A \hookrightarrow B$ ($A \subset B$) is an injective ring homomorphism and that $(f \otimes 1)^* : \text{Spec}(B \otimes_A C) \to \text{Spec}(C)$ is a closed mapping for any A-algebra C; further, assume that B is an integral domain. Let K be the field of fractions of B so that $A \subset B \subset K$. A is contained by local rings in K (e.g., localizations at a prime ideal). By Exercise 5.27, the set of local rings of K have maximal elements, and these maximal elements are valuation rings, so there is a valuation ring A' of K containing A. By Corollary 5.22, the integral closure of A in K is the intersection of valuation rings of K containing A. Since A' is an arbitrary valuation ring containing A, if we can show that A' contains B, then B will be a subset of this intersection and thus integral over A.

Since A' is an A-algebra, $(f \otimes 1)^* : \text{Spec}(B \otimes_A A') \to \text{Spec}(A')$ is a closed mapping, by assumption. We define a homomorphism $g : B \otimes_A A' \to K$ on simple tensors by $g : b \otimes a' \mapsto ba'$.

Since $(f \otimes 1)^*$ is a closed mapping and A' is a valuation ring in its field of fractions K, we have by Exercise 5.34 that $g(B \otimes_A A') = A'$. Since $1 \in A'$, for any $b \in B$, we have $b = b \cdot 1 = g(b \otimes 1) \in A'$. Thus, $B \subset A'$ for any valuation ring A' of K containing A, so B is integral over A.

Let's prove the converse again, but instead of assuming B is an integral domain, we will assume that B has only finitely many minimal prime ideals (e.g., if B is Noetherian). Let $\mathfrak{p}$ be a minimal prime ideal of B. Consider the composite homomorphism $\varphi : A \to B \to B/\mathfrak{p}$. If D is a $B/\mathfrak{p}$-algebra, then it is an A-algebra via φ. Then, $\psi^* : \operatorname{Spec}(B \otimes_A D) \to \operatorname{Spec}(D)$ is a closed map by assumption. By Exercise 5.3, $B \otimes_A D \to B/\mathfrak{p} \otimes_A D$ is integral, so by the first result, $\operatorname{Spec}(B/\mathfrak{p} \otimes_A D) \to \operatorname{Spec}(B \otimes_A D)$ is a closed mapping. Then, $\operatorname{Spec}(B/\mathfrak{p} \otimes_A D) \to \operatorname{Spec}(B \otimes_A D) \to \operatorname{Spec}(D)$ is a composition of closed mapppings and is thus a closed mapping. Therefore, since $B/\mathfrak{p}$ is an integral domain, $\varphi : A \to B/\mathfrak{p}$ satisfies the properties of the previous converse, so φ is integral. Thus, for all minimal primes $\mathfrak{p}_i$ of B, $A \to B \to B/\mathfrak{p}_i$ is integral.

Since there are finitely many minimal primes, and each of $B/\mathfrak{p}_i$ is an integral A-algebra, we have by Exercise 5.6 that $\prod_{i=1}^n B/\mathfrak{p}_i$ is an integral A-algebra. Note that we have a canonical ring homomorphism $B \to \prod_{i=1}^n B/\mathfrak{p}_i$. We want to show that this map is surjective. It will be sufficient to show that the generating elements $(0, \dots, 0, 1, 0, \dots, 0)$ are all in the image. Then, for each i, there is some $x_i \in B$ such that $x_i \mapsto (0, \dots, 0, 1, 0, \dots, 0)$ (with 1 in the factor $B/\mathfrak{p}_i$). If $(b_1, \dots, b_n)$ is an arbitrary element, there is some $a_i \in B$ such that $\overline{a_i} = b_i$ for each i so that $a_1 x_1 + \cdots + a_n x_n \mapsto (b_1, \dots, b_n)$.

By symmetry, it is sufficient to prove that there is some $x \in B$ with image $(0, \dots, 0, 1)$, i.e. $x \in \bigcap_{i=1}^{n-1} \mathfrak{p}_i$ but $\overline{x} = 1 \in B/\mathfrak{p}_n$. If $\mathfrak{p}_n \subset \bigcap_{i=1}^{n-1} \mathfrak{p}_i$, then $\mathfrak{p}_n \subset \mathfrak{p}_1$, contradicting minimality. Thus, the image of the ideal $\bigcap_{i=1}^{n-1} \mathfrak{p}_i$ under $B \to B/\mathfrak{p}_n$ is (1). Hence, there is some element $x \in \bigcap_{i=1}^{n-1} \mathfrak{p}_i$ such that $\overline{x} = 1 \in B/\mathfrak{p}_n$, as desired. Therefore, $B \to \prod_{i=1}^n B/\mathfrak{p}_i$ is surjective.

If $x \in B$ is in the kernel, $x \in \bigcap_{i=1}^n \mathfrak{p}_i$. But since $\{\mathfrak{p}_i\}$ are the minimal primes, this intersection is identical to the intersection over all primes, i.e. the nilradical $\mathfrak{N}$, so $x \in \mathfrak{N}$. Conversely, if $x \in \mathfrak{N}$, then $x \in \mathfrak{p}$ for every prime $\mathfrak{p}$, so $x \mapsto 0$. Therefore, $B/\mathfrak{N} \cong \prod_{i=1}^n B/\mathfrak{p}_i$. Since $\prod_{i=1}^n B/\mathfrak{p}_i$ is an integral A-algebra, $B/\mathfrak{N}$ is an integral A-algebra.

Suppose $b \in B$. Then, $\overline{b} \in B/\mathfrak{N}$ is integral over A. Thus, it satisfies some monic polynomial

$$\overline{b}^n + a_{n-1}\overline{b}^{n-1} + \cdots + a_0 = 0,$$

where $a_i \in A$. Since $B \to B/\mathfrak{N}$ is a surjective homomorphism, we can lift this to B:

$$b^n + a_{n-1}b^{n-1} + \cdots + a_0 \in \mathfrak{N}.$$

Since this element is in the nilradical, there is some power $k > 0$ that annihilates it, so

$$0 = (b^n + a_{n-1}b^{n-1} + \cdots + a_0)^k = b^{nk} + \cdots + a_0^k.$$

Thus, b satisfies a monic polynomial over A. Hence, $A \to B$ is integral. $\square$

6 Chain Conditions (Done)

6.1 Exercise

Proof. (i) $u : M \to M$ is a surjective module homomorphism. Then, $\ker u$ is a submodule of M. In fact, we have an ascending chain of submodules $\ker u \subset \ker u^2 \subset \cdots \subset \ker u^n \subset \cdots$. Since M is a Noetherian A-module, this chain becomes stationary so that $\ker u^n = \ker u^{n+1}$ for some $n > 0$. Since u is surjective, u^n is surjective, so every $m \in M$ is of the form $m = u^n(m')$ for some $m' \in M$. If $m \neq 0$, $m' \notin \ker u^n = \ker u^{n+1}$. Thus, $u(m) = u^{n+1}(m') \neq 0$, so $m \notin \ker u$. Thus, $\ker u = 0$, so u is an isomorphism.

(ii) $u : M \to M$ is an injective module homomorphism. Then, $\operatorname{im} u$ is a submodule of M. We have a descending chain of submodules $\operatorname{im}(u) \supset \operatorname{im}(u^2) \supset \cdots$. Since M is an Artinian A-module, this chain becomes stationary so that $\operatorname{im} u^n = \operatorname{im} u^{n+1}$ for some $n > 0$. Since u is injective, u^n is injective for all $n > 0$. Let $m \in M$. Then, $u^n(m) \in \operatorname{im} u^n = \operatorname{im} u^{n+1}$, so there is some $m' \in M$ such that $u^{n+1}(m') = u^n(m)$. Since u^n is injective, $u^n(m) = u^n(u(m'))$ implies that $m = u(m')$. Thus, $m \in \operatorname{im} u$ so that $\operatorname{im} u = M$. Hence, u is an isomorphism. □

6.2 Exercise

Proof. Let N be an arbitrary submodule of M. Let Σ be the set of finitely generated submodules of N. This is a non-empty set of finitely generated submodules of M, so by assumption, it has a maximal element N_0. N_0 is finitely generated and $N_0 \subset N$. Suppose $x \in N$. Then, $N_0 + Ax \subset N$ and $N_0 + Ax$ is a finitely generated submodule, so $N_0 + Ax \in \Sigma$. Since N_0 is maximal and $N_0 \subset N_0 + Ax$, we must have $x \in N_0$. Thus, $N = N_0$ so that N is finitely generated. By Proposition 6.2, since every submodule of M is finitely generated, M is Noetherian. □

6.3 Exercise

Proof. Since M/N_1 and M/N_2 are both Noetherian (resp. Artinian), $M/N_1 \oplus M/N_2$ is Noetherian (resp. Artinian) via Corollary 6.4. We have a module homomorphism $\varphi : M/(N_1 \cap N_2) \to M/N_1 \oplus M/N_2$ since $M/(N_1 \cap N_2) \to M/N_1$ and $M/(N_1 \cap N_2) \to M/N_2$ are both well-defined homomorphisms. Suppose $\varphi(m) = 0$. Then, $m \in N_1$ and $m \in N_2$, so $m \in N_1 \cap N_2$. Hence, $m = 0$, so φ is injective. Thus, we have a short exact sequence of A-modules

$$0 \to M/(N_1 \cap N_2) \xrightarrow{\varphi} M/N_1 \oplus M/N_2 \to Q \to 0,$$

where Q is the quotient module. By Proposition 6.3, since $M/N_1 \oplus M/N_2$ is Noetherian (resp. Artinian), $M/(N_1 \cap N_2)$ is Noetherian (resp. Artinian). □

6.4 Exercise

Proof. Since M is a Noetherian A-module and $\mathfrak{a} = \operatorname{Ann}(M)$, M is a faithful, Noetherian $A/\mathfrak{a}$-module since all $A/\mathfrak{a}$-submodules are A-submodules. Thus, to show that $A/\mathfrak{a}$ is Noetherian, it is sufficient to show that whenever M is a Noetherian A-module such that $\operatorname{Ann}(A) = 0$, then A is a Noetherian ring.

Since M is Noetherian, it is finitely generated by Proposition 6.2. Let $\{m_i\}_{i=1}^n$ be a finite generating set of M. Since M is Noetherian M^n is Noetherian by Corollary 6.4. We have an A-module homomorphism $a \mapsto (am_1, \ldots, am_n) \subset M^n$. Suppose that $a \mapsto 0$ so that for any

$m = a_1m_1 + \cdots + a_nm_n \in M$, we have $am = 0$. Since $\text{Ann}(M) = 0$, $a = 0$, so the map is injective. Since there is an A-module injection of A into the Noetherian A-module M^n, A is Noetherian by Proposition 6.3. □

Remark. *Note that this does not hold when we replace "Noetherian" with "Artinian". Consider the Prüfer p-group $\mathbb{Z}/p^\infty\mathbb{Z}$, which is an Artinian $\mathbb{Z}$-module. If $a \in \text{Ann}(\mathbb{Z}/p^\infty\mathbb{Z})$, then $a \cdot 1 \in (p)^n$ for all $n > 0$. Clearly then, $a = 0$. However, $\mathbb{Z}/(0) \cong \mathbb{Z}$ is not Artinian since $(p) \supset (p^2) \supset \cdots$ is an infinite, strictly descending chain of ideals of $\mathbb{Z}$.*

6.5 Exercise

Proof. Let S be a subspace of X, and suppose we have a descending chain of closed sets $C_1 \supset C_2 \supset \cdots$ of S. Then, $C_i = S \cap C_i'$, where C_i' is a closed set of X. Let $K_i = \bigcap_{j \leq i} C_j'$, a closed set of X. Clearly, $(K_i)_i$ forms a descending chain of closed sets of X. Since X is Noetherian with respect to the open sets, X is Artinian with respect to the closed sets, so there is some $n > 0$ such that $K_k = K_{k+1}$ for all $k \geq n$.

So, let $k \geq n$. Then,

$$C_k = S \cap C_k' = S \cap \bigcap_{i \leq k} C_i' = S \cap K_k = S \cap K_{k+1} = S \cap \bigcap_{i \leq k+1} C_i' = S \cap C_{k+1}' = C_{k+1},$$

so $(C_i)_i$ stabilizes. Thus, S is Artinian with respect to closed sets, so S is Noetherian with respect to open sets, as desired.

Finally, suppose $\{U\}$ is an open cover of X. If the cover is finite, it is itself a finite subcover, so suppose $\{U\}$ is infinite. Then, let $U_1 \in \{U\}$. If $x \in X - U_1$, there is some $U_2 \in \{U\}$ such that $x \in U_2$. Let $V_2 = U_2 \cup U_1$. If there is some $x \in X - V_2$, there is some $U_3 \in \{U\}$ such that $x \in U_3$. Define $V_3 = U_3 \cup V_2$. If we can continue this process indefinitely, then we have a strictly ascending chain of open sets $V_1 \subset V_2 \subset \cdots$ of X. Since X is Noetherian, this cannot happen. Thus, for some $n > 0$, $V_n = X$ so that $\bigcup_{i=1}^n U_n = X$. Thus, $\{U\}$ has a finite subcover, so X is quasi-compact. □

6.6 Exercise

Proof. (i) $\implies$ (iii). If X is Noetherian and S is a subspace, then by Exercise 6.5, S is a Noetherian space. Then, also by Exercise 6.5, S is quasi-compact.

(iii) $\implies$ (ii). Clear. All open subspaces are subspaces.

(ii) $\implies$ (i). Let $U_1 \subset U_2 \subset \cdots$ be an ascending chain of open sets of X. Then, $S = \bigcup_i U_i$ is an open set, so by assumption, the open subspace S is quasi-compact. Note that $\{U_i\}$ is an open covering of S, so there is a finite subcover $\{U_i\}_{i \in I}$. Thus, I has a maximum index $n > 0$, so $S = \bigcup_{i \in I} U_i = U_n$ since $U_i \subset U_n$ for all $i \leq n$. Thus, $U_n = \bigcup_i U_i \supset U_j$ for all j. Since $U_n \subset U_j$ for all $j \geq n$, $U_n = U_j$ for all $j \geq n$, so the chain stabilizes. Thus, X is a Noetherian space. □

6.7 Exercise

Proof. Let Σ be the set of closed subsets of X that are not finite unions of irreducible closed subspaces. Suppose to the contrary that X is not a finite union of irreducible closed subspaces so that $X \in \Sigma$, i.e. Σ is non-empty. Order Σ such that $C \leq C'$ if $C \subset C'$. Let $\mathcal{T}$ be a chain of Σ. Let $C_0 \in \mathcal{T}$. Then, we have a descending chain $\cdots \subset C_1 \subset C_0$ of closed subsets of X. Since X is

Noetherian, the set of closed sets of X is Artinian, so there is some $n > 0$ such that $C_{j+1} = C_j$ for all $j \geq n$. Thus, C_n is a minimal element of $\mathcal{T}$. By Zorn's lemma, Σ has minimal elements.

Let C be a minimal element of Σ. C is non-empty since $\emptyset \notin \Sigma$; $\emptyset$ is the empty union of irreducible closed subspaces. So, let $x \in C$ and let C_x be the closure of $\{x\}$. If C were irreducible, then C is just a finite union of closed irreducible closed subspaces since it is itself a closed irreducible subspace. Therefore, C is reducible so that $C = C_1 \cup C_2$, where C_1 and C_2 are closed subsets and $C_1 \neq C$ and $C_2 \neq C$. By the minimality of C, C_1 and C_2 are finite unions of irreducible closed subspaces. But then, $C = C_1 \cup C_2$ is a finite union of irreducible closed subspaces, yielding our desired contradiction. Thus, X is a finite union of irreducible closed subspaces.

By Exercise 1.20, the maximal irreducible subspaces of X are closed and cover X; these are the "irreducible components". Since X is a finite union of irreducible closed subspaces and each irreducible subspace is contained in a maximal irreducible subset by definition, X is covered by finitely many irreducible components $\{C_i\}$. Let Y be any irreducible component, so it is closed. Then,

$$Y = Y \cap X = Y \cap (C_1 \cup \cdots \cup C_n) = (Y \cap C_1) \cup \cdots \cup (Y \cap C_n),$$

a union of closed sets. By the irreducibility of Y, we must have $Y = Y \cap C_i$ for some i, so $Y \subset C_i$. Since Y is maximal, we must have $Y = C_i$. Hence, X has finitely many irreducible components. □

6.8 Exercise

Proof. Closed sets of $\mathrm{Spec}(A)$ are of the form $V(\mathfrak{a})$, where $\mathfrak{a}$ is an ideal of A. Thus, a descending chain of closed ideals of $\mathrm{Spec}(A)$ has the form $V(\mathfrak{a}_1) \supset V(\mathfrak{a}_2) \supset \cdots$. Note that $V(\mathfrak{a}_i) = V(r(\mathfrak{a}_i))$ and $r(\mathfrak{a}_i)$ is the intersection of prime ideals containing $\mathfrak{a}_i$. Then, since $V(r(\mathfrak{a}_i)) \supset V(r(\mathfrak{a}_{i+1}))$, we have $r(\mathfrak{a}_i) \subset r(\mathfrak{a}_{i+1})$ since all the primes in the intersection of $r(\mathfrak{a}_{i+1})$ are in the intersection of $r(\mathfrak{a}_i)$. Therefore, we have an ascending chain of ideals $r(\mathfrak{a}_1) \subset r(\mathfrak{a}_2) \subset \cdots$. Since A is Noetherian, this chain stabilizes so that $r(\mathfrak{a}_i) = r(\mathfrak{a}_n)$ for all $i \geq n$ for some $n > 0$. Then, $V(\mathfrak{a}_i) = V(r(\mathfrak{a}_i)) = V(r(\mathfrak{a}_n)) = V(\mathfrak{a}_n)$, so the chain of closed sets also stabilizes. Thus, $\mathrm{Spec}(A)$ is a Noetherian topological space. □

The converse is not true in general. Let $A = k[x_1, x_2, \ldots]/(x_1^2, x_2^2, \ldots)$. In A, each x_i is nilpotent, so every prime ideal of A contains each x_i. Thus, any prime $\mathfrak{p}$ contains $(x_1, x_2, \ldots)$. If $a \in \mathfrak{p}$ such that $a \notin (x_1, x_2, \ldots)$, then $a = l + f$, where $f \in (x_1, x_2, \ldots)$ and $l \in k^\times$. Then, $a, f \in \mathfrak{p}$ so that $l \in \mathfrak{p}$. But, $l^{-1} \in A$, so $1 = l^{-1}l \in \mathfrak{p}$, a contradiction. Thus, there is only one prime $\mathfrak{p}$ and $\mathfrak{p} = (x_1, x_2, \ldots)$. Since $\mathrm{Spec}(A)$ is a singleton, it is clearly a Noetherian topological space. However, $(x_1) \subset (x_1, x_2) \subset (x_1, x_2, x_3) \subset \cdots$ is an infinite, strictly ascending chain of ideals of A, so A is not Noetherian, even though $\mathrm{Spec}(A)$ is a Noetherian space.

6.9 Exercise

Proof. Since A is Noetherian, $\mathrm{Spec}(A)$ is a Noetherian topological space by Exercise 6.8. Then, by Exercise 6.7, $\mathrm{Spec}(A)$ has finitely many irreducible components. By Exercise 1.20, the irreducible components of $\mathrm{Spec}(A)$ are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of A. Thus, there are only finitely many closed sets of $\mathrm{Spec}(A)$ of the form $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal. If $V(\mathfrak{p}) = V(\mathfrak{q})$, then $\mathfrak{p}$ and $\mathfrak{q}$ are in both sets, so $\mathfrak{p} \subset \mathfrak{q}$ and vice-versa so that $\mathfrak{p} = \mathfrak{q}$. Thus, the mapping from minimal primes of A to closed sets, $\mathfrak{p} \mapsto V(\mathfrak{p})$, is injective, so A has finitely many minimal prime ideals. □

6.10 Exercise

Proof. Since M is a Noetherian A-module, it is finitely generated by Proposition 6.2. Then, by Exercise 3.19, $\operatorname{Supp}(M) = V(\operatorname{Ann}(M))$. Thus, $\operatorname{Supp}(M)$ is a closed subspace of $\operatorname{Spec}(A)$. By Exercise 6.4, $A/\operatorname{Ann}(M)$ is a Noetherian ring since M is Noetherian. By Exercise 6.8, $\operatorname{Spec}(A/\operatorname{Ann}(M))$ is a Noetherian space. Via the surjective ring homomorphism $\phi : A \to A/\operatorname{Ann}(M)$, we have a homeomorphism of $\operatorname{Spec}(A/\operatorname{Ann}(M))$ onto $V(\operatorname{Ann}(M))$ (Exercise 1.21). Hence, $\operatorname{Supp}(M) = V(\operatorname{Ann}(M))$ is Noetherian since it is homeomorphic to the Noetherian space $\operatorname{Spec}(A/\operatorname{Ann}(M))$. □

6.11 Exercise

Proof. $f : A \to B$ is a ring homomorphism, and $\operatorname{Spec}(B)$ is a Noetherian space. If $f^* : \operatorname{Spec}(B) \to \operatorname{Spec}(A)$ is a closed mapping, then by Exercise 5.10, f has the going-up property. Conversely, suppose f has the going-up property. To show that f^* is a closed mapping, we must show that for any radical ideal $\mathfrak{b}$ of B, $f^*(V(\mathfrak{b}))$ is closed.

We start with a prime ideal $\mathfrak{q}$ of B. By Exercise 1.21, $overline f^*(V(\mathfrak{q})) = V(\mathfrak{q}^c)$. To show that $f^*(V(\mathfrak{q}))$ is closed, it is then sufficient to show that $f^*(V(\mathfrak{q})) = V(\mathfrak{q}^c)$. We have, by the same exercise, that $f^*(V(\mathfrak{q})) \subset V(\mathfrak{q}^c)$, so suppose $\mathfrak{p}' \in V(\mathfrak{q}^c)$. Note that $\mathfrak{p} = \mathfrak{q}^c$ is a prime ideal of A, so we have $\mathfrak{q}^c = \mathfrak{p}$ and $\mathfrak{p}' \supset \mathfrak{p}$. By the going-up property, there is some prime ideal $\mathfrak{q}'$ of B such that $\mathfrak{q}' \supset \mathfrak{q}$ and $\mathfrak{p}' = \mathfrak{q}'^c$. Thus, $\mathfrak{q}' \in V(\mathfrak{q})$, so $\mathfrak{p}' = f^*(\mathfrak{q}') \in f^*(V(\mathfrak{q}))$. For any prime ideal $\mathfrak{q}$ of B, we thus have $f^*(V(\mathfrak{q})) = V(\mathfrak{q}^c)$.

Now, let $\mathfrak{b}$ be an arbitrary radical ideal of B, so $\mathfrak{b} = \bigcap_{\mathfrak{q} \in V(\mathfrak{b})} \mathfrak{q}$. Since $\operatorname{Spec}(B)$ is Noetherian, $V(\mathfrak{b})$ is Noetherian by Exercise 6.5. By Exercise 1.21, $V(\mathfrak{b})$ is homeomorphic to $\operatorname{Spec}(B/\mathfrak{b})$, so $\operatorname{Spec}(B/\mathfrak{b})$ is Noetherian. Thus, $\operatorname{Spec}(B/\mathfrak{b})$ has finitely many irreducible components by Exercise 6.7. By Exercise 1.20, the irreducible components of $\operatorname{Spec}(B/\mathfrak{b})$ are the closed sets $V(\mathfrak{p})$, where $\mathfrak{p}$ is a minimal prime ideal of $B/\mathfrak{b}$; hence $B/\mathfrak{b}$ has finitely many minimal prime ideals. By the one-to-one, order-preserving correspondence between prime ideals of $B/\mathfrak{b}$ and primes of B containing $\mathfrak{b}$, the set of primes of B containing $\mathfrak{b}$ has finitely many minimal elements $\{q_i\}$. Thus, we can write $\mathfrak{b} = \bigcap_{\mathfrak{q} \in V(\mathfrak{b})} \mathfrak{q} = \bigcap_{i=1}^n \mathfrak{q}_i$.

Finally, by Exercise 1.15, since the intersection is finite,

$$V(\mathfrak{b}) = V\left(\bigcap_{i=1}^n \mathfrak{q}_i\right) = \bigcup_{i=1}^n V(\mathfrak{q}_i)$$

so that

$$f^*(V(\mathfrak{b})) = f^*\left(\bigcup_{i=1}^n V(\mathfrak{q}_i)\right) = \bigcup_{i=1}^n f^*\left(V(\mathfrak{q}_i)\right) = \bigcup_{i=1}^n V(\mathfrak{q}_i^c),$$

which is closed since it is a finite union of closed sets. Thus, f^* is a closed mapping, as desired. □

6.12 Exercise

Proof. Let $\mathfrak{p}_1 \subset \mathfrak{p}_2 \subset \cdots$ be an ascending chain of prime ideals of A. Then, we have a descending chain of closed sets $V(\mathfrak{p}_1) \supset V(\mathfrak{p}_2) \supset \cdots$. Since $\operatorname{Spec}(A)$ is Noetherian, we must have $V(\mathfrak{p}_n) = V(\mathfrak{p}_i)$ for all $i \geq n$ for some $n > 0$. But then, $\mathfrak{p}_n \in V(\mathfrak{p}_i)$, so $\mathfrak{p}_n \supset \mathfrak{p}_i$. Thus, for all $i \geq n$, $\mathfrak{p}_n = \mathfrak{p}_i$. Therefore, the set of prime ideals of A satisfies the ascending chain condition. □

The converse does not hold in general. Let $A = \prod_{i=1}^\infty K$, where K is a field. Then, we have ideals of A of the form $\prod_{i=1}^\infty I_i$, where I_i is 0 or K. Suppose I is an ideal such that the ith factor

is 0, the rest being K. If $xy \in I$, then $x_i y_i = 0$, so either $x_i = 0$ or $y_i = 0$ since K is an integral domain. Thus, $x \in I$ or $y \in I$, so I is a prime ideal. We denote this prime ideal by $\mathfrak{p}_i$.

Let $\mathfrak{p}$ be an arbitrary prime ideal of A. For every $x = (x_1, x_2, \ldots) \in A$, we can define a "pseudo-inverse" by letting $y_i = x_i^{-1}$ if $x_i \neq 0$ and $y_i = 0$ if $x_i = 0$; let $y = (y_1, y_2, \ldots)$. Then, $(xy)_i = 1$ if $x_i \neq 0$ and $(xy)_i = 0$ if $x_i = 0$. Then, $(x^2 y)_i = x_i$ if $x_i \neq 0$ and $(x^2 y)_i = 0$ if $x_i = 0$. Thus, for all i, $(x^2 y)_i = x_i$, so $x^2 y = x$. Now, let $\overline{x}$ be an arbitrary element of $A/\mathfrak{p}$. Then, if $x \in A$ is a lift of $\overline{x}$, x has a pseudo-inverse $y \in A$, so $\overline{x}^2 \overline{y} = \overline{x}$. If $\overline{x} \neq 0$, then since $A/\mathfrak{p}$ is an integral domain, $\overline{xy} = 1$, so $A/\mathfrak{p}$ is a field. Thus, every prime ideal of A is maximal, so no two distinct primes of A are comparable. Therefore, the prime ideals of A satisfy the ascending chain condition trivially.

Let I_i be the ideal such that the first i factors are K and the remaining factors are 0. Then, $V(I_n) \supset \{\mathfrak{p}_{n+1}, \mathfrak{p}_{n+2}, \ldots\}$ and $\mathfrak{p}_n \notin V(I_n)$. Thus, we have an infinite, descending chain of closed sets $V(I_1) \supset V(I_2) \supset \cdots$. Note that all inclusions are strict since for all n, $\mathfrak{p}_{n+1} \in V(I_n) - V(I_{n+1})$, so the closed sets do not satisfy the descending chain condition. Thus, $\mathrm{Spec}(A)$ is not Noetherian.

7 Noetherian Rings (Done)

7.1 Exercise

Proof. Let Σ be the set of ideals in A which are not finitely generated. Since A is not Noetherian, Σ is non-empty. Let $\mathcal{T}$ be a chain in Σ. Define $\mathfrak{a} = \bigcup_{\mathfrak{b}\in\mathcal{T}} \mathfrak{b}$. If $x_1, x_2 \in \mathfrak{a}$, then $x_1 \in \mathfrak{b}_1$ and $x_2 \in \mathfrak{b}_2$ for ideals $\mathfrak{b}_1, \mathfrak{b}_2 \in \mathcal{T}$. Since $\mathcal{T}$ is a chain, these ideals are comparable, so relabelling if necessary, we may assume that $\mathfrak{b}_1 \subset \mathfrak{b}_2$. Then, $x_1, x_2 \in \mathfrak{b}_2$, so $x_1 + x_2 \in \mathfrak{b}_2 \subset \mathfrak{a}$. Thus, $\mathfrak{a}$ is additively closed. If $a \in A$ and $x \in \mathfrak{a}$, there is some $\mathfrak{b} \in \mathcal{T}$ such that $x \in \mathfrak{b}$. Then, $ax \in \mathfrak{b} \subset \mathfrak{a}$, so $\mathfrak{a}$ is an ideal of A.

Suppose to the contrary that $\mathfrak{a}$ is finitely generated. Then, there are finitely many generating elements $x_1, \ldots, x_n$. Each x_i is in some $\mathfrak{b}_i \in \mathcal{T}$. Since $\mathcal{T}$ is a chain, there is some $j \in \{1, \ldots, n\}$ such that $\mathfrak{b}_i \subset \mathfrak{b}_j$ for all i. Then, $x_i \in \mathfrak{b}_i \subset \mathfrak{b}_j$ for all i. Since $\mathfrak{b}_j \subset \mathfrak{a}$ and $\mathfrak{a} = \sum_i Ax_i \subset \sum_i \mathfrak{b}_j = \mathfrak{b}_j$, we have $\mathfrak{a} = \mathfrak{b}_j$. But since $\mathfrak{b}_j \in \Sigma$, $\mathfrak{a} = \mathfrak{b}_j$ is not finitely generated, a contradiction. Thus, $\mathfrak{a} \in \Sigma$, so all chains in Σ have maximal elements in Σ. By Zorn's lemma, Σ has maximal elements.

Now, suppose $\mathfrak{a}$ is a maximal element of Σ. Suppose there exist $x \notin \mathfrak{a}$ and $y \notin \mathfrak{a}$ such that $xy \in \mathfrak{a}$. Thus, the ideal $\mathfrak{a} + (x)$ is an ideal that properly contains $\mathfrak{a}$. Since $\mathfrak{a}$ is maximal in Σ, $\mathfrak{a} + (x)$ is finitely generated. In particular, there are elements $x_1, \ldots, x_n$ that generate $\mathfrak{a} + (x)$ as an A-module. Then, for each x_i, there is some $a_i \in A$ such that $x_i - a_i x \in \mathfrak{a}$. Let $\mathfrak{a}_0$ be the ideal generated by $\{x_i - a_i x\}_i$, so clearly, $\mathfrak{a}_0 \subset \mathfrak{a}$ and $\mathfrak{a}_0$ is finitely generated. Further, $x_i \in \mathfrak{a}_0 + (x)$ for all i since $x_i - a_i x \in \mathfrak{a}_0$ and $a_i x \in (x)$. Therefore, $\mathfrak{a}_0 + (x) = \mathfrak{a} + (x)$.

By the definition of $(\mathfrak{a} : x)$, $x(\mathfrak{a} : x) \subset \mathfrak{a}$. Further, $\mathfrak{a} \subset (\mathfrak{a} : x)$; this inclusion is proper since $y \in (\mathfrak{a} : x)$ but $y \notin \mathfrak{a}$. Since $\mathfrak{a}$ is maximal in Σ, $(\mathfrak{a} : x)$ is finitely generated; hence, $x(\mathfrak{a} : x)$ is finitely generated. Therefore, $\mathfrak{a}_0 + x(\mathfrak{a} : x) \subset \mathfrak{a}$. Suppose $a \in \mathfrak{a}$. Then, $a = a' + bx$, where $a' \in \mathfrak{a}_0$ and $b \in A$ since $\mathfrak{a}_0 + (x) = \mathfrak{a} + (x)$. Thus, $bx = a - a' \subset \mathfrak{a}$, so $b \in (\mathfrak{a} : x)$. Hence, $bx \in x(\mathfrak{a} : x)$ so that $a = a' + bx \in \mathfrak{a}_0 + x(\mathfrak{a} : x)$. Therefore, we have equality: $\mathfrak{a}_0 + x(\mathfrak{a} : x) = \mathfrak{a}$.

Since $\mathfrak{a}_0$ and $x(\mathfrak{a} : x)$ are both finitely generated, their sum, $\mathfrak{a}$, is also finitely generated. This is a clear contradiction, so we cannot have $x \notin \mathfrak{a}$ and $y \notin \mathfrak{a}$ such that $xy \in \mathfrak{a}$; hence, $\mathfrak{a}$ is prime. Thus, maximal elements of Σ is prime. In particular, if A is a ring in which all prime ideals are finitely generated, then Σ must be empty, so every ideal of A is finitely generated; in other words, A is Noetherian. □

7.2 Exercise

Proof. If $\sum_i a_i x^i \in A[[x]]$ is nilpotent, then by Exercise 1.5, each $a_i \in A$ is nilpotent. Conversely, suppose $a_i \in A$ is nilpotent for all i. Since A is Noetherian, we have by Corollary 7.15 that the nilradical is nilpotent. Thus, $\mathfrak{N}^m = 0$ for some $m > 0$. Expanding $(\sum_i a_i x^i)^m$, we obtain a sum of elements of the form $a_{i_1} \cdots a_{i_m} x^{i_1 + \cdots + i_m}$. Since $a_{i_k} \in \mathfrak{N}$ for all k, $a_{i_1} \cdots a_{i_m} \in \mathfrak{N}^m = 0$ so that each term in the sum vanishes. Thus, $(\sum_i a_i x^i)^m = 0$, as desired. □

7.3 Exercise

Proof. (i) $\implies$ (iii). By assumption, $\mathfrak{a}$ is primary. It is sufficient to show that in the quotient $A/\mathfrak{a}$, $(0 : x^n)$ is stationary for all $x \in A/\mathfrak{a}$. Let $x \in A/\mathfrak{a}$. Suppose $(0 : x) = 0$ and let $m > 1$ be minimal such that $(0 : x^m) \neq 0$. Then, $x^m y = 0$ for some $y \neq 0$, so $x(x^{m-1}y) = 0$. Because $(0 : x) = 0$, $x^{m-1}y = 0$, so $(0 : x^{m-1}) \neq 0$, violating the minimality of m. Thus, $(0 : x^n) = 0$ for all $n > 0$, so $(0 : x^n)$ is clearly stationary. On the other hand, suppose $(0 : x) \neq 0$ so that there

is some non-zero $y \in (0 : x)$. Then, x is a zero divisor. Since $\mathfrak{a}$ is primary, $x \in A/\mathfrak{a}$ is nilpotent, so there is some $m > 0$ such that $x^m = 0$. Then, $(0 : x^n) = A/\mathfrak{a}$ for all $n \geq m$, so $(0 : x^n)$ is stationary.

(iii) $\implies$ (i). By assumption the sequence $(\mathfrak{a} : x^n)$ is stationary for every $x \in A$. Passing to the quotient $A/\mathfrak{a}$, $(0 : x^n)$ is stationary for every $x \in A/\mathfrak{a}$. Suppose $x, y \in A/\mathfrak{a}$ such that $y \neq 0$ and $xy = 0$. Then, $y \in (0 : x)$. We have that $(0 : x) \subset (0 : x^2) \subset \cdots$ is stationary, so $(0 : x^n) = (0 : x^{n+1}) = \cdots$ for some $n > 0$. If $a \in (y)$, then $ax = 0$, and if $a \in (x^n)$ also, then $a = bx^n$. Hence, $bx^{n+1} = ax = 0$, so $b \in (0 : x^{n+1}) = (0 : x^n)$. Thus, $a = bx^n = 0$. In particular, $(y) \cap (x^n) = 0$. Since $\mathfrak{a}$ is irreducible, 0 is irreducible in $A/\mathfrak{a}$. Since $y \neq 0$, we must then have $(x^n) = 0$. Thus, every zero divisor x is nilpotent in $A/\mathfrak{a}$, so $\mathfrak{a}$ is primary.

(i) $\implies$ (ii). Since $\mathfrak{a}$ is primary, $\mathfrak{p} = r(\mathfrak{a})$ is prime. Let S be an arbitrary multiplicatively closed set of A. If $S \cap \mathfrak{p} = \emptyset$, then by Proposition 4.8, $(S^{-1}\mathfrak{a})^c = \mathfrak{a} = (\mathfrak{a} : 1)$. As required, $1 \in S$. On the other hand, if $S \cap \mathfrak{p} \neq \emptyset$, there is some $x \in S \cap \mathfrak{p}$. Since $\mathfrak{p} = r(\mathfrak{a})$, there is some $n > 0$ such that $x^n \in \mathfrak{a}$. Since S is multiplicatively closed and $x \in S$, $x^n \in S \cap \mathfrak{a}$. We then have $(\mathfrak{a} : x^n) = (1)$. By Proposition 4.8, $S^{-1}\mathfrak{a} = S^{-1}A$, so $(S^{-1}\mathfrak{a})^c = (1) = (\mathfrak{a} : x^n)$, where $x^n \in S$. Thus, for any multiplicatively closed set S of A, $(S^{-1}\mathfrak{a})^c = (\mathfrak{a} : x)$ for some $x \in S$.

(ii) $\implies$ (iii). We'll prove the contrapositive, so suppose there is some $x \in A$ such that $(\mathfrak{a} : x^n)$ is not stationary. Let $S_n = \{1, x^n, x^{n+1}, \ldots\}$, which is clearly multiplicatively closed and $S_{n+1} \subset S_n$ for all $n > 0$. Now, suppose to the contrary that for all n, we can choose some $s \in S_n$ such that $(S_n^{-1}\mathfrak{a})^c = (\mathfrak{a} : s)$. If $s = 1$, then $\mathfrak{a} = (\mathfrak{a} : 1) = (S_n^{-1}\mathfrak{a})^c$. By Proposition 3.11, S_n meets $\mathfrak{a}$, so for some $k > 0$, $x^k \in \mathfrak{a}$. Thus, for all $m \geq n$, $x^m \in \mathfrak{a}$ so that $(\mathfrak{a} : x^m) = A$, contradicting the fact that $(\mathfrak{a} : x^n)$ is not stationary. Thus, $s \neq 1$ for any $n > 0$.

Therefore, for all $n > 0$, there is some $s \in S_n$ $(s \neq 1)$ such that $(S_n^{-1}\mathfrak{a})^c = (\mathfrak{a} : s)$. Thus, there is some $m(n) \geq n$ such that $s = x^{m(n)}$. Choose $n > 0$. Then, since $S_1 S_n = S_1$, the homomorphism $A \to S_1^{-1}A$ factors through $A \to S_n^{-1}A \to S_1^{-1}A$. Hence, $S_n^{-1}\mathfrak{a} \subset S_1^{-1}\mathfrak{a}$ so that $(S_n^{-1}\mathfrak{a})^c \subset (S_1^{-1}\mathfrak{a})^c$. Thus, $(\mathfrak{a} : x^{m(n)}) \subset (\mathfrak{a} : x^{m(1)})$ so that $m(n) \leq m(1)$ since $(\mathfrak{a} : x^n)$ is not stationary. Then, for all n, $n \leq m(n) \leq m(1)$. This gives us the desired contradiction, as there are n such that $n > m(1)$. $\square$

7.4 Exercise

(i) The ring of rational functions $\mathbb{C}(z)$ without poles at z for any $|z| = 1$ is a Noetherian ring.

Proof. By Hilbert's basis theorem, $\mathbb{C}[z]$ is Noetherian. Let $a \in \mathbb{C}$. We have a ring homomorphism $\psi_a : \mathbb{C}[z] \to \mathbb{C}$ via $f \mapsto f(a)$. This is clearly surjective, so $\mathbb{C} \cong \mathbb{C}[z]/\ker\psi_a$. Since $\ker\psi_a$ is an ideal and $\mathbb{C}$ is a field, $\mathfrak{m}_a = \ker\psi_a$ is a maximal ideal.

Define $S = \mathbb{C}[z] - \bigcup_{a:|a|=1} \mathfrak{m}_a$. Since $\mathfrak{m}_a$ is prime for all a, if $x, y \in S$, $x, y \notin \mathfrak{m}_a$ for any a, so $xy \notin \mathfrak{m}_a$ for any a, so $xy \in S$. Thus, S is multiplicatively closed.

The ring of rational functions $\mathbb{C}(z)$ without poles at z for any $|z| = 1$ is $S^{-1}\mathbb{C}[z]$. By Proposition 7.3, $S^{-1}\mathbb{C}[z]$ is Noetherian since $\mathbb{C}[z]$ is Noetherian. $\square$

(ii) The ring of power series of $\mathbb{C}[[z]]$ with a positive radius of convergence, $\mathbb{C}\{z\}$, is Noetherian.

Proof. First, note that (z) is an ideal of the ring $\mathbb{C}\{z\}$. Now, suppose $f \in \mathbb{C}\{z\} - (z) \subset \mathbb{C}[[z]]$. Then, f has a non-zero constant term, so by Exercise 1.5, f has an inverse g in $\mathbb{C}[[z]]$. Since $f \in \mathbb{C}\{z\}$, $R_f > 0$ (the radius of convergence of f), and since $f(0) \neq 0$, $g(0) \neq 0$ since $fg = 1$. Since f is holomorphic on the disk $D(0, R_f)$ and $f(0) \neq 0$, then $1/f$ is holomorphic on $D(0, R_f)$ wherever it does not vanish. Since $f(0) \neq 0$, there is some $\epsilon > 0$ such that $1/f$ is holomorphic on

$D(0,\epsilon)$ by the continuity of f. Thus, $R_g \geq \epsilon > 0$, so $g \in \mathbb{C}\{z\}$. Thus, for any $f \in \mathbb{C}\{z\} - (z)$, f is a unit. By Proposition 1.6, $\mathbb{C}\{z\}$ is a local ring and (z) is its maximal ideal.

Now, let $\mathfrak{a}$ be an arbitrary ideal of $\mathbb{C}\{z\}$. Suppose $\mathfrak{a} \neq 0$ and $\mathfrak{a} \neq \mathbb{C}\{z\}$. Then, $\mathfrak{a}$ is contained in some maximal ideal, so since there is only one, $(z) \supset \mathfrak{a}$. Let $n > 0$ be the minimal n such that $(z)^n \supset \mathfrak{a}$; there is such an n since $(z)^1 \supset \mathfrak{a}$. Thus, $(z)^n \supset \mathfrak{a}$, but $(z)^{n+1} \not\supset \mathfrak{a}$. Thus, there is some $f \in \mathfrak{a}$ of the form $f = z^n(a_0 + a_1 z + \cdots)$, where $a_0 \neq 0$. Let $a = a_0 + a_1 z + \cdots$; note that $R_a = R_f > 0$, so $a \in \mathbb{C}\{z\}$. Further, since $a_0 \neq 0$, $a^{-1} \in \mathbb{C}\{z\}$, as we demonstrated above. Since $f = az^n \in \mathfrak{a}$, we have $z^n = a^{-1}f \in \mathfrak{a}$, so $(z^n) = (z)^n \subset \mathfrak{a}$. Thus, $\mathfrak{a} = (z)^n$. In particular, all non-trivial ideals of $\mathbb{C}\{z\}$ are of the form (z^n) for some $n > 0$.

Every ideal is principal and thus finitely generated, so $\mathbb{C}\{z\}$ is Noetherian. □

(iii) The ring of power series of $\mathbb{C}[[z]]$ with an infinite radius of convergence, $\mathcal{O}(\mathbb{C})$, is not Noetherian.

Proof. Every $f \in \mathcal{O}(\mathbb{C})$ can be evaluating at any point in $\mathbb{C}$. Let I_n be the subset of $\mathcal{O}(\mathbb{C})$ with zeros at $\{n, n+1, \ldots\}$. Then clearly, these are ideals. By the Weierstrass Factorization Theorem, since $\{n, n+1, \ldots\}$ does not have a limit point in $\mathbb{C}$, there exists some $f \in \mathcal{O}(\mathbb{C})$ with precisely these zeros. Thus, the ascending chain of ideals $I_1 \subset I_2 \subset \cdots$ is strictly ascending; hence, $\mathcal{O}(\mathbb{C})$ is not Noetherian. □

(iv) The ring of polynomials in z whose first k derivatives vanish at the origin, $R_k = \mathbb{C} + z^{k+1}\mathbb{C}[z]$, is Noetherian.

Proof. Note that $\mathbb{C}$ is Noetherian and that $\mathbb{C}$ is a subring of R_k. In fact, R_k is generated as a $\mathbb{C}$ algebra by $z^{k+1}, \ldots, z^{2k+1}$, so R_k is a finitely generated $\mathbb{C}$-algebra. By Proposition 7.2, R_k is Noetherian. □

(v) The ring of polynomials in $\mathbb{C}[w, z]$ whose partial derivatives with respect to w vanish for $z = 0$, $R = \mathbb{C} + z\mathbb{C}[w, z]$, is not Noetherian.

Proof. Let $I_n = (z, wz, \ldots, w^n z)$ for each $n > 0$. Suppose to the contrary that $w^{n+1}z \in I_n$. Then, $w^{n+1}z = f_0 z + \cdots f_n w^n z$, where $f_i \in \mathbb{C} + z\mathbb{C}[w, z]$. Then, since $\mathbb{C}[w, z]$ is an integral domain, we have (in $\mathbb{C}[w, z]$) $w^{n+1} = f_0 + \cdots + f_n w^n$. Mapping this relation $\mathbb{C}[w, z] \to \mathbb{C}[w]$ via $z \mapsto 0$, we have $w^{n+1} = \overline{f_0} + \cdots + \overline{f_n} w^n$. Since $f_i \in \mathbb{C} + z\mathbb{C}[w, z]$, under the map $z \mapsto 0$, $\overline{f_i} \in \mathbb{C}$. Thus, $w^{n+1} = \overline{f_0} + \cdots + \overline{f_n} w^n$ contradicts the fact that w is transcendental in $\mathbb{C}[w]$. Hence, the inclusion $I_n \subset I_{n+1}$ is proper for all $n > 0$. Therefore, the chain of ideals $I_1 \subset I_2 \subset \cdots$ is strictly ascending, so R is not Noetherian. □

7.5 Exercise

Proof. First of all, by Exercise 5.12, B is integral over B^G. We have that $A \subset B^G \subset B$, A is Noetherian, and B is a finitely generated A-algebra. By Proposition 7.8, B^G is finitely generated as an A-algebra. □

7.6 Exercise

Proof. Let K be a finitely generated ring that is also a field. If K has characteristic 0, we have $\mathbb{Z} \subset \mathbb{Q} \subset K$. Since K is finitely generated over $\mathbb{Z}$, it is finitely generated over $\mathbb{Q}$. Then, by Proposition 7.9, K is a finite algebraic extension of $\mathbb{Q}$, so K is finitely generated as a $\mathbb{Q}$-module. Since $\mathbb{Z}$ is Noetherian, we have by Proposition 7.8 that $\mathbb{Q}$ is a finitely generated $\mathbb{Z}$-algebra. Let $p_i/q_i \in \mathbb{Q}$ be the finite generators of $\mathbb{Q}$ as a $\mathbb{Z}$-algebra. Since there are finitely many $\{q_i\}$, finitely many primes divide each, and there are infinitely many primes of $\mathbb{Z}$, there is a prime p of $\mathbb{Z}$ that does not divide any q_i. In particular, $p_i/q_i \in \mathbb{Z}_{(p)}$ for all i. Since $\mathbb{Z}_{(p)}$ is a $\mathbb{Z}$-algebra that contains each generator of $\mathbb{Q}$ as a $\mathbb{Z}$-algebra, we must have $\mathbb{Q} \subset \mathbb{Z}_{(p)}$; this is a clear contradiction as $1/p \in \mathbb{Q}$ but $1/p \notin \mathbb{Z}_{(p)}$.

Thus, K has characteristic $p > 0$, so it is finitely generated as a $\mathbb{Z}/(p)$-algebra. Since $\mathbb{Z}/(p)$ is a field, we have by Proposition 7.9 that K is a finite algebraic extension of $\mathbb{Z}/(p)$. Thus, K is a finitely generated $\mathbb{Z}/(p)$-module. In particular, for some $n > 0$, we have a surjective $\mathbb{Z}/(p)$-module homomorphism $(\mathbb{Z}/(p))^n \to K$. Since $(\mathbb{Z}/(p))^n$ has p^n elements, K has at most p^n elements, so K is a finite field. □

7.7 Exercise

Proof. Let X be an affine algebraic variety given by a family of equations $f_\alpha(t_1, \ldots, t_n) = 0$; this defines an ideal $I(X)$ of $k[t_1, \ldots, t_n]$. k is a field and thus Noetherian, so $k[t_1, \ldots, t_n]$ is Noetherian by Corollary 7.6. Therefore, $I(X)$ is a finitely generated ideal, so let I_0 be a finite set of generators. Let $x = (x_1, \ldots, x_n) \in X$. Then, $f_\beta(x_1, \ldots, x_n) = 0$ for all $f_\beta \in I_0$, so $x \in V(I_0)$. Conversely, suppose $x = (x_1, \ldots, x_n) \in V(I_0)$ so that $f_\beta(x_1, \ldots, x_n) = 0$ for all $f_\beta \in I_0$. Then, since each $f_\alpha \in I(X)$ is generated by elements $f_\beta \in I_0$, $f_\alpha(x_1, \ldots, x_n) = 0$. Thus, $X = V(I_0)$. □

7.8 Exercise

If $A[x]$ is Noetherian, then A is Noetherian.

Proof. We have a surjective ring homomorphism $A[x] \to A$ defined by mapping $a \mapsto a$ for $a \in A$ and $x \mapsto 0$. By Proposition 7.1, A is Noetherian. □

7.9 Exercise

Proof. The ideal 0 is clearly finitely generated, so let $\mathfrak{a} \neq 0$ be an ideal in A. Then, there is some non-zero $x_0 \in \mathfrak{a}$. By property (2), there are finitely many maximal ideals that contain x_0. Thus, there are finitely many maximal ideals that contain $\mathfrak{a}$. Order them such that $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$ are the maximal ideals containing $\mathfrak{a}$ and $\mathfrak{m}_1, \ldots, \mathfrak{m}_{r+s}$ are the maximal ideals containing x_0.

Since $\mathfrak{m}_{r+1}, \ldots, \mathfrak{m}_{r+s}$ do not contain $\mathfrak{a}$, there exist $x_j \in \mathfrak{a} - \mathfrak{m}_{r+j}$ for each j $(1 \leq j \leq s)$. By property (1), $A_{\mathfrak{m}_i}$ is Noetherian for all i, so the extension of $\mathfrak{a}$ to $A_{\mathfrak{m}_i}$ is finitely generated. Pulling back these finitely many sets of finitely many generating elements, we have elements $x_{s+1}, \ldots, x_t$ in $\mathfrak{a}$ whose image in each $A_{\mathfrak{m}_i}$ generate $A_{\mathfrak{m}_i}\mathfrak{a}$ for all i $(1 \leq i \leq r)$. Let $\mathfrak{a}_0 = (x_0, \ldots, x_t)$. Since $x_i \in \mathfrak{a}$ for all i $(0 \leq i \leq t)$, $\mathfrak{a}_0 \subset \mathfrak{a}$.

Now, let $\mathfrak{m}$ be any maximal ideal of A. If $\mathfrak{m}$ is one of $\mathfrak{m}_1, \ldots, \mathfrak{m}_r$, then $A_\mathfrak{m}\mathfrak{a}$ is generated by $x_{s+1}, \ldots, x_t \in A_\mathfrak{m}\mathfrak{a}_0$, so $A_\mathfrak{m}\mathfrak{a} = A_\mathfrak{m}\mathfrak{a}_0$. If $\mathfrak{m}$ is one of $\mathfrak{m}_{r+1}, \ldots, \mathfrak{m}_{r+s}$, then $x_j \in A - \mathfrak{m}_{r+j}$, so by Proposition 3.11, an ideal of A that contains x_j extends to (1) in $A_{\mathfrak{m}_{r+j}}$. But $x_j \in \mathfrak{a}_0 \subset \mathfrak{a}$, so $A_{\mathfrak{m}_{r+j}}\mathfrak{a} = A_{\mathfrak{m}_{r+j}} = A_{\mathfrak{m}_{r+j}}\mathfrak{a}_0$. Finally, suppose $\mathfrak{m}$ is not any of $\mathfrak{m}_1, \ldots, \mathfrak{m}_{r+s}$. Then, $x_0 \in A - \mathfrak{m}$.

Since $x_0 \in \mathfrak{a}_0 \subset \mathfrak{a}$, we have by Proposition 3.11 that $A_\mathfrak{m}\mathfrak{a} = A_\mathfrak{m} = A_\mathfrak{m}\mathfrak{a}_0$. Thus, for all maximal ideals $\mathfrak{m}$ of A, $A_\mathfrak{m}\mathfrak{a} = A_\mathfrak{m}\mathfrak{a}_0$.

The inclusion $\mathfrak{a}_0 \hookrightarrow \mathfrak{a}$ is an A-module homomorphism. For all maximal ideals $\mathfrak{m}$ of A, the localization of this map, $A_\mathfrak{m}\mathfrak{a}_0 \to A_\mathfrak{m}\mathfrak{a}$, is surjective. Thus, by Proposition 3.9, the original map is surjective, so $\mathfrak{a}_0 = \mathfrak{a}$. Therefore, every ideal of A is finitely generated, so A is Noetherian. □

7.10 Exercise

Proof. Let $N \neq 0$ be an $A[x]$-submodule of $M[x]$. Define N_i to be the set of elements $m \in M$ that occur as the coefficient of the leading term of a polynomial of degree i in N, including 0. In particular, if $m \in N_i$, there is some polynomial $p(x) \in N$ with leading term mx^i. Since N is an $A[x]$-module, $xp(x) \in N$, which has leading term mx^{i+1}; thus, $m \in N_{i+1}$. Then, $N_i \subset N_{i+1}$ for all i, giving us an ascending chain of A-submodules of M. Since M is a Noetherian A-module, this chain becomes stationary for some $n \geq 0$.

Since M is a Noetherian A-module, each A-submodule N_i is finitely generated. Thus, for each i, we have a finite collection of A-module generators $\{m_{i,1}, \ldots, m_{i,k_i}\} \subset N_i$. Then, by the definition of N_i, we can associate to each $m_{i,j}$ an element $f_{i,j} \in N$ such that $f_{i,j}$ has leading term $m_{i,j}x^i$. Let N' be the $A[x]$-submodule generated by $\{f_{i,j}\}$, where $0 \leq i \leq n$ and $1 \leq j \leq k_i$. Then clearly, $N' \subset N$ and N' is finitely generated.

We prove that $N \subset N'$ by induction on degree. If $f \in N$ has degree 0, then $f = m \in M$, so $f = m \in N_0$. Since we're in degree 0, $m_{0,j} = f_{0,j}$ for all $1 \leq j \leq k_0$. Thus, $N_0 \subset N'$ so that $f \in N'$. Now, suppose that all polynomials $f \in N$ of degree less than r are in N'. If $f \in N$ has degree r, then f has leading term mx^r, where $m \in N_r$ by definition.

First, suppose $r \leq n$. Then, $m = \sum_{j=1}^{k_r} a_j m_{r,j}$ since $\{m_{r,j}\}$ generate N_r, so the element $p(x) = \sum_{j=1}^{k_r} a_j f_{r,j} \in N' \subset N$ has leading term mx^r. Thus, $f - p(x) \in N$ is 0 or has degree less than r. Using the inductive hypothesis, $f - p(x) \in N'$. Thus, $f = p(x) + (f - p(x)) \in N'$, as desired. Now, suppose $r > n$. Then, $N_r = N_n$, so $m \in N_n$. Thus, $m = \sum_{j=1}^{k_n} a_j m_{n,j}$ so that $p(x) = \sum_{j=1}^{k_n} a_j f_{n,j} \in N' \subset N$ has leading term mx^n. Since N' is an $A[x]$-module, $x^{r-n}p(x) \in N' \subset N$ and has leading term mx^r. Then, $f - x^{r-n}p(x) \in N$ is 0 or has degree less than r, so using the inductive hypothesis, $f - x^{r-n}p(x) \in N'$, so $f = x^{r-n}p(x) + (f - x^{r-n}p(x)) \in N'$, as desired. By induction, $N = N'$. Hence, every $A[x]$-submodule N of $M[x]$ is finitely generated, so $M[x]$ is a Noetherian $A[x]$-module by Proposition 6.2. □

7.11 Exercise

If A is a ring such that $A_\mathfrak{p}$ is Noetherian for all $\mathfrak{p} \in \operatorname{Spec}(A)$, then A is not necessarily Noetherian.

Proof. Let $A = \prod_{i=1}^{\infty} \mathbb{F}_2$. Then, if $x = (x_i)_i \in A$, $x^2 = (x_i^2)_i = (x_i)_i = x$ since $x_i \in \{0, 1\}$, so A is a Boolean ring. Consider the ideal $I = (0, \mathbb{F}_2, \mathbb{F}_2, \ldots)$. If $x, y \notin I$, then $x_1 = y_1 = 1$, so $(xy)_1 = x_1 y_1 = 1$ and thus $xy \notin I$. Therefore, I is a prime ideal, so $\operatorname{Spec}(A)$ is non-empty.

Let $\mathfrak{p} \in \operatorname{Spec}(A)$. Then, for any $a/s \in A_\mathfrak{p}$, we have $(a/s)^2 = a^2/s^2 = a/s$, so every element is idempotent. Since $A_\mathfrak{p}$ is a local ring, the only idempotents are 0 and 1 by Exercise 1.12, so $A_\mathfrak{p} \cong \mathbb{F}_2$, which is clearly Noetherian.

Let I_i be the ideal such that the first i factors are $\mathbb{F}_2$ and the remaining factors are 0. Then clearly, $I_1 \subset I_2 \subset \cdots$ is an infinite, strictly ascending chain of ideals, so A is not Noetherian. □

7.12 Exercise

Proof. Let $\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots$ be an ascending chain of ideals of A. Thus, $\mathfrak{a}_1^e \subset \mathfrak{a}_2^e \subset \cdots$ is an ascending chain of ideals of B. Since B is Noetherian, this chain becomes stationary, so $\mathfrak{a}_1^{ec} \subset \mathfrak{a}_2^{ec} \subset \cdots$ also becomes stationary. Since B is a faithfully flat A-algebra, we have by Exercise 3.16 that $\mathfrak{a} = \mathfrak{a}^{ec}$ for all ideals $\mathfrak{a}$ of A. Thus, the last chain is identical to the first, so $\mathfrak{a}_1 \subset \mathfrak{a}_2 \subset \cdots$ becomes stationary. Hence, A is Noetherian. □

7.13 Exercise

Proof. By Exercise 3.21, if $\mathfrak{p} \in \operatorname{Spec}(A)$, then the fiber $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p})$. Since $f : A \to B$ is of finite type, B is a finitely generated A-algebra. Thus, $B_\mathfrak{p}$ is a finitely generated $A_\mathfrak{p}$-algebra, so $B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$ is a finitely generated $A_\mathfrak{p}$-algebra. Under this map, $\mathfrak{p}A_\mathfrak{p} \to 0$, so $B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$ is a finitely generated $A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$-algebra. But $k(\mathfrak{p}) = A_\mathfrak{p}/\mathfrak{p}A_\mathfrak{p}$ is the residue field of $A_\mathfrak{p}$, so $B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$ is finitely generated as a $k(\mathfrak{p})$-algebra. By Corollary 7.7, every finitely generated algebra over a field is Noetherian, so $B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p}$ is Noetherian. By Exercise 6.8, $\operatorname{Spec}(B_\mathfrak{p}/\mathfrak{p}B_\mathfrak{p})$ is a Noetherian topological space. Hence, the fibers of f^* are Noetherian subspaces of $\operatorname{Spec}(B)$. □

7.14 Exercise

Proof. Let $\mathfrak{a}$ be an ideal of $A = k[t_1, \ldots, t_n]$, and let V be the variety in k^n defined by $\mathfrak{a}$. Let $I(V)$ be the ideal of A defined by V. Note that if $f \in \mathfrak{a}$, then for any $x \in V$, $f(x) = 0$. Suppose $g \in r(\mathfrak{a})$ so that $g^m = f$ for some $m > 0$. Then, for any $x \in V$, $(g(x))^m = f(x) = 0$. Since k is an integral domain, $g(x) = 0$, so $g \in I(V)$. Thus, $r(\mathfrak{a}) \subset I(V)$.

Conversely, suppose $f \notin r(\mathfrak{a})$. Since $r(\mathfrak{a})$ is the intersection of primes containing $\mathfrak{a}$, there is some prime $\mathfrak{p}$ containing $\mathfrak{a}$ but not f. Then, $\overline{f} \in B = A/\mathfrak{p}$ is non-zero. Let $C = B_{\overline{f}} = B[1/\overline{f}]$, and let $\mathfrak{m}$ be a maximal ideal of C. Since C is a finitely generated k-algebra, $C/\mathfrak{m}$ is a finite algebraic extension of k by Proposition 7.9. Since k is algebraically closed, $C/\mathfrak{m} \cong k$. Note that we have a composition of maps

$$A = k[t_1, \ldots, t_n] \to B = A/\mathfrak{p} \to C = B[1/\overline{f}] \to k = C/\mathfrak{m}.$$

Thus, we have $t_i \mapsto x_i \in k$ for each i, yielding a point $x = (x_i)_i \in k^n$.

If $g \in \mathfrak{a}$, then $g \mapsto g(x_1, \ldots, x_n)$. However, since $\mathfrak{a} \subset \mathfrak{p}$, $g \mapsto 0$ under $A \to A/\mathfrak{p}$, so $g(x) = 0$ for all $g \in \mathfrak{a}$. Thus, $x \in V$. Finally, we just have to show that $f \mapsto f(x)$ is non-zero. Since C contains $\overline{f}$ and $1/\overline{f}$, $\overline{f}$ is a unit of C. Therefore, $\overline{f} \notin \mathfrak{m}$, so $\overline{f}$ maps to a non-zero element of $k \cong C/\mathfrak{m}$. Thus, $f(x) \neq 0$ so that $f \notin I(V)$. $I(V) = r(\mathfrak{a})$, as desired. □

Remark. *This is the strong form of the Nullstellensatz.*

7.15 Exercise

Proof. (i) $\implies$ (ii). If M is free, then since it is also finitely generated, we have $M \cong \bigoplus_{i=1}^n A$ as A-modules. Now, let

$$0 \to N' \to N \to N'' \to 0$$

be a short exact sequence of A-modules. Then, we have the following short exact sequences:

$$0 \to \bigoplus_{i=1}^n N' \to \bigoplus_{i=1}^n N \to \bigoplus_{i=1}^n N'' \to 0$$

$$0 \to \bigoplus_{i=1}^{n}(A \otimes N') \to \bigoplus_{i=1}^{n}(A \otimes N) \to \bigoplus_{i=1}^{n}(A \otimes N'') \to 0$$

$$0 \to \left(\bigoplus_{i=1}^{n} A\right) \otimes N' \to \left(\bigoplus_{i=1}^{n} A\right) \otimes N \to \left(\bigoplus_{i=1}^{n} A\right) \otimes N'' \to 0$$

$$0 \to M \otimes N' \to M \otimes N \to M \otimes N'' \to 0.$$

By Proposition 2.19, M is flat.

(ii) $\implies$ (iii). If M is flat, then since $\mathfrak{m} \hookrightarrow A$ is injective, we have by Proposition 2.19 that $\mathfrak{m} \otimes M \to A \otimes M$ is injective.

(iii) $\implies$ (iv). From the short exact sequence

$$0 \to \mathfrak{m} \to A \to k \to 0,$$

we have the exact sequence

$$\operatorname{Tor}_1^A(k, M) \to \mathfrak{m} \otimes M \to A \otimes M \to k \otimes M \to 0.$$

Since $\mathfrak{m} \otimes M \to A \otimes M$ is injective, $\operatorname{Tor}_1^A(k, M) = 0$.

(iv) $\implies$ (i). Since M is finitely generated, the k-vector space $M/\mathfrak{m}M \cong k \otimes M$ has a k-basis. Let $x_1, \ldots, x_n$ be elements in M whose image forms a k-basis of $k \otimes M$. By Proposition 2.8, these elements generate M. Let F be a free A-module with A-basis $e_1, \ldots, e_n$. Define $\phi : F \to M$ by $\phi : e_i \mapsto x_i$, extending A-linearly to all of F. Let $E = \ker \phi$ so that we have a short exact sequence of A-modules $0 \to E \to F \to M \to 0$. Then, we have

$$\operatorname{Tor}_1^A(k, M) \to k \otimes_A E \to k \otimes_A F \xrightarrow{1 \otimes \phi} k \otimes_A M \to 0.$$

By assumption, $\operatorname{Tor}_1^A(k, M) = 0$. Since $1 \otimes \phi : k \otimes_A F \to k \otimes_A M$ is a surjection of k-vector spaces of the same dimension, $1 \otimes \phi$ is an isomorphism. Putting these together, $k \otimes_A E \to k \otimes_A F$ is injective with image 0, so $k \otimes_A E = 0$.

$k \otimes_A E \cong E/\mathfrak{m}E$, so $E = \mathfrak{m}E$. Since F is a finitely generated A-module and A is Noetherian, F is a Noetherian A-module by Proposition 6.5. Since E is an A-submodule of the Noetherian A-module F, we have by Proposition 6.2 that E is finitely generated. Since A is a local ring with maximal ideal $\mathfrak{m}$, $E = \mathfrak{m}E$, and E is finitely generated, we have by Nakayama's lemma that $E = 0$. Thus, $F \cong M$ as A-modules, so M is a free A-module. □

7.16 Exercise

Proof. (i) $\implies$ (ii). For every prime $\mathfrak{p}$, $M_\mathfrak{p}$ is a flat $A_\mathfrak{p}$-module since M is a flat A-module (Proposition 3.10). Since A is Noetherian, each $A_\mathfrak{p}$ is a Noetherian local ring (Proposition 7.3). Further, since M is a finitely generated A-module, $M_\mathfrak{p}$ is a finitely generated $A_\mathfrak{p}$-module. Hence, $M_\mathfrak{p}$ and $A_\mathfrak{p}$ satisfy the equivalent conditions of Exercise 7.15, so $M_\mathfrak{p}$ is a free $A_\mathfrak{p}$-module.

(ii) $\implies$ (iii). Clear. All maximal ideals are prime ideals.

(iii) $\implies$ (i). Since M is a finitely generated A-module, $M_\mathfrak{m}$ is a finitely generated $A_\mathfrak{m}$-module for all maximal ideals $\mathfrak{m}$ of A. Since A is Noetherian, $A_\mathfrak{m}$ is a Noetherian local ring (Proposition 7.3). Thus, by Exercise 7.15, $M_\mathfrak{m}$ is a flat $A_\mathfrak{m}$-module for all maximal ideal $\mathfrak{m}$. By Proposition 3.10, M is a flat A-module. □

Remark. *If M is a finitely generated A-module (A Noetherian), M is flat $\iff$ M is locally free.*

7.17 Exercise

Proof. First, we show that any A-submodule N of a Noetherian A-module M is a finite intersection of irreducible submodules.

Since M is a Noetherian A-module, N is a finitely generated A-module (Proposition 6.2). Suppose to the contrary that N is not a finite intersection of irreducible submodules of M. Then, the set of A-submodules of M that are not a finite intersection of irreducible submodules is not empty. Since M is Noetherian, this set of submodules has a maximal element, say N'. Then, N' is reducible, so we can write $N' = N_1 \cap N_2$, where both N_1 and N_2 are A-submodules of M that strictly contain N'. Since N' was chosen to be maximal, N_1 and N_2 are finite intersections of irreducible A-submodules of M, and thus, $N' = N_1 \cap N_2$ is also a finite intersection of irreducible A-submodules of M. This is a contradiction, so N is a finite intersection of irreducible submodules.

Now, we show that any irreducible A-submodule Q of M is primary. Suppose $m \in M - Q$ and $x \in A$ such that $xm \in Q$. In other words, x is a zero-divisor of M/Q. Note that M/Q is a Noetherian A-module since M is (Proposition 6.3). Then, $\ker(x) \neq 0$ is an A-submodule of M/Q. Further, we have an ascending chain $\ker(x) \subset \ker(x^2) \subset \cdots$ of submodules of M/Q. Since M/Q is Noetherian, we must have $\ker(x^n) = \ker(x^{n+1})$ for some $n > 0$.

We wish to show that $A\overline{m} \cap x^n(M/Q) = 0$. Choose some arbitrary $a\overline{m} \in A\overline{m}$. Further, assume $a\overline{m} \in x^n(M/Q)$. Then, $a\overline{m} = x^n d$ for some $d \in M/Q$. Since $x(a\overline{m}) = a\overline{xm} = 0$ ($xm \in Q$ by assumption), $x(x^n d) = x^{n+1}d = 0$. Thus, $d \in \ker(x^{n+1}) = \ker(x^n)$, so $a\overline{m} = x^n d = 0$. Thus, $A\overline{m} \cap x^n(M/Q) = 0$. Lifting to M, we have $(Am + Q) \cap (x^n M + Q) = Q$. Since $m \notin Q$, $Am + Q$ strictly contains Q. Thus, since Q is irreducible, we have $x^n M + Q = Q$, so $x^n M \subset Q$. Thus, x^n is the zero endomorphism on M/Q, so Q is primary.

Putting these together, we see that any A-submodule of a Noetherian A-module M is a finite intersection of primary A-submodules of M. □

7.18 Exercise

Proof. (i) $\implies$ (ii). If $\mathfrak{p}$ belongs to 0 in M, then by Exercise 4.22, there is some $m \in M$ such that $\mathfrak{p} = r(0 : m) = r(\mathrm{Ann}(m))$. If $\mathrm{Ann}(m) \neq r(\mathrm{Ann}(m))$, then choose some $x \in r(\mathrm{Ann}(m)) - \mathrm{Ann}(m)$. Then, $m_1 = xm$ is non-zero and $\mathrm{Ann}(m) \subset \mathrm{Ann}(m_1) \subset r(\mathrm{Ann}(m))$. Since $x \notin \mathrm{Ann}(m)$ but $x \in r(\mathrm{Ann}(m))$, there is some minimal $n > 1$ such that $x^n m = 0$. Thus, $x^{n-1} \in \mathrm{Ann}(m_1)$ but $x^{n-1} \notin \mathrm{Ann}(m)$, so the containment is strict.

Continuing in this way, choosing $m_k \in r(\mathrm{Ann}(m)) - \mathrm{Ann}(m_{k-1})$, we obtain an ascending chain $\mathrm{Ann}(m) \subset \mathrm{Ann}(m_1) \subset \cdots$ of ideals of A. Since A is Noetherian, this chain eventually stabilizes. Thus, there is some non-zero $m_n \in M$ such that we can no longer choose some $x \in r(\mathrm{Ann}(m)) - \mathrm{Ann}(m_n)$; hence, $\mathrm{Ann}(m_n) = r(\mathrm{Ann}(m)) = \mathfrak{p}$, as desired.

(ii) $\implies$ (i). By assumption, there is some $m \in M$ such that $\mathfrak{p} = \mathrm{Ann}(m)$. Thus, $\mathfrak{p} = r(\mathfrak{p}) = r(\mathrm{Ann}(m)) = r(0 : m)$. By Exercise 4.22, $\mathfrak{p}$ belongs to 0 in M.

(ii) $\implies$ (iii). By assumption, there is some $m \in M$ such that $\mathfrak{p} = \mathrm{Ann}(m)$. Note that we have an A-submodule Am of M. We also have a surjective A-module homomorphism $A \to Am$ ($a \mapsto am$) with kernel $\mathrm{Ann}(m) = \mathfrak{p}$. Thus, $A/\mathfrak{p} \cong Am$.

(iii) $\implies$ (ii). Suppose there is some A-submodule N of M such that $N \xrightarrow{\phi} A/\mathfrak{p}$ is an A-module isomorphism. Let $m = \phi^{-1}(1)$, and let $x \in A$. Then,

$$xm = 0 \iff \phi(xm) = 0 \iff \overline{x} = 0 \iff x \in \mathfrak{p}.$$

Thus, $\mathfrak{p} = \mathrm{Ann}(m)$. Therefore, conditions (i), (ii), and (iii) are equivalent.

Finally, since M is a finitely generated A-module and A is Noetherian, M is a Noetherian A-module (Proposition 6.5). By Exercise 7.17, 0 has a primary decomposition in M. By Exercise 4.22, there are finitely many prime ideals $\mathfrak{p}_1, \ldots, \mathfrak{p}_r$ belonging to 0.

Let Σ be the set of submodules N of M that possess a chain $0 = M_0 \subset M_1 \subset \cdots \subset M_k = N$ such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ for some $\mathfrak{p}_i \in \operatorname{Spec}(A)$. Note that Σ is non-empty since $0 \in \Sigma$. Actually, by the equivalent conditions above, each of the prime ideals $\mathfrak{p}_i$ belonging to 0 have an associated submodule N_i of M such that $N_i \cong A/\mathfrak{p}_i$. Thus, each N_i is in Σ. In any case, Σ is non-empty.

Since M is a Noetherian A-module, Σ has a maximal element M'. If $M' \neq M$, then M/M' is a non-zero Noetherian A-module. By Exercise 7.17, the submodule 0 of M/M' has a primary decomposition (non-trivial since $M/M' \neq 0$) and thus has some associated prime $\mathfrak{p}$. By the equivalent conditions, there is a submodule U/M' of M/M' that is isomorphic to $A/\mathfrak{p}$ as an A-module. Then, we have a chain $0 = M_0 \subset \cdots \subset M_r = M' \subset U$ such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ for all $i \leq r$ and $U/M' \cong A/\mathfrak{p}$. Thus, $U \in \Sigma$ and U strictly contains M', violating the maximality of M'. Thus, $M = M'$, so M can be expressed as a chain $0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$, where $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ with $\mathfrak{p}_i \in \operatorname{Spec}(A)$ for all $i \leq r$. □

Remark. *Such a chain is called a "prime filtration".*

7.19 Exercise

Proof. An ideal of A is an A-submodule of A, a primary ideal of A is a primary submodule of A, an irreducible ideal of A is an irreducible submodule of A, $r_A(\mathfrak{a}) = r(\mathfrak{a})$, and A is a Noetherian ring if it is Noetherian as an A-module. Thus, it will be enough to prove only the general result for a submodule N of a Noetherian A-module M.

Let $\bigcap_{i=1}^{r} P_i = \bigcap_{j=1}^{s} Q_j$ be two minimal irreducible decompositions of $N \subset M$. Relabelling if necessary, we may suppose $r \leq s$. Let $P_k' = \bigcap_{i \neq k} P_i$ for each k. Now, fix k and define $N_j = P_k' \cap Q_j$. Since all of the intersecting modules contain N, $N_j \supset N$; we now show that there exists some j such that $N_j \subset N$, yielding equality. Since $\{Q_j\}$ have intersection equal to N, we also have $\bigcap_j N_j = N$. Let $\pi_i : M \to M/P_i$ be the canonical projection maps, and $\phi = (\phi_1, \ldots, \phi_r) : M \to \prod_i M/P_i$. Then, $\ker \phi = \bigcap_i P_i = N$. For each j and each $i \neq k$, we have $N_j \subset P_k' \subset P_i$, so $\pi_i(N_j) = 0$. Therefore, the only potential non-zero coordinate of $\phi(N_j)$ is the kth. For each j, we have a submodule (by surjectivity) $O_j = \pi_k(N_j)$ of M/P_k. Since $\phi(N) = 0$, $0 = \phi_k(N) = \phi_k(\bigcap_j N_j) = \bigcap_j O_j$. Since $P_k \subset M$ is irreducible, $0 \subset M/P_k$ is irreducible. Therefore, there is some $j(k)$ such that $O_{j(k)} = 0$. Then, $\pi_k(N_{j(k)}) = 0$ so that $\phi(N_{j(k)}) = 0$, giving $N_{j(k)} \subset N$ and thus $N_{j(k)} = N$.

For each $k \in \{1, \ldots, r\}$, we have an irreducible decomposition $P_1 \cap \cdots \cap Q_{j(k)} \cap \cdots \cap P_r$ of N. Thus, we have an irreducible decomposition $Q_{j(1)} \cap P_2 \cap \cdots \cap P_r$. The argument of the preceding paragraph did not rely on the decomposition being minimal, so we may apply the same argument to this new irreducible decomposition. Then, we have an irreducible decomposition of N as $Q_{j(1)} \cap Q_{j(2)} \cap P_3 \cap \cdots \cap P_r$. Continuing recursively, we obtain an irreducible decomposition of N as $Q_{j(1)} \cap \cdots \cap Q_{j(r)}$. Since the decomposition $\bigcap_{j=1}^{s} Q_j$ of N was minimal, $k \mapsto j(k)$ must be surjective so that $r \geq s$. Since we assumed $r \leq s$, we have $r = s$, as desired.

We want to show that there is a bijection $f : \{Q_i\} \to \{P_j\}$ such that $r_M(f(Q_i)) = r_M(Q_i)$ for all i. We've already shown that $r = s$, so it will suffice to show that each $\mathfrak{p}$ belonging to N in M has an equal number of Q_i and P_j that are $\mathfrak{p}$-primary.

By Exercise 7.17 (7.12), since M is a Noetherian A-module, every irreducible submodule of M is primary. Thus, our irreducible decompositions of N are primary decompositions of M. By Exercise 4.22 (4.5), the sets $\{r_M(P_i)\}$ and $\{r_M(Q_j)\}$ are equal. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_n$ be the associated

primes. There is at least one isolated prime, so let $t \geq 1$ be the number of isolated primes. Reorder them such that for $k \leq t$, $\mathfrak{p}_k$ is isolated. For $k > t$, we order the primes such that whenever $\mathfrak{p}_j \subset \mathfrak{p}_i$, $j < i$.

For each prime $\mathfrak{p}$, let $P^{\mathfrak{p}}$ and $Q^{\mathfrak{p}}$ be the intersection of $\mathfrak{p}$-primary submodules in $\{P_j\}$ and $\{Q_j\}$, respectively. Then,

$$N = \bigcap_{\mathfrak{p} \in \mathrm{Ass}(N)} P^{\mathfrak{p}} = \bigcap_{\mathfrak{p} \in \mathrm{Ass}(N)} Q^{\mathfrak{p}}$$

are minimal primary decompositions of N since, by Exercise 4.21 (4.3), an intersection of $\mathfrak{p}$-primary submodules is $\mathfrak{p}$-primary.

Let p_k and q_k be the number of $\{P_i\}$ and $\{Q_j\}$ that are $\mathfrak{p}_k$-primary. At this point, it clearly suffices to show that $p_k = q_k$ for all k. We'll do this by induction. If $k = 1$, $\mathfrak{p}_1$ is isolated by our ordering. By Exercise 4.23 (4.10), $P^{\mathfrak{p}_1} = Q^{\mathfrak{p}_1}$. Thus, the intersection of $\{P_i\}$ and $\{Q_j\}$ that are $\mathfrak{p}_1$-primary yield a minimal irreducible decomposition of $P^{\mathfrak{p}_1} = Q^{\mathfrak{p}_1}$ since they are contained in a minimal irreducible decomposition of N. By the first result of this proof, $p_1 = q_1$, completing the base case.

Assume inductively that $p_j = q_j$ for all $j < i$. We have our minimal irreducible decomposition $N = \bigcap P_j = \bigcap Q_j$. Localizing at $\mathfrak{p}_i$, we obtain

$$N_{\mathfrak{p}_i} = \bigcap_k P_{k,\mathfrak{p}_i} = \bigcap_k Q_{k,\mathfrak{p}_i}.$$

By Exercise 4.23 (4.8), if P_k or Q_k is $\mathfrak{p}$-primary and $\mathfrak{p} \not\subset \mathfrak{p}_i$, then the localized module is $M_{\mathfrak{p}_i}$ and can thus be ignored in the intersection. Thus, we have a minimal irreducible decomposition

$$N_{\mathfrak{p}_i} = \bigcap_{j:\mathfrak{p}_j \subset \mathfrak{p}_i} P_{j,\mathfrak{p}_i} = \bigcap_{j:\mathfrak{p}_j \subset \mathfrak{p}_i} Q_{j,\mathfrak{p}_i}.$$

Note that these modules remain irreducible since all submodules of $M_{\mathfrak{p}_i}$ come from localizing submodules of M, and localization respects containment and intersections. Since M is a Noetherian A-module, $M_{\mathfrak{p}_i}$ is a Noetherian $A_{\mathfrak{p}_i}$-module (Proposition 7.3). Then, invoking the preceding result from this proof, the total number of irreducible modules in the decompositions are equal: $\sum_{j:\mathfrak{p}_j \subset \mathfrak{p}_i} p_j = \sum_{j:\mathfrak{p}_j \subset \mathfrak{p}_i} q_j$. Since $\mathfrak{p}_j \subset \mathfrak{p}_i$ implies $j < i$ by our ordering, for all $j < i$ such that $\mathfrak{p}_j \subset \mathfrak{p}_i$, we can subtract $p_j = q_j$ from both sides. This leaves behind the equality $p_i = q_i$. By induction, $p_k = q_k$ for all k, as desired. □

7.20 Exercise

(i) Proof. Suppose E is a finite union of sets of the form $U \cap C$, where U is open and C is closed. Then, C^c is open, so U and C^c are in $\mathcal{F}$. Thus, $C \in \mathcal{F}$, so $U \cap C \in \mathcal{F}$ for any open set U and closed set C. Now, suppose $E_1, E_2 \in \mathcal{F}$. Then, $E_1 \cup E_2 = (E_1^c \cap E_2^c)^c \in \mathcal{F}$ since $\mathcal{F}$ is closed under finite intersections and complements; thus, $\mathcal{F}$ is closed under finite unions as well. Since E is a finite union of sets of the form $U \cap C$ ($\in \mathcal{F}$), we have $E \in \mathcal{F}$.

Conversely, note that $\mathcal{F}$ is generated by the open sets of X under finite intersections and complements. Every open set is trivially a finite union of sets of the form $U \cap C$, where U is open and C is closed since an open set U is $U \cap X$ (noting that X is closed). Thus, it suffices to show that sets of this form are closed under complements and finite intersections. Suppose we have two

sets of this form: $\bigcup_i U_i \cap C_i$ and $\bigcup_j V_j \cap K_j$. Then,

$$\left(\bigcup_i U_i \cap C_i\right) \cap \left(\bigcup_j V_j \cap K_j\right) = \bigcup_{(i,j)} (U_i \cap C_i) \cap (V_j \cap K_j) = \bigcup_{(i,j)} (U_i \cap V_j) \cap (C_i \cap K_j).$$

Since $U_i \cap V_j$ is open and $C_i \cap K_j$ is closed, the intersection has the desired form. Moving to complements, we have

$$\left(\bigcup_{i=1}^n U_i \cap C_i\right)^c = \bigcap_{i=1}^n U_i^c \cup C_i^c = \bigcup_{A \in \mathcal{P}(\{1,\ldots,n\})} \left(\bigcap_{i \in A} C_i^c \cap \bigcap_{i \notin A} U_i^c\right).$$

This is a finite union, $\bigcap C_i^c$ is open (even if X, i.e. empty indexing set), and $\bigcap U_i^c$ is closed (even if X, i.e. empty indexing set). Thus, complements preserve the form. Hence, a set E of X is in $\mathcal{F}$ if and only if it is a finite union of sets of the form $U \cap C$, where U is open and C is closed. □

(ii) Proof. Suppose E contains a non-empty open set U of X. Since X is irreducible, $X = \overline{U} \subset \overline{E} \subset X$, so E is dense in X.

Conversely, suppose $E \in \mathcal{F}$ is dense in X. By (i), $E = \bigcup_{i=1}^n U_i \cap C_i$, where each U_i is open and each C_i is closed. If $U_i \cap C_i$ is empty, we may exclude it from the union; thus, we can assume $U_i \cap C_i$ is non-empty for all i. Then, $E \subset \bigcup_{i=1}^n C_i$, a closed set, so $X = \overline{E} \subset \bigcup_{i=1}^n C_i \subset X$, giving equality throughout. Taking complements, $\emptyset = \bigcap_{i=1}^n C_i^c$. Since X is irreducible and $\bigcap_{i=1}^n C_i^c$ is an empty intersection of finitely many open sets, there is some i such that C_i^c is empty. Thus, $C_i = X$ so that $E \supset U_i \cap C_i = U_i \cap X = U_i$. Since $U_i \cap C_i$ is non-empty and equal to U_i, U_i is a non-empty open subset of E, as desired. □

7.21 Exercise

Proof. Suppose $E \in \mathcal{F}$, and let X_0 be an irreducible closed set of X. By Exercise 7.20, $E = \bigcup_{i=1}^n U_i \cap C_i$, where the U_i are open and the C_i are closed. Let $E_0 = E \cap X_0$, and suppose $\overline{E_0} = \overline{E \cap X_0} = X_0$. Since X_0 is an irreducible closed set of X, X_0 is an irreducible topological space. Let $\mathcal{F}_0$ be the collection of X_0 defined in Exercise 7.20. Note that

$$E_0 = X_0 \cap \bigcup_{i=1}^n U_i \cap C_i = \bigcup_{i=1}^n X_0 \cap (U_i \cap C_i) = \bigcup_{i=1}^n (X_0 \cap U_i) \cap (X_0 \cap C_i).$$

Since $X_0 \cap U_i$ are open in X_0 and $X_0 \cap C_i$ are closed in X_0, both by definition, we see by Exercise 7.20 (i) that $E_0 \in \mathcal{F}_0$. Then, since X_0 is irreducible and E_0 is dense in X_0, we have by Exercise 7.20 (ii) that E_0 contains a non-empty open subset of X_0. Therefore, we've shown that if $E \in \mathcal{F}$, then $\overline{E \cap X_0} \neq X_0$ or $E \cap X_0$ contains a non-empty open subset of X_0.

Conversely, suppose $E \notin \mathcal{F}$. Then, the collection of closed sets $X' \subset X$ such that $E \cap X' \notin \mathcal{F}$ is not empty (X is such a closed set). This collection has a minimal element X_0 since X is a Noetherian space. Suppose X_0 is reducible, i.e. there are two closed sets C_1 and C_2 such that $X_0 = C_1 \cup C_2$ and X_0 properly contains each closed set. Since X_0 is minimal in the collection, $E \cap C_1, E \cap C_2 \in \mathcal{F}$. Since $\mathcal{F}$ is closed under finite unions (as we saw in the proof of Exercise 7.20 (i)), $E \cap X_0 = (E \cap C_1) \cup (E \cap C_2) \in \mathcal{F}$, a contradiction. Thus, X_0 is closed and irreducible.

Suppose to the contrary that $\overline{E \cap X_0} \neq X_0$. Then, X_0 properly contains the closed set $\overline{E \cap X_0}$, so $E \cap \overline{E \cap X_0} \in \mathcal{F}$ by the minimality of X_0 in the aforementioned collection. Since X_0 is a closed set, $X_0 \in \mathcal{F}$, so we have

$$X_0 \cap (E \cap \overline{E \cap X_0}) = (E \cap X_0) \cap \overline{E \cap X_0} = E \cap X_0 \in \mathcal{F}.$$

This contradicts the definition of X_0.

Now, suppose to the contrary that $E \cap X_0$ contains a non-empty open set $X_0 \cap U$. We can thus write

$$E \cap X_0 = (E \cap X_0 \cap U) \cup (E \cap X_0 \cap U^c).$$

Since $X_0 \cap U^c$ is closed and properly contained in X_0 (since $X_0 \cap U$ is non-empty), by the minimality of X_0 in the collection, $E \cap (X_0 \cap U^c) \in \mathcal{F}$. On the other hand, since $E \cap X_0$ contains $U \cap X_0$, $E \cap X_0 \cap U = X_0 \cap U$. Since X_0 is closed and U is open, we have by Exercise 7.20 (i) that $X_0 \cap U \in \mathcal{F}$. Thus, $E \cap X_0$ is the union of two sets in $\mathcal{F}$, so $E \cap X_0 \in \mathcal{F}$, a contradiction. □

Remark. *The sets belonging to $\mathcal{F}$ are called the "constructible" subsets of X.*

7.22 Exercise

Proof. Suppose E is an open set of X. Let X_0 be an irreducible closed subset of X. If $E \cap X_0 \neq \emptyset$, then $E \cap X_0$ is a non-empty open subset of X_0. Thus, if E is an open set of X, then for every irreducible closed subset X_0 of X, either $E \cap X_0 = \emptyset$ or $E \cap X_0$ contains a non-empty open subset of X_0.

Conversely, let E be a subset of X that is not open. Then, $E \cap X$ is trivially not open; thus, the collection of closed sets C of X such that $E \cap C$ is not open in C is non-empty. Since X is Noetherian, this collection has a minimal element X_0. Suppose that X_0 is reducible so that $X_0 = C_1 \cup C_2$ and X_0 properly contains both closed sets C_1 and C_2. Then, by minimality, $E \cap C_1$ is open in C_1 and $E \cap C_2$ is open in C_2. Thus, $E \cap C_i = V_i \cap C_i$ for open sets V_1, V_2 of X. Let $U_i = V_i \cup C_i^c$, so U_i is open and $U_i \cap C_i = (V_i \cup C_i^c) \cap C_i = V_i \cap C_i = E \cap C_i$. Further,

$$E = (E \cap C_i) \cup (E \cap C_i^c) \subset (E \cap C_i) \cup C_i^c = (U_i \cap C_i) \cup C_i^c = (U_i \cap C_i) \cup (U_i \cap C_i^c) = U_i.$$

Thus, $E \subset U_1 \cap U_2$ so that $E \cap C_i \subset (U_1 \cap U_2) \cap C_i \subset U_i \cap C_i = E \cap C_i$, so we have equality throughout. In particular, the open set $U = U_1 \cap U_2$ satisfies $U \cap C_i = E \cap C_i$ for both $i = 1, 2$. Then,

$$E \cap X_0 = (E \cap C_1) \cup (E \cap C_2) = (U \cap C_1) \cup (U \cap C_2) = U \cap (C_1 \cup C_2) = U \cap X_0,$$

so $E \cap X_0$ is open in X_0, a contradiction. Thus, X_0 is closed and irreducible.

$E \cap X_0 \neq \emptyset$ since $\emptyset$ is open. To finish the contrapositive then, we must show that $E \cap X_0$ does not contain a non-empty open subset. So, suppose to the contrary that there is some open set U of X such that $U \cap X_0 \neq \emptyset$ and $U \cap X_0 \subset E \cap X_0$. Since $U \cap X_0 \neq \emptyset$, X_0 strictly contains the closed set $U^c \cap X_0$. By the minimality of X_0, $E \cap (U^c \cap X_0)$ is open and thus open in X_0. Since $U \cap X_0 \subset E \cap X_0$, $E \cap U \cap X_0 = U \cap X_0$, which is an open set of X_0. Thus,

$$E \cap X_0 = (E \cap X_0 \cap U) \cup (E \cap X_0 \cap U^c)$$

is the union of two open sets of X_0, so $E \cap X_0$ is an open set of X_0, the desired contradiction. □

7.23 Exercise

Proof. By Exercise 7.20, a constructible set has the form $\bigcup_{i=1}^n U_i \cap C_i$, where U_i are open sets and C_i are closed. Then, $f^* \left(\bigcup_{i=1}^n U_i \cap C_i\right) = \bigcup_{i=1}^n f^*(U_i \cap C_i)$. Since constructible sets are closed under finite unions, it is enough to show that $f^*(U \cap C)$ is constructible for every open set U and closed set C.

Since C is closed in $\mathrm{Spec}(B)$, we have $C = V(\mathfrak{b})$ for some ideal $\mathfrak{b}$ of B. Then, we can extend our homomorphism f to $\overline{f} : A \to B \to B/\mathfrak{b}$. By Exercise 1.21, $V(\mathfrak{b})$ is homeomorphic to $\mathrm{Spec}(B/\mathfrak{b})$ so that $\overline{f}^* : \mathrm{Spec}(B/\mathfrak{b}) \cong V(\mathfrak{b}) \hookrightarrow \mathrm{Spec}(B) \to \mathrm{Spec}(A)$. Since $U \cap C = U \cap V(\mathfrak{b}) \subset V(\mathfrak{b})$, $f^*(U \cap C) = \overline{f}^*(U \cap V(\mathfrak{b}))$. But $U \cap V(\mathfrak{b})$ is open in the subspace $V(\mathfrak{b})$, by definition. Since f is a homomorphism of finite type, $\overline{f} : A \to B/\mathfrak{b}$ is a homomorphism of finite type. Therefore, if we can show that for all homomorphisms $f : A \to B$ of finite type, f^* maps open sets to constructible sets, we'd be done.

We've reduced the problem to showing that for A a Noetherian ring and $f : A \to B$ a homomorphism of finite type, f^* maps open sets to constructible sets. Let U be an open set of $\mathrm{Spec}(B)$. First of all, since f is a homomorphism of finite type, B is also Noetherian (Corollary 7.7). Thus, by Exercise 6.8, $\mathrm{Spec}(B)$ is a Noetherian topological space. Hence, by Exercise 6.6, U is quasi-compact in $\mathrm{Spec}(B)$. Since the open sets $V(g)^c$ ($g \in B$) form a basis of open sets of $\mathrm{Spec}(B)$ (Exercise 1.17), U is the union of sets of this form. But U is quasi-compact, so it is the union of finitely many open sets of the form $V(g)^c$. Thus, we write $U = \bigcup_{i=1}^n V(g_i)^c$. Let $f_i : B \to B_{g_i}$ be the canonical homomorphism. By Exercise 3.21, $f_i^* : \mathrm{Spec}(B_{g_i}) \to V(g_i)^c$ is a homeomorphism. Since $f : A \to B$ is of finite type and $f_i : B \to B_{g_i}$ is of finite type, $f_i \circ f : A \to B_g$ is of finite type. Suppose that we have shown that for all homomorphisms $\psi : A \to B'$ of finite type, $\psi * (\mathrm{Spec}(B'))$ is a constructible set. Then,

$$f^*(U) = f^*\left(\bigcup_{i=1}^n V(g_i)^c\right) = \bigcup_{i=1}^n f^*(f_i^*(\mathrm{Spec}(B_{g_i}))) = \bigcup_{i=1}^n (f_i \circ f)^*(\mathrm{Spec}(B_{g_i}))$$

is constructible since it is a finite union of constructible sets.

Hence, we have reduced the problem to showing that for A a Noetherian ring and $f : A \to B$ a homomorphism of finite type, $f^*(\mathrm{Spec}(B))$ is a constructible set. Since A is Noetherian, $\mathrm{Spec}(A)$ is a Noetherian topological space (Exercise 6.8). Thus, to show that $f^*(\mathrm{Spec}(B))$, we can use the criterion of Exercise 7.21: for each irreducible closed set $X_0 \subset \mathrm{Spec}(A)$, we must show that $\overline{f^*(\mathrm{Spec}(B)) \cap X_0} \neq X_0$ or else $f^*(\mathrm{Spec}(B)) \cap X_0$ contains a non-empty open subset of X_0. If $\overline{f^*(\mathrm{Spec}(B)) \cap X_0 \neq X_0}$ for all X_0, we're done, so suppose we have an irreducible closed subspace X_0 such that $\overline{f^*(\mathrm{Spec}(B)) \cap X_0} = X_0$.

We'll show now that $f^*(\mathrm{Spec}(B)) \cap X_0 = f^*(f^{*-1}(X_0))$. If $\mathfrak{p} \in f^*(\mathrm{Spec}(B)) \cap X_0$, then there is some $\mathfrak{q} \in \mathrm{Spec}(B)$ such that $\mathfrak{p} = f^{-1}(\mathfrak{q}) \in X_0$. Then, $\mathfrak{q} \in f^{*-1}(X_0)$ since $\mathfrak{p} \in X_0$ so that $\mathfrak{p} \in f^*(f^{*-1}(X_0))$. Conversely, if $\mathfrak{p} \in f^*(f^{*-1}(X_0))$, then clearly, $\mathfrak{p} \in f^*(\mathrm{Spec}(B))$. Further, there is some $\mathfrak{q} \in f^{*-1}(X_0)$ such that $f^{-1}(\mathfrak{q}) = \mathfrak{p}$. But since $f^{-1}(\mathfrak{q}) \in X_0$, $\mathfrak{p} \in X_0$, so $\mathfrak{p} \in f^*(\mathrm{Spec}(B)) \cap X_0$. Thus, we need to show that $f^*(f^{*-1}(X_0))$ contains a non-empty open subset of X_0.

Since X_0 is a closed, $X_0 = V(\mathfrak{a})$ for some radical ideal $\mathfrak{a}$. Since A is Noetherian, $A/\mathfrak{a}$ is Noetherian (Proposition 7.1). Thus, by Exercise 6.9, $A/\mathfrak{a}$ has finitely many minimal primes. Since $\mathfrak{a}$ is a radical ideal, it is the intersection of prime ideals containing it. This intersection is clearly identical to the intersection over the subset of minimal elements, which, as we just saw, is finite. Thus, $\mathfrak{a}$ is a finite intersection of the prime ideals minimal over $\mathfrak{a}$: $\mathfrak{a} = \mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_n$. If there are no primes, then $\mathfrak{a} = A$ so that $X_0 = V(A) = \emptyset$, a violation of irreducibility. Thus, $n \geq 1$. Suppose $n \geq 2$. By Exercise 1.15, $V(\mathfrak{a}) = V(\mathfrak{p}_1 \cap \cdots \cap \mathfrak{p}_n) = V(\mathfrak{p}_1) \cup V(\mathfrak{p}_2 \cap \cdots \cap \mathfrak{p}_n)$. Since $V(\mathfrak{a})$ is irreducible, we must have $V(\mathfrak{a}) = V(\mathfrak{p}_1)$ or $V(\mathfrak{a}) = V(\mathfrak{p}_2 \cap \cdots \cap \mathfrak{p}_n)$. In the first case, we have $\mathfrak{p}_2 \in V(\mathfrak{a}) = V(\mathfrak{p}_1)$, so $\mathfrak{p}_2 \supset \mathfrak{p}_1$, violating the fact that $\mathfrak{p}_2$ is a minimal prime over $\mathfrak{a}$. In the second case, $\mathfrak{p}_1 \in V(\mathfrak{a}) = V(\mathfrak{p}_2 \cap \cdots \cap \mathfrak{p}_n)$, so $\mathfrak{p}_1 \supset \mathfrak{p}_2 \cap \cdots \cap \mathfrak{p}_n$. By Proposition 1.11, $\mathfrak{p}_1 \supset \mathfrak{p}_i$ for some $i = 2, \ldots, n$. But this violates the fact that $\mathfrak{p}_1$ is a minimal prime over $\mathfrak{a}$. Thus, we have a contradiction, so $n = 1$. In other words, $X_0 = V(\mathfrak{p})$ for some $\mathfrak{p} \in \mathrm{Spec}(A)$.

Returning to the central task, we must show that $f^*(\mathrm{Spec}(B)) \cap V(\mathfrak{p}) = f^*(f^{*-1}(V(\mathfrak{p})))$ contains a non-empty open subset of $V(\mathfrak{p})$. By Exercise 1.21, $f^*{-}1(V(\mathfrak{p})) = V(\mathfrak{p}^e)$, so by Exercise 3.21, f^* restricts to a map $\overline{f}^* : \mathrm{Spec}(B/\mathfrak{p}^e) \to \mathrm{Spec}(A/\mathfrak{p})$. Then, $f^*(f^{*-1}(V(\mathfrak{p}))) = f^*(V(\mathfrak{p}^e)) = \overline{f}^*(\mathrm{Spec}(B/\mathfrak{p}^e))$. Thus, we want to show that $\overline{f}^*(\mathrm{Spec}(B/\mathfrak{p}^e))$ contains a non-empty open subset of $\mathrm{Spec}(A/\mathfrak{p})$. Since $f : A \to B$ is of finite type and A and B are Noetherian, $\overline{f} : A/\mathfrak{p} \to B/\mathfrak{p}^e$ is of finite type, and $A/\mathfrak{p}$ and $B/\mathfrak{p}^e$ are Noetherian (Proposition 7.1). Note that $A/\mathfrak{p}$ is an integral domain. Finally, recall that we assumed $V(\mathfrak{p}) = \overline{f^*(\mathrm{Spec}(B)) \cap V(\mathfrak{p})} = \overline{\overline{f}^*(\mathrm{Spec}(B/\mathfrak{p}^e))}$, and note that $\mathrm{Spec}(A/\mathfrak{p})$ is irreducible by the irreducibility of $V(\mathfrak{p})$.

Thus, we've reduced the problem again. For a homomorphism $f : A \to B$ of finite type, where A is a Noetherian integral domain, $\mathrm{Spec}(A)$ is irreducible, and $\overline{f^*(\mathrm{Spec}(B))} = \mathrm{Spec}(A)$, we must show that $f^*(\mathrm{Spec}(B))$ contains a non-empty open subset of $\mathrm{Spec}(A)$. Since A is Noetherian, B is Noetherian (Corollary 7.7), so by Exercise 6.8, $\mathrm{Spec}(B)$ is a Noetherian space. By Exercise 6.7, $\mathrm{Spec}(B)$ is a finite union of irreducible closed subspaces $Y_1, \ldots, Y_n$. Thus, it is sufficient to show that for some i, $f^*(Y_i)$ contains a non-empty open set of $\mathrm{Spec}(A)$. We have

$$\mathrm{Spec}(A) = \overline{f^*(\mathrm{Spec}(B))} = \bigcup_{i=1}^{n} \overline{f^*(Y_i)}.$$

Since $\mathrm{Spec}(A)$ is irreducible, we must have $\overline{f^*(Y)} = \mathrm{Spec}(A)$ for some $Y = Y_i$. As we saw earlier, an irreducible closed subspace Y must have the form $Y = V(\mathfrak{q})$ for some $\mathfrak{q} \in \mathrm{Spec}(B)$, so we have $\mathrm{Spec}(A) = \overline{f^*(V(\mathfrak{q}))}$. By Exercise 1.21, $V(\mathfrak{q})$ is homeomorphic to $\mathrm{Spec}(B/\mathfrak{q})$ so that under the composition $g : A \to B \to B/\mathfrak{q}$, $\overline{g^*(\mathrm{Spec}(B/\mathfrak{q}))} = \overline{f^*(V(\mathfrak{q}))} = \mathrm{Spec}(A)$. By Exercise 1.21, since $g^*(\mathrm{Spec}(B/\mathfrak{q}))$ is dense in $\mathrm{Spec}(A)$, $\ker g \subset \mathfrak{N}_A = 0$, so g is injective. Note that $B/\mathfrak{q}$ is an integral domain, and note that g is of finite type since it is a composition of two homomorphisms of finite type.

We've reduced the problem one last time: $f : A \to B$ is an injective homomorphism of finite type, where A and B are integral domains, and we want to show that $f^*(\mathrm{Spec}(B))$ contains a non-empty open subset of $\mathrm{Spec}(A)$. We're now in the situation of Exercise 5.21, so there exists a non-zero $s \in A$ such that given an algebraically closed field Ω and homomorphism $\phi : A \to \Omega$ (where $\phi(s) \neq 0$), we can extend ϕ to a homomorphism $B \to \Omega$. The image of both homomorphisms are subrings of a field, hence integral domains, so the kernels $\mathfrak{p}$ and $\mathfrak{q}$ are prime with $\mathfrak{p} = f^*(\mathfrak{q})$. We have $\phi(s) \neq 0$ if and only if $s \notin \ker \phi = \mathfrak{p}$, so for all $\mathfrak{p} \in V(s)^c$, we have $\mathfrak{p} \in f^*(\mathrm{Spec}(B))$ so that $V(s)^c$ is an open subset of $f^*(\mathrm{Spec}(B))$. Since A is an integral domain, s is not nilpotent. Thus, by $V(s)^c$ is non-empty by Exercise 1.17, as desired. □

Remark. *This is "Chevalley's Theorem": If A is a Noetherian ring and $f : A \to B$ is a homomorphism of finite type, f^* sends constructible sets to constructible sets.*

7.24 Exercise

Proof. By Exercise 5.10, if f^* is an open mapping, f has the going-down property. Conversely, suppose f has the going-down property. To show that f^* is an open mapping, it is clearly sufficient to show $f^*(Y_s)$ is open for an arbitrary basic open set $Y_s \subset Y = \mathrm{Spec}(B)$ (Exercise 1.17), where $s \in B$. By Proposition 3.11, the canonical map $B \to B_s$ has the going-down property and $\mathrm{Spec}(B_s) \cong Y_s$ (Exercise 3.21). Note that $B \to B_s$ is of finite type, so $g : A \to B \to B_s$ is of finite type. Then, $g^*(\mathrm{Spec}(B_s)) = f^*(Y_s)$.

Thus, for A Noetherian, $f : A \to B$ a homomorphism of finite type, it is enough to show that $f^*(Y)$ is open, where $Y = \mathrm{Spec}(B)$. Let X_0 be an irreducible closed subset of $X = \mathrm{Spec}(A)$. By

Exercise 6.8, $\mathrm{Spec}(A)$ is a Noetherian space. Then, in light of Exercise 7.22, we must show that either $f^*(Y) \cap X_0 = \emptyset$ or $f^*(Y) \cap X_0$ contains a non-empty open subset of X_0. So, assume X_0 is such that $f^*(Y) \cap X_0 \neq \emptyset$. Thus, there is some prime $\mathfrak{p}' \in f^*(Y) \cap X_0$. By the proof of Exercise 1.21, since X_0 is closed and irreducible, $X_0 = V(\mathfrak{p})$ for some prime ideal $\mathfrak{p} \in \mathrm{Spec}(A)$. Then, $\mathfrak{p}' \supset \mathfrak{p}$. Since f has the going-down property and $\mathfrak{p}' \in f^*(Y)$, $\mathfrak{p} \in f^*(Y)$. Thus, $\mathfrak{p} \in V(\mathfrak{p}) \cap f^*(Y)$. Thus, $V(\mathfrak{p}) = \overline{V(\mathfrak{p}) \cap f^*(Y)}$.

By Exercise 7.23, $f^*(Y)$ is a constructible set since Y is open and thus constructible. Thus, $f^*(Y) \cap V(\mathfrak{p})$ is a constructible set of the irreducible space $V(\mathfrak{p})$. By Exercise 7.20, since $f^*(Y) \cap V(\mathfrak{p})$ is dense in $V(\mathfrak{p})$, $f^*(Y) \cap V(\mathfrak{p})$ contains a non-empty open set, as desired. □

7.25 Exercise

Proof. Since $f : A \to B$ is flat, f has the going-down property by Exercise 5.11. Then, since A is Noetherian and $f : A \to B$ is of finite type, we have by Exercise 7.24 that f^* is an open mapping. □

7.26 Exercise

(i) Proof. Let λ be an additive function on the class of finitely generated A-modules with values in an abelian group G. If a homomorphism $\lambda_0 : K(A) \to G$ exists such that $\lambda(M) = \lambda_0(\gamma(M))$ for all M, then since the elements $\gamma(M)$ generate $K(A)$, this map is clearly unique. Thus, we must construct λ_0. Let C be the free abelian group of isomorphism classes of finitely generated A-modules, and let $n_1(M_1)+\cdots+n_t(M_t)$ be an arbitrary element of C. Then, we define $\lambda' : C \to G$ by $\lambda'(n_1(M_1) + \cdots + n_t(M_t)) = n_1\lambda(M_1) + \cdots + n_t\lambda(M_t)$. Clearly, λ' is a homomorphism. Let $0 \to M' \to M \to M'' \to 0$ be an arbitrary short exact sequence. Since λ is an additive function, $\lambda(M') - \lambda(M) + \lambda(M'') = 0$. Thus, $\lambda'((M') - (M) + (M'')) = 0$. Therefore, if D is the subgroup of C generated by these elements, $\lambda'(D) = 0$. Hence, λ' lifts to a well-defined homomorphism $\lambda_0 : K(A) \to G$ since $K(A) = C/D$. Then, $\lambda_0(\gamma(M)) = \lambda'(M) = \lambda(M)$, as desired. □

(ii) Proof. Since A is Noetherian, if M is a finitely-generated A-module, we have by Exercise 7.18 that there exists a chain of submodules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_r = M$$

such that $M_i/M_{i-1} \cong A/\mathfrak{p}_i$ for prime ideals $\mathfrak{p}_i$ of A. For each i then, we have a short exact sequence

$$0 \to M_{i-1} \to M_i \to A/\mathfrak{p}_i \to 0$$

so that $\gamma(M_{i-1}) - \gamma(M_i) + \gamma(A/\mathfrak{p}_i) = 0$, or rather, $\gamma(M_i) = \gamma(M_{i-1}) + \gamma(A/\mathfrak{p}_i)$. For $i = 1$, we have $\gamma(M_1) = \gamma(M_0) + \gamma(A/\mathfrak{p}_1) = \gamma(0) + \gamma(A/\mathfrak{p}_1) = \gamma(A/\mathfrak{p}_1)$. By a simple inductive argument, we see that

$$\gamma(M) = \gamma(M_r) = \gamma(A/p_r) + \cdots + \gamma(A/\mathfrak{p}_1).$$

Thus, $K(A)$ is generated by elements of the form $\gamma(A/\mathfrak{p})$ for $\mathfrak{p} \in \mathrm{Spec}(A)$. □

(iii) Proof. Let A be a principal ideal domain. Using the structure theorem for finitely generated modules over a PID, every finitely generated A-module M can be decomposed as

$$M \cong A^n \oplus A/(a_1) \oplus \cdots \oplus A/(a_k)$$

for elements $a_i \in A$. Thus, $\gamma(M) = n\gamma(A) + \gamma(A/(a_1)) + \cdots + \gamma(A/(a_k))$. Now, let $a \in A$. Then, we have a short exact sequence

$$0 \to A \xrightarrow{\cdot a} A \to A/(a) \to 0.$$

Thus, $\gamma(A) - \gamma(A) + \gamma(A/(a)) = 0$, so $\gamma(A/(a)) = 0$ for every $a \in A$. Hence, $\gamma(M) = n\gamma(A)$. Finally then, $K(A)$ is a free abelian group with one generator, $\gamma(A)$, so $K(A) \cong \mathbb{Z}$. □

(iv) Proof. Let $f : A \to B$ be a finite homomorphism. By restriction of scalars, B-modules are A-modules. In fact, since f is finite, finitely generated B-modules are finitely generated A-modules, so if $C(R)$ is the free abelian group of isomorphism classes of R-modules, we have a homomorphism $C(B) \to C(A)$. Thus, we have a homomorphism $C(B) \to K(A)$. If we have a short exact sequence of B-modules, $0 \to M' \to M \to M'' \to 0$, this is a short exact sequence of A-modules, as the maps are the same. Thus, $D(B) \subset \ker(C(B) \to K(A))$, so the homomorphism $C(B) \to K(A)$ lifts to a homomorphism $f_! : K(B) \to K(A)$.

Now, suppose we have another finite homomorphism $B \to C$. To show that $(g \circ f)_! = f_! \circ g_!$, first note the commutativity of the following diagram

$$\begin{array}{ccccc} C(C) & \longrightarrow & C(B) & \longrightarrow & C(A) \\ \downarrow & & \downarrow & & \downarrow \\ K(C) & \xrightarrow{g_!} & K(B) & \xrightarrow{f_!} & K(A). \end{array}$$

Then, it is sufficient to see that the composition $C(C) \to C(B) \to C(A)$ is the map produced by $g \circ f$. This is clear; if M is a C-module, then it is a B-module via g, so it is an A-module via f. The A-module structure of M is thus produced by the map $g \circ f$, as desired. □

7.27 Exercise

(i) Proof. By Exercise 2.8, If M and N are flat, finitely generated A-modules, then $M \otimes_A N$ is a flat A-module. Also, since M and N are finitely generated by elements $\{m_i\}$ and $\{n_j\}$, $M \otimes_A N$ is generated as an A-module by elements $m_i \otimes n_j$, so $M \otimes_A N$ is also finitely generated. Thus, for $\gamma_1(M), \gamma_1(N) \in K_1(A)$, $\gamma_1(M \otimes N) \in K_1(A)$. Thus, define $\gamma_1(M) \cdot \gamma_1(N) = \gamma_1(M \otimes N)$. The commutativity of the product is inherited from the commutativity of the tensor product.

For all $\gamma_1(M) \in K_1(A)$, we have $\gamma_1(A) \cdot \gamma_1(M) = \gamma_1(A \otimes M) = \gamma_1(M)$ since $A \otimes_A M \cong M$ as A-modules. Thus, $\gamma_1(A)$ is the multiplicative identity. Since $K_1(A)$ is already an abelian additive group, to see that $K_1(A)$ has the structure of a commutative ring, the only thing left to show is the distributive property. Let M, N, P be flat, finitely generated A-modules. Then,

$$\gamma_1(P) \cdot (\gamma_1(M) + \gamma_1(N)) = \gamma_1(P) \cdot \gamma_1(M \oplus N)$$

$$= \gamma_1(P \otimes (M \oplus N)) = \gamma_1((P \otimes M) \oplus (P \otimes N))$$

$$= \gamma_1(P \otimes M) + \gamma_1(P \otimes N) = \gamma_1(P) \cdot \gamma_1(M) + \gamma_1(P) \cdot \gamma_1(N),$$

as desired. □

(ii) Proof. We wish to show that $K(A)$ has a $K_1(A)$-module structure defined by $\gamma_1(M) \cdot \gamma(N) = \gamma(M \otimes N)$. Since M and N are both finitely generated A-modules, $M \otimes N$ is a finitely generated A-module, so $\gamma(M \otimes N) \in K(A)$, as required.

With $1 = \gamma_1(A) \in K_1(A)$, we have

$$1 \cdot \gamma(N) = \gamma_1(A) \cdot \gamma(N) = \gamma(A \otimes N) = \gamma(N),$$

so the identity acts appropriately. Now, we show that the action is compatible with the additive structure of $K(A)$. Let P be flat, finitely generated A-module, and let M, N be finitely generated A-modules. Then,

$$\begin{aligned}\gamma_1(P) \cdot (\gamma(M) + \gamma(N)) &= \gamma_1(P) \cdot \gamma(M \oplus N) \\ &= \gamma(P \otimes (M \oplus N)) = \gamma((P \otimes M) \oplus (P \otimes N)) \\ &= \gamma(P \otimes M) + \gamma(P \otimes N) = \gamma_1(P) \cdot \gamma(M) + \gamma_1(P) \cdot \gamma(N),\end{aligned}$$

as required.

Let M, N be flat, finitely generated A-modules, and let P be a finitely generated A-module. Then,

$$\begin{aligned}(\gamma_1(M) + \gamma_1(N)) \cdot \gamma(P) &= \gamma_1(M \oplus N) \cdot \gamma(P) \\ &= \gamma((M \oplus N) \otimes P) = \gamma((M \otimes P) \oplus (N \otimes P)) \\ &= \gamma(M \otimes P) + \gamma(N \otimes P) = \gamma_1(M) \cdot \gamma(P) + \gamma_1(N) \cdot \gamma(P),\end{aligned}$$

as required.

Finally, suppose M, N are flat, finitely generated A-modules and P is a finitely generated A-module. Then,

$$\begin{aligned}&\gamma_1(M) \cdot (\gamma_1(N) \cdot \gamma(P)) = \gamma_1(M) \cdot (\gamma(N \otimes P)) = \gamma(M \otimes (N \otimes P)) \\ &= \gamma((M \otimes N) \otimes P) = \gamma_1(M \otimes N) \cdot \gamma(P) = (\gamma_1(M) \cdot \gamma_1(N)) \cdot \gamma(P)\end{aligned}$$

since, as in the proof of (i), $M \otimes N$ is a flat and finitely generated A-module. Thus, $K(A)$ has a $K_1(A)$-module structure in this way. □

(iii) Proof. Suppose A is a Noetherian local ring. Then, if M is a flat, finitely generated A-module, then M is free by Exercise 7.15. Thus, $M \cong A^n$ as an A-module. Hence, $\gamma_1(M) = \gamma_1(A^n) = n\gamma_1(A)$. Thus, $K_1(A)$ is a free abelian group with one generator, $\gamma_1(A)$, so $K_1(A) \cong \mathbb{Z}$. □

(iv) Proof. Let $f : A \to B$ be a ring homomorphism, where A and B are Noetherian. Let M be a flat, finitely generated A-module. Then, if $\{m_i\}$ are the finitely many generators, then $\{1 \otimes m_i\}$ generate $M_B = B \otimes_A M$ as a B-module, so M_B is finitely generated. Suppose $N_1 \to N_2$ is an injection of B-modules. Then, we can view N_1 and N_2 as A-modules via f, so since M is flat, $M \otimes_A N_1 \to M \otimes_A N_2$ is injective. However, we can rewrite this map:

$$\begin{aligned}M \otimes_A (B \otimes_B N_1) &\to M \otimes_A (B \otimes_B N_2) \\ (M \otimes_A B) \otimes_B N_1 &\to (M \otimes_A B) \otimes_B N_2 \\ M_B \otimes_B N_1 &\to M_B \otimes_B N_2.\end{aligned}$$

Thus, the final map, being identical to $M \otimes_A N_1 \to M \otimes_A N_2$, is also injective, so M_B is a flat, finitely generated B-module. Thus, we have a homomorphism $C_1(A) \to C_1(B)$ from the free abelian group of isomorphism classes of flat, finitely generated A-modules to those of B.

This becomes a homomorphism $C_1(A) \to K_1(B)$. Now, suppose we have a short exact sequence

$$0 \to N' \to N \to N'' \to 0$$

of flat, finitely generated A-modules. This gives rise to a short exact sequence

$$\mathrm{Tor}_1^A(B, N'') \to N'_B \to N_B \to N''_B \to 0.$$

By Exercise 2.24, $\mathrm{Tor}_1^A(B, N'') = 0$ since N'' is flat. Thus, short exact sequences map to short exact sequences, so $D_1(A) \subset \ker(C_1(A) \to K_1(A))$. Hence, our homomorphism lifts to a homomorphism $f^! : K_1(A) \to K_1(B)$.

Finally, suppose $g : B \to C$ is another ring homomorphism (with C Noetherian). To see that $(g \circ f)^! = g^! \circ f^!$, first note the following commutative diagram:

$$\begin{array}{ccccc} C_1(A) & \longrightarrow & C_1(B) & \longrightarrow & C_1(C) \\ \downarrow & & \downarrow & & \downarrow \\ K_1(A) & \xrightarrow{f^!} & K_1(B) & \xrightarrow{g^!} & K_1(C). \end{array}$$

Thus, it is sufficient to show that the composite map $C_1(A) \to C_1(B) \to C_1(C)$ is identical to the map $C_1(A) \to C_1(C)$ coming from $g \circ f$. Let M be an A-module. Under f, this gives rise to a B-module $B \otimes_A M$. Under g, this gives rise to a C-module $C \otimes_B (B \otimes_A M)$. But we have the isomorphism

$$C \otimes_B (B \otimes_A M) \cong (C \otimes_B B) \otimes_A M \cong C \otimes_A M,$$

which is precisely the extension of scalars of M for A straight to C via $g \circ f$. □

(v) Proof. Let $f : A \to B$ be a finite ring homomorphism. Then, B is a finitely generated A-module so that $\gamma_A(B) \in K(A)$. Now, suppose $M \in K_1(A)$ and $N \in K(B)$. We have

$$f_!(f^!(\gamma_{1,A}(M)) \cdot \gamma_B(N)) = f_!(\gamma_{1,B}(M_B) \cdot \gamma_B(N)) = f_!(\gamma_B(M_B \otimes_B N))$$

$$= \gamma_A(M_B \otimes_B N) = \gamma_A((M \otimes_A B) \otimes_B N) = \gamma_A(M \otimes_A (B \otimes_B N))$$

$$= \gamma_A(M \otimes_A N_A) = \gamma_{1,A}(M) \cdot \gamma_A(N_A) = \gamma_{1,A}(M) \cdot f_!(\gamma_B(N)).$$

In other words, regarding $K(B)$ as a $K_1(A)$-module by restriction of scalars, the homomorphism $f^!$ is a $K_1(A)$-module homomorphism. □

8 Artin Rings (Done)

8.1 Exercise

Proof. Let $0 = \mathfrak{q}_1 \cap \cdots \cap \mathfrak{q}_n$ be a minimal primary decomposition of zero in a Noetherian ring A, where $\mathfrak{q}_i$ is $\mathfrak{p}_i$-primary. Localizing to $A_{\mathfrak{p}_i}$, if $S = A - \mathfrak{p}_i$, then since $S \cap \mathfrak{p}_i = \emptyset$, we have by Proposition 4.8 that $S^{-1}\mathfrak{q}_i$ is $S^{-1}\mathfrak{p}_i$-primary, and its contraction is $\mathfrak{q}_i$.

Since A is Noetherian, $A_{\mathfrak{p}_i}$ is also (Corollary 7.4), so by Proposition 7.14, $S^{-1}\mathfrak{q}_i \supset (S^{-1}\mathfrak{p}_i)^r = S^{-1}\mathfrak{p}_i^{r_i}$ for some $r_i > 0$. Contracting, we obtain $\mathfrak{q}_i \supset \mathfrak{p}_i^{(r_i)}$.

Suppose $\mathfrak{q}_i$ is an isolated primary component. Since there is some $r_i > 0$ such that $\mathfrak{q}_i \supset \mathfrak{p}_i^{(r_i)}$, we can replace $\mathfrak{q}_i$ with the $\mathfrak{p}_i$-primary ideal $\mathfrak{p}_i^{(r_i)}$ (Exercise 4.13) to obtain another minimal primary decomposition of 0. By Corollary 4.11 (2nd uniqueness theorem for primary decompositions), $\mathfrak{q}_i = \mathfrak{p}_i^{(r_i)}$. Likewise, if $m > r_i$, then $\mathfrak{p}_i^{(m)} \subset \mathfrak{p}_i^{(r_i)} = \mathfrak{q}_i$, so by the same argument, $\mathfrak{q}_i = \mathfrak{p}_i^{(m)}$ so that $\mathfrak{p}_i^{(r_i)} = \mathfrak{p}_i^{(m)}$. By the one-to-one correspondence of Proposition 4.8, since $\mathfrak{p}_i^{(m)} = \mathfrak{p}_i^{(r_i)}$, $S^{-1}\mathfrak{p}_i^m = S^{-1}\mathfrak{p}_i^{r_i}$ for all $m > r_i$. Note that $A_{\mathfrak{p}_i}$ is a Noetherian local ring, so since $S^{-1}\mathfrak{p}_i$ is the unique maximal ideal and its powers stablize, we have by Proposition 8.6 that $S^{-1}\mathfrak{p}_i^m = 0$ for all m such that $m > m_i$ for some $m_i > 0$ and $A_{\mathfrak{p}_i}$ is an Artinian local ring. In particular, if $m > \max\{m_i, r_i\}$, then $\mathfrak{q}_i = \mathfrak{p}_i^{(m)} = (S^{-1}\mathfrak{p}_i^m)^c = (S^{-1}0)^c$.

Now, suppose $\mathfrak{q}_i$ is an embedded primary component. Again, $A_{\mathfrak{p}_i}$ is a local Noetherian ring. However, since $\mathfrak{q}_i$ is embedded, there is some prime ideal $\mathfrak{p}_j \neq \mathfrak{p}_i$ belonging to 0 such that $\mathfrak{p}_j \subset \mathfrak{p}_i$. Localizing to $A_{\mathfrak{p}_i}$, since $\mathfrak{p}_j \cap A - \mathfrak{p}_i = \emptyset$, we have by Proposition 4.8 that $S^{-1}\mathfrak{p}_j$ and $S^{-1}\mathfrak{p}_i$ are unique prime ideals. So, we have a proper chain of primes $S^{-1}\mathfrak{p}_j \subset S^{-1}\mathfrak{p}_i$. In particular, $\dim A_{\mathfrak{p}_i} \geq 1$, so by Theorem 8.5, $A_{\mathfrak{p}_i}$ is not Artinian.

Since $A_{\mathfrak{p}_i}$ is a local Noetherian ring that is not Artinian and $S^{-1}\mathfrak{p}_i$ is the unique maximal ideal, we have by Proposition 8.6 that $S^{-1}\mathfrak{p}_i^n \neq S^{-1}\mathfrak{p}_i^{n+1}$ for all $n > 0$. Thus, by Proposition 4.8, their contractions are all distinct so that we have proper inclusions $\mathfrak{p}_i^{(n+1)} \subset \mathfrak{p}_i^{(n)}$ for all $n > 0$. However, we have still have $\mathfrak{q}_i \supset \mathfrak{p}_i^{(r_i)}$ for some $r_i > 0$. Since $\mathfrak{p}_i^{(n)}$ is $\mathfrak{p}_i$-primary by Exercise 4.13, for any $n \geq r_i$, we can replace $\mathfrak{q}_i$ with the $\mathfrak{p}_i$-primary ideal $\mathfrak{p}_i^{(n)}$ to obtain another minimal primary decomposition of 0. Therefore, there are infinitely many minimal primary decompositions of 0 which differ only in the $\mathfrak{p}_i$-primary component. □

8.2 Exercise

Proof. (i) $\implies$ (ii). If A is Artinian, all prime ideals are maximal (Proposition 8.1), so all singletons of $\mathrm{Spec}(A)$ are closed. By Proposition 8.3, A has finitely many maximal ideals and thus prime ideals, so $\mathrm{Spec}(A)$ is finite. Let S be a subset of $\mathrm{Spec}(A)$, so it is also a finite set. Then, $S = \bigcup_{x \in S}\{x\}$ is a finite union of closed sets, so S is closed. Thus, $\mathrm{Spec}(A)$ is discrete and finite.

(ii) $\implies$ (iii). Trivial.

(iii) $\implies$ (i). Suppose $\mathrm{Spec}(A)$ is discrete. Thus, every singleton of $\mathrm{Spec}(A)$ is closed, so all prime ideals of A are maximal. Hence, $\dim A = 0$. Since A is Noetherian, we have by Theorem 8.5 that A is Artinian. □

8.3 Exercise

Proof. Let k be a field and A a finitely generated k-algebra. Since k is Noetherian, A is also Noetherian (Proposition 7.2).

(i) $\implies$ (ii). If A is Artinian, then by Theorem 8.7, A is isomorphic to a finite direct product of Artin local rings, $\prod_{i=1}^{n} A_i$. Then, each A_i is a finitely generated k-algebra, so to show that A is a finite k-algebra, it is enough to show that each factor A_i is a finite k-algebra. Thus, we've reduced to the case where A is an Artin local ring and a finitely generated k-algebra. Let $\mathfrak{m}$ be the unique maximal ideal of A. By Corollary 7.10, $B = A/\mathfrak{m}$ is a finite algebraic extension of k. Thus, B is finitely generated as a k-module, i.e. B is a finite-dimensional k-vector space.

Now, since A is local and Artinian, its only prime ideal is its unique maximal ideal $\mathfrak{m}$. Since A is Noetherian, 0 has a primary decomposition, so since $\mathfrak{m}$ is the only prime, $\mathfrak{m}$ belongs to 0. Since A is clearly a finitely generated A-module, we have by Exercise 7.18 that there exists a chain of submodules

$$0 = M_0 \subset M_1 \subset \cdots \subset M_r = A$$

such that $M_i/M_{i-1} \cong A/\mathfrak{m} = B$ for all i. Then, $M_1 \cong A/\mathfrak{m}$ so that $\dim_k M_1 = \dim_k B$. For all i, we have

$$0 \to M_{i-1} \to M_i \to B \to 0,$$

so $\dim_k M_i = \dim_k M_{i-1} + \dim_k B$. Thus, by induction, $\dim_k A = \dim_k M_r = r \dim_k B$, so A is a finite k-algebra.

(ii) $\implies$ (i). Suppose A is a finite k-algebra. Then, A is a finite-dimensional k-vector space. By Proposition 6.10, A satisfies the descending chain condition, so A is Artinian. $\square$

8.4 Exercise

Proof. Let $f : A \to B$ be a homomorphism of finite type.

(i) $\implies$ (ii). If f is finite, then B is finitely generated as an A-module. Let $\mathfrak{p} \in \operatorname{Spec}(A)$. Then, $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ by Exercise 3.21, where $k(\mathfrak{p})$ is the residue field of $A_{\mathfrak{p}}$. Since B is a finitely generated A-module, $B \otimes_A k(\mathfrak{p})$ is a finitely generated $k(\mathfrak{p})$-module. By Proposition 6.10, $B \otimes_A k(\mathfrak{p})$ is both Noetherian and Artinian, so by Exercise 8.2, $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ is discrete. Thus, the fibers of f^* are discrete subspaces of $\operatorname{Spec}(B)$.

(ii) $\implies$ (iii). Suppose the fibers of f^* are discrete subspaces of $\operatorname{Spec}(B)$. For each prime $\mathfrak{p} \in \operatorname{Spec}(A)$, $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ by Exercise 3.21, so $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ is discrete. Since $f : A \to B$ is of finite type, $B \otimes_A k(\mathfrak{p})$ is a finitely generated $k(\mathfrak{p})$-algebra. Thus, $B \otimes_A k(\mathfrak{p})$ is Noetherian by Proposition 7.3 since $k(\mathfrak{p})$ is Noetherian. Since $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ is finite and $B \otimes_A k(\mathfrak{p})$ is Noetherian, we have by Exercise 8.2 that $B \otimes_A k(\mathfrak{p})$ is Artinian. Thus, by Exercise 8.3, $B \otimes_A k(\mathfrak{p})$ is a finite $k(\mathfrak{p})$-algebra.

(iii) $\implies$ (ii). Suppose $B \otimes_A k(\mathfrak{p})$ is a finite $k(\mathfrak{p})$-algebra for each prime $\mathfrak{p} \in \operatorname{Spec}(A)$. By Exercise 8.3, $B \otimes_A k(\mathfrak{p})$ is Artinian, so by Exercise 8.2, $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ is discrete. Since $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ by Exercise 3.21, the fibers of f^* are discrete subspaces of $\operatorname{Spec}(B)$.

(iii) $\implies$ (iv). Suppose $B \otimes_A k(\mathfrak{p})$ is a finite $k(\mathfrak{p})$-algebra for each prime $\mathfrak{p} \in \operatorname{Spec}(A)$. By Exercise 8.3, $B \otimes_A k(\mathfrak{p})$ is Artinian, so by Exercise 8.2, $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ is discrete and finite. Since $f^{*-1}(\mathfrak{p})$ is homeomorphic to $\operatorname{Spec}(B \otimes_A k(\mathfrak{p}))$ by Exercise 3.21, the fibers of f^* are finite. $\square$

If f is integral and the fibers of f^* are finite, then f is not necessarily finite.

Proof. Let k be a field of characteristic 2 (e.g., $\mathbb{F}_2$), let $A = k[x_1^2, x_2^2, \ldots]$, and let $B = k[x_1, x_2, \ldots]$, so we have a map $f : A \hookrightarrow B$.

B is integral over A. For each $x_i \in B$, x_i solves the monic polynomial $x^2 - x_i^2 \in A[x]$. Since the integral elements of B over A form a subring of B and since B is generated as a ring by elements x_i and elements of $k \subset A$, we see that f is integral.

Let $\mathfrak{p}$ be a prime ideal of A. Suppose $\mathfrak{q} \in B$ such that $\mathfrak{p} = \mathfrak{q} \cap A$. For $b \in \mathfrak{q}$, we can write $b = \sum b_\alpha x_{i_1}^{e_i} \cdots x_{i_n}^{e_n}$. Then, since $x \mapsto x^2$ is a ring homomorphism (k has characteristic 2),

$$b^2 = \left(\sum b_\alpha x_{i_1}^{e_i} \cdots x_{i_n}^{e_n}\right)^2 = \sum b_\alpha^2 x_{i_1}^{2e_i} \cdots x_{i_n}^{2e_n} \in A.$$

Thus, for all $b \in \mathfrak{q}$, $b^2 \in A \cap \mathfrak{q} = \mathfrak{p}$. Conversely, if $b \in B$ such that $b^2 \in \mathfrak{p}$, then $b^2 \in \mathfrak{p} \subset \mathfrak{q}$; since $\mathfrak{q}$ is prime, $b \in \mathfrak{q}$. Thus, $\mathfrak{q} = \{b \in B \mid b^2 \in \mathfrak{p}\}$. In particular, $\mathfrak{q}$ depends only on $\mathfrak{p}$ so is uniquely determined. Hence, $f^{*-1}(\mathfrak{p})$ is either empty or a singleton, so f^* clearly has finite fibers.

B is not a finitely generated A-module. Suppose $m_1, \ldots, m_n$ are a finite generating set of B as an A-module. Then, each m_i is an expression of B containing finitely many x_i. Let S be the finite set of x_i occuring in the expressions $m_1, \ldots, m_n$. Thus, there is some $x_r \notin S$. Write $x_r = a_1 m_1 + \cdots + a_n m_n$, where $a_i \in A$. Now, we have a ring homomorphism $\psi : B \mapsto k[x_r]$ achieved by sending $x_i \mapsto 0$ for all $i \neq r$. Under this map, we have $x_r = \psi(x_r) = \psi(a_1)\psi(m_1) + \cdots + \psi(a_n)\psi(m_n)$. Since m_i does not include x_r for any i, $\psi(m_i) \in k$. Further, for all a_i, $\psi(a_i) \in k[x^2]$. Thus, the right hand side is an element of $k[x_r^2]$. This is a contradiction since $x_r \notin k[x_r^2]$, so f is not finite. □

8.5 Exercise

Proof. Let's briefly recall the context of Exercise 5.16 and harvest some details from the proof. X is an affine algebraic variety of k^n, where k is an algebraically closed field. L is an r-dimensional linear subspace of k^n. We have an integral map $f : P(L) \hookrightarrow P(X)$ which corresponds to a surjective regular map $\psi : X \to L$. Both $P(L)$ and $P(X)$ are isomorphic as k-algebras to quotients of $k[y_1', \ldots, y_n']$, and f preserves the images of each y_i'. Thus, $P(X)$ is finitely generated over $P(L)$, so f is integral and of finite type. In light of Corollary 5.2, f is finite. By Exercise 8.4, f^* has finite fibers.

By Exercise 5.17, maximal ideals of $k[y_1', \ldots, y_n']$ have the form $\mathfrak{m}_a = (y_1' - a_1, \ldots, y_n' - a_n)$ for $a = (a_i)_i \in k^n$. Further, $\ell \in L$ if and only if $\mathfrak{m}_\ell$ is a maximal ideal of $P(L)$. Since f is integral, if $\mathfrak{p} \in \operatorname{Spec}(P(X))$ is such that $f^{-1}(\mathfrak{p}) = \mathfrak{m}_\ell$, then $\mathfrak{p}$ is maximal by Corollary 5.8. Thus, $\mathfrak{p} = \mathfrak{m}_x$ for some $x \in X$. Thus, to show that the fibers of ψ have bounded size (so that X is a finite cover of L), it is enough to show that the fibers of f^* have bounded size.

Since f is finite, $P(X)$ is finitely generated as a $P(L)$-module, so suppose $P(X)$ has n generators as a $P(L)$-module. Then, for $\mathfrak{p} \in \operatorname{Spec}(P(L))$, $B = P(X) \otimes_{P(L)} k(\mathfrak{p})$ has $k(\mathfrak{p})$-dimension at most n. By Exercise 3.21, $f^{*-1}(\mathfrak{p}) \cong \operatorname{Spec}(B)$, so it is enough to show that $\operatorname{Spec}(B)$ has bounded size. Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ be distinct primes of $\operatorname{Spec}(B)$. Then, $B \to \prod_i B/\mathfrak{p}_i$ is surjective. Since B is a $k(\mathfrak{p})$-algebra, the $B/\mathfrak{p}_i$ are also $k(\mathfrak{p})$-vector spaces. Then,

$$n \geq \dim_{k(\mathfrak{p})} B \geq \sum_{i=1}^{m} \dim_{k(\mathfrak{p})}(B/\mathfrak{p}_i) \geq m,$$

so $\operatorname{Spec}(B)$ has at most n elements for any choice of $\mathfrak{p} \in \operatorname{Spec}(A)$. □

8.6 Exercise

Proof. Let A be a Noetherian ring, and let $\mathfrak{q}$ be a $\mathfrak{p}$-primary ideal. Suppose we have a strict chain of $\mathfrak{p}$-primary ideals

$$\mathfrak{q} = \mathfrak{q}_0 \subset \mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_r = \mathfrak{p}.$$

Let $S = A - \mathfrak{p}$. By Proposition 4.8, we have the following strict chain of $S^{-1}\mathfrak{p}$-primary ideals of the Noetherian ring $A_\mathfrak{p}$ (Corollary 7.4):

$$S^{-1}\mathfrak{q} = S^{-1}\mathfrak{q}_0 \subset S^{-1}\mathfrak{q}_1 \subset \cdots \subset S^{-1}\mathfrak{q}_r = S^{-1}\mathfrak{p}.$$

Since $r(S^{-1}\mathfrak{q}) = S^{-1}\mathfrak{p}$ and $A_\mathfrak{p}$ is a Noetherian ring, we have by Proposition 7.14 that there is some $m > 0$ such that $(S^{-1}\mathfrak{p})^m \subset S^{-1}\mathfrak{q}$.

Passing to the ring $A_\mathfrak{p}/S^{-1}\mathfrak{q}$ (Noetherian by Proposition 7.1), we have a strict (Proposition 1.1) chain of ideals

$$0 \subset S^{-1}\mathfrak{q}_1/S^{-1}\mathfrak{q} \subset \cdots \subset S^{-1}\mathfrak{q}_r/S^{-1}\mathfrak{q} = S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}.$$

By the order-preserving correspondence of Proposition 1.1, $S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}$ is the unique maximal ideal of $A_\mathfrak{p}/S^{-1}\mathfrak{q}$, so $A_\mathfrak{p}/S^{-1}\mathfrak{q}$ is a Noetherian local ring. However,

$$(S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q})^m = (S^{-1}\mathfrak{p})^m/S^{-1}\mathfrak{q} \subset S^{-1}\mathfrak{q}/S^{-1}\mathfrak{q} = 0,$$

so by Theorem 8.6, $A_\mathfrak{p}/S^{-1}\mathfrak{q}$ is an Artinian local ring.

Consider $A_\mathfrak{p}/S^{-1}\mathfrak{q}$ as a $A_\mathfrak{p}/S^{-1}\mathfrak{q}$-module. Then, it is an Artinian and Noetherian module, so it has finite length by Proposition 6.8. Thus, $S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}$ has finite length $n = \ell(S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q})$. Since the chain of submodules (ideals) above is strict, $r \leq n$, so the length of $\mathfrak{p}$-primary chains between $\mathfrak{p}$ and $\mathfrak{q}$ is bounded. By Proposition 6.7, every composition series of $S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}$ has length n and every chain of length less than n can be extended to one of length n. Therefore, to show that maximal chains of $\mathfrak{p}$-primary ideals between $\mathfrak{p}$ and $\mathfrak{q}$ have length n, it is enough to show that arbitrary composition series of $S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}$ can be lifted to chains of $\mathfrak{p}$-primary ideals between $\mathfrak{p}$ and $\mathfrak{q}$.

Suppose we have a composition series of $A_\mathfrak{p}/S^{-1}\mathfrak{q}$:

$$0 = \mathfrak{b}_0 \subset \mathfrak{b}_1 \subset \cdots \subset \mathfrak{b}_n = S^{-1}\mathfrak{p}/S^{-1}\mathfrak{q}.$$

By Proposition 1.1, each $\mathfrak{b}_i$ lifts to a unique ideal of $A_\mathfrak{p}$ containing $S^{-1}\mathfrak{q}$, and this lift preserves ordering. Since by Proposition 3.11 all ideals of $A_\mathfrak{p}$ have the form $S^{-1}\mathfrak{q}$ for some ideal $\mathfrak{q}$ of A, we lift to a strict chain

$$S^{-1}\mathfrak{q} = S^{-1}\mathfrak{q}_0 \subset S^{-1}\mathfrak{q}_1 \subset \cdots \subset S^{-1}\mathfrak{q}_n = S^{-1}\mathfrak{p}.$$

Since $S^{-1}\mathfrak{q}$ is $S^{-1}\mathfrak{p}$-primary,

$$S^{-1}\mathfrak{p} = r(S^{-1}\mathfrak{q}) \subset r(S^{-1}\mathfrak{q}_i) \subset r(S^{-1}\mathfrak{p}) = S^{-1}\mathfrak{p}$$

for all i. $S^{-1}\mathfrak{p}$ is maximal in $A_\mathfrak{p}$, so by Proposition 4.2, $S^{-1}\mathfrak{q}_i$ is $S^{-1}\mathfrak{p}$-primary for all i. By Proposition 4.8 (and Proposition 3.11), this lifts to a strict chain

$$\mathfrak{q} = \mathfrak{q}_0 \subset \mathfrak{q}_1 \subset \cdots \subset \mathfrak{q}_n = \mathfrak{p}$$

of $\mathfrak{p}$-primary ideals $\mathfrak{q}_i$, as desired. □

9 Discrete Valuation Rings and Dedekind Domains (Done)

9.1 Exercise

Proof. Since A is a Dedekind domain, it is a Noetherian domain of dimension 1. Thus, $S^{-1}A$ is a Noetherian domain (Proposition 7.3). Since all the prime ideals of $S^{-1}A$ are extended prime ideals of A (Proposition 3.11), a chain of prime ideals $S^{-1}\mathfrak{p}_0 \subset S^{-1}\mathfrak{p}_1 \subset S^{-1}\mathfrak{p}_2$ pulls back to a chain of prime ideals $\mathfrak{p}_0 \subset \mathfrak{p}_1 \subset \mathfrak{p}_2$ of A. Then, since A has dimension 1, $S^{-1}A$ has dimension at most 1.

Suppose $S^{-1}A$ has dimension 0. Since $S^{-1}A$ is an integral domain, 0 is a prime. Every $\mathfrak{p} \in \operatorname{Spec}(S^{-1}A)$ satisfies $0 \subset \mathfrak{p}$, so since $S^{-1}A$ has dimension 0, $\mathfrak{p} = 0$. Thus, the only primes of $S^{-1}A$ are 0, so $S^{-1}A$ is a field. In particular, every prime ideal of A intersects S. Let $\mathfrak{a}$ be an arbitrary non-zero ideal of A. Since A is a domain, $\mathfrak{a}^e \neq 0$, so $\mathfrak{a}^e = (1)$, implying $\mathfrak{a}$ intersects S by Proposition 3.11. Now, let $a \in A$ and let $b \in A$ be such that $b \neq 0$; this yields an arbitrary element $a/b \in K$. Since (b) is nonzero, (b) intersects S. Thus, there is some $x \in A$ such that $xb \in S$. Thus, $\frac{a}{b} = \frac{xa}{xb} \in S^{-1}A$. Since $A \subset S^{-1}A \subset K$, we see that $S^{-1}A = K$.

Alternatively, $S^{-1}A$ has dimension 1, so it is a Noetherian domain of dimension 1. Since A is integrally closed in its field of fractions K, and $A \subset S^{-1}A \subset K$, we have by Proposition 5.12 that $S^{-1}A$ is closed in its field of fractions, also K. Thus, $S^{-1}A$ is integrally closed, so by the equivalent conditions of Theorem 9.3, $S^{-1}A$ is a Dedekind domain. □

Proof. Let H and H' be the ideal class groups of A and $S^{-1}A$, respectively. If $S^{-1}A$ is a field, then $H' = 1$, so there is clearly a surjective homomorphism $H \to H'$. Otherwise, by the previous proof, $S^{-1}A$ is a Dedekind domain.

Note that I', the group of fractional ideals of $S^{-1}A$, is generated by the prime ideals of $S^{-1}A$. The ring homomorphism $A \to S^{-1}A$ naturally gives rise to a group homomorphism $\mathfrak{a} \to \mathfrak{a}^e$. By Proposition 3.11, all prime ideals of $S^{-1}A$ are extended, so since the prime ideals generate I', the homomorphism $I \to I'$ is surjective.

Similarly, if $xA \in P$, then $xA \mapsto (x/1)S^{-1}A \in P'$, so the map $I \to I'$ restricted to P gives a homomorphism $P \to P'$. Thus, our homomorphism lifts to $H = I/P \to I'/P' = H'$, and it is surjective since $I \to I'$ is surjective. □

9.2 Exercise

Proof. For a polynomial $f = \sum_i a_i x^i$ with coefficients in a Dedekind domain, we define the "content" of f to be the ideal $c(f) = (a_i)_i$ of A, the ideal generated by its coefficients. Now, suppose $f, g \in A[x]$, where $f = \sum a_i x^i$ and $g = \sum b_j x^j$. If $c(fg) = 0$, then $fg = 0$ so that either $f = 0$ or $g = 0$; A is an integral domain, so $A[x]$ is also. Then, $c(f) = 0$ or $c(g) = 0$ so that $c(fg) = c(f)c(g)$. If $c(f)c(g) = 0$, then by containment, $c(fg) = 0$. Thus, we can assume that $c(fg)$, $c(f)$, and $c(g)$ are nonzero ideals. Then, since A is a Dedekind domain, each is uniquely described as a product of nonzero prime ideals. In particular, to show that $c(fg) = c(f)c(g)$, we must show that for each nonzero prime ideal $\mathfrak{p}$, the exponent of $\mathfrak{p}$ of $c(fg)$ is the sum of its exponents in $c(f)$ and $c(g)$.

Let's localize at an arbitrary nonzero prime ideal $\mathfrak{p}$ of A. Since A is Dedekind, every nonzero prime ideal is maximal. In particular, any nonzero prime that is not $\mathfrak{p}$ extends to all of $A_\mathfrak{p}$. Thus, under $A \mapsto A_\mathfrak{p}$, the factorizations of $c(fg)$, $c(f)$, and $c(g)$ map to $(S^{-1}\mathfrak{p})^{v(fg)}$, $(S^{-1}\mathfrak{p})^{v(f)}$, and $(S^{-1}\mathfrak{p})^{v(g)}$, respectively. These exponents are the same as the exponents on $\mathfrak{p}$ in the original factorizations. Thus, we must show that $v(fg) = v(f) + v(g)$. By Theorem 9.3, $A_\mathfrak{p}$ is a discrete valuation ring since A is a Dedekind ring. Thus, we've reduced the problem to the following: Let

$f = \sum_i a_i x^i$, $g = \sum_j b_j x^j \in A[x]$ be nonzero, where A is a discrete valuation ring. Then, we must show that $c(fg) = c(f)c(g)$.

In this new setting, since A is a discrete valuation ring, there is some $\pi \in A$ such that every nonzero ideal has the form $(\pi)^k$, $k \geq 0$. Thus, $c(f) = (\pi)^{v(f)}$ and $c(g) = (\pi)^{v(g)}$. Since $c(f) = (a_i)_i$ and $c(g) = (b_j)_j$, for all a_i and b_j, $a_i \in (\pi)^{v(f)}$ and $b_j \in (\pi)^{v(g)}$. Thus, $a_i' = a_i/\pi^{v(f)} \in A$ and $b_j' = b_j/\pi^{v(g)} \in A$. Let $f_0 = \sum a_i' x^i$ and $g_0 = \sum b_j' x^j$. Then, $f_0 = f/\pi^{v(f)}$ and $g_0 = g/\pi^{v(g)}$ so that $c(f_0) = (a_i')_i = (a_i/\pi^{v(f)})_i = (a_i)_i/(\pi)^{v(f)} = (\pi)^{v(f)}/(\pi)^{v(f)} = (1)$; similarly, $c(g_0) = (1)$. By Exercise 1.2, $c(f_0 g_0) = (1)$. Finally, $fg = \pi^{v(f)} f_0 \pi^{v(g)} g_0 = \pi^{v(f)+v(g)} f_0 g_0$. Then,

$$c(fg) = (\pi)^{v(f)+v(g)} c(f_0 g_0) = (\pi)^{v(f)+v(g)} = (\pi)^{v(f)}(\pi)^{v(g)} = c(f)c(g),$$

as desired. □

9.3 Exercise

Proof. If A is a discrete valuation ring, then A is Noetherian by definition. Conversely, suppose A is a Noetherian valuation ring that is not a field. Then, A is an integral domain, by definition, and by Proposition 5.18, A is a local ring that is integrally closed. Supposing for a moment that A has dimension one, then A is a local Noetherian domain of dimension one that is integrally closed, so by Proposition 9.2, A is a discrete valuation ring. Therefore, we just have to show that all nonzero prime ideals of A are maximal.

Since A is a local ring, it has a unique maximal ideal $\mathfrak{m}$. Let $\mathfrak{p}$ be an arbitrary prime ideal of A so $\mathfrak{p} \subset \mathfrak{m}$. Using the construction of Exercise 5.30, we have a valuation $v : K^\times \to K^\times/U$, where K is the field of fractions of A and U is the group of units of A.

Now, we will construct an ascending chain of ideals. Let $y \in \mathfrak{p}$ and $\mathfrak{a}_1 = \{a \in A \mid v(a) \geq v(y)\}$. If $a \in A$ is such that $v(a) \geq v(y)$, then $ay^{-1} \in A$, so $a = ay^{-1} \cdot y \in \mathfrak{p}$. Thus, $\mathfrak{a}_1 \subset \mathfrak{p}$. Suppose there is some $y' \in \mathfrak{p} - \mathfrak{a}_1$. We define $\mathfrak{a}_2 = \{a \in A \mid v(a) \geq v(y')\}$. We have a proper inclusion $\mathfrak{a}_1 \subset \mathfrak{a}_2$ since $y' \in \mathfrak{a}_2 - \mathfrak{a}_1$, implying $v(y') < v(y)$. By the same argument as before, $\mathfrak{a}_2 \subset \mathfrak{p}$. Continue in this fashion; since A is Noetherian, there is some n such that the strictly ascending chain of ideals terminates. Then, $\mathfrak{p} - \mathfrak{a}_n$ is empty so that $\mathfrak{p} = \mathfrak{a}_n$. Thus, there is some $y \in A$ such that $\mathfrak{p} = \{a \in A \mid v(a) \geq v(y)\}$.

Suppose to the contrary that $\mathfrak{p}$ is not $\mathfrak{m}$, so there is some $x \in \mathfrak{m} - \mathfrak{p}$, which is clearly nonzero. Since $x \in \mathfrak{m}$, $v(x) > 0$. Because $x \notin \mathfrak{p}$ and $y \in \mathfrak{p}$, $x \notin (y)$ so $xy^{-1} \notin A$. Since A is a valuation ring, we must then have $yx^{-1} \in A$. Further, $v(yx^{-1}) = v(y) - v(x) < v(y)$ since $v(x) > 0$ so $yx^{-1} \notin \mathfrak{p}$. Since $x, yx^{-1} \in A - \mathfrak{p}$ and $\mathfrak{p}$ is prime, we have $y = x \cdot yx^{-1} \notin \mathfrak{p}$. This is a clear contradiction. Thus, the only nonzero prime ideal of A is the unique maximal ideal, so all nonzero prime ideals of A are maximal, as desired. □

9.4 Exercise

Proof. Let A be a local domain (not a field) in which the maximal ideal $\mathfrak{m}$ is principal and $\bigcap_{i=1}^{\infty} \mathfrak{m}^n = 0$. Suppose for a moment that $\mathfrak{m}$ is the only nonzero prime ideal. Then, every prime ideal is finitely generated since $\mathfrak{m}$ is principal, so by Exercise 7.1, A is Noetherian. Further, it has dimension one since $0 \subset \mathfrak{m}$ is the only strict chain of primes. Then, since A is a Noetherian local domain of dimension one with a maximal ideal that is principal, we have by Proposition 9.2 that A is a discrete valuation ring.

Therefore, we just have to show that any nonzero prime ideal $\mathfrak{p}$ of A is equal to $\mathfrak{m}$. Since $\mathfrak{m}$ is principal, $\mathfrak{m} = (\pi)$ for some $\pi \in A$. Now, let $y \in \mathfrak{p}$. If $y = 0$, then $y \in \mathfrak{m}$, so suppose $y \neq 0$. By

assumption, $\bigcap_{i=1}^{\infty} \mathfrak{m}^n = 0$, so $y \notin \bigcap_{i=1}^{\infty} \mathfrak{m}^n$. Thus, there is some minimal $n(y) > 1$ (since $y \in \mathfrak{m}$) such that $y \notin \mathfrak{m}^{n(y)}$. Then, $\{n(y) \mid y \in \mathfrak{p} - 0\}$ is a non-empty subset of integers greater than 1, so it has a minimum $N > 1$. Thus, there is some $b \in \mathfrak{p}$ such that $b \notin \mathfrak{m}^N$. Since $b \in \mathfrak{p} \subset \mathfrak{m} = (\pi)$, we can write $b = a\pi$, where $a \in A$.

If $a \in \mathfrak{m}^{N-1} = (\pi^{N-1})$, then $b = a\pi \in (\pi^N)$, contradicting our choice of b, so $a \notin \mathfrak{m}^{N-1}$. Since $b \in \mathfrak{p}$, either $\pi \in \mathfrak{p}$ or $a \in \mathfrak{p}$. But $a \notin \mathfrak{p}$; otherwise, $a \in \mathfrak{p}$ and $a \notin \mathfrak{m}^{N-1}$ together violate the minimality of N. Thus, $\pi \in \mathfrak{p}$ so that $\mathfrak{m} = (\pi) \subset \mathfrak{p}$, implying that $\mathfrak{p} = \mathfrak{m}$, as desired. □

9.5 Exercise

Proof. Let M be a finitely generated module over a Dedekind domain. Since M is finitely generated and A is Noetherian, we have by Exercise 7.16 that M is flat if and only if $M_{\mathfrak{m}}$ is free for all maximal ideals $\mathfrak{m}$ of A. By Exercise 3.13, M is torsion-free if and only if $M_{\mathfrak{m}}$ is torsion-free for all maximal ideals $\mathfrak{m}$ of A. Thus, to show that M is flat if and only if M is torsion-free, it suffices to show that for all maximal ideals $\mathfrak{m}$ of A, $M_{\mathfrak{m}}$ is torsion-free if and only if $M_{\mathfrak{m}}$ is free.

Note that if $M_{\mathfrak{m}} = 0$, the result is trivial, so suppose $M_{\mathfrak{m}} \neq 0$. Since $M_{\mathfrak{m}}$ is finitely generated, if $M_{\mathfrak{m}}$ is free, $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}^n$ as an $A_{\mathfrak{m}}$-module for some $n > 0$. Let $(a_1, \ldots, a_n) \in A_{\mathfrak{m}}^n$, and let $x \in A_{\mathfrak{m}}$ be non-zero. If $x(a_1, \ldots, a_n) = 0$, then $(xa_1, \ldots, xa_n) = (0, \ldots, 0)$. Since $A_{\mathfrak{m}}$ is an integral domain and $x \neq 0$, $a_i = 0$ for all i, so $(a_1, \ldots, a_n) = (0, \ldots, 0)$. Thus, $T(M_{\mathfrak{m}}) = 0$.

Conversely, suppose $T(M_{\mathfrak{m}}) = 0$. Since A is a Dedekind domain, we have by Theorem 9.3 that $A_{\mathfrak{m}}$ is a discrete valuation ring. Thus, by Proposition 9.2, $A_{\mathfrak{m}}$ is a principal ideal domain. Hence, by the structure theorem for finitely generated modules over a principal ideal domain,

$$M_{\mathfrak{m}} \cong A_{\mathfrak{m}}^n \oplus A_{\mathfrak{m}}/\mathfrak{a}_1 \oplus \cdots \oplus A_{\mathfrak{m}}/\mathfrak{a}_r$$

for nonzero ideals $\mathfrak{a}_i$ of $A_{\mathfrak{m}}$, where $r \geq 0$. If $r \geq 1$, $M_{\mathfrak{m}}$ has an $A_{\mathfrak{m}}$-submodule M' isomorphic to $A_{\mathfrak{m}}/\mathfrak{a}_1$. Let $m \in M'$ be the preimage of 1 in the isomorphism $M' \cong A_{\mathfrak{m}}/\mathfrak{a}_1$, and let $a \in \mathfrak{a}_1$ be a nonzero element. Then, $am = 0$, violating the fact that $M_{\mathfrak{m}}$ is torsion-free. Thus, $r = 0$ so that $M_{\mathfrak{m}} \cong A_{\mathfrak{m}}^n$, as desired. □

9.6 Exercise

Proof. Let $\mathfrak{p}$ be a nonzero prime ideal of A. Since A is a Dedekind domain, $A_{\mathfrak{p}}$ is a discrete valuation ring by Theorem 9.3. Thus, $A_{\mathfrak{p}}$ is a principal ideal domain by Proposition 9.2. Suppose $M_{\mathfrak{p}} \neq 0$. By the structure theorem for principal ideal domains,

$$M_{\mathfrak{p}} \cong A_{\mathfrak{p}}^n \oplus A_{\mathfrak{p}}/\mathfrak{a}_1 \oplus \cdots \oplus A_{\mathfrak{p}}/\mathfrak{a}_r,$$

where $\mathfrak{a}_i$ are nontrivial ideals of $A_{\mathfrak{p}}$. By Exercise 3.13, $T(M_{\mathfrak{p}}) = T(M)_{\mathfrak{p}} = M_{\mathfrak{p}}$. If $n \geq 1$, there is a submodule $M' \cong A_{\mathfrak{p}}$ of $M_{\mathfrak{p}}$. Since $A_{\mathfrak{p}}$ is an integral domain, M' contains non-torsion elements, contradicting $T(M_{\mathfrak{p}}) = M_{\mathfrak{p}}$. Thus, $n = 0$ so that

$$M_{\mathfrak{p}} \cong A_{\mathfrak{p}}/\mathfrak{a}_1 \oplus \cdots \oplus A_{\mathfrak{p}}/\mathfrak{a}_r.$$

Since $A_{\mathfrak{p}}$ is a discrete valuation ring with maximal ideal $\mathfrak{p}$, we have $\mathfrak{a}_j = \mathfrak{p}^{n_j}$ for some $n_j > 0$ (Proposition 9.2). Thus,

$$M_{\mathfrak{p}} \cong A_{\mathfrak{p}}/\mathfrak{p}^{n_1} \oplus \cdots \oplus A_{\mathfrak{p}}/\mathfrak{p}^{n_r}.$$

Note that this decomposition is unique (modulo reordering, of course) by the structure theorem. Let $n(\mathfrak{p}) = \max\{n_1, \dots, n_r\}$. Then clearly, $\text{Ann}(M_\mathfrak{p}) = \mathfrak{p}^{n(\mathfrak{p})}$. In particular, note that $\text{Ann}(M_\mathfrak{p}) \neq 0$ whenever $M_\mathfrak{p} \neq 0$.

Since M is finitely generated, by Proposition 3.14, $\text{Ann}(M)_\mathfrak{p} = \text{Ann}(M_\mathfrak{p}) = \mathfrak{p}^{n(\mathfrak{p})}$. If $\text{Ann}(M) = 0$, then $\text{Ann}(M_\mathfrak{p}) = 0$ so that $M_\mathfrak{p} = 0$ for all prime ideals $\mathfrak{p}$. Thus, by Proposition 3.8, $M = 0$. So, if $M \neq 0$, $\text{Ann}(M)$ is a nonzero ideal. Since $\text{Ann}(M)$ is a nonzero ideal of a Dedekind domain A, $\text{Ann}(M)$ has a unique decomposition as a product of prime powers (Corollary 9.4). If $\mathfrak{p}$ is a nonzero prime that is not in the decomposition, then since A is Dedekind, $\mathfrak{p}$ does not contain any of the primes in the decomposition since all nonzero primes all maximal. Thus, $0 = \text{Ann}(M)_\mathfrak{p} = \text{Ann}(M_\mathfrak{p})$, so $M_\mathfrak{p} = 0$. Thus, $\text{Supp}(M)$ is finite.

Let $\{\mathfrak{p}_1, \dots, \mathfrak{p}_k\} = \text{Supp}(M)$. For each $\mathfrak{p}_i$, we have, as above,

$$M_{\mathfrak{p}_i} \cong A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_{i,1}} \oplus \cdots \oplus A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_{i,r_i}}.$$

Note that we have an A-module homomorphism $\phi : M \to \bigoplus_{i=1}^k \bigoplus_{j=1}^{r_i} A/\mathfrak{p}_i^{n_{i,j}}$. If $\mathfrak{p} \notin \text{Supp}(M)$, then we have $\phi_\mathfrak{p} : M_\mathfrak{p} \to \bigoplus_{i=1}^k \bigoplus_{j=1}^{r_i} A_\mathfrak{p}/\mathfrak{p}_i^{n_{i,j}}$. For all $\mathfrak{p}_i \in \text{Supp}(M)$, $\mathfrak{p}$ does not contain $\mathfrak{p}_i$, so $A_\mathfrak{p}\mathfrak{p}_i^{n_{i,r_i}} = A_\mathfrak{p}$ so that all quotients on the right-hand side vanish. Further, $M_\mathfrak{p} = 0$, so $\phi_\mathfrak{p} : 0 \to 0$, which is clearly an isomorphism. On the other hand, let $\mathfrak{p}_s \in \text{Supp}(M)$. Then, $\phi_{\mathfrak{p}_s} : M_{\mathfrak{p}_s} \to \bigoplus_{i=1}^k \bigoplus_{j=1}^{r_i} A_{\mathfrak{p}_s}/\mathfrak{p}_i^{n_{i,j}}$. If $i \neq s$, then $\mathfrak{p}_s$ does not contain $\mathfrak{p}_i$, so $A_{\mathfrak{p}_s}\mathfrak{p}_i^{n_{i,j}} = A_{\mathfrak{p}_s}$. In particular, $A_{\mathfrak{p}_s}/\mathfrak{p}_i^{n_{i,j}} = 0$ for $i \neq s$. Thus, we have $\phi_{\mathfrak{p}_s} : M_{\mathfrak{p}_s} \to \bigoplus_{j=1}^{r_s} A_{\mathfrak{p}_s}/\mathfrak{p}_s^{n_{s,j}}$, which is an isomorphism by construction. By Proposition 3.9, ϕ is an isomorphism, so

$$M \cong \bigoplus_{i=1}^{k} \bigoplus_{j=1}^{r_i} A/\mathfrak{p}_i^{n_{i,j}}.$$

To see that this decomposition is unique, note that two decompositions of M must disagree for some prime $\mathfrak{p}$, i.e. the two compositions do not have either the same number of factors or the powers of $\mathfrak{p}$ do not match. But then, they would disagree under localization to $M_\mathfrak{p}$, which violates the uniqueness guaranteed by the structure theorem. □

9.7 Exercise

Proof. Note that $A/\mathfrak{a}$ is a finitely generated A-module, as it is generated by the element $1 + \mathfrak{a}$. Further, let $a + \mathfrak{a} \in A/\mathfrak{a}$. Then, since $\mathfrak{a} \neq 0$, there is some nonzero $x \in \mathfrak{a}$. Then, $x(a + \mathfrak{a}) = xa + \mathfrak{a} = 0 + \mathfrak{a}$, so $A/\mathfrak{a}$ is a finitely generated torsion module. Since A is a Dedekind domain, we have by Exercise 9.6 that $A/\mathfrak{a} \cong \prod_{i=1}^m A/\mathfrak{p}_i^{n_i}$ as A-modules, where $\mathfrak{p}_i \in \text{Spec}(A)$ and $n_i > 0$.

Since A is a Dedekind domain, $A_{\mathfrak{p}_i}$ is a discrete valuation ring for each $\mathfrak{p}_i$ by Theorem 9.3. Thus, by Proposition 9.2, $A_{\mathfrak{p}_i}$ is a principal ideal domain, so all ideals of $A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$ are principal. Thus, all ideals of $A/\mathfrak{p}_i^{n_i} \cong A_{\mathfrak{p}_i}/\mathfrak{p}_i^{n_i}A_{\mathfrak{p}_i}$ are principal for each $\mathfrak{p}_i$.

Now, recall $A/\mathfrak{a} \cong \prod_{i=1}^m A/\mathfrak{p}_i^{n_i}$. Let I be an arbitrary ideal of $A/\mathfrak{a}$. Then, under the surjective homomorphisms $\pi_j : \prod_{i=1}^m A/\mathfrak{p}_i^{n_i} \to A/\mathfrak{p}_j^{n_j}$, $\pi_j(I)$ is an ideal for each j. Thus, I has the form $\prod_{i=1}^m I_i$, where I_i is an ideal of $A/\mathfrak{p}_i^{n_i}$ for each i. Since every ideal of $A/\mathfrak{p}_i^{n_i}$ is principal, $I_i = (d_i)$ for some $d_i \in A/\mathfrak{p}_i^{n_i}$. Then, I is generated by the element $(d_1, \dots, d_n)$, so I is a principal ideal. Thus, when $\mathfrak{a} \neq 0$, every ideal of $A/\mathfrak{a}$ is principal.

Now, let $\mathfrak{a}$ be any nonzero ideal of a Dedekind domain A, and let $x \in \mathfrak{a}$ be nonzero. Then, $(x) \subset \mathfrak{a}$. By the result, under the canonical map $A \mapsto A/(x)$, $\overline{\mathfrak{a}} = (y')$ for some $y' \in A/(x)$ since every ideal of $A/(x)$ is principal. Thus, there is some $y \in A$ such that $\overline{y} = y'$. Then, (x, y) is an

ideal containing (x), so we have $\overline{(x,y)} = \overline{(y)} = (\overline{y}) = (y') = \overline{\mathfrak{a}}$. By Proposition 1.1, the ideals of $A/(x)$ are in a one-to-one correspondence with ideals of A containing (x), so $\mathfrak{a} = (x,y)$. Thus, every ideal of a Dedekind domain is generated by at most two elements. □

Remark. *In fact, the result is a bit stronger. In the proof, if $\mathfrak{a}$ is a nonzero ideal of a Dedekind domain A, then for any choice of nonzero $x \in \mathfrak{a}$, there is some $y \in \mathfrak{a}$ such that $\mathfrak{a} = (x,y)$.*

9.8 Exercise

Proof. Let $\mathfrak{a}$, $\mathfrak{b}$, and $\mathfrak{c}$ be three ideals in a Dedekind domain A. Suppose $f(\mathfrak{a},\mathfrak{b},\mathfrak{c})$ is a finite composition of sums and intersections of the three ideals. Then, by Proposition 3.11, for every prime ideal $\mathfrak{p}$ of A, $f(\mathfrak{a},\mathfrak{b},\mathfrak{c})_\mathfrak{p} = f(\mathfrak{a}_\mathfrak{p},\mathfrak{b}_\mathfrak{p},\mathfrak{c}_\mathfrak{p})$. If $g(\mathfrak{a},\mathfrak{b},\mathfrak{c})$ is another ideal such that $g(\mathfrak{a},\mathfrak{b},\mathfrak{c})_\mathfrak{p} = g(\mathfrak{a}_\mathfrak{p},\mathfrak{b}_\mathfrak{p},\mathfrak{c}_\mathfrak{p})$ and $f(\mathfrak{a}_\mathfrak{p},\mathfrak{b}_\mathfrak{p},\mathfrak{c}_\mathfrak{p}) = g(\mathfrak{a}_\mathfrak{p},\mathfrak{b}_\mathfrak{p},\mathfrak{c}_\mathfrak{p})$ for all prime ideals $\mathfrak{p}$, then $f(\mathfrak{a},\mathfrak{b},\mathfrak{c}) = g(\mathfrak{a},\mathfrak{b},\mathfrak{c})$. To see this, suppose that I and J are two ideals that are identical under localization at every prime. Then, the A-module $(I+J)/(I\cap J)$ is 0 under localization for all primes, so by Proposition 3.8, $(I+J)/(I\cap J) = 0$ so that $J \subset I+J = I\cap J \subset I$ and similarly $I \subset J$, yielding $I = J$.

Now, we want to prove the equalities

$$\mathfrak{a}\cap(\mathfrak{b}+\mathfrak{c}) = (\mathfrak{a}\cap\mathfrak{b})+(\mathfrak{a}\cap\mathfrak{c}) \quad \text{and} \quad \mathfrak{a}+(\mathfrak{b}\cap\mathfrak{c}) = (\mathfrak{a}+\mathfrak{b})\cap(\mathfrak{a}+\mathfrak{c}).$$

Observe that if any of the ideals are 0 or (1), the equalities are easily verified, so we may assume each ideal is nonzero. By the previous paragraph, it is sufficient to show that the equalities are true under localization at every nonzero prime ideal $\mathfrak{p}$. Since A is a Dedekind domain, $A_\mathfrak{p}$ is a discrete valuation ring (Theorem 9.3), so by Proposition 9.2, every nontrivial ideal of $A_\mathfrak{p}$ has the form $(\pi)^j$ for a fixed $\pi \in A_\mathfrak{p}$ and some $j > 0$. If under localization, any of $\mathfrak{a}_\mathfrak{p}$, $\mathfrak{b}_\mathfrak{p}$, and $\mathfrak{c}_\mathfrak{p}$ are 0 or $A_\mathfrak{p}$, then again, the equalities easily hold. Thus, we may assume they are all nontrivial so that $\mathfrak{a}_\mathfrak{p} = (\pi)^i$, $\mathfrak{b}_\mathfrak{p} = (\pi)^j$, and $\mathfrak{c}_\mathfrak{p} = (\pi)^k$ for some $i,j,k > 0$.

Now, we just have to verify that the equalities hold in this case. We have

$$\mathfrak{a}_\mathfrak{p}\cap(\mathfrak{b}_\mathfrak{p}+\mathfrak{c}_\mathfrak{p}) = (\pi)^i\cap((\pi)^j+(\pi)^k) = (\pi)^i\cap(\pi)^{\min\{j,k\}} = (\pi)^{\max\{i,\min\{j,k\}\}}$$

and

$$(\mathfrak{a}_\mathfrak{p}\cap\mathfrak{b}_\mathfrak{p})+(\mathfrak{a}_\mathfrak{p}\cap\mathfrak{c}_\mathfrak{p}) = ((\pi)^i\cap(\pi)^j)+((\pi)^i\cap(\pi)^k) = (\pi)^{\max\{i,j\}}+(\pi)^{\max\{i,k\}} = (\pi)^{\min\{\max\{i,j\},\max\{i,k\}\}}.$$

Since $\min\{\max\{i,j\},\max\{i,k\}\} = \max\{i,\min\{j,k\}\}$, the first equality holds. For the second,

$$\mathfrak{a}_\mathfrak{p}+(\mathfrak{b}_\mathfrak{p}\cap\mathfrak{c}_\mathfrak{p}) = (\pi)^i+((\pi)^j\cap(\pi)^k) = (\pi)^i+(\pi)^{\max\{j,k\}} = (\pi)^{\min\{i,\max\{j,k\}\}}$$

and

$$(\mathfrak{a}_\mathfrak{p}+\mathfrak{b}_\mathfrak{p})\cap(\mathfrak{a}_\mathfrak{p}+\mathfrak{c}_\mathfrak{p}) = ((\pi)^i+(\pi)^j)\cap((\pi)^i+(\pi)^k) = (\pi)^{\min\{i,j\}}\cap(\pi)^{\min\{i,k\}} = (\pi)^{\max\{\min\{i,j\},\min\{i,k\}\}}.$$

Since $\max\{\min\{i,j\},\min\{i,k\}\} = \min\{i,\max\{j,k\}\}$, the second equality also holds. □

9.9 Exercise

Proof. Let $\mathfrak{a}_1,\ldots,\mathfrak{a}_n$ be ideals of a Dedekind domain A. We'll prove first that the sequence of A-modules

$$A \xrightarrow{\phi} \prod_{i=1}^{n} A/\mathfrak{a}_i \xrightarrow{\psi} \prod_{i<j} A/(\mathfrak{a}_i+\mathfrak{a}_j)$$

is exact, where $\phi_i(x) = x + \mathfrak{a}_i$ and $\psi_{i,j}(x_1 + \mathfrak{a}_1, \ldots, x_n + \mathfrak{a}_n) = x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j$. This requires us to show that $\ker \psi = \operatorname{im} \phi$. It will suffice to show that for all nonzero prime ideals $\mathfrak{p}$ of A (the maximal ideals since A is a Dedekind domain), $\ker \psi_{\mathfrak{p}} = (\ker \psi)_{\mathfrak{p}} = (\operatorname{im} \phi)_{\mathfrak{p}} = \operatorname{im} \phi_{\mathfrak{p}}$. Since A is a Dedekind domain, $A_{\mathfrak{p}}$ is a discrete valuation ring (Theorem 9.3).

Therefore, it suffices to show that

$$A \xrightarrow{\phi} \prod_{i=1}^{n} A/\mathfrak{a}_i \xrightarrow{\psi} \prod_{i<j} A/(\mathfrak{a}_i + \mathfrak{a}_j)$$

is exact when A is a discrete valuation ring. By Proposition 9.2, nontrivial ideals have the form $(\pi)^j$ for some fixed $\pi \in A$ and $j > 0$. Now, choose some arbitrary $(x + \mathfrak{a}_k)_k \in \operatorname{im} \phi$. Then, $\psi((x + \mathfrak{a}_k)_k) = (x - x + \mathfrak{a}_i + \mathfrak{a}_j)_{i,j} = (0 + \mathfrak{a}_i + \mathfrak{a}_j)_{i,j}$, so $(x + \mathfrak{a}_k)_k \in \ker \psi$. Thus, $\operatorname{im} \phi \subset \ker \psi$.

Conversely, suppose we have $(x_k + \mathfrak{a}_k)_k \in \ker \psi$ so that $0 = \psi_{i,j}((x_k + \mathfrak{a}_k)_k) = x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j$ for all $i \neq j$, i.e. $x_i - x_j \in \mathfrak{a}_i + \mathfrak{a}_j$. As we showed above, $\mathfrak{a}_i = (\pi)^{n_i}$ for some $n_i \geq 0$; allow $n_i = \infty$ for $\mathfrak{a}_i = 0$ and $n_i = 0$ for $\mathfrak{a}_i = A$. Then, for all $i \neq j$, $x_i - x_j \in (\pi)^{n_i} + (\pi)^{n_j} = (\pi)^{\min\{n_i, n_j\}}$. Now, let $m = \max_i\{n_i\}$, so there is some s such that $\mathfrak{a}_s = (\pi)^m$. For any i, we then have $x_i - x_s \in (\pi)^{\min\{n_i, n_s\}} = (\pi)^{\min\{n_i, m\}} = (\pi)^{n_i}$. Thus, $(x_k + \mathfrak{a}_k)_k = ((x_s + \mathfrak{a}_k)_k) \in \operatorname{im} \phi$. Hence, $\ker \psi = \operatorname{im} \phi$ for any discrete valuation ring A, so as argued above, for any Dedekind domain A.

Now, suppose $x_1, \ldots, x_n \in A$. If there is some $x \in A$ such that $x \equiv x_i \bmod \mathfrak{a}_i$ for all i, then $(x_i + \mathfrak{a}_i)_i = (x + \mathfrak{a}_i)_i \in \operatorname{im} \phi = \ker \psi$. Hence, for all $i \neq j$, $0 = \psi_{i,j}((x_i + \mathfrak{a}_i)_i) = x_i - x_j + \mathfrak{a}_i + \mathfrak{a}_j$ so that $x_i \equiv x_j \bmod \mathfrak{a}_i + \mathfrak{a}_j$. Conversely, suppose $x_i \equiv x_j \bmod \mathfrak{a}_i + \mathfrak{a}_j$ for all $i \neq j$. Then, $(x_i + \mathfrak{a}_i)_i \in \ker \psi = \operatorname{im} \phi$, so there is some $x \in A$ such that $(x + \mathfrak{a}_i)_i = (x_i + \mathfrak{a}_i)_i$. Then, $x \equiv x_i \bmod \mathfrak{a}_i$ for all i, as desired. □

Remark. *This is the "Generalized Chinese Remainder Theorem."*

10 Completions (Done)

10.1 Exercise

Proof. Let A be the $\mathbb{Z}$-module $\bigoplus_{i=1}^{\infty}(\mathbb{Z}/p\mathbb{Z})$. Then, $pA = 0$ and thus $p^nA = 0$ for all $n \geq 1$. Then, we have $\varprojlim A/p^nA = \varprojlim A = A$.

Let B be the $\mathbb{Z}$-module $\bigoplus_{i=1}^{\infty}(\mathbb{Z}/p^i\mathbb{Z})$. If $\alpha_i : \mathbb{Z}/p\mathbb{Z} \to \mathbb{Z}/p^i\mathbb{Z}$ is the group homomorphism sending $1 \mapsto p^{i-1}$, then let $\alpha : A \to B$ be the direct sum of the α_i. This is a $\mathbb{Z}$-module homomorphism simply because it is a homomorphism of abelian groups. Note that B also has a natural p-filtration. For $n > 0$, $p^nB = \bigoplus_{i=1}^{\infty} p^n\mathbb{Z}/p^i\mathbb{Z}$. Thus, $\alpha_i^{-1}(p^nB) = \mathbb{Z}/p\mathbb{Z}$ if $i > n$, but $\alpha_i^{-1}(p^nB) = 0$ if $i \leq n$. Thus, the p-filtration of A induced by the p-filtration of B is $A_n = \bigoplus_{i=1}^{n} 0 \oplus \bigoplus_{i=n+1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. Hence, $A/A_n = \bigoplus_{i=1}^{n} \mathbb{Z}/p\mathbb{Z} = \prod_{i=1}^{n} \mathbb{Z}/p\mathbb{Z}$.

Note that the canonical homomorphism $\theta_{n+1} : A/A_{n+1} \to A/A_n$ is the one that projects the first n factors isomorphically onto $A/A_n = \prod_{i=1}^{n} \mathbb{Z}/p\mathbb{Z}$, while sending the final $\mathbb{Z}/p\mathbb{Z}$ factor to 0. As in the proof of Proposition 10.2, we can let $M = \prod_{r=1}^{\infty} A/M_r$ and $d : M \to M$ be $(d(a))_n = a_n - \theta_{n+1}(a_{n+1})$. Then, $\varprojlim A/A_n \cong \ker d$. Thus, $a = (a_n)_n \in \ker d$ if and only if $\theta_{n+1}(a_{n+1}) = a_n$ for all n. By the characterization of θ_{n+1}, this means that the n components of a_n match the first n components of a_{n+1}. Since this is true for all n, it's clear that $\ker d \cong \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. Hence, $\varprojlim A/A_n \cong \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$.

Letting $C = B/\alpha(A)$, we have an exact sequence

$$A \xrightarrow{\alpha} B \to C \to 0.$$

For the p-adic completion functor to be right exact, we'd need $\operatorname{im}(\varprojlim A/p^n \to \hat{B}) = \ker(\hat{B} \to \hat{C})$. Since we showed $A = \varprojlim A/p^n$, we would need $\operatorname{im}(A \to \hat{B}) = \ker(\hat{B} \to \hat{C})$.

As defined above, $A_n = \alpha^{-1}(p^nB)$, so we have an exact sequence of p-filtrations

$$0 \to A/A_n \to B/p^nB \to C/p^nC \to 0.$$

Since $A/A_{n+1} \to A/A_n$ is clearly surjective for all n, we have by Proposition 10.2 that

$$0 \to \varprojlim A/A_n \to \hat{B} \to \hat{C} \to 0$$

is exact. Then, $\operatorname{im}(\varprojlim A/A_n \to \hat{B}) = \ker(\hat{B} \to \hat{C})$. From above, $\varprojlim A/A_n \cong \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. Since $A = \bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ is a strict subgroup of $\prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ and $\varprojlim A/A_n \to \hat{B}$ is injective, $\operatorname{im}(A \to \hat{B})$ is a strict subgroup of $\operatorname{im}(\varprojlim A/A_n \to \hat{B}) = \ker(\hat{B} \to \hat{C})$, so the sequence

$$\varprojlim A/p^nA \xrightarrow{\hat{\alpha}} \varprojlim B/p^nB \to \varprojlim C/p^nC \to 0$$

is not exact. Hence, p-adic completion is not a right exact functor on the category of $\mathbb{Z}$-modules. □

10.2 Exercise

Proof. As in Exercise 10.1, define $A_n := \alpha^{-1}(p^nB)$. We have the short exact sequence

$$0 \to A_n \to A \to A/A_n \to 0.$$

From the proof of Exercise 10.1, we have $A_n = \bigoplus_{i=n+1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ and $A/A_n = \prod_{i=1}^{n} \mathbb{Z}/p\mathbb{Z}$. We have the natural embeddings $A_{n+1} \to A_n$ and natural homomorphisms $A/A_{n+1} \to A/A_n$ that map the $n+1$ component to 0. Thus, we have commutative diagrams with exact rows:

$$\begin{array}{ccccccccc} 0 & \longrightarrow & A_{n+1} & \longrightarrow & A & \longrightarrow & A/A_{n+1} & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & A_n & \longrightarrow & A & \longrightarrow & A/A_n & \longrightarrow & 0. \end{array}$$

So,

$$0 \to \{A_n\} \to \{A\} \to \{A/A_n\} \to 0$$

is an exact sequence of inverse systems. By Proposition 10.2, we have an exact sequence

$$0 \to \varprojlim A_n \to \varprojlim A \to \varprojlim A/A_n.$$

Then, $\varprojlim A = A = \bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ and, by Exercise 10.1, $\varprojlim A/A_n \cong \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. Now, clearly $\bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z} \to \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ is not surjective, so $\varprojlim$ is not right exact.

Let $M = \prod_{k=0}^{\infty} A_k$ ($A_0 = A$) and define $d : M \to M$ via $(d(a))_n = a_n - \theta_{n+1}(a_{n+1})$. Then, we have $\varprojlim^1 A_n = \operatorname{coker} d$, by definition. Let $x \in \operatorname{im}(d)$. Then, $x_n = (d(a))_n = a_n - \theta_{n+1}(a_{n+1})$. Since θ_{n+1} is just the identification of A_{n+1} as a subgroup of A_n, we can write $\theta_{n+1}(a_{n+1})$ as just a_{n+1} so that $x_n = a_n - a_{n+1}$. Thus,

$$\sum_{i=0}^{n} x_i = \sum_{i=0}^{n} (a_i - a_{i+1}) = a_0 - a_{n+1}$$

for all n, so $a_{n+1} = a_0 - \sum_{i=0}^{n} x_i \in A_{n+1}$ for all n. Hence, for any $j \leq n+1$,

$$0 = (a_{n+1})_j = (a_0)_j - \left(\sum_{i=0}^{n} x_i \right)_j.$$

Since $x_i \in A_i$, $(x_i)_j = 0$ whenever $i \geq j$, so we have $(a_0)_j = \left(\sum_{i=0}^{j} x_i \right)_j$ for all j. Now, since $a_0 \in A_0 = \bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$, $(a_0)_j = 0$ for sufficiently large j. Thus, for sufficiently large j,

$$\left(\sum_{i=0}^{j} x_i \right)_j = (a_0)_j = 0$$

so that $\sum_{i=0}^{j} (x_i)_j = 0$. Since $x_n \in A_n$ for all n, the sum $X = \sum_{i=0}^{\infty} x_i$ is well-defined since it becomes stationary in every coordinate. Then, $X_j = (\sum_{i=0}^{\infty} x_i)_j = \sum_{i=0}^{j} (x_i)_j$. Thus, for sufficiently large j, $X_j = 0$. Hence, $X \in A$.

Conversely, suppose for $(x_i)_i \in M$, $X = \sum_{i=0}^{\infty} x_i$ (well-defined since each coordinate becomes stationary) lies in A. Then, let $a_0 = X \in A = A_0$ and define $a_{n+1} = a_0 - \sum_{i=0}^{n} x_i$ for $n \geq 0$. Then, for $j \leq n+1$,

$$(a_{n+1})_j = (a_0)_j - \left(\sum_{i=0}^{n} x_i \right)_j = \left(\sum_{i=0}^{\infty} x_i \right)_j - \left(\sum_{i=0}^{n} x_i \right)_j = \left(\sum_{i=0}^{n} x_i \right)_j - \left(\sum_{i=0}^{n} x_i \right)_j = 0,$$

so $a_{n+1} \in A_{n+1}$. We used the fact that for all $i \geq n+1 \geq j$, $(x_i)_j = 0$, thus reducing the sum to the first $n+1$ terms. Finally, for any n,

$$a_n - a_{n+1} = a_0 - \sum_{i=0}^{n-1} x_i - \left(a_0 - \sum_{i=0}^{n} x_i \right) = x_n,$$

so $(x_n) = d((a_i))$ and thus $(x_n) \in \operatorname{im}(d)$. Hence, $(x_n) \in \operatorname{im}(d)$ if and only if $\sum_{i=0}^{\infty} x_i \in A$.

Now, let $S : M \to \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ be the summation map defined above. Clearly, it is a $\mathbb{Z}$-module homomorphism. Then, we've shown that for $z \in M$, $z \in \operatorname{im}(d)$ if and only if $S(z) \in \bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. Note that S is surjective since for $y \in \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$, we can define $z \in M$ to be such that for all $n \geq 0$, $(z_n)_j = 0$ when $j \neq n+1$ and $(z_n)_{n+1} = y_{n+1}$. Then clearly, $S(z) = y$. Thus, the map

$$\overline{S} : M \to \frac{\prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}}{\bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}}$$

is surjective with kernel equal to $\bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$. We showed above that this is equal to $\operatorname{im}(d)$. Thus, $\varprojlim^1 A_n = M/\operatorname{im}(d) \cong (\prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z})(\bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z})$, completing the proof. □

Remark. *In retrospect, one could have simply noted that $d_A : \prod_{i=0}^{\infty} A \to \prod_{i=0}^{\infty} A$ is surjective so that we have an exact sequence $A \to \varprojlim A/A_n \to \varprojlim^1 A_n \to 0$. Then, from Exercise 10.1, $A = \bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ and $\varprojlim A/A_n \cong \prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z}$ giving us $\varprojlim^1 A_n \cong (\prod_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z})/(\bigoplus_{i=1}^{\infty} \mathbb{Z}/p\mathbb{Z})$.*

10.3 Exercise

Proof. Since A is Noetherian and M is a finitely generated A-module, we have by Krull's Theorem (Theorem 10.17) that $m \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = \ker(M \to \hat{M})$ if and only if m is annihilated by an element of $1 + \mathfrak{a}$.

Suppose $m \in \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$ so that there is some $a \in \mathfrak{a}$ so that $(1+a)m = 0$. Then, for any maximal ideal $\mathfrak{m}$ containing $\mathfrak{a}$, $1 + a \in 1 + \mathfrak{a} \subset 1 + \mathfrak{m}$. In particular, $1 + a \in A - \mathfrak{m}$ so that $1 + a$ is a unit in $A_{\mathfrak{m}}$. Since $(1+a)m = 0$ in M, we have the same relation in the $A_{\mathfrak{m}}$-module $M_{\mathfrak{m}}$. Because $1+a$ is a unit in $A_{\mathfrak{m}}$, we have $0 = (1+a)^{-1}0 = (1+a)^{-1}(1+a)m = m$ so that $m \in \ker(M \to M_{\mathfrak{m}})$. Thus, $m \in \bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$.

Conversely, let $N = \bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$. Then, $N_{\mathfrak{m}} = 0$ for all maximal ideals $\mathfrak{m}$ containing $\mathfrak{a}$. By Exercise 3.14, $N = \mathfrak{a}N$. Since A is Noetherian, and M is a finitely generated A-module, M is Noetherian by Proposition 6.5. Since N is an A-submodule of M, N is then finitely generated by Proposition 6.2. Thus, by Corollary 2.5, since $N = \mathfrak{a}N$ and N is finitely generated, there is some $x \in 1 + \mathfrak{a}$ such that $xN = 0$. Thus, every $m \in N$ is annihilated by $x \in 1 + \mathfrak{a}$, so $N \subset \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$. Hence, we have equality: $\bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}}) = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M$.

If $\hat{M} = 0$, then $M = \ker(M \to \hat{M}) = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = \bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}})$. Thus, $M_{\mathfrak{m}} = 0$ for each $\mathfrak{m}$ containing $\mathfrak{a}$, so $\operatorname{Supp}(M)$ does not include any $\mathfrak{m}$ over $\mathfrak{a}$. In fact, for any prime ideal $\mathfrak{p}$ such that $\mathfrak{m} \supset \mathfrak{p} \supset \mathfrak{a}$, the map $M \to M_{\mathfrak{p}}$ factors as $M \to M_{\mathfrak{m}} \to M_{\mathfrak{p}}$, so $\ker(M \to M_{\mathfrak{p}}) \supset \ker(M \to M_{\mathfrak{m}}) = M$. Thus, for any $\mathfrak{p} \in V(\mathfrak{a})$, $\mathfrak{p} \notin \operatorname{Supp}(M)$, so $\operatorname{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$.

Conversely, if $\operatorname{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$, then $\ker(M \to M_{\mathfrak{m}}) = M$ for all maximal ideals $\mathfrak{m}$ containing $\mathfrak{a}$. Thus, $M = \bigcap_{\mathfrak{m} \supset \mathfrak{a}} \ker(M \to M_{\mathfrak{m}}) = \bigcap_{n=1}^{\infty} \mathfrak{a}^n M = \ker(M \to \hat{M})$. Thus, $\hat{M} = 0$. Hence, $\hat{M} = 0$ if and only if $\operatorname{Supp}(M) \cap V(\mathfrak{a}) = \emptyset$. □

10.4 Exercise

Proof. If $x \in A$ is not a zero divisor, then $A \xrightarrow{\cdot x} A$ is injective, thereby giving a short exact sequence

$$0 \to A \xrightarrow{\cdot x} A \to A/xA \to 0$$

of A-modules. Since A is Noetherian, A and thus A/xA are Noetherian A-modules. Thus, each A-module in the sequence is finitely generated (Proposition 6.2), so by Proposition 10.12, the sequence $0 \to \hat{A} \xrightarrow{\cdot \hat{x}} \hat{A}$ is exact. Hence, $\hat{A} \xrightarrow{\cdot \hat{x}} \hat{A}$ is injective, so $\hat{x}$ is not a zero divisor of $\hat{A}$. □

Even if A is a Noetherian domain, $\hat{A}$ is not necessarily an integral domain.

Proof. Let $\mathfrak{a}$, $\mathfrak{b}$ be coprime ideals of a Noetherian domain A. Then, $\mathfrak{a}^n$ and $\mathfrak{b}^n$ are coprime for all $n \geq 1$. By Proposition 1.10 (Chinese Remainder Theorem), $A/(\mathfrak{ab})^n \cong A/\mathfrak{a}^n \times A/\mathfrak{b}^n$. Hence, we have a commutative diagram

$$\begin{array}{ccc} A/(\mathfrak{ab})^{n+1} & \xrightarrow{\sim} & A/\mathfrak{a}^{n+1} \times A/\mathfrak{b}^{n+1} \\ \downarrow & & \downarrow \\ A/(\mathfrak{ab})^{n} & \xrightarrow{\sim} & A/\mathfrak{a}^{n} \times A/\mathfrak{b}^{n}. \end{array}$$

For an ideal I, let A^I denote the completion in the I-adic topology. Then, the inverse limit $A^{\mathfrak{ab}} = \varprojlim A/(\mathfrak{ab})^n$ is equal to $\varprojlim(A/\mathfrak{a}^n \times A/\mathfrak{b}^n)$ by Proposition 10.2.

We have a map

$$\varprojlim(A/\mathfrak{a}^n \times A/\mathfrak{b}^n) \to \varprojlim A/\mathfrak{a}^n \times \varprojlim A/\mathfrak{b}^n$$

defined by the projections $A/\mathfrak{a}^n \times A/\mathfrak{b}^n \to A/\mathfrak{a}^n$ and $A/\mathfrak{a}^n \times A/\mathfrak{b}^n \to A/\mathfrak{b}^n$. These are clearly compatible with the transition homomorphisms, so they induce the stated maps on inverse limits. Conversely, if $(x_n) \in \varprojlim A/\mathfrak{a}^n$ and $(y_n) \in \varprojlim A/\mathfrak{b}^n$, (x_n, y_n) is a compatible sequence in $\prod_{n=1}^{\infty}(A/\mathfrak{a}^n \times A/\mathfrak{b}^n)$, giving the inverse map. Hence,

$$\varprojlim(A/\mathfrak{a}^n \times A/\mathfrak{b}^n) \xrightarrow{\sim} \varprojlim A/\mathfrak{a}^n \times \varprojlim A/\mathfrak{b}^n$$

so that $A^{\mathfrak{a}} \times A^{\mathfrak{b}} \cong \varprojlim A/\mathfrak{a}^n \times A/\mathfrak{b}^n \cong A^{\mathfrak{ab}}$. Since $A^{\mathfrak{a}} \times A^{\mathfrak{b}}$ is not an integral domain, $A^{\mathfrak{ab}}$ is not an integral domain, thus proving the counterexample.

For instance, $\mathbb{Z}$ is a Noetherian domain with coprime ideals (2) and (3), so with respect to the (6)-adic topology, we have $\hat{\mathbb{Z}} \cong \mathbb{Z}_2 \times \mathbb{Z}_3$, which is not an integral domain, as $(1,0) \cdot (0,1) = 0$. □

10.5 Exercise

Proof. For a Noetherian ring A, an ideal I, and an A-module M, let M^I denote the I-adic completion. Suppose that M is finitely generated and $\mathfrak{a}$, $\mathfrak{b}$ are two ideals of A. Since M is finitely generated and A is Noetherian, M is a Noetherian A-module (Proposition 6.5). Thus,

$$0 \to \mathfrak{b}^m M \to M \to M/\mathfrak{b}^m M \to 0$$

is a short exact sequence of finitely generated A-modules. Then, by Proposition 10.12, the sequence of $\mathfrak{a}$-adic completions is also exact:

$$0 \to (\mathfrak{b}^m M)^{\mathfrak{a}} \to M^{\mathfrak{a}} \to (M/\mathfrak{b}^m M)^{\mathfrak{a}} \to 0.$$

By Proposition 10.13, if N is any finitely generated A-module, then since A is Noetherian, $N^{\mathfrak{a}} = A^{\mathfrak{a}} \otimes_A N$. Thus, the map $(\mathfrak{b}^m M)^{\mathfrak{a}} \to M^{\mathfrak{a}}$ factors as

$$(\mathfrak{b}^m M)^{\mathfrak{a}} \xrightarrow{\sim} A^{\mathfrak{a}} \otimes_A (\mathfrak{b}^m M) \to A^{\mathfrak{a}} \otimes_A M \xrightarrow{\sim} M^{\mathfrak{a}}$$

and is given by $(b_m) \otimes x \mapsto (b_m) \otimes (x) \mapsto (b_m x)$. Therefore, the image of the map is $\mathfrak{b}^m M^{\mathfrak{a}}$. Then, applying this to our exact sequence gives $(M/\mathfrak{b}^m M)^{\mathfrak{a}} \cong M^{\mathfrak{a}}/\mathfrak{b}^m M^{\mathfrak{a}}$.

Therefore, we have

$$(M^{\mathfrak{a}})^{\mathfrak{b}} = \varprojlim_m (M^{\mathfrak{a}}/\mathfrak{b}^m M^{\mathfrak{a}}) \cong \varprojlim_m (M/\mathfrak{b}^m M)^{\mathfrak{a}} \cong \varprojlim_m \varprojlim_n (M/\mathfrak{b}^m M)/\left(\frac{\mathfrak{a}^n M + \mathfrak{b}^m M}{\mathfrak{b}^m M}\right)$$

since $\mathfrak{a}^n(M/\mathfrak{b}^m M) \cong (\mathfrak{a}^n M + \mathfrak{b}^m M)/\mathfrak{b}^m M$ (Proposition 2.1) is the image of $\mathfrak{a}^n$ under the map $M \to M/\mathfrak{b}^m M$. By Proposition 2.1,

$$(M/\mathfrak{b}^m M)/\left(\frac{\mathfrak{a}^n M + \mathfrak{b}^m M}{\mathfrak{b}^m M}\right) \cong M/(\mathfrak{a}^n M + \mathfrak{b}^m M),$$

which we will denote by $M_{n,m}$.

Now, note that an element $\varprojlim_n M_{n,m}$ gives coherent sequences of the form $x_{n+1,m} \mapsto x_{n,m}$. Then, $\varprojlim_m \varprojlim_n M_{n,m}$ gives coherent sequences (x_m), where $(x_{n,m+1})_n = x_{m+1} \mapsto x_m = (x_{n,m})_n$ yields compatible, coherent maps $x_{n,m+1} \mapsto x_{n,m}$ for all n. Then, if $n \le n'$ and $m \le m'$, we have a commutative diagram

$$\begin{array}{ccc} M_{n',m'} & \longrightarrow & M_{n,m'} \\ \downarrow & & \downarrow \\ M_{n',m} & \longrightarrow & M_{n,m}. \end{array}$$

In particular, an element of the double inverse limit of $M_{n,m}$ is a doubly coherent sequence (with commutativity) $(x_{n,m})_{n,m}$. Then, given such a double inverse limit, we can look at the diagonal $(x_{r,r})_r$, which is itself a coherent sequence. This defines a map $\varprojlim_m \varprojlim_n M_{n,m} \to \varprojlim_r M_{r,r}$. Suppose $(x_{r,r})_r = 0$ so that $x_{r,r} = 0$ for all r. For all fixed (n, m), there is some r such that $n, m \le r$, so we have $x_{r,r} \mapsto x_{r,m} \mapsto x_{n,m}$ and thus $x_{n,m} = 0$. Hence, the map is injective. For surjectivity, suppose we have a coherent sequence $(y_r)_r \in \varprojlim_r M_{r,r}$. Define $x_{n,m}$ by choosing some $r \ge n, m$; then, $x_{n,m}$ is the image of y_r under $M_{r,r} \to M_{r,m} \to M_{n,m}$. By commutativity, this does not depend on r, so this defines a coherent sequence in $\varprojlim_m \varprojlim_n M_{n,m}$, so the map is surjective and hence bijective. The map clearly preserves the A-module structure, so it is an isomorphism.

Therefore, we have

$$(M^{\mathfrak{a}})^{\mathfrak{b}} \cong \varprojlim_m \varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^m M) \cong \varprojlim_n M/(\mathfrak{a}^n M + \mathfrak{b}^n M) = \varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^n)M.$$

Now, note that $(\mathfrak{a} + \mathfrak{b})^{2n} \subset \mathfrak{a}^n + \mathfrak{b}^n \subset (\mathfrak{a} + \mathfrak{b})^n$. Thus, the filtration $(\mathfrak{a}^n + \mathfrak{b}^n)M$ defines the same topology as $(\mathfrak{a} + \mathfrak{b})^n M$, so the completions are isomorphic. Hence,

$$(M^{\mathfrak{a}})^{\mathfrak{b}} \cong \varprojlim_n M/(\mathfrak{a}^n + \mathfrak{b}^n)M \cong \varprojlim_n M/(\mathfrak{a} + \mathfrak{b})^n M \cong M^{\mathfrak{a}+\mathfrak{b}},$$

as desired. □

10.6 Exercise

Proof. Note that $\mathfrak{a}$ lies in the Jacobson radical if and only if $\mathfrak{m} \supset \mathfrak{a}$ for all maximal ideals $\mathfrak{m}$ of A. Thus, it is sufficient to show that $\mathfrak{m}$ is closed in the $\mathfrak{a}$-topology if and only if $\mathfrak{m} \supset \mathfrak{a}$.

If $\mathfrak{m} \supset \mathfrak{a}$, then for every $x \notin \mathfrak{m}$, $x + \mathfrak{a} \subset x + \mathfrak{m}$ and $(x + \mathfrak{m}) \cap \mathfrak{m} = \emptyset$. Since $x + \mathfrak{a}$ is open, their union over $x \notin \mathfrak{m}$ is open. Note that

$$A - \mathfrak{m} \subset \bigcup_{x \notin \mathfrak{m}} (x + \mathfrak{a}) \subset \bigcup_{x \notin \mathfrak{m}} (x + \mathfrak{m}) \subset A - \mathfrak{m},$$

so $A - \mathfrak{m} = \bigcup_{x \notin \mathfrak{m}} (x + \mathfrak{m})$ is open, and thus, $\mathfrak{m}$ is closed.

Suppose that $\mathfrak{m} \not\supset \mathfrak{a}$, so there is some $x \in \mathfrak{a} - \mathfrak{m}$. Since $x \notin \mathfrak{m}$, x descends to a unit in $A/\mathfrak{m}$, so there is some $y \in A$ such that $\overline{xy} = 1$. Since $x \in \mathfrak{a}$, $xy \in \mathfrak{a}$ and $xy \in 1 + \mathfrak{m}$. Thus, replace x with xy, so we have $x \in \mathfrak{a} - \mathfrak{m}$ such that $x \in 1 + \mathfrak{m}$. Then, $x^n \in \mathfrak{a}^n$ and $x^n \in 1 + \mathfrak{m}$. Thus, $1 - x^n \in (1 + \mathfrak{a}^n) \cap \mathfrak{m}$ for all $n > 0$. In the $\mathfrak{a}$-topology, $(1 - x^n)_{n>0}$ converges to 1. Since $1 \notin \mathfrak{m}$, even though $(1 - x^n)_{n>0} \subset \mathfrak{m}$, $\mathfrak{m}$ is not closed. □

Remark. *A Noetherian ring whose topology is defined by an ideal contained in the Jacobson radical is called a "Zariski ring."*

10.7 Exercise

Proof. Since A is a Noetherian ring, $\hat{A}$ is a flat A-algebra by Proposition 10.14. By Exercise 3.16, $A \to \hat{A}$ is faithfully flat if and only if for every A-module M, $M \to \hat{A} \otimes_A M$ is injective. If $M \to \hat{A} \otimes_A M$ is not injective for some A-module M, then there is some nonzero $m \in M$ such that $m \mapsto 0$. Hence, $Am \to \hat{A} \otimes_A Am \to \hat{A} \otimes_A M$ is not injective. Since $\hat{A}$ is flat, and $Am \hookrightarrow M$ is injective, $\hat{A} \otimes_A Am \to \hat{A} \otimes_A M$ is injective, so $Am \to \hat{A} \otimes_A Am$ is not injective. Thus, if there is an A-module M such that $M \to \hat{A} \otimes_A M$ is not injective, there is a finitely generated module N such that $N \to \hat{A} \otimes_A N$ is not injective.

Therefore, to show that $M \to \hat{A} \otimes_A M$ is injective for every A-module M, it is sufficient to prove that $M \to \hat{A} \otimes_A M$ is injective for every finitely generated A-module M. In the case of finitely generated A-modules M, we have $\hat{A} \otimes_A M \cong \hat{M}$ since A is Noetherian (Proposition 10.13). Therefore, we've reduced the proof to showing that A is a Zariski ring if and only if $M \to \hat{M}$ is injective for all finitely generated A-modules M.

Suppose A is a Zariski ring, i.e. the ideal $\mathfrak{a}$ defining its topology is contained in the Jacobson radical. Then, for any finitely generated A-module M, we have by Corollary 10.19 that $\bigcap \mathfrak{a}^n M = 0$. This is the kernel of $M \to \hat{M}$ (Theorem 10.17), so $M \to \hat{M}$ is injective.

Conversely, suppose for every finitely generated A-module M, $M \to \hat{M}$ is injective. For each maximal ideal $\mathfrak{m}$ of A, $A/\mathfrak{m}$ is finitely generated, so $A/\mathfrak{m} \to \widehat{A/\mathfrak{m}}$ is injective. By Theorem 10.17, $\bigcap_n \mathfrak{a}^n(A/\mathfrak{m}) = 0$. Suppose to the contrary that $\mathfrak{m} \not\supset \mathfrak{a}$. Then, there is some $x \in \mathfrak{a} - \mathfrak{m}$. Then, since $\mathfrak{m}$ is maximal, $A/\mathfrak{m}$ is a field. Since $\overline{x} \neq 0$, there is some $y \in A$ such that $\overline{xy} = 1$. Since $x \in \mathfrak{a}$, $z = xy \in \mathfrak{a}$ and $z \in 1 + \mathfrak{m}$. Thus, $z^n \in \mathfrak{a}$ and $z^n \equiv 1^n \equiv 1 \bmod \mathfrak{m}$. In particular, for every $n > 0$, $1 \in \mathfrak{a}^n(A/\mathfrak{m})$ since $z^n \cdot 1 = 1$, so $1 \in \bigcap_n \mathfrak{a}^n(A/\mathfrak{m}) = 0$, a contradiction. Thus, for all maximal ideals $\mathfrak{m}$ of A, $\mathfrak{a} \subset \mathfrak{m}$, so $\mathfrak{a}$ is contained in the Jacobson radical. Hence, A is a Zariski ring. □

10.8 Exercise

Proof. Let A be the local ring of the origin in $\mathbb{C}^n$, i.e. the ring of all rational functions $f/g \in \mathbb{C}(z_1, \ldots, z_n)$ with $g(0) \neq 0$. Let B be the ring of power series in $z_1, \ldots, z_n$ which converge in some neighborhood of the origin, and let C be the ring of formal power series in $z_1, \ldots, z_n$. Then clearly, $A \subset B \subset C$.

First, we show that B is a local ring. Let $\mathfrak{m}$ be the ideal of B defined by $f \in B$ such that $f(0) = 0$. If $g \notin \mathfrak{m}$, then $g(0) \neq 0$. Then, there is some neighborhood of the origin U such that g is nonzero and converges, so $1/g$ is a power series in $z_1, \ldots, z_n$ that converges in some neighborhood, so $1/g \in B$. Thus, g is a unit in B. By Proposition 1.6, B is a local ring and $\mathfrak{m}$ is its maximal ideal.

Now, we show that $\hat{A} \cong \hat{B} \cong C$, where completion is taken in the $\mathfrak{m}$-topology. First, note that $\mathfrak{m} = (z_1, \ldots, z_n)$ is the maximal ideal for both A and B, so $\hat{B} \cong \varprojlim_k B/(z_1, \ldots, z_n)^k$ and

$\hat{A} \cong \varprojlim_k A/(z_1,\dots,z_n)^k$. Note that for all k, we have inclusions

$$A/(z_1,\dots,z_n)^k \hookrightarrow B/(z_1,\dots,z_n)^k \hookrightarrow C/(z_1,\dots,z_n)^k.$$

To show that we have isomorphisms throughout, it is sufficient to show that $A/(z_1,\dots,z_n)^k \to C/(z_1,\dots,z_n)^k$ is surjective for all k. Any element in $C/(z_1,\dots,z_n)^k$ has a preimage in C, so let $f \in C$. We can decompose f into its homogeneous parts: $f = f_0 + f_1 + \cdots$, where f_i is the homogeneous part of degree i. Let $F = f_0 + \cdots + f_{k-1}$ so that $f = F + g$, where $g \in (z_1,\dots,z_n)^k$. Then, F and f have the same image under $C \to C/(z_1,\dots,z_n)^k$. Further, $F \in \mathbb{C}[z_1,\dots,z_n] \subset A$, so $\overline{F} \mapsto \overline{f}$ in $A/(z_1,\dots,z_n)^k \to C/(z_1,\dots,z_n)^k$, so we have surjectivity. Hence, $A/(z_1,\dots,z_n)^k \cong B/(z_1,\dots,z_n)^k \cong C/(z_1,\dots,z_n)^k$ for all k, so $\hat{A} \cong \hat{B} \cong \varprojlim_k C/(z_1,\dots,z_n)^k$. Note that $C = \mathbb{C}[[z_1,\dots,z_n]] = \varprojlim_k \mathbb{C}[z_1,\dots,z_n]/(z_1,\dots,z_n)^k$, so C is complete with respect to the ideal $(z_1,\dots,z_n)$. Thus, by Proposition 10.5, $\varprojlim_k C/(z_1,\dots,z_n)^k \cong C$ so that $\hat{A} \cong \hat{B} \cong C$.

By Exercise 7.4, B is Noetherian. Since B is a local ring, its maximal ideal $\mathfrak{m}$ is the Jacobson radical, so B is a Zariski ring. Thus, by Exercise 10.7, $\hat{B} \cong C$ is faithfully flat over B. Further, note that A is the localization of $\mathbb{C}[z_1,\dots,z_n]$ at $(z_1,\dots,z_n)$. Since $\mathbb{C}[z_1,\dots,z_n]$ is Noetherian by Hilbert's basis theorem (Theorem 7.5), A is Noetherian by Proposition 7.3. Thus, by Proposition 10.14, $\hat{A} \cong C$ is flat over A. Hence, we have the composition $A \overset{f}{\hookrightarrow} B \overset{g}{\hookrightarrow} C$, where $g \circ f$ is flat and g is faithfully flat. By Exercise 3.17, f is flat, so B is a flat A-algebra. □

10.9 Exercise

Proof. Let A be a local ring, $\mathfrak{m}$ its maximal ideal, and suppose A is $\mathfrak{m}$-adically complete. For any $f \in A[x]$, let $\overline{f}(x) \in (A/\mathfrak{m})[x]$ be the reduction of f modulo $\mathfrak{m}$. We aim to show: if $f(x)$ is monic of degree n and if there exist coprime monic polynomials $\overline{g}(x)$, $\overline{h}(x) \in (A/\mathfrak{m})[x]$ of degrees r, $n-r$ with $\overline{f}(x) = \overline{g}(x)\overline{h}(x)$, then there are lifts to monic polynomials $g(x)$, $h(x) \in A[x]$ such that $f(x) = g(x)h(x)$.

Assume $f(x) \in A[x]$ is monic of degree n and there are coprime monic polynomials $\overline{g}(x)$, $\overline{h}(x) \in (A/\mathfrak{m})[x]$ of degrees r, $n-r$ with $\overline{f}(x) = \overline{g}(x)\overline{h}(x)$. Let $g_1(x)$, $h_1(x) \in A[x]$ be coefficient-wise lifts of $\overline{g}(x)$ and $\overline{h}(x)$; clearly, we can choose the lifts to be monic. Then, $\overline{g_1(x)h_1(x) - f(x)} = \overline{g}(x)\overline{h}(x) - \overline{f}(x) = 0$, so $g_1(x)h_1(x) - f(x) \in \mathfrak{m}^1 A[x]$, and $g_1(x)$ and $h_1(x)$ are monic of degrees r and $n-r$, respectively. We aim to inductively construct coherent sequences $(g_k)_k$ and $(h_k)_k$ such that $f - g_k h_k \in \mathfrak{m}^k A[x]$ for all $k > 0$. We've established the $k = 1$ base case.

Now, suppose we've constructed the coherent sequences up to j, so we have $g_j(x), h_j(x) \in A[x]$ such that $g_j(x)h_j(x) - f(x) \in \mathfrak{m}^j A[x]$, where $g_j(x)$ and $h_j(x)$ are monic of degrees r and $n-r$ respectively, and $\overline{g_j}(x) = \overline{g}(x)$ and $\overline{h_j}(x) = \overline{h}(x)$. Since $\overline{g_j}(x) = \overline{g}(x)$ and $\overline{h_j}(x) = \overline{h}(x)$ are coprime and monic, there are polynomials $\overline{a_p}(x), \overline{b_p}(x) \in (A/\mathfrak{m})[x]$ such that $\overline{a_p}(x)\overline{g}(x) + \overline{b_p}(x)\overline{h}(x) = x^p$ for all $0 \le p < n$. Suppose $\deg \overline{a_p}(x) \ge n-r$, and let ax^m be its leading term. Then, if $k = m-(n-r)$, $ax^k\overline{h}(x)$ has leading term ax^m since $\overline{h_j}(x)$ is monic. Then, we have

$$\begin{aligned} x^p &= \overline{a_p}(x)\overline{g}(x) + \overline{b_p}(x)\overline{h}(x) \\ &= \overline{a_p}(x)\overline{g}(x) + \overline{b_p}(x)\overline{h}(x) + (ax^k\overline{g}(x)\overline{h}(x) - ax^k\overline{g}(x)\overline{h}(x)) \\ &= (\overline{a_p}(x) - ax^k\overline{h}(x))\overline{g}(x) + (\overline{b_p}(x) + ax^k\overline{g}(x))\overline{h}(x), \end{aligned}$$

so making the replacements $\overline{a_p}(x) \mapsto \overline{a_p}(x) - ax^k\overline{h}(x)$ and $\overline{b_p}(x) \mapsto \overline{b_p}(x) + ax^k\overline{g}(x)$, $\deg \overline{a_p}(x)$ is strictly lessened. Continuing, we can ensure $\deg \overline{a_p}(x) < n-r$. So, choose $\overline{a_p}(x)$ and $\overline{b_p}(x)$ so

that $\deg \overline{a_p}(x) < n - r$. Now, suppose to the contrary that $\deg \overline{b}_p(x) \geq r$. Then, since $\overline{g}(x)$ and $\overline{h}(x)$ are monic, $\deg(\overline{b_p}(x)\overline{h}(x)) = \deg(\overline{b_p}(x)) + n - r \geq r + (n - r) = n$ and $\deg(\overline{a_p}(x)\overline{g}(x)) = \deg(\overline{a_p}(x)) + r < n - r + r = n$. Thus, the degree of $\overline{b_p}(x)\overline{h}(x)$ is strictly greater than the degree of $\overline{a_p}(x)\overline{g}(x)$ so that

$$\begin{aligned}\deg(x^p) &= \deg(\overline{a_p}(x)\overline{g}(x) + \overline{b_p}(x)\overline{h}(x)) \\ p &= \deg(\overline{b_p}(x)\overline{h}(x)) \\ p &\geq n,\end{aligned}$$

violating the fact that $n > p$. Therefore, $\overline{b_p}(x)$ has degree $< r$ so that $\deg \overline{a_p}(x) < n - r$ and $\deg \overline{b_p}(x) < r$ simultaneously.

By monicity, $\deg(g_j(x)h_j(x)) = \deg(g_j(x)) + \deg(h_j(x)) = r + (n - r) = n$, so the leading term of $f(x)$ and $g_j(x)h_j(x)$ is x^n. Thus, $f(x) - g_j(x)h_j(x)$ has degree less than n, so we can write $f(x) - g_j(x)h_j(x) = \sum_{p=0}^{n-1} m_p x^p$. Since $f(x) - g_j(x)h_j(x) \in \mathfrak{m}^j[x]$, $m_p \in \mathfrak{m}^j$ for all p. Since the degree of each $\overline{a_p}(x)$ is $< n - r$ and the degree of each $\overline{b_p}(x)$ is $< r$, we can choose lifts $a_p(x), b_p(x) \in A[x]$ such that $\deg a_p(x) < n - r$ and $\deg b_p(x) < r$ for all p. Then, $a(x) := \sum_{p=0}^{n-1} m_p a_p(x)$ has degree $< n - r$ and $b(x) := \sum_{p=0}^{n-1} m_p b_p(x)$ has degree $< r$. Since $m_p \in \mathfrak{m}^j$ for all p, $a(x), b(x) \in \mathfrak{m}^j[x]$.

Now, let $g_{j+1}(x) = g_j(x) + b(x)$ and $h_{j+1}(x) = h_j(x) + a(x)$. Since g_j is monic of degree r and b has degree $< r$, g_{j+1} is monic of degree r. By the same argument, h_{j+1} is monic of degree $n - r$. Then,

$$\begin{aligned}f(x) - g_{j+1}(x)h_{j+1}(x) &= f(x) - g_j(x)h_j(x) - a(x)g_j(x) - b(x)h_j(x) - a(x)b(x) \\ &= \sum_{p=0}^{n-1} m_p x^p - \sum_{p=0}^{n-1} m_p a_p(x)g_j(x) - \sum_{p=0}^{n-1} m_p b_p(x)h_j(x) - a(x)b(x) \\ &= \sum_{p=0}^{n-1} m_p[x^p - a_p(x)g_j(x) - b_p(x)h_j(x)] - a(x)b(x).\end{aligned}$$

By construction, $x^p \equiv a_p(x)g_j(x) + b_p(x)h_j(x) \bmod \mathfrak{m}[x]$, so since $m_p \in \mathfrak{m}^j$ for all p, $\sum_{p=0}^{n-1} m_p[x^p - a_p(x)g_j(x) - b_p(x)h_j(x)] \in \mathfrak{m}^{j+1}[x]$. Since $a(x), b(x) \in \mathfrak{m}^j[x]$, $-a(x)b(x) \in \mathfrak{m}^{2j}[x] \subset \mathfrak{m}^{j+1}[x]$ ($2j \geq j + 1$ since $j \geq 1$). Finally, $f(x) - g_{j+1}(x)h_{j+1}(x) \in \mathfrak{m}^{j+1}[x]$, where g_{j+1} is monic of degree r and h_{j+1} is monic of degree $n - r$. This continues the coherent sequence since $g_{j+1} = g_j + b \in g_j + \mathfrak{m}^j[x]$, so $g_{j+1} \equiv g_j \bmod \mathfrak{m}^j[x]$. By the same argument, $h_{j+1} \equiv h_j \bmod \mathfrak{m}^j[x]$.

By induction, we have coherent sequences $(g_k)_k$ and $(h_k)_k$ such that $f - g_k h_k \in \mathfrak{m}^k[x]$ for all k. Note that the sequence $(g_k)_k$ is a sequence of monic polynomials of degree r such that $g_{k+1} \equiv g_k \bmod \mathfrak{m}^k[x]$ for all $k > 0$. Since A is complete in the $\mathfrak{m}$-topology, $(g_k)_k$ converges coefficient-wise to a degree r monic polynomial $g(x) \in A[x]$ such that $g \equiv g_k \bmod \mathfrak{m}^k[x]$ for all $k > 0$. We define h similiarly to be a degree $n - r$ monic polynomial. Thus, for all $k > 0$, $f - gh \equiv f - g_k h_k \equiv 0 \bmod \mathfrak{m}^k[x]$. Thus, every coefficient of $f - gh$ lies in $\mathfrak{m}^n$ for all $n > 0$. By Corollary 10.19, $\bigcap_n \mathfrak{m}^n = 0$, so $f = gh$, as desired. □

Remark. *This is a general form of "Hensel's lemma."*

10.10 Exercise

(i) Proof. Suppose $\overline{f}(x) \in (A/\mathfrak{m})[x]$ has a simple root $\alpha \in A/\mathfrak{m}$. Then, we can write $\overline{f}(x) = (x-\alpha)\overline{g}(x)$ for some $\overline{g}(x) \in (A/\mathfrak{m})[x]$. Since α is a simple root, $(x-\alpha)$ does not divide $\overline{g}(x)$, so $(x-\alpha)$ and $\overline{g}(x)$ are coprime. By Exercise 10.9, there are polynomials $x-a \in A[x]$ and $g(x) \in A[x]$ such that $f(x) = (x-a)g(x)$; we used the fact that the lifts are monic and preserve degree. Further, $\overline{x-a} = x-\alpha$ and $\overline{g(x)} = \overline{g}(x)$. Therefore, a is a root of f and $a \equiv \alpha \bmod \mathfrak{m}$. Finally, suppose to the contrary that a is not a simple root of f so that $g(x) = (x-a)h(x)$ for some $h(x) \in A[x]$. Then, $\overline{g}(x) = (x-\alpha)\overline{h}(x)$, a clear contradiction as $(x-\alpha)$ and $\overline{g}(x)$ are coprime. Thus, a is a simple root. □

(ii) Proof. Let $\mathbb{Z}_{(7)}$ be the localization of $\mathbb{Z}$ at the prime ideal (7). Since $\mathbb{Z}$ is Noetherian and (7) is a prime ideal of $\mathbb{Z}$, $\mathbb{Z}_{(7)}$ is a Noetherian local ring with maximal ideal (7). The (7)-adic completion of $\mathbb{Z}_{(7)}$ is $\mathbb{Z}_7$, so by Proposition 10.16, $\mathbb{Z}_7$ is a local ring with maximal ideal (7) that is complete (7)-adically. Thus, we can apply (i) to the ring $\mathbb{Z}_7$.

Let $f(x) = x^2 - 2 \in \mathbb{Z}_7[x]$, and consider the reduction $x^2 - 2 \in (\mathbb{Z}_7/(7))[x]$. Note that $-2 \equiv -9 \bmod (7)$, so we have $x^2 - 2 \equiv x^2 - 9 \equiv (x-3)(x+3) \bmod (7)$. Thus, $\overline{f}(x)$ has simple roots ± 3 since $3 \not\equiv -3 \bmod (7)$. By (i), these simple roots lift to simple roots of $f(x)$. But a root r of $f(x)$ gives $r^2 - 2 = 0$, so 2 is a square in $\mathbb{Z}_7$. □

(iii) Proof. Let $f(x,y) \in k[x,y]$, where k is a field, and assume that $f(0,y)$ has $y = a_0$ as a simple root. Note that $A = k[[x]]$ is a complete, local ring with maximal ideal (x). Further, $A/(x) \cong k$. Then, we can consider $f(x,y)$ to be an element of $A[y]$. Reducing to the residue field, we have $\overline{f(x,y)} = f(0,y) \in (A/(x))[y] \cong k[y]$. By assumption, $f(0,y)$ has a simple root a_0. By (i), we have some $y(x) \in k[[x]]$ such that $f(x, y(x)) = 0$. Further, $\overline{y(x)} = a_0 \in A/(x) \cong k$, so we can write $y(x) = \sum_{n=0}^{\infty} a_n x^n$ with constant term a_0. □

Remark. *This gives the "analytic branch" of the curve $f = 0$ through the point $(0, a_0)$.*

10.11 Exercise

We show that even if A is local and $\hat{A}$ is a finitely-generated A-module, $\hat{A}$ may be Noetherian while A is not Noetherian. This shows that the converse of Theorem 10.26 is false.

Proof. Let $A = C_0^\infty(\mathbb{R})$ be the germs $[f]$ at $0 \in \mathbb{R}$ for real-valued and C^∞ functions f. To see that A is a local ring, note that $[f] \mapsto f(0)$ is surjective onto $\mathbb{R}$, so the kernel is a maximal ideal $\mathfrak{m}$. Now, suppose $[f] \notin \mathfrak{m}$. Then, $[f]$ has a representative f such that $f(x) \neq 0$ on a neighborhood of 0, so there is some $[g] \in A$ such that $[f][g] = [1]$, so $[f]$ is a unit. By Proposition 1.6, since every element of A not in $\mathfrak{m}$ is a unit, A is a local ring with maximal ideal $\mathfrak{m}$.

Let $\mathfrak{a}_n$ be the ideal of $[f] \in A$ such that $f^{(j)}(0) = 0$ for $0 \leq n < j$. Thus, $\mathfrak{a}_1 = \mathfrak{m}$. We wish to show that $\mathfrak{a}_n = \mathfrak{m}^n$ for all $n > 0$. Suppose we have $\mathfrak{a}_k = \mathfrak{m}^k$ for all $k < n$. If $[f] \in \mathfrak{m}^n$, then f is the sum of elements of the form gh, where $[g] \in \mathfrak{m}$ and $[h] \in \mathfrak{m}^{n-1}$. By the product rule, $(gh)^{(n)} = \sum_{j=0}^{n} \binom{n}{j} g^{(j)} h^{(n-j)} = g^{(0)} h^{(n)}$ since $h \in \mathfrak{m}^{n-1} = \mathfrak{a}_{n-1}$, so $h^{(i)} = 0$ for all $i < n$. However, $g \in \mathfrak{m}$, so $g^{(0)} = 0$ and thus $(gh)^{(n)} = 0$. Hence, $f^{(n)} = 0$ since it is the sum of elements with zero nth derivative, so $[f] \in \mathfrak{a}_n$. Conversely, if $[f] \in \mathfrak{a}_n$, then we can use Taylor's theorem. Let f be a representative of $[f]$ and let U be an open neighborhood such that $f \in C^\infty(U)$. Then, $f(x) = \sum_{i=0}^{n-1} \frac{1}{i!} f^{(i)}(0) x^i + g_n(x) x^n$ on U, where $g_n \in C^\infty(U)$. Since $f^{(i)} = 0$ for all $i < n$, we have $f(x) = x^n g_n(x)$ so that $\mathfrak{a}_n \subset (x^n) \subset \mathfrak{m}^n$. By induction, $\mathfrak{a}_n = \mathfrak{m}^n = (x^n)$.

Again, let f be a representative of $[f] \in A$ and let U be an open neighborhood such that $f \in C^\infty(U)$. Then, by Taylor's theorem, $f = f_0 + x^n g(x)$, where $f_0 \in \mathbb{R}[x]$ and $[x^n g(x)] \in \mathfrak{m}^n$. Thus, $[f] \to A/\mathfrak{m}^n$ has the same image as $[f_0] \to A/\mathfrak{m}^n$. In particular, we have an isomorphism $\mathbb{R}[x]/(x^n) \xrightarrow{\sim} A/\mathfrak{m}^n$ that clearly commutes with the projections $\mathbb{R}[x]/(x^{n+1}) \to \mathbb{R}[x]/(x^n)$ and $A/\mathfrak{m}^{n+1} \to A/\mathfrak{m}^n$. Thus, $\hat{A} = \varprojlim A/\mathfrak{m}^n \cong \varprojlim \mathbb{R}[x]/(x^n) = \mathbb{R}[[x]]$, which is Noetherian by Theorem 7.5 since $\mathbb{R}$ is a field (and thus Noetherian). By Borel's theorem, every power series in $\mathbb{R}[[x]]$ is the Taylor series of some function in $C^\infty(U)$, where U is some neighborhood of the origin, so $A \to \hat{A}$ is surjective. In particular, $\hat{A}$ is a finitely-generated A-module.

Collecting everything, A is a local ring, $\hat{A}$ is a Noetherian ring, and $\hat{A}$ is a finitely-generated A-module. Suppose to the contrary that A is Noetherian. Then, A is a Noetherian local ring, so by Theorem 10.17, $\bigcap_n \mathfrak{m}^n = 0$. Thus, $\bigcap_n \mathfrak{a}_n = 0$. However, note that $e^{-1/x^2} \in C^\infty(\mathbb{R})$ and $(e^{-1/x^2})^{(n)} = 0$ for all $n \geq 0$. Thus, $e^{-1/x^n} \in \mathfrak{a}_n$ for all $n \geq 0$, violating $\bigcap_n \mathfrak{a}_n = 0$, so A is not a Noetherian ring. □

10.12 Exercise

Proof. By Exercise 2.8, if B is a flat A-algebra and C is a flat B-algebra, then C is a flat A-algebra. In other words, a composition of flat maps is flat. Now, by Exercise 2.5, $A[x]$ is a flat A-algebra. Thus, we have a composition of flat maps

$$A \to A[x_1] \to A[x_1][x_2] \to \cdots \to A[x_1]\cdots[x_n] \cong A[x_1, \ldots, x_n],$$

so $A[x_1, \ldots, x_n]$ is a flat A-algebra. Since A is Noetherian, $A[x_1, \ldots, x_n]$ is Noetherian by Theorem 7.5. In the $(x_1, \ldots, x_n)$-topology of $A[x_1, \ldots, x_n]$, $\hat{A} = \varprojlim_k A[x_1, \ldots, x_n]/(x_1, \ldots, x_n)^k = A[[x_1, \ldots, x_n]]$. Since $A[x_1, \ldots, x_n]$ is Noetherian, $A[[x_1, \ldots, x_n]]$ is a flat $A[x_1, \ldots, x_n]$-algebra by Proposition 10.14. Then, since $A[x_1, \ldots, x_n]$ is a flat A-algebra, $A[[x_1, \ldots, x_n]]$ is a flat A-algebra. Under the map $A \hookrightarrow A[[x_1, \ldots, x_n]]$, an ideal $\mathfrak{a}$ of A extends to $\mathfrak{a}[[x_1, \ldots, x_n]]$. Then, $\mathfrak{a}[[x_1, \ldots, x_n]]^c = A \cap \mathfrak{a}[[x_1, \ldots, x_n]] = \mathfrak{a}$, so $\mathfrak{a} = \mathfrak{a}^{ec}$ for all ideals $\mathfrak{a}$ of A. Thus, since $A[[x_1, \ldots, x_n]]$ is a flat A-algebra, we have by Exercise 3.16 that $A[[x_1, \ldots, x_n]]$ is faithfully flat over A. □

11 Dimension Theory

11.1 Exercise - Complete

Proof. Let $f \in k[x_1, \ldots, x_n]$ be an irreducible polynomial, where k is an algebraically closed field, and let $A = k[x_1, \ldots, x_n]/(f)$. Let V be the irreducible affine variety defined by $f(x) = 0$. For each point P on V, there is the associated maximal ideal $\mathfrak{m}_P = (x_1 - P_1, \ldots, x_n - P_n)$ by the Nullstellensatz. Since $f(P) = 0$ for any P in V, $(f) \subset \mathfrak{m}_P$, so $\mathfrak{m} = \mathfrak{m}_P/(f)$ is a maximal ideal of A. We aim to show that a point P in V is non-singular if and only if $A_\mathfrak{m}$ is a regular local ring.

We now show that $f \in \mathfrak{m}_P^2$ if and only if f is singular at P. If $f \in \mathfrak{m}_P^2 = (x_1 - P_1, \ldots, x_n - P_n)^2$, we can write $f = \sum_{i,j} g_{i,j}(x)(x_i - P_i)(x_j - P_j)$, where $g_{i,j} \in k[x_1, \ldots, x_n]$. Then,

$$\frac{\partial f}{\partial x_s} = \sum_{i,j} \left(\frac{\partial}{\partial x_s} g_{i,j}(x) \right) (x_i - P_i)(x_j - P_j) + \sum_{i \neq s} (g_{i,s}(x) + g_{s,i}(x))(x_i - P_i) + 2g_{s,s}(x)(x_s - P_s),$$

which vanishes for all s at P since $(x_i - P_i) = 0$ at P for all i. Thus, f is singular at P. Conversely, suppose f is singular at P. Note that $f \in (x_1 - P_1, \ldots, x_n - P_n)$, so we can write $f = \sum_i g_i(x)(x_i - P_i)$. Then, we have

$$0 = \frac{\partial f}{\partial x_j}(P) = \frac{\partial}{\partial x_j}\bigg|_P \sum_i g_i(x)(x_i - P_i) = g_j(P) + \sum_i \frac{\partial g_i}{\partial x_j}(P)(P_i - P_i) = g_j(P).$$

Thus, $g_i \in \mathfrak{m}_P$ and $(x_i - P_i) \in \mathfrak{m}_P$ for all i, so f is a sum of terms in $\mathfrak{m}_P^2$, hence $f \in \mathfrak{m}_P^2$.

Therefore, we've reduced the problem to showing that $A_\mathfrak{m}$ is a regular local ring if and only if $f \notin \mathfrak{m}_P^2$. Since $A = k[x_1, \ldots, x_n]/(f)$ and the map $k[x_1, \ldots, x_n] \to A$ takes $S_{\mathfrak{m}_P} \to S_\mathfrak{m}$ we have by Corollary 3.4 ($\mathfrak{m}_P$ is finitely generated since $k[x_1, \ldots, x_n]$ is Noetherian) and Exercise 3.4 that $k[x_1, \ldots, x_n]_{\mathfrak{m}_P}/(f)_{\mathfrak{m}_P} \cong (k[x_1, \ldots, x_n]/(f))_\mathfrak{m} = A_\mathfrak{m}$. Since $f \in \mathfrak{m}_P$ is nonzero and $k[x_1, \ldots, x_n]_{\mathfrak{m}_P}$ is an integral domain (since $k[x_1, \ldots, x_n]$ is), we have by Corollary 11.18 that $\dim A_\mathfrak{m} = \dim k[x_1, \ldots, x_n]_{\mathfrak{m}_P}/(f)_{\mathfrak{m}_P} = \dim k[x_1, \ldots, x_n]_{\mathfrak{m}_P} - 1 = n - 1$. By the third isomorphism theorem (Proposition 2.1), $k \cong k[x_1, \ldots, x_n]/\mathfrak{m}_P \cong (k[x_1, \ldots, x_n]/(f))/(\mathfrak{m}_P/(f)) = A/\mathfrak{m} \cong A_\mathfrak{m}/\mathfrak{m}A_\mathfrak{m}$. By Theorem 11.22, $A_\mathfrak{m}$ is a regular ring if and only if $\dim A_\mathfrak{m} = \dim_k \mathfrak{m}A_\mathfrak{m}/\mathfrak{m}^2A_\mathfrak{m}$. But $n - 1 = \dim A_\mathfrak{m}$ and, by Corollary 3.4, $\mathfrak{m}/\mathfrak{m}^2 \cong \mathfrak{m}A_\mathfrak{m}/\mathfrak{m}^2A_\mathfrak{m}$, so $A_\mathfrak{m}$ is a regular ring if and only if $\dim_k \mathfrak{m}/\mathfrak{m}^2 = n - 1$.

Hence, the problem has been reduced to showing that $\dim_k \mathfrak{m}/\mathfrak{m}^2 = n-1$ if and only if $f \notin \mathfrak{m}_P^2$. Finally, $\mathfrak{m} = \mathfrak{m}_P/(f)$, and $\mathfrak{m}^2$ is the image of $\mathfrak{m}_P^2$ under $k[x_1, \ldots, x_n] \to A = k[x_1, \ldots, x_n]/(f)$. The image is also $(\mathfrak{m}_P^2 + (f))/(f)$, so $\mathfrak{m}^2 = (\mathfrak{m}_P^2 + (f))/(f)$. Thus, we have

$$\mathfrak{m}/\mathfrak{m}^2 \cong (\mathfrak{m}_P/(f))/((\mathfrak{m}_P^2 + (f))/(f)) \cong \mathfrak{m}_P/(\mathfrak{m}_P^2 + (f))$$

by Proposition 2.1. Thus, we must show $n - 1 = \dim_k \mathfrak{m}_P/(\mathfrak{m}_P^2 + (f))$ if and only if $f \notin \mathfrak{m}_P^2$. If $f \in \mathfrak{m}_P^2$, $\mathfrak{m}_P/(\mathfrak{m}_P^2 + (f)) \cong \mathfrak{m}_P/\mathfrak{m}_P^2$, which has dimension $n \neq n-1$. On the other hand, if $f \notin \mathfrak{m}_P^2$, then $\mathfrak{m}_P^2 + (f)$ strictly contains $\mathfrak{m}_P^2$, so the surjective k-vector space homomorphism

$$\mathfrak{m}_P/\mathfrak{m}_P^2 \to \mathfrak{m}_P/((f) + \mathfrak{m}_P^2)$$

is not injective, implying $\dim_k \mathfrak{m}_P/((f) + \mathfrak{m}_P^2) < \dim_k \mathfrak{m}_P/\mathfrak{m}_P^2 = n$. Since $A_\mathfrak{m}$ is a Noetherian local ring, we have by Corollary 11.15 that $\dim A_\mathfrak{m} \leq \dim_k \mathfrak{m}/\mathfrak{m}^2$, so $n - 1 \leq \dim_k \mathfrak{m}_P/((f) + \mathfrak{m}_P^2) < n$, so $\dim_k \mathfrak{m}_P/((f) + \mathfrak{m}_P^2) = n - 1$, as desired. □

11.2 Exercise - Complete

Proof. Suppose A is complete in the $\mathfrak{m}$-topology, that $k \subset A$ is a field satisfying $k \cong A/\mathfrak{m}$, and $x_1, \dots, x_d$ is a system of parameters of A. Let $\mathfrak{q} = (x_1, \dots, x_d)$ be the $\mathfrak{m}$-primary ideal generated by the elements. By Corollary 11.21, $x_1, \dots, x_d$ are algebraically independent over k. Thus, if we define the homomorphism $\varphi : k[t_1, \dots, t_d] \to A$ via $t_i \mapsto x_i$, then φ is injective; $f \mapsto f(x_1, \dots, x_d)$, so if $f(x_1, \dots, x_d) = 0$, then $f = 0$ by the algebraic independence of $x_1, \dots, x_d$ over k. Now, note that $\varphi((t_1, \dots, t_d)^n) = (x_1, \dots, x_d)^n = \mathfrak{q}^n$, so by injectivity, $\varphi^{-1}(\mathfrak{q}^n) = (t_1, \dots, t_d)^n$. Thus, φ projects to injective homomorphisms $\varphi_n : k[t_1, \dots, t_d]/(t_1, \dots, t_d)^n \to A/\mathfrak{q}^n$ for all $n > 0$.

Note that φ_n is natural in the sense that we have commutative diagrams

$$\begin{array}{ccc} k[t_1, \dots, t_d]/(t_1, \dots, t_d)^{n+1} & \xrightarrow{\varphi_{n+1}} & A/\mathfrak{q}^{n+1} \\ \downarrow & & \downarrow \\ k[t_1, \dots, t_d]/(t_1, \dots, t_d)^n & \xrightarrow{\varphi_n} & A/\mathfrak{q}^n. \end{array}$$

Thus, $(\varphi_n)_n$ is compatible with the respective inverse systems. Since φ_n is injective for each n, we have an exact sequence of inverse systems

$$0 \to \{k[t_1, \dots, t_d]/(t_1, \dots, t_d)^n\}_n \xrightarrow{(\varphi_n)_n} \{A/\mathfrak{q}^n\}_n.$$

By Proposition 10.2, we have an exact sequence

$$0 \to k[[t_1, \dots, t_d]] \xrightarrow{\hat{\varphi}} \varprojlim_n A/\mathfrak{q}^n.$$

Since $\mathfrak{q}$ is $\mathfrak{m}$-primary and A is $\mathfrak{m}$-complete, $A = \varprojlim_n A/\mathfrak{q}^n$, so $\hat{\varphi} : k[[t_1, \dots, t_d]] \to A$ defined by $t_i \mapsto x_i$ is injective.

Let $B = k[[t_1, \dots, t_d]]$ with $\mathfrak{b} = (t_1, \dots, t_d)$, so B is complete in the $\mathfrak{b}$-topology. Note that since we have $\hat{\varphi} : B \to A$, A is a B-module. Note that $\mathfrak{b}(\mathfrak{q}^n) = \mathfrak{q}\mathfrak{q}^n = \mathfrak{q}^{n+1}$, so $(\mathfrak{q}^n)_n$ is a $\mathfrak{b}$-filtration of A. Further, since A is $\mathfrak{m}$-complete and $\mathfrak{q} \subset \mathfrak{m}$, $\bigcap_n \mathfrak{q}^n \subset \bigcap_n \mathfrak{m}^n = 0$, so A is Hausdorff in the $\mathfrak{b}$-filtration topology. Therefore, by Proposition 10.24, to show that A is a finitely generated B-module, it is sufficient to show that $G(A)$ is a finitely generated $G(B)$-module.

We have $G(A) = \bigoplus_{n=0}^{\infty} \mathfrak{q}^n/\mathfrak{q}^{n+1}$ and $G(B) = \bigoplus_{n=0}^{\infty} \mathfrak{b}^n/\mathfrak{b}^{n+1}$. For every $n > 0$, each class in $\mathfrak{q}^n/\mathfrak{q}^{n+1}$ is represented by some $q \in \mathfrak{q}^n = (x_1, \dots, x_d)^n = \hat{\varphi}(\mathfrak{b}^n)$. So, choose $f \in \mathfrak{b}^n$ such that $\hat{\varphi}(f) = q$. Then, $[f] \in \mathfrak{b}^n/\mathfrak{b}^{n+1}$ gives $[q] = [f] \cdot [1]$. Thus, $G(A)$ is generated as a $G(B)$-module by its degree-zero part, $A/\mathfrak{q}$. Thus, if we can show that $A/\mathfrak{q}$ is finitely generated as a k-module, then $G(A)$ will be finitely generated as a $G(B)$-module since $k \cong k[[t_1, \dots, t_d]]/(t_1, \dots, t_d) = G(B)_0$. Note that $k \cong A/\mathfrak{m}$, and since A is Noetherian, we have by Proposition 7.14 that there is some $j > 0$ such that $\mathfrak{m}^j \subset \mathfrak{q}$. In particular, we have a surjective A-module homomorphism $A/\mathfrak{m}^j \to A/\mathfrak{q}$. Because k is a subring of A, $A/\mathfrak{m}^j \to A/\mathfrak{q}$ is also a surjective k-module homomorphism.

Note that we have a finite $\mathfrak{m}$-filtration $A/\mathfrak{m}^j \supset \mathfrak{m}/\mathfrak{m}^j \supset \cdots \supset \mathfrak{m}^{j-1}/\mathfrak{m}^j \supset 0$, where the quotients are the A-modules $\mathfrak{m}^i/\mathfrak{m}^{i+1}$. Since A is Noetherian, these quotients are finitely generated. Further, since $\mathfrak{m}$ annihilates each $\mathfrak{m}^i/\mathfrak{m}^{i+1}$, $\mathfrak{m}^i/\mathfrak{m}^{i+1}$ is a finitely generated k-vector space since $k \cong A/\mathfrak{m}$. Thus, $A/\mathfrak{m}^j$ is a finite-dimensional k-vector space, so since the k-module homomorphism $A/\mathfrak{m}^j \to A/\mathfrak{q}$ is surjective, $A/\mathfrak{q}$ is also a finite-dimensional k-vector space, as desired. $\square$

11.3 Exercise

11.4 Exercise

11.5 Exercise

11.6 Exercise

11.7 Exercise